Phase 3 Algebraic Structure: Obstruction, Null, Alignment

Contents

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This section reports a measurement programme carried out entirely inside phase 3 (steps 120–200 of a 200-step run on the standard corpus, P = 4,330,240 parameters, VOCAB = 1017, nnz = 1347). Its purpose is to determine which algebraic structure, if any, the seven-node decomposition

$$\mathcal{N} = \{\text{EMB, LN, FF}, W_Q, W_K, W_V, W_O\}$$

carries. The programme is organised so that every positive claim is stated against an explicit null, since the central methodological finding is that several earlier claims in this document were reported without one.

1.1 Trajectory geometry of the phase

Phase 3 begins where the block gradient-profile drift settles (0.130 at steps 21–40, falling to 0.038 by step 100 and flat thereafter). Within the phase the step size is constant at $\|d_t\| \approx 0.60$ and the discrete curvature is constant at $\kappa \approx 0.70$–0.74, giving radius of curvature $R \approx 1.35$–1.46 against $\|\theta\| \approx 93$, i.e.

$$R/\|\theta\| \;=\; 0.015. \tag{1}$$

The tangent autocorrelation decays with no negative lobe at any lag ($k = 1$: 0.863, $k = 4$: 0.513, $k = 8$: 0.271, $k = 16$: 0.058, $k = 32$: 0.002, $k = 64$: 0.002), so the turning is diffusive, not rotational. There is no coherent frame precession and no 32-step period.

Observation 1.1 (Chord–path dichotomy). *The useful chord over the phase is $\|\Delta\| = 19.6 = 0.22\|\theta\|$, which is $\sim 14R$. The trajectory is therefore not a stiff arc, yet the straight chord from θ_{120} to θ_{200} is a monotone descent path (recovery* 40%, 73%, 93%, 100% *at chord fractions* 0.25, 0.5, 0.75, 1.0*). The basin is chord-convex while the path is diffusive. Consequently the base coordinate cannot be arc length; it must be chord fraction toward the attractor.*

1.2 The random-walk law and the persistence order parameter

Let S_n denote the parameter displacement accumulated over n steps. For a driftless random walk $\cos(S_n, S_N) = \sqrt{n/N}$ identically. Measured on $120 \to 200$:

n	$\cos(S_n, S_{80})$	$\sqrt{n/80}$
5	0.2968	0.250
10	0.3807	0.354
20	0.4919	0.500
30	0.6040	0.612
40	0.7042	0.707
60	0.8829	0.866

Agreement is within 0.02 at every n. Phase 3 is statistically indistinguishable from a driftless walk at the 80-step scale. This retrospectively explains every failed extrapolation experiment in this document: there is no persistent direction to estimate, and the ceiling on any local probe is set by the walk rather than by the estimator.

Define the *persistence excess*

$$E(W) \;=\; \cos\big(S_{W/2},\, S_W\big) \;-\; \tfrac{1}{\sqrt{2}}, \tag{2}$$

which vanishes identically for a driftless walk, giving a parameter-free null.

W	E on $40 \to 120$	E on $120 \to 200$	s.e.m.
4	+0.2523	+0.2491	0.002
8	+0.2103	+0.2098	0.002
16	+0.1510	+0.1448	0.003
32	+0.1325	+0.0739	0.003
64	+0.1621	+0.0117	0.003

The phases are indistinguishable at short windows (that part is optimizer smoothing) and separate cleanly at $W = 64$: +0.162 against +0.012, a $\sim 50\sigma$ gap. $E(64)$ is therefore an order parameter for the drift/walk transition, requiring three parameter snapshots and no gradient evaluations. It replaces the hand-calibrated `drift_thr` of the compiler with a quantity whose null is analytic.

Observation 1.2 (Gated extrapolation). *Applying $\theta \leftarrow \theta_t + \beta(\theta_t - \theta_{t-40})$ followed by k corrector steps: at $t = 80$ ($E(64) = +0.16$), $\beta = 0.5$ with $k = 10$ reaches a value requiring 20 ordinary steps, a net saving of 10; at $t = 120$ the saving is 0; at $t = 160$ ($E(64) = +0.01$) every configuration loses. The jump alone is uphill ($0.3274 \to 0.3485$) and finishes below baseline only after correction, confirming the lift-then-select mechanism. Optimal $\beta = 0.5$, not 1.0; $\beta = 2$ is catastrophic throughout.*

1.3 Cellular sheaf on the phase: $H^0 = 0$

Partition $120 \to 200$ into four cells of 20 steps and let the stalk over a cell be the local displacement subspace. Effective ranks are 3.26, 3.25, 3.63, 3.36 (CV 5%): *no rank jump occurs anywhere in the phase*, so there is no index event in the interior and the family $\{A_b\}$ is trivialisable there. Principal angles between adjacent cells at $k = 6$ give exactly one surviving direction (35–41°) with the remaining five at 82–90°, overlap 0.10–0.12.

Proposition 1.3 (No global section). *The intersection of all four cell subspaces is trivial: the mean projector $\frac{1}{4}\sum_i P_i$ has top eigenvalue 0.43–0.51 for every k tested, so $\dim H^0 = 0$ at any tolerance above $\cos > 0.5$. The net displacement projects onto each individual cell at 0.48–0.66 but lies in no intersection: it is a quasi-section, present in every stalk and exact in none.*

The reduced dynamics therefore exists locally in every cell and globally nowhere, which is the precise sense in which phase 3 is simultaneously measurable and unpredictable.

1.4 Node interactions and the sign flip

Over $120 \to 200$, per-parameter cancellation is 62.5%. Single-node oracle jumps recover: EMB 88.4%, LN 21.9%, FF 31.8%, W_Q 6.4%, W_K −5.3%, W_V 23.9%, W_O 29.0%, summing to 196.2% against a joint recovery of 100%.

Observation 1.4 (Additivity sign flip). *On $40 \to 120$ the skeleton is* super-*additive (singles sum 26%, joint 76%); on $120 \to 200$ the nodes are* sub-*additive (singles sum 196%, joint 100%). The sign of the additivity defect is a threshold-free phase discriminator.*

Pairwise synergies $S_{ij} = \mathrm{rec}(i,j) - \mathrm{rec}(i) - \mathrm{rec}(j)$ reproduce the architecture: W_V–$W_O = -35.6$ (multiplicative composition in the output path, so joint motion overshoots), W_Q–$W_K = +9.1$ (composition in the score, so each needs the other), and W_K is a pure connector — solo −5.3% but positive with every partner, edge sum +72.5. Its contribution lives on edges, not on its vertex.

The Gram matrix of functional changes $\cos(\delta f_i, \delta f_j)$ exhibits four near-orthogonal blocks: $\{\mathrm{EMB}, \mathrm{LN}\}$ (internal 0.331), $\{W_Q, W_K\}$ (0.576), $\{W_V, W_O\}$ (0.612), $\{\mathrm{FF}\}$, with cross-block overlaps 0.02–0.16 and EMB $\leftrightarrow$attention at 0.02. Eigenvalues 1.854, 1.437, 1.343, 0.902, 0.653, 0.424, 0.387: the pairing is nondegenerate. Functional overlap predicts interaction sign, $\mathrm{corr}(\cos(\delta f_i, \delta f_j), S_{ij}) = -0.37$.

1.5 The null, and what it does to the closure claim

Each node acts almost exactly linearly on its own ($\|\delta f(1.0\, d_g)\|/2\|\delta f(0.5\, d_g)\| \in [0.916, 1.007]$), but the nodes do not superpose: $\|\delta f(\mathrm{full}) - \sum_g \delta f(g)\|$ is 49.1% of $\|\delta f(\mathrm{full})\|$. All non-additivity is therefore bilinear, defining a symmetric product

$$m(u, v) \;=\; f(\theta + u + v) - f(\theta + u) - f(\theta + v) + f(\theta). \tag{3}$$

Since m maps $P \otimes P \to F$ with $F \neq P$, closure is only meaningful after lowering the index with the Jacobian adjoint, $\tilde{m} = J^* \circ m : P \otimes P \to P$.

generator set	mean in-span	vs. null
node cuts of Δ (same segment)	0.695%	4,299×
random directions per node, norm-matched	0.007%	42×
random directions, second seed	0.006%	38×
node cuts of an independent segment	1.575%	9,742×

The null for a random direction against a 7-dimensional span in $\mathbb{R}^{4{,}330{,}240}$ is $7/P = 1.62 \times 10^{-6}$.

Observation 1.5 (Alignment without closure)**.** *The product is* $\sim 10^3$–10^4 *times more aligned with its own generators than chance, and is* more *aligned when the generators come from an independent segment (*9,742×*) than from the segment used to build it (*4,299×*), ruling out inheritance. Random directions with identical node support give only* $\sim$40×*, so the alignment is a property of on-trajectory directions and not of the block decomposition. Nevertheless the product does not close:* 99.3% *of its energy lies outside the generator span.*

The verdict is scale-independent. Writing $A(t) = m_t/t^2$, the raw product at $t = 1, \frac{1}{2}, \frac{1}{4}$, the two-point Richardson $H = 2A(t/2) - A(t)$ and the three-point Richardson $H = \frac{1}{3}A(t) - 2A(t/2) + \frac{8}{3}A(t/4)$ give in-span fractions 0.69%, 0.68%, 0.67%, 0.67%, 0.67% respectively (4,299× to 4,130× null), with pairwise scaling ratios 3.29–3.90 against the ideal 4. Higher-order contamination therefore does not account for the escape. The connected triple obstruction satisfies $\|m_3\|/\|m_2\| = 0.339$ and $\|m_3\|/\|m_1\| = 0.184$ — the expansion converges geometrically — but only 20.7% of m_3 lies in $\mathrm{span}(m_2)$. Effective ranks by order are $2.88 \to 5.75 \to 8.24$, i.e. $\sim$2.5 new directions per order: linear growth, far below the free bound 7^k, but never closing.

1.6 m_1 from the flow, and why the A_∞ relations fail

Take $\Phi(\theta) = \theta - \eta \nabla L(\theta)$ with a fixed batch, and set

$$m_0 = \Phi(\theta) - \theta, \qquad m_1(v) = \Phi(\theta + v) - \Phi(\theta) - v, \tag{4}$$
$$m_2(u, v) = \Phi(\theta{+}u{+}v) - \Phi(\theta{+}u) - \Phi(\theta{+}v) + \Phi(\theta),$$

with m_3 the connected third difference. *Numerical warning:* differencing Φ directly cancels $\|\theta\| \approx 93$ down to 10^{-5} and is pure single-precision roundoff — diagnosed by m_3 scaling as 1.08 instead of the required 8. Differencing the gradient (where the θ terms cancel analytically) in double precision restores validity: m_2 scales 3.32 (expect 4), m_3 scales 12.04 (expect 8).

With valid numerics both curved relations fail:

relation	scale	residual	relative
n=1: $m_1^2 x + 2m_2(m_0, x)$, $x = \mathrm{EMB}$	$4.64{\times}10^{-7}$	$4.46{\times}10^{-7}$	0.961
n=1: $x = \mathrm{FF}$	$1.69{\times}10^{-6}$	$1.45{\times}10^{-6}$	0.856
n=3 Stasheff, EMB-FF-W_Q	—	—	$\cos = 0.005$
n=3 Stasheff, W_Q-W_K-W_V	—	—	$\cos = -0.287$

Proposition 1.6 (No A_∞ structure in the gradient flow). *For Φ a gradient map, every m_k is a derivative of the scalar potential L: $m_1 = -\eta H$, $m_2 = -\eta\nabla^3 L$, $m_3 = -\eta\nabla^4 L$. These are fully symmetric tensors. A fully symmetric family admits no associativity to correct, so the associator is not the boundary of anything and the Stasheff relations cannot hold. The measured failure is a consequence of the category, not of the measurement.*

This closes the symmetric route. Any antisymmetric structure must therefore arise from an operation that is not a derivative of L.

1.7 The Lie structure: node-restricted fields nearly close

Let P_i be the coordinate projector onto node i and

$$V_i(\theta) \;=\; -\,P_i\nabla L(\theta), \qquad [V_i, V_j] \;=\; -P_j H V_i + P_i H V_j. \tag{5}$$

Although H is symmetric, $P_i H P_j$ and $P_j H P_i$ occupy *different* blocks and cannot cancel; the bracket is nonvanishing. Antisymmetry is structural and the Jacobi identity is automatic for vector fields, so the only substantive question is closure. All 21 brackets follow from 8 gradient evaluations.

	symmetric product	Lie bracket
mean in-span	0.7%	**53.3**%
proper null	$1.62{\times}10^{-6}$	$2.62{\times}10^{-6}$
ratio to null	4,299×	**351,064**×
effective-rank flow	$3.76 \to 5.50$ (expands)	$2.40 \to 1.82$ (contracts)
energy outside span	99.3%	36.4%

Brackets are 0.469× the magnitude of the generators, so this is a leading-order structure. The control is decisive: projecting the *same* brackets onto random node-supported fields of matched norm collapses the in-span fraction to 0.0078% (28.8× null), a signal-to-control ratio of 6,879.

Observation 1.7 (Contraction versus expansion). *The symmetric product expands rank and escapes almost completely; the antisymmetric bracket contracts rank (*$1.82 < 2.40$*) and retains half its energy in the generator span, escaping in only* ~ 2.07 *effective directions. This is the behaviour of a derived subalgebra* $[\mathfrak{g}, \mathfrak{g}] \subseteq \mathfrak{g}$*, missing closure by about two directions rather than by everything.*

1.8 Level two: the tower closes better than it opens

The bracket fields $[V_i, V_j]$ are themselves vector fields, so the derived series is computable. Let the level-1 span be $\mathcal{G}_1 = \mathrm{span}\{V_i\} \oplus \mathrm{span}\{P_j\,\mathrm{D}g_i\}$ (56 dimensions, a superset of $\mathrm{span}\{V, [V, V]\}$), and evaluate $[V_k, [V_i, V_j]]$ by finite differences of the bracket field.

$[V_k, [V_i, V_j]]$	$\|\cdot\|$	in span$\{V\}$	in $\mathcal{G}_1$	vs. null
[EMB, $[W_Q, W_K]$]	0.02814	18.4%	91.3%	70,561×
[FF, $[W_Q, W_K]$]	0.01445	19.1%	91.5%	70,731×
$[W_Q, [W_Q, W_K]]$	0.00140	16.3%	90.7%	70,135×
[EMB, $[W_V, W_O]$]	0.03162	6.2%	39.5%	30,518×
[FF, $[W_V, W_O]$]	0.01980	10.5%	58.1%	44,918×
$[W_Q, [W_V, W_O]]$	0.00121	9.2%	50.2%	38,813×
mean		12.8%	67.7%	52,342×

Observation 1.8 (Root-space behaviour of W_Q, W_K)**.** *The pair that fails hardest to close at level one closes hardest at level two.* $[W_Q, W_K]$ *retains only* 34.1% *in its own span, but every double bracket* $[V_k, [W_Q, W_K]]$ *returns to* $\mathcal{G}_1$ *at* 90.7–91.5%*, and does so uniformly across* k*. Conversely* $[W_V, W_O]$*, the most closed pair at level one, generates level-two elements that escape (*39.5–58.1%*). A generator pair whose bracket leaves the span but whose adjoint orbit returns to it is the behaviour of a root vector:* W_Q, W_K *generate a new graded piece which is then stable under the adjoint action, while* W_V, W_O *behave as a solvable block that does not.*

1.9 The algebra is a stalk, not a global object

Computing the brackets at step 120 and projecting them onto the fields $V_i(\theta_c)$ at other checkpoints tests whether the closure is a property of the phase or of the base point.

fields from	mean in-span	min	max	vs. null
step 40	0.11%	0.03	0.24	678×
step 80	0.14%	0.00	0.36	871×
step 120 (own)	44.97%	19.97	58.08	278,197×
step 160	1.53%	0.35	3.90	9,471×
random node fields	0.00%	0.00	0.00	3×

Proposition 1.9 (Locality of the Lie structure)**.** *Closure falls by a factor of* ~ 30 *at* $+40$ *steps and* ~ 300 *at* -40 *steps, while remaining* 10^2–10^4 *times above the random control (*3×*). The Lie structure is therefore a local object with a decaying, not vanishing, transport between checkpoints, and it is asymmetric in time: the future (*9,471×*) is better aligned than the past (*871×*), consistent with convergence toward the attractor.*

This forces the algebraic reading into the sheaf-theoretic frame already established in §1: what exists is not one Lie algebra but a *field of Lie algebras over the base*, one per stalk, with weak restriction maps. The measured transport (0.034× own over 40 steps) is quantitatively consistent with the principal-angle result that adjacent cells share one direction out of six. The vanishing of H^0 for the transition laws and the locality of the bracket closure are two readings of the same fact.

As a robustness note, the own-checkpoint closure evaluated at two independently reached step-120 states gives 44.97% and 53.31%, so the figure is stable to about ±8 points across runs.

1.10 Skeleton → dense: the transition, measured

Evaluating (5) at two checkpoints:

subalgebra	$\|[\cdot,\cdot]\|_{80}$	$\|[\cdot,\cdot]\|_{160}$	growth	in-span$_{80}$	in-span$_{160}$
all seven	0.00838	0.03828	4.57×	36.6%	55.9%
skeleton EMB,LN,FF	0.02456	0.13420	5.47×	31.8%	63.0%
attention Q, K, V, O	0.00152	0.00490	3.22×	31.8%	51.7%
W_Q, W_K only	0.00047	0.00100	2.11×	15.6%	34.1%
W_V, W_O only	0.00534	0.01424	2.67×	50.5%	60.5%

Every bracket grows and every subalgebra closes better as the run proceeds. This is the skeleton-to-dense transition expressed algebraically: while the skeleton carries the updates the Lie structure is weak and open; as attention comes online the brackets strengthen by 2–5× and closure rises from ∼32% to ∼63%.

Observation 1.10 (W_Q, W_K generate; W_V, W_O close)**.** *W_V, W_O is the most closed pair at both checkpoints (*50.5% → 60.5%*), consistent with their multiplicative composition and functional overlap* 0.612*: they behave as a nearly solvable piece. W_Q, W_K is the least closed (*15.6% → 34.1%*), with*

$$[V_Q, V_K] = +0.0203\, V_Q - 0.0140\, V_K + r, \qquad \|r\|/\|[V_Q, V_K]\| = 0.812$$

at step 160*, and normalised bracket strength rising* 1.263 → 1.696*. A generator pair whose bracket lands overwhelmingly outside its own span, with opposite-sign coefficients, is the signature of a rank-2 system generating new root spaces rather than of a closed subalgebra.*

1.11 The layer grading and the culling mechanism

Grading by depth rather than by parameter role sharpens every result. Let $V_\ell = -P_\ell \nabla L$ with ℓ ranging over EMB and the six blocks. Bracket closure rises to 62.2% in-span (384,812× null), above the 53.3% obtained from the role decomposition, and decays monotonically with depth separation:

$\lvert\ell - m\rvert$	pairs	mean in-span	mean $\|[\cdot,\cdot]\|$
1	5	63.7%	0.03117
2	4	62.1%	0.03024
3	3	59.8%	0.03099
4	2	57.7%	0.02952
5	1	56.8%	0.02373
EMB–∗	6	64.6%	0.1127

The mechanism is a rank collapse. Probing block ℓ with all seven independent directions $\widehat{V}_m$ and collecting the responses $P_\ell H \widehat{V}_m$:

block	eff-rank	rank(5%)	normalised spectrum				
EMB	1.07	1	1.00	0.02	0.01	0.00	0.00
L_0	1.10	1	1.00	0.03	0.01	0.01	0.00
L_2	1.11	1	1.00	0.02	0.02	0.01	0.01
L_5	1.13	1	1.00	0.04	0.01	0.00	0.00

Proposition 1.11 (Rank-one curvature response)**.** *Within every block the second-order response is rank one to within* 2–4%*: $P_\ell H v \approx u_\ell\,(w_\ell \cdot v)$ for a fixed unit u_ℓ. This is the Jacobian structure of LayerNorm (a projection removing mean and scale) composed with softmax (whose Jacobian* $\mathrm{diag}(p) - pp^\top$ *annihilates the constant direction).*

The proposition explains the algebraic results rather than accompanying them. Bracket closure is a single cosine per block, not a subspace overlap, which is why it sits near 60% rather than near 0 or 1; and adjoint stability at level two is automatic, since once a bracket lies along u_ℓ every further bracket does too. The 90.7–91.5% closure of the W_Q, W_K orbit is the culling acting after the direction has been established.

1.12 The obstruction to exploiting it

Rank-one response invites a reduced Newton step: solve $g + Hd = 0$ on the K block coefficients, at a cost of $K+1$ gradient evaluations. Measured against AdamW from the same state:

step	out-of-u_ℓ residual	cond(C)	equivalent AdamW steps	verdict
40	72.0%	49.6	1	no gain
80	61.9%	1.55×10^3	1	no gain
120	70.8%	477	4	no gain
160	58.6%	1.47×10^3	1	no gain

Observation 1.12 (Fredholm obstruction, quantified)**.** *Rank-one response means the image of the block-restricted curvature operator is one-dimensional, so its cokernel is of codimension one in each block. Between* 59% *and* 72% *of the gradient lies in that cokernel. By Corollary* **??** *(the Fredholm alternative) the equation is unsolvable for that component by any block-restricted update, and the reduced Newton step correspondingly buys nothing. The culling that renders the algebra tractable is the same structure that renders the gradient unreachable.*

What the rank-one result does justify is the phase-2 cull already validated empirically: if $P_\ell H v \approx u_\ell(w_\ell \cdot v)$ to 2–4%, a *scalar* step size per block is not an approximation but the correct curvature model, and maintaining a full second moment on those blocks is wasted state. This is the mechanism behind $W_V, W_O \to$ scalar SGD and $W_Q, W_K \to$ frozen diagonal being free within noise. It yields a generalisable test: measure the response spectrum per block and cull those exhibiting the 1.00/0.02 gap. The depth decay above adds a second prediction: inter-block coupling weakens with $|\ell - m|$, so layer-local optimizer state and pipeline-parallel schedules are cheap exactly in proportion to the measured off-diagonal of the coupling matrix C.

1.13 Forward/backward alignment does not occur

If the forward map (inter-token meaning) and the backward map (loss-driven correction) came into alignment as training converged, the gradient would progressively enter the reachable image and the out-of-u_ℓ residual would decay to zero. Measured across a full run at 20-step resolution:

step	val	$\|g\|$	mean residual	eff-rank	cond(C)
20	2.1348	0.5689	76.7%	1.99	44.3
60	0.6466	0.3165	67.5%	1.36	87.8
100	0.1980	0.1982	61.9%	1.18	189
140	0.0856	0.2891	69.4%	1.11	3.44×10^4
160	0.0704	0.2010	88.4%	1.25	229
200	0.0768	0.2221	53.9%	1.08	778
220	0.0985	0.3717	80.9%	1.09	4.38×10^4

Proposition 1.13 (The obstruction is permanent)**.** *The mean residual over the first half of the run is* 70.1% *and over the second half is* 70.1%*; a linear fit gives slope* -0.0049% *per step, extrapolating to zero at step* 14,401. *The run attains its loss minimum at step* 160 *with the residual at its* largest *value (*88.4%*). Forward image and backward demand therefore do not*

align, and the Fredholm obstruction of the previous subsection is a permanent feature rather than a transient of early training.

One quantity is monotone. The effective rank of the block response falls $1.99 \to 1.64 \to 1.36 \to 1.11$ over steps 20–80 and remains in $[1.08, 1.25]$ thereafter. The rank-one culling is thus not a fixed property of the architecture but *crystallises* during skeleton formation, completing at the skeleton-to-dense boundary. This dates the mechanism of Proposition 1.11 and explains why the algebraic closure results strengthen after step 80.

1.14 The projection path and its resolution scale

The directions u_ℓ define rank-one projections $e_\ell = u_\ell u_\ell^\top$. Two projections at principal angle ϑ satisfy $\|e - f\| = \sin\vartheta$, and $\|e - f\| < 1$ is the standard criterion for them to determine the same K_0 class. Sampling every step over a window in phase 3:

lag (steps)	mean cos	$\|e - f\|$
1	0.891	0.454
2	0.733	0.680
4	0.402	0.916
8	0.074	0.997

Observation 1.14 (Continuity is a matter of sampling). *At* 1*-step spacing* $\|e - f\| = 0.454$*, well inside the criterion, so the projections form a continuous path and the* K_0 *class is defined along it. The decorrelation time is* ≈ 4 *steps; sampling at* 20*-step intervals (as in the preceding subsection) lies far beyond it and spuriously suggests a discontinuous path. The second singular value satisfies* $S_2/S_1 = 0.204$*, so* u_ℓ *is separated from its successor by a factor of five and the path is not an artefact of a degenerate spectrum.*

The path is nevertheless diffusive rather than rotational. The per-step angle is $\arccos(0.891) = 27.1°$, accumulating $\approx 434°$ over sixteen steps, while the net displacement over the same interval is $\arccos(0.158) = 80.9°$: a path-length to net-displacement ratio of 5.4. There is therefore a well-defined path of projections but no coherent rotation, and a training run — which does not close a loop — carries no winding. Extracting a K_1 element would require constructing a loop deliberately, for instance by a cyclic schedule returning to the same basin, and measuring the holonomy of the transport around it. That construction is the natural next step for the operator-theoretic programme and has not been attempted here.

2 The Tile Complex: One Space, Two Orientations

The preceding sections measured algebraic structure on a decomposition by parameter role. This section replaces that decomposition with a refinement hierarchy and finds that the resulting object is operator-algebraic rather than Lie-theoretic, with a topology that is forced by the construction.

2.1 Construction

Each layer's parameter vector is partitioned, per group, into overlapping tiles along its flat index. The resolution ladder is assigned by depth,

$$\text{forward}\quad L_0{:}16,\ L_1{:}32,\ L_2{:}64,\ L_3{:}256,\ L_4{:}512,\ L_5{:}1024,$$

with the backward assignment mirrored (L_5:16 through L_0:1024). Overlap is gated by the persistence excess of §1: 10% while $E(64) > 0.05$, 25% below. On a 200-step run the gate fired unprompted at step 176, where $E(64)$ crossed from $+0.060$ to $+0.043$. Forward tiles record the weights after each step; backward tiles record the applied update $\theta_t - \theta_{t-1}$, both reduced to mean and $|\max|$. Groups LN, FF, EMB are tracked from step 1 and W_Q, W_K, W_V, W_O from step 121.

Observation 2.1 (One complex, two orientations)**.** *Forward and backward are not two spaces. They share the layer axis and the index set and differ only in the orientation of the refinement maps: forward branches toward the output, backward toward the input. The formal difference* $[\mathcal{H}_f] - [\mathcal{H}_b]$ *is therefore taken on a single complex. Ranks multiply to a constant at every depth,*

$$16 \cdot 1024 = 32 \cdot 512 = 64 \cdot 256 = 256 \cdot 64 = 512 \cdot 32 = 1024 \cdot 16 = 2^{14},$$

while the graded difference $\operatorname{rank} \mathcal{H}_f - \operatorname{rank} \mathcal{H}_b$ *takes the values* -1008, -480, -192, $+192$, $+480$, $+1008$*: antisymmetric about the middle, summing to zero, and* never *zero at any layer. The fixed point would be* $\sqrt{2^{14}} = 128$*, which the ladder omits. The pairing is consequently nondegenerate at every layer and the graded index changes sign between* L_2 *and* L_3 *by a jump rather than through a zero.*

Within a layer the tiles are overlapping intervals on a line, so the nerve of the cover is a path graph, all intersections are contractible, and by the Nerve Lemma each layer is homotopy equivalent to a point. All topology therefore resides in the vertical direction. As a CW complex, per group and orientation: 1904 zero-cells, 1898 horizontal and 1888 vertical one-cells, and 875 plaquettes, giving $\chi = -1007$.

2.2 Stalks: a log power law in the loss

Over each tile the stalk is the pair (α, β) fitted from

$$\log\bigl|u_{\text{tile}}(t)\bigr| \;=\; \alpha \,\log\bigl(L(t) - H\bigr) \;+\; \beta, \qquad H = 0.062, \tag{6}$$

where u is the tile's $|\max|$ channel and H is the corpus entropy floor. Taking logarithms is what makes the stalk a vector space: addition and scalar multiplication are defined, so restriction maps can be linear and cellular sheaf cohomology is available. The metric on each stalk is supplied by the diagonal Fisher information, computed with labels sampled from the model, aggregated over the tile's index span.

FF forward (weights)				FF backward (updates)			
layer	tiles	α	R^2	layer	tiles	α	R^2
L_0	16	-0.052	0.729	L_5	16	$+0.049$	0.733
L_1	32	-0.055	0.776	L_4	32	$+0.054$	0.697
L_2	64	-0.057	0.765	L_3	64	$+0.058$	0.658
L_3	256	-0.062	0.774	L_2	256	$+0.066$	0.596
L_4	512	-0.065	0.782	L_1	512	$+0.071$	0.556
L_5	1024	-0.068	0.781	L_0	1024	$+0.078$	0.530

The exponents are mirror-symmetric between the two orientations, $\alpha_f \approx -\alpha_b$, both monotone in depth: weights contract as the loss falls while updates grow. LN and W_K fit poorly ($R^2 = 0.02$–0.32), so the power law in the loss is specific to FF.

2.3 Restriction maps as pants coproducts

A refinement rung takes one parent tile to r children with $r \in \{2, 4\}$: one input, several outputs. This is a coproduct $\Delta : \mathcal{S}(p) \to \bigoplus_i \mathcal{S}(c_i)$, and the reverse merge is the corresponding product. Fitting a single 2×2 map $M : \mathcal{S}(p) \to \mathcal{S}(c)$ per rung by Fisher-weighted least squares:

group/side	rung	M	rel. residual
FF bwd	$L_5 \to L_4$	$\begin{bmatrix} 0.240 & -0.006 \\ 0.598 & 1.008 \end{bmatrix}$	0.001
FF bwd	$L_4 \to L_3$	$\begin{bmatrix} 0.067 & -0.008 \\ -0.251 & 1.003 \end{bmatrix}$	0.001
FF bwd	$L_3 \to L_2$	$\begin{bmatrix} 0.194 & -0.008 \\ -0.470 & 1.004 \end{bmatrix}$	0.002
FF bwd	$L_2 \to L_1$	$\begin{bmatrix} 0.262 & -0.008 \\ -1.300 & 0.992 \end{bmatrix}$	0.003
FF bwd	$L_1 \to L_0$	$\begin{bmatrix} 0.310 & -0.008 \\ -1.503 & 0.990 \end{bmatrix}$	0.003

Residuals are 0.1–0.8%: the coproduct is essentially exact. Every map has the same spectral shape, one eigenvalue near unity and one strongly contracted. Across all rungs, groups and orientations the near-unit eigenvalue lies in [0.990, 1.045] and the contracted one in [0.005, 0.53].

2.4 The β line is a global section

Comparing the eigenvalue-one eigenvector across every group and orientation:

	FF/f	FF/b	LN/f	LN/b	W_K/f	W_K/b
FF/f	1.0000	0.9993	0.9999	0.9994	0.9998	0.9996
FF/b	0.9993	1.0000	0.9997	1.0000	0.9999	1.0000
LN/f	0.9999	0.9997	1.0000	0.9998	1.0000	0.9999
W_K/b	0.9996	1.0000	0.9999	1.0000	1.0000	1.0000

Proposition 2.2 (Universality of the fixed line within this cover)**.** *All pairwise cosines exceed* 0.9993*, and within-group agreement across rungs exceeds* 0.9985*. The eigenvalue-one eigenspace is a single line shared by every stalk of every group in both orientations. Within the refinement system, an observable is fixed by every restriction map exactly when it lies on this line.*

This is a statement about the tile refinement and not about the network: §2.10 shows that β does not survive a change of cover, so it should not be identified with a basic observable of the optimisation dynamics in the sense of §1.

2.5 The inductive limit selects β

Composing all five rungs:

group/side	eigenvalues	$\lvert\lambda_2\rvert/\lvert\lambda_1\rvert$	$\cos(\text{top}, \beta)$
FF/fwd	(0.0000, 1.1169)	2×10^{-5}	1.0000
FF/bwd	(0.0002, 1.0257)	2×10^{-4}	1.0000
LN/fwd	(0.0037, 1.2065)	3×10^{-3}	1.0000
LN/bwd	(0.8615, 1.0909)	0.790	0.9998
W_K/fwd	(0.0000, 1.0538)	$< 10^{-5}$	1.0000
W_K/bwd	(0.0001, 1.0435)	1×10^{-4}	1.0000

Proposition 2.3 (β is the Perron–Frobenius observable)**.** *Five of six composites are rank one to within* 10^{-5}–10^{-3}*, with surviving eigenvector equal to the* β *line at cosine* 1.0000*. Refinement discards every observable except* β*, which is therefore not merely basic but the unique attracting observable of the refinement dynamics.*

Two qualifications. The dominant eigenvalue exceeds unity everywhere (1.026–1.207), so the fixed *line* is exact while the fixed *scale* is not: β is weakly expanding under refinement. And LN backward does not reduce, with $\lvert\lambda_2\rvert/\lvert\lambda_1\rvert = 0.790$; it is also the group with the strongest depth attenuation and the most tile-heterogeneous (CV 0.40 against FF's 0.095). §2.6 shows this is not a fitting artefact and identifies it as a second slow observable rather than an exception. Across two seeds the eigenvalue *ratio* replicates (LN 0.31–0.60, FF 0.0000–0.0155) while the eigen*vector* does not (LN_w gives $(+0.999, +0.051)$ then $(+0.491, +0.871)$), so the stable object is the spectral gap and not the eigenspace. Since the stalk is two-dimensional, the surviving second direction is α itself: for LN the loss-response exponent survives refinement, for FF it is annihilated. Determinant and λ_2 are not independent evidence, as $\lambda_1 \approx 1.09$ throughout makes $\det \approx 1.09\lambda_2$; no claim of an SL_2 or cluster-algebraic structure is warranted.

2.6 Validation: the collapse is not regression attenuation

The restriction maps of the previous subsections are least-squares fits of the child stalk on the parent stalk. When a predictor is measured with error, the fitted coefficient is biased toward zero by the reliability ratio $\sigma^2_{\text{signal}}/(\sigma^2_{\text{signal}} + \sigma^2_{\text{noise}})$. Since β is a log-magnitude intercept (precisely estimated) while α is a slope fitted against $\log(L - H)$ (less so), attenuation alone would produce a near-unit eigenvalue along β and an eigenvalue equal to the reliability of α along α — reproducing the observed spectral shape without any refinement geometry. This is a complete alternative explanation of Propositions above and must be excluded.

Fitting (α, β) independently on odd and even steps and correlating across tiles gives the split-half reliability, Spearman–Brown corrected.

group/side	rel(α)	rel(β)	mean λ_α	$\lambda_\alpha - \text{rel}(\alpha)$
FF/fwd	0.999	1.000	0.125	−0.875
LN/fwd	0.999	1.000	0.392	−0.607
W_K/fwd	0.952	0.983	0.095	−0.857
FF/bwd	0.887	0.975	0.208	−0.680
LN/bwd	0.961	0.994	0.915	−0.045
W_K/bwd	0.584	0.627	0.318	−0.266

Proposition 2.4 (The contraction is geometric, not statistical)**.** *Over* 30 *rungs, mean* $\text{rel}(\alpha) =$ 0.897 *against a mean contracted eigenvalue of* 0.342*; the correlation between reliability and eigenvalue is* +0.20 *and the mean absolute gap is* 0.571*. Under the attenuation null these two quantities would coincide. In the forward bundles α is estimated at reliability* 0.95–0.999 *and is nonetheless contracted to* 0.10–0.39*, so refinement discards a well-measured coordinate. The collapse onto β is a property of the refinement system.*

Two consequences for the preceding results.

First, LN backward is promoted from exception to finding. Its α is measured at reliability 0.961 — comparable to the forward bundles — yet its restriction eigenvalue is 0.915, and three of its five rungs exceed unity (1.016, 1.045, 1.017). Refinement does not contract α there; it mildly expands it. Under the attenuation null the least reliable branch should contract most, so this runs opposite to the null in the informative direction. LN backward therefore carries a second slow observable that β does not capture, and the inductive system is not completely reducible on that branch.

Second, W_K backward must be excluded rather than interpreted. Its split-half reliabilities are 0.584 and 0.627, with one rung at −0.02 — the two halves are uncorrelated. Its stalks are unfitted parameters. This is the same group whose Fisher state is worst conditioned (max / min $\approx 5 \times 10^6$, normalised entropy 0.667), and the two facts are plausibly one: Fisher mass concentrated on a few tiles leaves the remainder with no signal to fit. The β-line consistency table above accordingly rests on five reliable columns rather than six, which does not affect its conclusion since those five agree at cosine ≥ 0.9993.

2.7 The observable algebra is AF, and K_1 vanishes by construction

The branching ratios $2, 2, 4, 2, 2$ make the ladder a Bratteli diagram $\mathbb{Z}^{16} \to \mathbb{Z}^{32} \to \mathbb{Z}^{64} \to \mathbb{Z}^{256} \to \mathbb{Z}^{512} \to \mathbb{Z}^{1024}$, each edge of multiplicity one and each child having a unique parent. The strand space is the path space of this diagram: a Cantor set carrying the 2-adic ultrametric $d(s, s') = 2^{-\mathrm{lca}(s,s')}$ truncated at depth ten, with 2^{10} strands. The associated algebra is commutative AF, $C(X)$ with X totally disconnected.

Proposition 2.5 (Forced vanishing of K_1)**.** *For any AF algebra $K_1 = 0$. Every negative winding result reported in this document is therefore a consequence of the observable algebra rather than of the optimisation dynamics: the diffusive projection path, the absence of coherent rotation, and the failure to extract a K_1 element were determined before measurement. The claim is that* the AF algebra generated by the refinement tree has trivial K_1, *not that optimisation has trivial K_1; enlarging the algebra, for instance by a crossed product with the optimisation dynamics or by adjoining time as a circle, can restore it.*

What survives is K_0, the dimension group of locally constant integer functions, which is purely combinatorial, together with the state on it, which is geometric. This is the α/β separation in operator-algebraic form: $\alpha_f + \alpha_b \approx 0$ is virtual cancellation, visible to K-theory and equal to zero there, while β is metric data that K-theory forgets.

2.8 Faithfulness and the weighted grading

group	tiles	zero-weight	max / min	entropy/max
FF	1904	0	9.13×10^2	0.851
LN	1904	0	1.37×10^3	0.833
W_K	1904	0	5.39×10^6	0.667
W_V	1904	0	1.48×10^2	0.890
W_Q	1904	0	2.32×10^6	0.791

No tile carries zero Fisher weight in any group, so the state has full support and the AF algebra admits no order ideal annihilated by it. The state is nevertheless ill-conditioned on the score pair: W_K and W_Q concentrate their Fisher mass with condition numbers above 10^6 and normalised entropy 0.67–0.79, against 0.85–0.89 for FF, W_V, W_O.

Backward magnitude increases monotonically with depth in every group, with L_5/L_0 ratios 1.25 (FF), 1.38–1.39 (attention), 1.83 (LN). The backward bundle is therefore attenuated toward the input, and the exchange J between orientations satisfies $J_w^2 = w$ rather than $J^2 = 1$. Measured on FF the profile is $w = (0.800, 0.838, 0.875, 0.936, 0.970, 1.000)$ across L_0 to L_5: a mild smooth deformation, so the ± 1 eigenspaces remain nearly orthogonal and the Hodge decomposition survives with respect to the Fisher-weighted inner product.

2.9 β is a magnitude field, and magnitude carries no value

β is the intercept of $\log |u_{\mathrm{tile}}|$: a field of magnitudes over tiles, with no directional content. Whether any scheme driven by such a field can generate useful updates is therefore bounded by how much of an update's value resides in its magnitude profile. Taking the true 20-step displacement d at a checkpoint and decomposing it:

arm	step 80	step 120	step 160
true update	100.0%	100.0%	100.0%
magnitude only (in-tile directions randomised)	−21.3%	−38.0%	+17.9%
direction only (magnitudes flattened per layer)	+82.9%	+154.7%	+224.3%
sign only (one global scale)	+91.2%	+163.9%	+205.3%
random vector, norm matched	−20.9%	−29.9%	−3.4%

Entries are the fraction of the true progress retained.

Proposition 2.6 (Ceiling on magnitude-driven updates)**.** *Preserving the exact per-tile magnitude profile while randomising directions within tiles performs indistinguishably from a random vector of matched norm at every checkpoint. Any update scheme driven by a per-tile magnitude field, β included, is therefore bounded above by random.*

Direction alone retains or exceeds the full progress, and sign alone suffices. Since $\|\operatorname{sign}(d)\cdot\overline{|d|}\|_2 = \|d\|_1/\sqrt{P} \le \|d\|_2$ by Cauchy–Schwarz, the sign step has *smaller* norm than the optimiser's own step while making more progress at two of three checkpoints. We record this as an observation warranting separate investigation rather than a claim: it is a single 20-step application at one scale, not a training run.

2.10 β does not survive a change of cover

The preceding sections compute β on one cover. Whether β is intrinsic to the parameters or an artefact of that cover is decided by computing the per-parameter ground truth — the cover by singletons — and asking how well each cover's regional β predicts it. On a 128×256 block of FF weights at L_2:

cover	regions	corr(β)	sd(β_{region})
rows (neurons)	128	0.256	0.056
columns (inputs)	256	0.188	0.039
contiguous tiles	64	0.176	0.037
contiguous tiles	512	0.314	0.069
random partition	128	0.052	0.011
random partition	512	0.121	0.023
singleton ground truth		1	0.226

Proposition 2.7 (Cover dependence of β)**.** *Structured covers correlate with the per-parameter β at* 0.18–0.31 *against* 0.05–0.12 *for random partitions, so a structured cover resolves roughly three times the random baseline and the signal is not nil. But the best cover attains* sd $= 0.069$ *against a singleton* sd *of* 0.226*: approximately* 9% *of the variance. Covers agree with one another only when one refines the other (*0.56–0.82 *among rows and contiguous tiles, which coincide under row-major flattening); covers partitioning along different axes share nothing. β is therefore predominantly a property of the cover rather than of the parameters.*

Reported R^2 values are negative (-2.3 to -2.6) for every cover, but this is a Jensen artefact rather than a failure of prediction: regional β is fitted to $\log(\overline{|u|})$ while the ground truth averages $\log|u|$, and $\log\overline{x} \ge \overline{\log x}$, so regional β is biased upward systematically. Correlation is the appropriate statistic.

Taken with §2.9, these results close the optimisation branch of the programme by two independent routes: β carries no directional information, and it is not intrinsic to the parameters. What survives is narrower and should be kept distinct from β: the spectral gap of the refinement operator (LN 0.31–0.60 against FF 0.0000–0.0155 across two seeds), the persistence excess $E(64)$, which is computed from whole-parameter displacements with no cover at all, and the rank-one curvature response, which is a property of the block projectors. None of these depends on β being intrinsic.

2.11 Strands and the p-adic metric

A strand is a root-to-leaf path through the refinement ladder. With branchings $2, 2, 4, 2, 2$ there are $2^{10} = 1024$ strands, each identified by its leaf, and each carrying the concatenated stalks

(α, β) over six layers — a point in $\mathbb{R}^{12}$, z-scored per layer. The leaf set carries the ultrametric $d(s, s') = p^{-\mathrm{lca}(s,s')}$.

Two remarks are needed before the measurements. First, the forward and backward strand sets are not canonically paired: forward paths refine toward the output while backward paths refine toward the input, so a forward strand and a backward strand coincide at a layer only through the indexing convention. The $\mathcal{H}_f/\mathcal{H}_b$ pairing must therefore be taken layer-wise, as it is throughout. Second, $p^{-\mathrm{lca}}$ is a monotone function of the integer lca, so $p = 2$, 3 and 5 induce identical rank orderings on the true tree; the choice of p is a labelling convention with no empirical content. Measured Spearman correlations for $p = 2, 3, 5$ agree to three decimals.

metric (Spearman vs. strand distance)	FF/fwd	FF/bwd	LN/fwd	LN/bwd
true tree ($p = 2, 3, 5$ identical)	0.243	0.243	0.270	0.307
2-adic on index	0.109	0.178	0.195	0.193
3-adic on index	0.145	0.239	0.355	0.249
5-adic on index	0.136	0.269	0.175	0.137
$\lvert i - j \rvert$ (plain contiguity)	0.189	0.237	0.195	0.092
permuted tree (shape kept)	0.001	−0.003	0.000	−0.006
partial, tree given $\log \lvert i - j \rvert$	0.074	0.061	0.099	0.196

Strand distance does fall monotonically with shared-ancestor depth, with shallow-to-deep ratios of 1.92–4.86, so strands are not independent noise vectors.

Proposition 2.8 (The strand hierarchy is contiguity)**.** *Retaining the tree's combinatorial shape while permuting which parameter index occupies which leaf reduces the correlation from* 0.24–0.31 *to* 0.000 ± 0.006 *in all four branches. The hierarchy therefore contributes nothing independently of its alignment with contiguous parameter blocks. Arbitrary p-adic trees on the index perform comparably to the true refinement and in one case better (*3*-adic, LN/fwd,* 0.355 *against* 0.270*). The ultrametric structure of the strand space is inherited from the contiguity of the tiling, not discovered in the network.*

2.12 What distinguishes LayerNorm, and what does not

Four measurements appeared to single out LN under refinement: the spectral gap, the near-unit determinant, the strand-distance ratio, and the partial correlation of §2.11. They do not survive equally.

The determinant is not independent evidence: $\lambda_1 \approx 1.09$ throughout, so $\det \approx 1.09\lambda_2$ and the two are one measurement.

The strand residual fails decomposition. Combined LN gives partial correlation 0.192 and 0.208 on two seeds, but splitting it yields LN_w at 0.127/0.083 and LN_b at 0.054/0.063 — both below the whole. A quantity exceeding both its parts locates the structure in the joining, and the combined group is `attn.ln.weight` | `attn.ln.bias` | `ff.n.weight` | `ff.n.bias`: four contiguous blocks alternating between gains near 1 and offsets near 0. The residual is the concatenation order. Moreover W_V reaches 0.103/0.135, matching or exceeding LN_w, so no LayerNorm signature remains after the split.

The spectral gap survives every control:

group	elements/tile	λ_2/λ_1 (seed 17)	(seed 23)
LN_w	0.5	0.604	0.516
LN_b	0.5	0.526	0.330
LN	1.0	0.464	0.313
FF	384.5	0.0008	0.0155
FF1024	1.0	0.0000	0.0001
W_V	64.0	0.0000	0.0000

FF1024, a 1024-element slice of FF tiled with the identical ladder, matches LN's occupancy exactly and behaves like FF, excluding occupancy. Both LN_w and LN_b show the gap independently, excluding concatenation. The separation is two to four orders of magnitude and replicates across seeds.

Observation 2.9 (The one surviving LayerNorm result). *Under this refinement, LayerNorm's transfer operator retains a second eigenvalue of order* 0.5 *while every other group annihilates it to order* 10^{-3}*. Since the stalk is two-dimensional, the surviving direction is* α*: the loss-response exponent survives refinement for* LN *and is annihilated for* FF*. This is consistent with LayerNorm's Jacobian being a projection,* $I - P_{\mathbf{1}} - P_x$*, and hence less dissipative than an unconstrained linear map, but the mechanism is inferred rather than measured.*

2.13 Summary, and what the construction does not establish

The cover is the computational architecture; the latent foliation is prior to it and is measured only through the cover, so cohomological statements constrain the cover rather than the space. Stalks carry $(\mathcal{H}_f, \mathcal{H}_b, D_{\text{sheaf}}, \beta)$ with $\mathcal{H}_f = \ell^2(\text{tiles}_{\text{fwd}}) \otimes \mathbb{R}^2$ of dimension 3808 per group, Fisher supplying the inner product, and $D_{\text{sheaf}} = \delta + \delta^*$ the Hodge–Dirac operator of the restriction system — distinct from the stochastic generator $-G \cdot \nabla + \frac{1}{2}\Sigma : \nabla^2$, which acts on functions over parameter space and not on stalks.

Four of the five structural claims are theorems about the construction rather than measurements of training. The hierarchy exists because it was built; the maps form an inductive system because any nested partition does; $K_1 = 0$ follows from the Bratteli diagram having unique parents and finite branching; and faithfulness fails only if the architecture carries structurally dead parameters. The fifth, the Perron–Frobenius collapse, passes its null test (§2.6) and is a genuine property of the refinement system, but β then fails cover-independence (§2.10) and carries no directional content (§2.9). The strand ultrametric reduces to contiguity (§2.11).

What survives controls is therefore narrow and should be stated without the operator-algebraic framing, on which none of it depends: the persistence excess $E(64)$, computed from whole-parameter displacements with no cover at all; the rank-one curvature response, a property of the block projectors, which derives the scalar-per-block preconditioner from measurement rather than convention; the Fredholm obstruction, which explains why block-local methods cannot close the gap; and the LayerNorm spectral gap of §2.12, which is a fact about the refinement operator and not about the network.

The vanishing of the topological invariants is correspondingly not evidence about optimisation. K_1 vanishes because an AF algebra admits no K_1; the index is forced by dimension counting in finite dimensions; the strand ultrametric is the tiling's own adjacency. Establishing whether any of this describes training rather than the refinement requires a cover that is not a nested partition, where the algebra need not be AF, and that has not been attempted.

3 The Sign–Magnitude Decomposition of the Update

The tile programme sought a reduced observable from which updates could be generated. §2.9 and §2.10 showed that β is not that observable. This section reports what the same experimental apparatus finds when the question is inverted: rather than asking which reduced quantity predicts the update, we ask how far the update can be compressed before training degrades.

3.1 Necessary motion, in hindsight

Over 200 steps the per-parameter path length exceeds the net displacement by a factor of 2.3: writing $\mathrm{net}_i = |\sum_t d_t^{(i)}|$ and $\mathrm{path}_i = \sum_t |d_t^{(i)}|$, the aggregate ratio is $\sum_i \mathrm{net}_i / \sum_i \mathrm{path}_i = 0.441$, so 55.9% of all motion cancels. Truncating the net displacement to its largest coordinates and applying the result to θ_0 gives:

top k% of |net|	% of $\sum$ |net|	val	improvement kept	random k% control
1%	4.1%	4.9560	−11.2%	0.7%
5%	16.0%	6.2832	−41.5%	3.0%
10%	27.8%	5.6599	−27.3%	5.6%
25%	53.9%	1.6289	64.7%	5.3%
50%	81.4%	0.0920	99.8%	5.5%
100%	100%	0.0754	100.0%	99.7%

Roughly half the coordinates are required, and partial application is *worse than doing nothing* below 25%: the 5% arm lands at val 6.28 against an initial 4.47. Moving a LayerNorm gain without its paired bias, or W_O without W_V, leaves the network in a state that is neither the initialisation nor the trained model — the non-monotonicity is the signature of the block coupling measured in §1 ($S_{W_V W_O} = -35.6$). Combining the two factors, $0.441 \times 0.814 = 35.9\%$ of the total motion is necessary in hindsight.

This necessary mass is not localised. Density (mass per parameter, normalised) ranges only over 0.87–1.28 across the seven components, with the embedding highest; across the 1024 strands the Gini coefficient is 0.095 with lag-one autocorrelation +0.995, so the mass field is a smooth plateau rather than a concentrated chunk, and the top 10% of strands carry 13.2% of it. Forward and backward ladders agree to three decimals. Nor does the corpus predict it: per-token displacement in the embedding correlates with token frequency at −0.103, with log frequency at −0.142, and with the number of distinct successors at −0.112, varying by 5% across frequency deciles.

Observation 3.1 (Cancellation is not removable)**.** *Rerunning from the same initialisation while zeroing, at every step, any update component opposing* $\mathrm{sign}(D_i)$ *discards* 41.4% *of the motion and reaches val* 0.0676 *against a baseline* 0.0677*. But retaining* only *the opposing components —* 42.4% *of the motion — also converges, to val* 0.0937 *(*99.4%*). Any cull that leaves the optimiser free to recompute gradients is not a constraint; the trajectory re-routes. The* 55.9% *cancellation is an accounting artefact of hindsight, not spendable waste.*

3.2 One constant per strand per step

Replace the applied update d at every step by a quantised version in which each of T contiguous chunks of the parameter vector receives a single constant. Two schemes: retain the per-parameter sign and supply one magnitude per chunk, $d' = \mathrm{sign}(d) \odot c_{\text{chunk}}$ with $c = \overline{|d|}$; or make the chunk uniform, $d' = \bar{d}_{\text{chunk}}$. Gradients are recomputed at each new point, so the trajectory is free to diverge.

T	numbers/step	scheme	val	improvement kept
—	4,330,240	baseline (full update)	0.0677	100.0%
1	1	$\mathrm{sign}(d) \times \overline{\|d\|}$	0.0646	100.1%
64	64	$\mathrm{sign}(d) \times \overline{\|d\|}$	0.0571	100.2%
1024	1024	$\mathrm{sign}(d) \times \overline{\|d\|}$	0.0572	100.2%
1024	1024	uniform signed constant	4.4228	0.9%

Proposition 3.2 (Sign carries the update; magnitude is one number)**.** *A single global scalar per step, applied with the per-parameter sign pattern, retains* 100.1% *of the loss improvement over a full* 200*-step run — a* 4.33×10^6 *compression of the magnitude field with no degradation. Making the constant uniform within a chunk, thereby discarding the per-parameter sign, collapses training to* 0.9%*. The update therefore decomposes into a* 4.33 *Mbit sign pattern carrying essentially all the content and a magnitude field compressible to a single float.*

The per-step payload falls from 139 Mbit to 4.33 Mbit plus 32 bits, a 32× information compression that is exact rather than approximate. This corroborates §2.9 from the opposite direction: there, retaining magnitudes while randomising directions performed indistinguishably from a random vector; here, retaining directions while flattening magnitudes to one global constant loses nothing. Both experiments locate the entire content of an update in its sign pattern, and the present one does so over a full training run rather than a single application.

Two qualifications. The compression is of *bandwidth*, not of computation: the backward pass is still required to obtain the signs, and costs the same whether one bit or thirty-two are retained per coordinate. The relevance is therefore to distributed training, where gradient exchange is the bottleneck and a one-bit-plus-scale payload is exactly what is transmitted; this is the known 1-bit SGD and signSGD family, and what is new here is only the measurement that on this problem the scheme is not a lossy approximation but marginally superior to AdamW (0.0572 against 0.0677). Second, this is one seed, one architecture, and a corpus consisting of a single repeated sentence, a regime in which sign-based methods are known to be favoured. The exportable claim is the decomposition, which two independent experiments establish, not the superiority.

3.3 Scope and limitations

All measurements are at a single corpus (one 1364-token base sequence, 300 loops), a single architecture, and one to four checkpoints per claim. The tile complex of §2 rests on one 200-step run; the power-law stalks are fitted on the $|\max|$ channel only, and the restriction maps are single 2×2 fits per rung rather than per parent, so their 0.1–0.8% residuals measure consistency of the rung, not of individual tiles. Per §2.6, the W_K backward branch fails split-half reliability and is excluded from interpretation throughout.

Four of the five structural claims about the tile complex are theorems about the construction rather than measurements of training: the hierarchy exists because it was built, the maps form an inductive system because any nested partition does, $K_1 = 0$ follows from the Bratteli diagram having unique parents and finite branching, and faithfulness fails only if the architecture carries structurally dead parameters. Only the Perron–Frobenius collapse carries empirical weight, which is why §2.6 tests it against its null. Relatedly, β and $\mathrm{tr}\,\Sigma$ should not be conflated: the former is a log-magnitude intercept of tile updates, the latter a trace of the gradient covariance, and the evidence for each is independent. Following §2.9 and §2.10, β should be described as the dominant eigendirection of a particular refinement system — a reproducible fact about that system — and not as an intrinsic observable of the network. The cover-dependence test covers one block of one layer of one group; extending it to LN, where the refinement operator behaves differently, is outstanding.

On the question the programme originally posed, the evidence assembled here is negative and the negatives are independent. Phase 3 obeys the driftless-walk law, so no local estimator can recover a persistent direction; the block-restricted curvature operator leaves 59–72% of the gradient in its cokernel at every checkpoint, largest at the loss minimum, so no block-local method can supply what is missing; and the reduced observable β carries magnitude only, which §2.9 shows is worth no more than a random vector. Gradient descent computes direction and magnitude at the same cost, and direction is the component that carries the value. The single measured lever is the $E(64)$-gated jump, saving $+2$, $+14$ and -2 steps across three seeds: positive in mean, unreliable per run, and the correct order of magnitude for a regime governed by a random walk. The observation developed in §3 — that the entire content of an update lies in its sign pattern, with the magnitude field compressible to a single scalar at no cost over a full run — did not arise from this framework and is the lead most worth pursuing. The seven fields are masks of a single gradient vector and have effective rank 2.40 of 7, so their mutual independence is modest. Finite differences use $h = 10^{-4}$ in double precision throughout; single precision is *not* sufficient and silently returns roundoff, as diagnosed above. The gated-jump saving of 10 steps is a single measurement on one seed, and validation noise near the entropy floor is $\pm 20\%$; seed replication remains outstanding. The level-two computation covers 9 of the 147 possible double brackets, chosen to contrast the W_Q, W_K and W_V, W_O blocks, and a full enumeration has not been performed.

The claims we regard as established are: (i) the phase-3 trajectory obeys the driftless-walk law $\cos(S_n, S_N) = \sqrt{n/N}$ to within 0.02, so no local estimator can recover the chord, and $E(64)$ separates the phases at $\sim 50\sigma$; (ii) the symmetric product built from node displacements is $\sim 4{,}200\times$ better aligned with its generators than chance yet closes at only 0.7%, a verdict stable under three-point Richardson extrapolation and reproduced with generators from an independent segment; and (iii) the node-restricted gradient fields carry a Lie structure closing at 45–53% in-span, $\sim 3 \times 10^5$ times its null and $\sim 7 \times 10^3$ times its matched control, which strengthens by 2–5$\times$ and closes from $\sim$32% to $\sim$63% across the skeleton-to-dense transition.

Three claims are refuted. The hypothesis that forward and backward maps come into alignment as training converges is contradicted: the out-of-image residual is flat at 70.1% across both halves of a run and is largest at the loss minimum. The reduced Newton step built on the rank-one response gives no speedup at any checkpoint, because 59–72% of the gradient lies in the cokernel of the block-restricted curvature operator; the structure is explanatory, not exploitable, as an optimizer. And the Lie structure is not a single global algebra: cross-checkpoint closure decays by $\sim 30\times$ over 40 steps, so the object is a sheaf of algebras over the base rather than one algebra, and any Kac–Moody presentation would have to be stalk-local.

Two computations would decide the remaining structure. First, a full enumeration of level-two brackets, testing whether the 90% closure observed for the W_Q, W_K orbit holds generally or is specific to that pair; if it holds, the graded pieces are stable under the adjoint action and structure constants can be extracted stalk by stalk. Second, transport of the bracket basis along the flow, to determine whether the $0.034\times$ decay over 40 steps is genuine loss of structure or merely a rotating frame — the distinction between a sheaf with weak restriction maps and one whose stalks are isomorphic under a connection that has not yet been identified.

4 The Sign Process and the Stability Field

The magnitude-versus-direction result of §3 left the sign pattern as the carrier of the update. This section characterises that pattern as a dynamical object and identifies the variable that organises it. It is a record of experiments in the order run, including those that failed, because the failures are what narrowed the object.

4.1 Sign persistence and stale-sign at matched compute

The step-to-step sign agreement of the applied update is 87.7%, flat across the run (88.6% first fifty steps, 87.5% last fifty, never below 77.8%). This suggests reusing a cached sign to skip backward passes. Refreshing the true sign every k steps and applying sign $\times$ scalar in between:

scheme	backward passes	val	vs. equal-compute baseline
stale-sign $k = 2$	100	0.1670	13% better than 100 full steps (0.1927)
stale-sign $k = 4$	50	0.6606	25% better than 50 full steps (0.8761)

Observation 4.1 (The first compute-matched saving)**.** *At equal backward passes, interleaving free stale-sign updates beats honest AdamW by* 13% *at* $k = 2$ *and* 25% *at* $k = 4$*. This is the only lever in the programme that survives a compute-matched control; the earlier gated jump and all magnitude schemes do not. It works because it reuses a measured direction rather than predicting one, sidestepping the driftless-walk obstruction.*

The sparse-event refinement of this idea fails. Sign flips are not rare: 12.3% of coordinates flip per step (530,540 of 4.33M), flat across the run, so "detect only boundary crossings" has no sparsity to exploit — the orthant label changes almost completely each step.

4.2 Momentum is the transport projection

The oscillatory component of the update is separable and its separator already exists: Adam's β_1 is a first-order low-pass filter. Sweeping it:

β_1	val	cancellation	path	$\cos(d_t, d_{t+1})$	flip rate
0.00	0.1122	91.3%	599.1	−0.093	51.1%
0.90	0.0677	67.4%	160.5	0.906	12.3%
0.99	0.0682	41.2%	116.9	0.987	4.5%

Every quantity moves monotonically with β_1. At $\beta_1 = 0$ the dynamics is a genuine walk (consecutive steps *anticorrelated*, half of all coordinates flipping per step); the $\sqrt{n/N}$ law of §1 is the filtered image of this. At $\beta_1 = 0.99$ the path shortens 27%, sign flips fall to a quarter, and val is unchanged within floor noise. Two consequences: the oscillation is real and largely removable but removing it buys only a shorter path to the same endpoint; and the 55.9% cancellation figure quoted throughout is a property of Adam at $\beta_1 = 0.9$, ranging 41–91% with the filter, not a property of the loss surface.

4.3 Stochasticity is the regulariser, not coverage

Suppressing the transverse motion by making the gradient deterministic converges to the wrong place, and the reason is not batch coverage.

arm	val	$\cos(d, \text{to-final})$	$\cos(d_t, d_{t+1})$	path
stochastic (baseline)	0.0677	0.356	0.906	160.5
fixed single batch	6.8069	0.843	0.996	68.2
deterministic full coverage	5.9505	0.847	0.997	86.8

Full coverage (1344 of 1364 base tokens) reproduces the fixed-batch geometry exactly and fails just as badly. Any deterministic gradient produces a smooth 2.4×-shorter path that memorises: train loss 10^{-4} against val 5.93. The regimes separate at step ~ 20. A rescue confirms the damage is reversible: 50 deterministic steps (landing at val 3.30) followed by 150 stochastic reach 0.0660, *beating* pure stochastic (0.0794). The transverse stochastic motion is the implicit regulariser; smoothing the flow removes exactly what makes the run generalise.

4.4 The flip hazard field

Sign flips are governed by a single variable already present in the optimiser state: $r_i = |m_i|/\sqrt{v_i}$, Adam's normalised update magnitude.

decile of r	flip rate	vs. mean
D1 (lowest)	32.0%	2.64×
D5	9.2%	0.76×
D9	0.8%	0.06×
D10 (highest)	0.2%	0.01×

The hazard is monotone across every decile, spans 160×, and has AUC 0.809 as a single-variable predictor. The curve is not exponential but a *hard threshold*: above the ∼70th percentile of r, flips vanish entirely (zero flips in the top three bins over 300k sampled coordinates). The flip mass concentrates where the normalised momentum is small, so the dominant mechanism is low signal-to-noise near zero, not persistent opposing gradients overwhelming a large momentum.

Observation 4.2 (Aggregation destroys the signal)**.** *The AUC* 0.809 *is per-coordinate. Aggregated to tiles it collapses to* 0.60–0.64*, and lead-time prediction is flat (AUC* 0.61 *at* $k = 1$ *and at* $k = 20$*) with tile activity autocorrelation* ∼0.66 *out to* 40 *steps and no decay. Tiles are static "hot regions," not dynamic boundary-approach regions. Near-boundary is a per-coordinate property; the tile mean is dominated by the frozen majority. The adaptive tile scheduler this was meant to enable does not work.*

4.5 Flips cluster, by neuron and by index

Per-tile flip fractions are over-dispersed relative to binomial independence. A cover comparison at matched tile count and size separates the causes:

cover	over-dispersion vs. binomial
random	1.00×
input axis (column)	1.36–1.57×
output axis (row)	1.95–2.58×
contiguous (large tiles)	12.35×

Random calibrates the null at exactly 1.00×. Output-axis (neuron) clustering exceeds input-axis by ∼1.7× in both a feed-forward and an attention matrix, so flips cluster by the neuron a weight feeds, a functional rather than storage grouping. The large 12.35× for 4228-parameter tiles is mostly cross-tensor heterogeneity; within a single matrix the effect is 2–3×. Strided and frieze covers give ∼3.5×, so roughly half the clustering is global-in-index and half is adjacency.

4.6 A periodic cover gives the first non-forced obstruction

Every earlier cohomology vanished by construction: contractible bases, path nerves, an AF Bratteli diagram. A frieze (periodic) cover makes the nerve a *cycle*, which need not be trivial. Fitting each of the 256 edge maps independently and transporting around the loop:

holonomy H	$\|H - I\|$	eigenvalues $\lvert\cdot\rvert$	det
shared M^{256}	0.4498	(1.000, 0.636)	0.636
per-edge $\prod_i M_i$	0.3424	(1.000, 0.729)	0.729

Proposition 4.3 (Genuine holonomy with a preserved line). *Independently-fitted edge maps accumulate a non-identity holonomy around the cycle, so it is not a shared-fit artefact. Each* M_i *is near-unimodular* ($\overline{|\det|} = 0.999$*, spread* 1.7%*) yet the product has* $\det = 0.73$*: the contraction accumulates from many tiny consistent deviations, the signature of a connection with small constant curvature. The loop map fixes one direction (eigenvalue* 1.000*) and contracts the transverse one* (0.73)*, so* $H^0 \cong \ker(H - I)$ *is one-dimensional — global sections exist in one stalk coordinate and fail in the other. This is the first cohomological obstruction in the programme that reports the data rather than the graph.*

The curvature is real but localised: per-plaquette, 88.2% have $|F_\square| < 0.05$, and curvature magnitude correlates with the variation of r around the plaquette (Spearman 0.590; 8.4× split between low- and high-∇r halves). The connection is flat almost everywhere with defects at hazard-field gradients.

4.7 The sign field is compressible; adjacency is nearly redundant

The conditional entropy of the update sign, in bits per parameter per step:

conditioning	bits/param/step	vs. raw
none (raw sign field)	1.000	1.000
previous step's sign	0.557	0.557
+ neuron (row) context	0.541	0.541
+ Adam $r = \lvert m\rvert/\sqrt{v}$	0.428	0.428
+ r and neuron context	0.405	0.405

The sign field carries 0.41 bits per parameter per step, not 1: a 79× compression against fp32 when combined with the single magnitude scalar. Temporal context does most of the work; r adds 23% and is free, already in the optimiser state; neuron adjacency adds only 3–5%. Despite the neuron clustering of §4.5, the sign domains are essentially r-level sets, so adjacency carries little information beyond the hazard variable. This is a bandwidth result, not a compute result: the backward pass is still needed to learn the signs.

4.8 The residual: Fisher-large, trajectory-null

Does $u \approx a \cdot \mathrm{sign}(u)$ discard anything that matters? Under diagonal Fisher the reconstruction retains $R = 0.64$, barely improving with 16,384× more magnitude parameters (0.636 → 0.697), while training retains 100.1%. The metric escape — that diagonal Fisher underweights the residual — fails in the informative direction:

metric	retention of $a \cdot \mathrm{sign}(u)$
diagonal Fisher	0.630 ± 0.020
full Fisher (finite-difference JVP)	0.417 ± 0.078

The true Fisher retains *less* (0.42), so the discarded residual $\varepsilon = u - a\,\mathrm{sign}(u)$ carries 58% of the true Fisher energy, concentrated in the correlated directions the diagonal cannot see — and training is unaffected. The resolution is geometric.

Proposition 4.4 (The residual is transverse to descent). $\cos^2(\varepsilon, g) = 0.031$ *(Euclidean) and* 0.034 *(Fisher): the residual is* 97% *perpendicular to the gradient. Fisher energy measures* $\varepsilon^\top G \varepsilon$ *and is large; first-order loss change measures* $\langle g, \varepsilon\rangle$ *and is near zero, because* ε *points across the level set rather than down it. The small parallel part* ($\langle g, \varepsilon\rangle / \langle g, u\rangle = +0.38$) *mildly* opposes *descent, so* $a\,\mathrm{sign}(u)$ *retains* 62% *of the descent pairing while shedding a slightly counterproductive residual. "58% of Fisher energy discarded" and "training unchanged" are therefore not in tension: they are the two faces of a transverse vector.*

The residual is structured, not noise: temporal autocorrelation 0.62 at lag one, decaying to noise by lag ten; it does not predict future gradients (≈ 0 at all future lags); and its magnitude correlates with r at 0.88. The energy is bimodal in r — 45% in the top decile (frozen coordinates, where one magnitude misfits a near-zero distribution) and 24% in the bottom two (active coordinates, where the flipping is itself the transverse motion), with a near-empty middle. The strong claim "$a \cdot \mathrm{sign}$ is the descent projection" is *not* supported: both u and its reconstruction are $\sim$95% transverse, and the sign step simply keeps enough of the descent sliver while the residual keeps almost none. What is established is weaker and more robust: Fisher-norm importance and gradient-alignment importance are different notions of importance, and the residual is large in the first and null in the second. Whether accumulated transverse motion has second-order consequences over long horizons is not settled by these instantaneous measurements.

4.9 Progression, and the object that survived

The programme began seeking a reduced observable from which weights could be computed without gradient descent. It ends with that goal refuted on two independent grounds and a narrower object in hand. The honest sequence:

Refuted, algebraic closures. The symmetric flow product does not close (0.7%, §1); no A_∞ correction exists (Stasheff residual 1.0); the Lie structure is local, decaying $30\times$ over 40 steps (§??); the tile data is not a vector bundle (cocycle fails 5–80%, plaquette holonomy 0.04–1.29); refinement and the Frobenius relation do not commute. Each was tested and failed, which is what excludes treating the surviving object as a static algebraic structure.

Refuted, the reduced-observable programme. β is cover-dependent ($\sim 9\%$ of per-parameter variance, §2.10) and carries no directional content (§2.9); the strand ultrametric is contiguity (permuted tree $\to$ 0.000, §2.11); the topological invariants vanish by construction ($K_1 = 0$ by Elliott, the index by dimension counting, $H^\bullet$ by contractible bases). The one closed-loop holonomy test in parameter space returned trivial ($w_1 = 0$, every block $+1$).

Established, and reproducible. The driftless-walk law and $E(64)$ ($\sim 50\sigma$, three seeds); the rank-one curvature response, which derives the scalar-per-block preconditioner; the Fredholm cokernel (59–72%, flat, largest at the minimum), which explains why block-local methods cannot close the gap; the LayerNorm spectral gap, surviving occupancy and weight/bias controls; and the frieze holonomy (§4.6), the sole non-forced obstruction.

The surviving object. It is not an algebra. It is a controlled stochastic process on the pair (r, s), where $r = |m|/\sqrt{v}$ is a scalar stability field and $s \in \{\pm 1\}^N$ the sign field, with the update reconstructed as $u_t \approx a_t\, s_t$. The dynamics live on (r, s), not on u. Five independently measured phenomena organise under r: flip hazard (AUC 0.809, §4.4), sign entropy (23% reduction, §4.7), plaquette curvature (localised at ∇r, §4.6), coordinate freezing (smooth saturation of the gap $1 - 2h(r)$), and residual magnitude (correlation 0.88, §4.8). Every attempt to give the object algebraic closure was measured and failed (Frobenius 0.7%, Stasheff residual 1.0, Lie decaying $30\times$, cocycle 5–80%, AU coherence); it is dynamical, not algebraic.

4.10 Controlled Sign Transport: the theory, and its regime

The measurements assemble into a single structure with a bounded small parameter. The transition law of the sign process factorises to leading order:

$$P(s_{t+1} \mid s_t, r_t) = K_{r_t}, \qquad K_r = K_r^{(0)} + \varepsilon\, K_r^{(1)},$$

where $K_r^{(0)} = \bigotimes_i \begin{pmatrix} 1-h(r_i) & h(r_i) \\ h(r_i) & 1-h(r_i) \end{pmatrix}$ is a product of independent two-state chains with hazard $h(r_i)$, and $K_r^{(1)}$ is a weak neuron-supported coupling. Each layer of this decomposition was reduced by a control, not proposed by analogy.

The backbone $K^{(0)}$ is trivial and carries most of the behaviour. Its spectral gap is $1 - 2h(r)$, which is the hazard curve rewritten, so no index-like transition beyond the freezing crossover exists there. The gap rises smoothly from 0.175 (lowest r-octile) to 0.974 (highest); freezing is a saturation, not a phase transition, and no critical r_c appears.

The coupling $K^{(1)}$ is the only non-factorised content. Conditioning on both coordinates' r, adjacent (same-neuron) coordinates still flip together 27% more than independence predicts, five times the residual of random distant pairs. Spectrally, 11 of 128 flip-covariance eigenvalues exceed the shuffled-independent null by 3σ (top mode 10.9σ), and their magnitude (23% excess) matches the pair coupling: the collective modes *are* the neuron coupling. The interesting operator theory is $\sigma(K^{(1)})$, not $\lambda_2(K)$.

The holonomy is abelian. The frieze transport maps commute to four decimals ($\|[M_i, M_j]\|/\|M_i\|\|M_j\| < 10^{-3}$), and the ordered loop product has the same eigenvalues as the order-free product of eigenvalues ($(1.000, 0.729)$ vs $(1.004, 0.726)$). The holonomy is therefore a rank-one dissipative phase — one fixed direction, one contraction to 0.73, one scalar per loop — and $F = dA$ with $A \wedge A \approx 0$: curvature comes from spatial variation of the scalar control, not from non-commutativity. Geometric language is available but optional; the object is one number attached to a loop.

Proposition 4.5 (The perturbative regime is the computational parameters)**.** *The coupling ε (relative joint-flip excess after conditioning on r) is small and stable across the transformer blocks and large in the embedding:*

tensor	ε early (steps 1–100)	ε late (101–200)
FF gate, L_2	24%	25%
FF out, L_2	11%	6%
W_Q, L_2	24%	27%
W_V, L_2	14%	12%
W_O, L_2	14%	12%
FF gate, L_0	24%	27%
FF gate, L_5	27%	31%
embedding	211%	267%

For the block parameters — the parameters implementing the function — $\varepsilon \in [0.06, 0.31]$, flat between early and late training, so the decomposition is genuinely perturbative and the sign process is dominated by r-controlled independent flips. For the embedding — the lookup table storing the single repeated corpus — $\varepsilon > 2$ and grows with training, so the coupling dominates and the independent-flip backbone does not describe the dynamics. The boundary was not imposed; it fell out of the sweep, and it separates distributed computation (perturbative) from associative memory (non-perturbative). Controlled Sign Transport holds in the former regime.

Generation versus transport. The representation (a, r, s) reconstructs the update but cannot be produced without the backward pass: r and s are outputs of backprop, not substitutes for it. This is the terminal statement of the programme. Computation separates into generation — which requires the gradient and remains the irreducible cost — and transport — which requires only (a, r, s) and is therefore cheap to store, transmit, and partially reuse. The two levers that survive a compute or bandwidth accounting are transport-side: stale-sign at matched backward passes (§4.1) and the 79× sign-field compression (§4.7). Neither reduces the cost of generation. The programme began seeking to replace gradient descent and ends having characterised, rather than removed, the one operation that cannot be replaced.

4.11 The backward pass: what is irreducible, and why

The generation–transport split of §4.10 leaves one question: can the generation step — the backward pass — itself be made cheaper? Eight experiments attack it from independent directions and return one wall.

Prediction and pruning fail. The next active frontier (coordinates whose sign is about to change) is predictable from pre-backward optimiser state only to AUC-equivalent recall 0.62 at 20% selection, and a linear model on $(r, \Delta r, |m|, v)$ adds nothing over r alone. Adding the current batch's forward activations — the most direct encoding of the incoming sample — leaves recall at chance for the batch feature, because a weight's gradient is $\delta_{\text{out}} \cdot \text{act}_{\text{in}}$ and the forward pass supplies only the second factor; the first is the backward pass itself.

Sketching fails on structured error. The update sign is invariant to i.i.d. gradient noise up to 25% (100% agreement) but breaks under rank-25% SVD truncation (82% agreement, 1.6× val cost) and under any low-rank surrogate. The enemy is correlated error, which is exactly what every compute-saving spatial approximation introduces. Neither a per-neuron mean-removal nor the graph-Laplacian basis of the flip-coupling graph recovers the signs, because that graph is dense and non-modular (degree 140 of 512, no spectral gap): there is no privileged basis to find.

Gating fails worse than random. Training under a rule that computes the exact gradient on a selected 20% and persists the strand elsewhere:

gate	granularity	val vs. full
random	coordinate	3.4×
r-gated	coordinate	6.9×
random	neuron-tile	3.4×
r-gated	neuron-tile	23×
momentum	neuron-tile	24×

Proposition 4.6 (Coupling-aligned gating is anti-optimal). *Random selection is the floor; every principled gate does strictly worse, and the damage scales with how tightly the gate concentrates the computed/persisted boundary onto the coupling structure. Neuron-tile gating, which aligns the cut with the between-neuron couplings, is worse than coordinate gating; both are worse than random, which has zero coupling-alignment. A coordinate's importance is non-local — set by its coupling to the coordinates one would skip — so any selection rule correlated with that coupling cuts precisely what must be preserved.*

The mechanism is the absence of scale separation. Multigrid and adaptive-mesh methods reduce cost when error is smooth with local corrections; the sign field is a densely coupled field with no hierarchy (no modularity, no spectral gap, no low-rank structure), which is exactly the regime where hierarchical and sparse methods provably do not apply.

Observation 4.7 (Terminal statement). *The strand — a temporally coherent, r-organised sign field — is the irreducible object of the optimisation. Backpropagation is the only known mechanism that generates it. Its cost cannot be reduced by sparsity in any form (coordinate, tile, predicted, or sketched), because importance is non-local: dense, non-hierarchical coupling without scale separation. What remains compressible is transport, not generation: stale-sign across steps (§4.1) and sign entropy across coordinates (§4.7), both operating on the already-measured field. Whether a different* dense *mechanism — feedback alignment through the recursion, forward-gradient, or a synthetic backward field — can generate the strand more cheaply while preserving weighted sign agreement above* $\sim$0.95 *through network depth under correlated error is the single question this programme leaves open. Sparsity-based economy is refuted; substitution is untested.*

The programme set out to ask whether the disc–strand decomposition could replace gradient descent. The answer is more precise than a refusal: disc and strand are a complete, compressible, transportable description of what gradient descent *produces*, and the reason they cannot replace what produces it is a measurable structural fact about the loss geometry — non-locality without scale separation — rather than a limitation of the representation.

5 Phase 4: Corpus Generality of the High-Dimensionality Result

Every measurement to §4.11 was taken on one corpus: a single sentence looped roughly 300 times. The central negative finding — that the strand field is high-dimensional and therefore does not compress across corpus states — was therefore open to the objection that it is an artefact of that triviality. A structured corpus might make the batch → strand map low-dimensional, which would reopen dictionary compression and make the sparsity refutation corpus-specific rather than general. This section settles that objection.

5.1 The structure axis and its confound

Six corpora at $V = 256$, 60,000 tokens each, spanning a structure axis validated two ways: gzip compression ratio, and the block-entropy gap at $L = 6$ against each corpus's own token shuffle. The shuffle controls are the only confound-free comparison available, since they hold vocabulary and unigram statistics fixed and vary order alone.

corpus	live vocab	gzip ratio	blockH gap
`iid`	256	1.001	0.000
`english_shuf`	27	0.631	−0.000
`english`	27	0.282	0.362
`code_shuf`	95	0.663	−0.000
`code`	95	0.259	0.436
`repeat`	97	0.007	1.494

The live-vocabulary column carries a confound that governs how the results may be read. Vocabulary size is not constant along the axis — it runs $256, 27, 95, 97$ — so the cross-corpus ordering conflates structure with alphabet size and unigram entropy. Only the paired contrasts (`english` versus `english_shuf`, `code` versus `code_shuf`) isolate structure. The cross-corpus trend is reported below but is not load-bearing.

5.2 Protocol, and two corrections to it

For each corpus: train to a checkpoint at 80 steps, freeze θ, sweep N_B batches, and record $s_B = \operatorname{sign}(g_B)$ on a fixed random subset of 20,000 block-parameter coordinates together with a batch embedding z_B (mean token embedding concatenated with the normalised token histogram). Predict s_B from z_B linearly, nonlinearly by random cosine features, and by k-nearest neighbours; report the participation ratio of the strand spectrum and the correlation between batch distance and strand dissimilarity.

Two corrections were necessary, and both change what the numbers mean.

First, the original comparison was between two interpolators. With z_B of dimension $D+V = 512$ and $N_B = 200$ giving 150 training rows, both the linear least-squares fit and the 800-dimensional random-feature fit are underdetermined minimum-norm solutions. The earlier reading — nonlinear 0.535, "overfits," against linear 0.699 — is therefore substantially a statement about minimum-norm behaviour at $p \gg n$, not about the intrinsic dimension of the map. Here $N_B = 600$, giving 360 training rows, and both model classes are placed on a ridge path with the penalty selected on a held-out split, so each is given its best available fit. Untuned numbers are reported alongside for continuity.

Second, the initialisation must not depend on the corpus. The compiler prefix constructs the embedding from the corpus bigram Laplacian via $\operatorname{eigsh}(L_{\text{sym}}, k = D{+}1)$. At $V \leq 256$ this is not merely inadvisable but impossible, since $k = 257$ exceeds the matrix dimension; more importantly it would make θ_0 a function of the corpus, and differentially so — `english` has 27 live tokens against `iid`'s 256, so the graph is degenerate to different degrees along the axis. A probe asking whether corpus structure lowers the dimension of the batch → strand map cannot have corpus structure pre-loaded into the initialisation at corpus-dependent strength. All runs therefore use $V = 256$ fixed and a plain seeded initialisation, identical across corpora. This forfeits comparability with the historical $V = 1017$ numbers, which is the correct trade for a cross-corpus comparison and the wrong one for continuity with earlier phases.

5.3 Result: the wall holds

Block parameters, $N_B = 600$, split 360/90/150, single seed.

corpus	lin (raw)	nl (raw)	lin (ridge)	nl (ridge)	gain	kNN_{15}	PR
`iid`	0.587	0.527	0.729	0.597	−0.133	0.646	208.0
`english_shuf`	0.719	0.579	0.842	0.663	−0.180	0.751	69.9
`english`	0.748	0.639	0.849	0.716	−0.134	0.787	65.1
`code_shuf`	0.670	0.590	0.817	0.672	−0.145	0.736	80.4
`code`	0.572	0.557	0.691	0.652	−0.039	0.663	243.6
`repeat`	0.745	0.661	0.865	0.756	−0.109	0.769	57.3

Observation 5.1 (The discriminator returns *no*)**.** *The nonlinear model does not beat the linear model on any corpus on the axis. The gain is negative throughout, from* −0.039 *to* −0.180*, and k-nearest neighbours — which requires no regularisation choice and is therefore the independent check — tracks the linear model from below everywhere. Two structurally different non-parametric estimators both fail to beat a linear map. The high-dimensionality of the batch → strand map is not an artefact of the degenerate corpus, and dictionary compression does not reopen.* Extended in §8.5*: the same gain, measured on the authentic corpus at five checkpoints from the drift-dominated regime to the memorised minimum, never approaches zero. The finding holds across corpora and across regimes.*

Ridge tuning raised the linear model substantially (0.748 → 0.849 on `english`) and raised the nonlinear model too, without closing the gap: the earlier "nonlinear overfits" reading was partly an artefact of the minimum-norm fit, and the conclusion survives correcting it.

The paired contrasts are where caution is required.

pair	lin (ridge)	nl (ridge) → gain	PR
`english_shuf→english`	0.842 → 0.849	−0.180 → −0.134	69.9 → 65.1
`code_shuf→code`	0.817 → 0.691	−0.145 → −0.039	80.4 → 243.6

Structure moves the gain toward compressibility in both pairs (+0.046, +0.106), so this is not a flat null; but the sign never reverses, and the two pairs disagree on the dimension measure. On `english` the participation ratio falls slightly with structure, as the dictionary hypothesis predicts. On `code` it triples, while linear predictability falls from 0.817 to 0.691 and the distance–dissimilarity correlation falls from 0.561 to 0.318. Real code structure makes the strand field *less* predictable and *higher*-dimensional — the opposite of a low-dimensional curved manifold, and the single most informative number in the table.

Observation 5.2 (Structured but not manifold-structured)**.** *Participation ratios of* 57–244 *against a ceiling of* 600 *are well below what independent sign vectors would give. The strand field carries real redundancy, consistent with the* 0.405 *bits/parameter/step of §4.7; it is simply not organised as a low-dimensional curved base. High-dimensional and compressible are not in tension — this is the transport/generation split of §4.11 seen from the corpus side.*

5.4 Replication

The four corpora entering the paired contrasts were rerun at a second seed.

pair	Δgain (s17)	Δgain (s23)	ΔPR (s17)	ΔPR (s23)
`english_shuf`→`english`	+0.046	+0.016	−4.8	−6.3
`code_shuf`→`code`	+0.106	+0.097	+163.3	+179.7

The sign of every gain replicates and no gain approaches zero, so Observation 5.1 is not seed-dependent. The two paired deltas behave differently under replication: the `code` delta is stable (+0.106, +0.097) while the `english` delta is not (+0.046, +0.016), so only the former should be treated as a measured effect. The divergence of the two pairs on the dimension measure replicates cleanly and is the more robust finding: `english` loses a few units of participation ratio with structure, `code` gains over 160.

Limitations. Two seeds, one checkpoint at 80 steps, one architecture. A subsequent audit (§8.1) established that at this checkpoint the model has learned only token frequencies on `english` and reaches its corpus floor only on `code` and `repeat`, so four of the six corpora were probed on a representation close to unigram. That concern is *superseded* by §8.5, which reruns the identical probe at five checkpoints spanning val 3.0 → 0.07 on the authentic corpus and finds the gain invariant in $[-0.171, -0.092]$. The result is corpus-general and regime-general. The negative result is a statement about one nonlinear family — 800 random cosine features — and the agreement of kNN is what licenses the broader reading. The embedding arm was measured separately and is reported nowhere in this section, because `head.weight` is tied to `te.weight` and its gradient is dominated by a nearly batch-independent softmax term: that arm measures weight tying, not associative memory, and testing the ε-table confound would require an untied model.

6 Phase 5: Symmetry, and the Stabilizer of the Strand

If no low-dimensional manifold organises the strand, a weaker and more robust object may still exist: a group of transformations under which the strand is approximately invariant. Invariance does not require low-dimensional coordinates, and a stabilizer subgroup would induce equivalence classes on batch space that are cheaper to specify than the strands themselves. This section constructs the apparatus to test that and reports what it returns.

6.1 Synthetic corpora with exact symmetries

Symmetry can only be tested where it is exact by construction, so the corpora are generated from an explicit parse tree over a byte-level expression grammar,

$$S \to E\,; \qquad E \to T \mid E{+}T \mid E{-}T \qquad T \to F \mid T{*}F \qquad F \to V \mid D \mid (E)$$

with $V \in \{a..h\}$ and $D \in \{0..9\}$. Three exact symmetries of the generator are available, with deliberately different statistical profiles:

transform	parse tree	token statistics	surface altered
`rename`	identical	identical	85%
`comm`	mirrored	≈ identical	78%
`ws`	identical	*changed* (gzip 0.503 → 0.336)	84%

`rename` applies a bijection to the variable alphabet; `comm` swaps the operands of the commutative operators; `ws` inserts whitespace around binary operators, leaving the parse tree untouched while altering the byte statistics substantially. Both `rename` and `comm` preserve length exactly, so paired batches align token for token. A structural ladder of known distance (one terminal, one operator, one subtree, full resample) and two controls complete the suite: `shuf` destroys structure while preserving the histogram, and `lexctl` carries `rename`'s histogram with its structure destroyed.

`lexctl` is the control that makes the experiment mean anything. Without it, statistical similarity is indistinguishable from symmetry.

6.2 The stratification, and why the prediction failed

The intended discriminator was stratification by the disc. Under the reading of §4.4, coordinates above the hazard threshold should have disc-determined sign and therefore agree between *any* two batches, while coordinates below it should agree at chance; only the middle band is batch-conditioned. A genuine symmetry invariance would then have to show its excess agreement in the middle band, and excess in the top band would indicate disc-determined agreement being misread as invariance.

Measured, the independent-batch floor is flat across r deciles: 0.587, 0.591, 0.588, 0.585, 0.582, 0.591, 0.595, The excess for every condition is largest in the *top* band, which is the signature the prediction identified as spurious.

The diagnostic explains this, and the explanation is a statement about regime rather than a rescue. On block parameters at this checkpoint the disc field has median $r = 0.001$, $r_{90} = 0.061$, $r_{99} = 0.247$, maximum 0.879, with $r > 0.5$ holding for 0.09% of coordinates. Since $r \to 1$ is the consistent-gradient limit, essentially every coordinate is noise-dominated: there is no deterministic regime for the top decile to occupy. The p_{90}/p_{10} spread is 159×, consistent with the 160× monotone range of §4.4, and agreement does rise monotonically with decile ($0.587 \to 0.610$).

Observation 6.1 (Assumptions unsatisfied, not prediction falsified)**.** *The experiment sampled one asymptotic regime of the theory. The two-phase prediction presupposes a deterministic population that this operating point does not contain, so it is untested rather than confirmed or refuted. What the data do show is that* $\partial_r P\left(s_i^{(A)} = s_i^{(B)}\right) > 0$ *already holds* inside *the purely stochastic regime — a weak monotone law, and the kind of quantitative statement a theory of the disc ought eventually to derive.* Resolved in §8.4*: the deterministic population is real but early. On the programme's own corpus* 35.5% *of block coordinates exceed* $r = 0.5$ *at step* 10*, falling to* 0.9% *by step* 200*, with* $\mathrm{corr}(val, r_{90}) = +0.749$ *pooled across corpora. This* 80*-step checkpoint sits at the depleted end, so the two-phase prediction was tested in the one regime where it has least purchase.*

6.3 Result: the stabilizer is trivial at this operating point

Twenty-four trials, each comparing a base batch against matched variants.

condition	agreement	excess	positional identity	histogram cosine
`comm`	0.887	+0.293	0.444	1.000
`pert1`	0.814	+0.220	0.901	0.997
`rename`	0.770	+0.176	0.778	0.995
`pert2`	0.767	+0.173	0.907	0.990
`lexctl`	0.747	+0.153	0.074	0.995
`pert3`	0.692	+0.098	0.150	0.990
`pert4`	0.607	+0.013	0.140	0.982
`indep`	0.594	0	0.133	0.981
`ws`	0.506	−0.088	0.037	0.361

Three readings, which agree with each other.

Observation 6.2 (Alpha-renaming is not a symmetry of the strand)**.** *`rename` preserves the parse tree exactly and changes* 85% *of the surface, giving agreement* 0.770*. Its structure-destroyed control at the same histogram gives* 0.747*. The apparent invariance decomposes into* 0.153 *of* *histogram effect and* 0.023 *of structure effect. What looked like structural invariance is low-order statistical similarity, which shuffling preserves.*

Observation 6.3 (The strand follows surface statistics, not syntax)**.** *`ws` leaves the parse tree identical and alters only surface statistics. Its agreement is* 0.506*, which is* 0.088 below *the unrelated-batch floor: a transformation that does not touch the grammar destroys strand agreement more thoroughly than an unrelated batch does.*

Observation 6.4 (The structural ladder is a surface-distance ladder)**.** *Across the nine conditions, agreement correlates* $+0.66$ *with positional identity and* $+0.66$ *with histogram cosine. The monotone decay along the perturbation ladder (*$0.814, 0.767, 0.692, 0.607$ *against a floor of* 0.594*) coincides with positional identity falling from* 0.90 *to* 0.15*. Once `comm` (*0.887 *at positional identity* 0.444*, histogram cosine* 1.000*) and `ws` (*0.506 *at* 0.037*,* 0.361*) are included, the pattern resolves without a residual structural term: histogram similarity dominates and positional identity modulates.* None of the conditions in this table is matched on surface perturbation*; §8.3 imposes that control and finds the resulting structural term to be zero early in training, so the orderings here should be read as surface-distance orderings and not as evidence about symmetry.*

6.4 Does the stabilizer grow with training? Inconclusive, and why

A stabilizer measured at one checkpoint is a snapshot of a possibly moving object. The natural follow-up is to hold the symmetry suite fixed and sweep it along a training trajectory, watching the gap `rename` − `lexctl` — structural invariance net of histogram — as a function of step. One run, checkpoints at $20, 40, 80, 160, 320, 640, 1280$, twelve trials each, measurement never calling an optimiser step.

step	val loss	r_{90}	$\Pr[r > 0.5]$	`rename`	`lexctl`	gap
20	2.934	0.187	0.0053	0.817	0.775	+0.042
40	2.926	0.057	0.0013	0.790	0.772	+0.018
80	2.928	0.040	0.0004	0.797	0.775	+0.022
160	2.917	0.056	0.0006	0.743	0.736	+0.007
320	2.922	0.148	0.0072	0.739	0.725	+0.014
640	2.913	0.085	0.0006	0.749	0.711	+0.039
1280	2.917	0.270	0.0240	0.762	0.695	+0.066

Observation 6.5 (The sweep is inconclusive by construction)**.** *The unigram entropy of the corpus is* 2.914 *nats and the validation loss sits at* 2.913–2.934 *for the entire run. The model learned the token frequencies and nothing further; it never acquired the grammar. A measurement of whether the stabilizer acquires grammatical invariance therefore had no opportunity to register one, and the apparent upward drift of the gap (*$+0.042 \to +0.066$*) sits within noise at twelve trials, where the per-condition standard deviation is approximately* 0.05.

The cause is a design fault in the grammar rather than a property of strand generation: terminals are drawn uniformly, so the syntax carries almost no predictive mass relative to the random leaves, and a model at the unigram entropy is already optimal against everything except the operator skeleton. The fix is a generator whose structure is load-bearing for the loss — skewed terminal distributions, or dependencies between leaves — so that val loss falls materially below unigram entropy and there exists a learned representation for a stabilizer to be a stabilizer of. Until that is run, the question of §7.6 is open.

Scope. All Phase 5 statements are conditioned on an operating point at which the model has demonstrably not learned the grammar. The correct reading of §6.3 is therefore not that the symmetry group of the strand is trivial, but that *at this operating point strand similarity is governed by low-order statistical similarity rather than grammatical equivalence.* Whether a model that had learned the grammar would exhibit grammatical invariance is untested. The methodological result — that without a histogram control one will mistake statistical similarity for symmetry — is not conditioned on the operating point and is the more portable finding.

7 Toward a Theory: What the Measurements Constrain

What follows is reformulation, not evidence. It reorganises measurements already reported and adds no independent confirmation of any of them. Its purpose is to record which mathematical descriptions are consistent with the measurements, which are excluded by them, and — for the parts that remain conjectural — what would decide them. Status is marked throughout.

7.1 Two retractions

Two claims made in the course of this programme's discussion were stronger than the evidence and are withdrawn.

"There is no geometry." Dense coupling without a spectral gap (§4.11) excludes *spatial locality in the parameter index.* It does not exclude geometry. Spin glasses, kernel methods, Gaussian processes and random matrix ensembles all combine dense interaction with rich spectral, statistical or information-geometric structure. The defensible statement is the narrow one: there is no useful metric on the parameter index.

"The correspondence with kinetic Ising is exact." Glauber dynamics is defined by an energy function, and no Gibbs form for K_r has been exhibited. The dictionary — strand to spin configuration, r to quenched site disorder, ε to weak coupling, the trivial product backbone to the non-interacting limit — is an *effective model*, useful for the predictions it generates and not an identification. Demonstrating that the measured transition probabilities admit an effective Hamiltonian is a specific open task, not a formality.

7.2 The state space

The measurements do not support a geometry on parameters alone. The strand depends jointly on model state, optimiser state and batch, so the object to be given structure is

$$\mathcal{X} \times \Theta \times \mathcal{O} \longrightarrow \{\pm 1\}^P,$$

with $\mathcal{X}$ the batch space, Θ the parameters and $\mathcal{O}$ the optimiser moments. In this factorisation the disc is a slowly varying field over $\Theta \times \mathcal{O}$, the corpus selects a trajectory through $\mathcal{X}$, and the strand is the response field. This accommodates directly what §5 and §6 measured: corpus identity strongly conditions strand identity, while the optimiser state governs churn.

7.3 The parameter index is a partition with a gauge group

Status: argued from architecture, with one testable consequence outstanding. The architecture carries exact permutation symmetries — the feed-forward hidden units, the attention heads, and the residual-stream coordinates may be relabelled with matched relabelling of the matrices that read and write them, leaving the represented function unchanged, and LayerNorm does not break this. Any statement of the form "index i neighbours index j" is therefore a property of the labelling and not of the network, and no observable may depend on it.

The sign map is equivariant under the group $\mathcal{D}$ generated by positive diagonal rescalings and signed permutations, and under no larger subgroup of $GL(P)$. This gives a compact reading of three results already recorded: sign survives 25% i.i.d. noise (a within-chart perturbation) while breaking under rank truncation (which leaves $\mathcal{D}$), and the permuted refinement ladder collapses to 0.000 (§2.11) because the permutation was not a signed permutation of the disc chart.

What replaces the lattice is a set with a nested architectural partition — network, layer, matrix, row, parameter — carrying a permutation gauge group acting within each cell. Nesting is gauge-invariant; adjacency within a level is not. The consequent reinterpretation of the contiguity result is that contiguity in the flat parameter vector is largely co-membership in a matrix and row, so that the 0.000 of the permuted ladder records the gauge symmetry appearing as a null rather than the destruction of a metric. *This is a prediction and it is untested*: conditioning on same-matrix and same-row membership should remove the residual contiguity effect. If it does not, the reinterpretation is wrong and there is genuine within-row geometry.

7.4 Disc Riemannian, strand Finsler

Status: derivation from the measured update law; consistent with all reported measurements, independently confirmed by none. Steepest descent under a norm constraint, $u = \arg\min_{\|u\|\le a}\langle g, u\rangle$, gives $u \propto -g$ under ℓ^2, $u \propto -G^{-1}g$ under a Riemannian metric G, and $u_i = -a\,\mathrm{sign}(g_i)$ under ℓ^∞, whose unit ball is the hypercube and whose extreme points are exactly the sign vectors. The measured law $u_t \approx a_t s_t$ of §3 is the third of these.

This is the reason a Riemannian formulation of the *strand* cannot be correct: a metric predicts a dense real-valued natural-gradient update, which is measurably not what the optimiser produces. It also explains why the strand is not a tensor — gradients are covectors, and $\mathrm{sign}((D\varphi)^{-T}g) \neq \mathrm{sign}(g)$ in general, so the object that survived the programme is precisely the object that does not transform covariantly.

The Fisher metric belongs instead to the disc, where it is already present implicitly: v_i estimates the diagonal empirical Fisher $\mathbb{E}_B[g_i^2]$, so $G_\theta = \mathrm{diag}(\sqrt{v} + \epsilon)$ is a metric on Θ and $r_i = |m_i|/\sqrt{v_i}$ is the drift-to-noise ratio of coordinate i measured in it. The hazard law of §4.4 then reads as a statement about signal-to-noise: flips occur where Fisher-normalised drift is dominated by Fisher-normalised noise, which predicts both the monotone flip rate and a threshold. *Partly refuted in §9.2*: the monotonicity holds robustly (AUC 0.81–0.86), but the threshold is not a drift/noise crossing — it sits an order of magnitude below $\rho = 1$ and is stationary in neither value nor rank. The signal-to-noise reading survives for the monotone part of the law and not for the threshold. The disc is not a companion to the strand but the structure that makes the strand well defined: $\mathcal{D}$ is the group preserving diagonal Fisher structure, and without it there is no sign field to speak of.

7.5 Frameworks excluded, and by which measurement

Status: exclusions, each traced to a specific measurement. Recording these prevents their recurrence.

framework	requires	excluded by
multigrid / refinement transport	scale separation	no spectral gap (§4.11)
persistent homology	filtration with separated scales	no spectral gap
lattice field theory	metric on the index	permutation gauge (§7.3)
trajectory homology	non-contractible complex	trajectory is a path
index theory	stability under compact perturbation	rank truncation breaks sign
K-theory	additive or exact category	free category on a path

Three of these fail for one reason. The index-theoretic case is worth stating separately: an index is invariant under compact perturbation, rank truncation is finite-rank hence compact, and §4.11 measured that the sign field breaks under rank truncation while surviving i.i.d. noise — so the persistent/frontier decomposition has the opposite stability property to an index. On K-theory, strands do not compose within strands, so the trajectory generates a free category with no additive structure and K_0 is undefined; passing to representations of the trajectory quiver gives $K_0(\operatorname{rep} A_n) \cong \mathbb{Z}^n$, which counts steps. The honest position is not that K-theory is dead but that no canonical object has appeared to which it applies.

The cohomological content that does exist is already computed. The sign process lives on $\{\pm 1\}^P = (\mathbb{Z}/2)^P$, which is abelian, so abelian holonomy is forced; the natural home for the non-forced part is $H^1(\cdot\,; \mathbb{Z}/2)$, and the frieze cover of §4.6 is the one non-forced class in it.

7.6 The stabilizer programme

Status: conjectural, and the one experiment attempted was inconclusive (§6.4).

Phase 5 sought a fixed symmetry group of the data generator and found none. The more defensible object is not fixed. The strand is a function of gradients, gradients are functions of activations, and activations depend on whatever statistics the representation has actually learned; so any invariance of the strand is an invariance of the *learned representation*, not of the generator. Early in training that is token statistics, which is exactly what §6.3 measured. Whether it later becomes syntax is undetermined.

This reframes the question. Rather than asking which transformations preserve the strand, ask how the stabilizer subgroup

$$\mathrm{Stab}_t = \{\sigma :\ s_{\sigma(B)} \approx s_B \text{ at } (\theta_t, \mathcal{O}_t)\}$$

evolves along the trajectory. This is a geometric object that changes with training, and it connects the observations without assuming a fixed group: early training dominated by low-order statistical invariants, later training possibly acquiring higher-level ones. It also carries an immediate falsifier — if Stab_t never grows on a corpus whose structure the model demonstrably learns, then something stronger holds about strand generation than any statement in this programme so far.

The prerequisite is a corpus on which the model learns structure at all. The sweep of §6.4 failed that prerequisite, and the suite — symmetries exact by construction, histogram control, structural ladder, r-stratification — is in hand and unchanged for the rerun.

7.7 What the programme now claims

The disc is the slow state: optimiser memory, a diagonal Fisher metric, accumulated statistics over $\Theta \times \mathcal{O}$. The strand is the fast binary response field generated from the current batch, living on the vertices of an ℓ^∞ ball, well defined only relative to the chart the disc supplies. The dynamics respect the permutation gauge group of the parameterisation, so the parameter index is a partition rather than a space. The corpus acts on the strand through whatever invariants the current representation has learned, and the symmetry structure is therefore a function of training time rather than a property of the data.

Two things are established across corpora rather than on one: that the batch $\to$ strand map is high-dimensional (§5), and that sparsity-based economy is refuted in consequence. The remaining mathematical target is the smallest symmetry-respecting state space on which disc and strand evolve — and the remaining empirical question is unchanged from §4.11: whether a dense substitution mechanism can generate the strand more cheaply. Sparsity-based economy is refuted; substitution is untested.

8 Phase 6: Training Adequacy, the Disc Regime, and a Surface-Matched Stabilizer

Three experiments were run to test the load-bearing conjectures of §7: the contiguity reinterpretation of §7.3, the stabilizer programme of §7.6, and the two-phase reading of the disc that §6.2 left unresolved. One returned a null on its own premise, one returned a decisive methodological correction with a first positive signal, and one returned a negative result that constrains §7.4.

8.1 An audit that reframes Phases 4 and 5

Achieved loss at the 80-step checkpoint, compared against the context-entropy floors of each corpus:

corpus	H_0	H_1	achieved	status
`repeat`	4.513	0.274	0.089	learned
`code`	3.267	2.380	2.233	learned
`english`	3.151	2.775	3.157	*at unigram entropy*
`cfg`	2.914	2.208	2.917	*at unigram entropy*
`iid`, shuffles	—	—	$\approx H_0$	correct; no structure

Training is not defective — `repeat` and `code` are learned, the latter below its own bigram floor — but on `english` and `cfg` the model at 80 steps has learned token frequencies and nothing more. *Four of the six corpora entering §5 were therefore probed on a model that had not learned them.* This was not known when that section was written and it conditions Observation 5.1: the high-dimensionality of the batch $\to$ strand map is established across corpora, but at an operating point where the representation is close to unigram on most of them. The result may well be robust to training — `code` and `repeat` were learned and behave no differently — but that is an inference, not a measurement.

8.2 Experiment 1: contiguity has nothing to condition

§7.3 predicted that conditioning on same-row membership would remove the residual distance effect underlying the contiguity result. At a trained checkpoint on `bind` (val 0.791, 2000 steps), across-batch sign correlation on `blocks.0.ff.v.weight` (512×256) as a function of flat-index distance:

d	1	2	4	8	16	32	64	128
corr	0.003	0.009	0.007	0.016	0.010	-0.008	0.001	-0.010

against a random-pair baseline of $+0.0009$. Within-row correlation over $d = 1 \ldots 256$ spans -0.018 to $+0.016$; same-column pairs at row-distance $1, 2, 4, 16, 64$ give $+0.002, +0.009, +0.005, -0.003, -0.006$. Everything is indistinguishable from zero.

Observation 8.1 (Null on the premise, and on the wrong quantity)**.** *There is no contiguity effect in the batch-conditioned sign field at this checkpoint, so the prediction of §7.3 is neither confirmed nor refuted. Separately, the experiment measures across-batch sign correlation at frozen θ, whereas §4.5 measured* flip *clustering between consecutive training steps. These are different objects, and the correct test of the contiguity claim is the latter. §7.3 remains untested.*

8.3 Experiment 2: the surface-matched control

Three separate results — the perturbation ladder of §6.3, the `rename`/`lexctl` gap, and the `break` ordering on `bind` — were governed by surface distance rather than structure. The defect is common to all three: the structure-preserving and structure-destroying conditions perturb different numbers of bytes, so agreement orderings track positional identity. The correction holds the perturbation exactly fixed.

On the `bind` grammar, records `let a=4;let f=6;a+f;` carry four variable slots. `rename` applies a fixed-point-free bijection to all four, changing four bytes and *preserving* the binding. `scramble` replaces each slot by an independent draw excluding the original letter, changing four bytes and *destroying* the binding. Positional identity is matched by construction; measured, it is 0.803 for both at every checkpoint. Their difference is structure with surface held fixed.

	step	val	rename	rename2	scramble	break	ws	indep	gap
	200	0.756	0.911	0.914	0.924	0.939	0.652	0.897	−0.013
	800	0.737	0.840	0.836	0.840	0.851	0.531	0.836	−0.000
	3200	0.723	0.755	0.737	0.699	0.686	0.492	0.594	+0.056
positional identity			0.803	0.803	0.803	0.902	0.179	0.706	

Observation 8.2 (The earlier gaps were surface artefacts). *At* 200 *and* 800 *steps the matched gap is* -0.013 *(*$z = -1.60$*) and* -0.000 *(*$z = -0.01$*): with surface distance held fixed, preserving the binding confers no advantage whatever. The large gaps reported for* `bind` *under the unmatched protocol (*$+0.21$ *to* $+0.44$*) do not survive the control, and neither does the* `rename`*/*`lexctl` *construction that produced them.* `lexctl` *is in any case the wrong baseline on a structured corpus, where shuffling drops agreement* below *the unrelated-batch floor (*0.456 *against* 0.811*).*

Observation 8.3 (A first positive signal for the stabilizer, at 3200 steps). *The gap is monotone in training —* -0.013, -0.000, $+0.056$ *— reaching* $+0.056 \pm 0.026$ *(*$z = +2.13$*) at* 3200 *steps. The stronger evidence is qualitative and independent of that* z*: at* 3200 *the ordering inverts against surface distance.* `break` *has* higher *positional identity (*0.902 *against* 0.803*) yet* lower *agreement (*0.686 *against* 0.755*), which no surface-distance account can produce. The second bijection* `rename2` *tracks* `rename` *(*0.737 *against* 0.755*), as a structural effect requires and an idiosyncratic one would not.*

This is the first measurement in the programme favouring §7.6: an invariance that is absent early, appears as the representation improves, and survives a surface-matched control. It should be held lightly. Three checkpoints, one seed, twenty trials, and marginal significance at the single point where the effect appears; the independent floor also moves sharply over the same interval ($0.897 \to 0.594$), so the excesses over floor ($+0.161$ for `rename`, $+0.104$ for `scramble`) rest on a shifting baseline. Replication at further seeds and checkpoints beyond 3200 is the obvious next step, and the prediction is specific: the gap should continue to grow as val loss falls below the binding-free floor of 0.695.

8.4 Experiment 3: the disc regime is a function of training progress

The two-phase reading of the disc requires a population of coordinates with r near unity. Measured along a training run on `bind` that does learn (val $2.776 \to 0.716$, passing $H_1 = 1.066$ by step 100):

step	val	r_{90}	$\Pr[r > 0.5]$	$\Pr[r > 0.9]$
50	1.002	0.252	0.0049	0.00000
100	0.809	0.255	0.0031	0.00000
200	0.763	0.308	0.0103	0.00005
400	0.753	0.277	0.0025	0.00000
800	0.731	0.312	0.0062	0.00000
1600	0.766	0.402	0.0307	0.00000
3200	0.716	0.153	0.0029	0.00000

On this evidence alone the deterministic regime appears unreachable. That reading was published in an earlier draft and is *withdrawn*. It is an artefact of where on the trajectory these checkpoints sit.

Reconstructing the programme's own corpus from `build_corpus.py` (VOCAB = 1017, base sequence 1364 tokens × 300 reps) and running the authentic prefix — at VOCAB = 1017 the spectral E_0 initialisation is well posed, since $k = D{+}1 = 257 < 1017$ — reproduces the reported operating point exactly: val 0.0652 at step 200 against the compiler's own anchor `val_floor`= 0.062. Along that trajectory:

step	val	r_{50}	r_{90}	$r_{\max}$	$\Pr[r > 0.5]$	$\Pr[r > 0.9]$
10	2.992	0.379	0.757	1.025	0.3547	0.0177
20	2.122	0.303	0.671	1.074	0.2469	0.0085
50	0.809	0.207	0.492	1.042	0.0943	0.0007
100	0.180	0.145	0.350	0.808	0.0156	0.0000
150	0.092	0.147	0.353	0.756	0.0173	0.0000
200	0.065	0.134	0.323	0.837	0.0086	0.0000

Observation 8.4 (The deterministic population exists, and is early)**.** *r decreases monotonically as the model learns. At step 10, 35.5% of block coordinates lie above $r = 0.5$, 1.8% above 0.9, and $r_{\max}$ exceeds unity; by step 200 these are 0.9%, zero, and 0.84. Pooling all thirteen checkpoints across both corpora, $\mathrm{corr}(val, r_{90}) = +0.749$: the disc regime is a function of training progress and not of corpus. The mechanism is immediate — at a minimum the systematic gradient component vanishes while batch noise does not, so $m \to 0$ with v held at the noise floor and $r = |m|/\sqrt{v} \to 0$. High r means strong consistent drift, which is a property of early training.*

The `bind` table above therefore samples the depleted end: on that corpus val collapses from 2.776 to 1.002 within fifty steps, so the drift-dominated phase had already passed before the first measurement. The correct statement is that the deterministic population exists transiently early in optimisation and does not induce a low-dimensional update manifold — regime-dependence, not absence. Observation 8.6 establishes the second half: the drift-dominated checkpoints are no more compressible than the noise-dominated ones.

The consequence for §7.4 is specific and should be recorded. Reading the hazard law as a drift-versus-noise threshold in the Fisher-normalised ratio predicts a threshold at $r \sim 1$; the measured threshold of §4.4 sits at the 70th percentile of the r distribution, and that percentile moves by more than a factor of two along the trajectory (r_{70} falls with r_{50} and r_{90} throughout). The derivation is therefore consistent with the measured monotonicity, and the threshold is well posed only relative to a checkpoint. Any statement of the hazard law that quotes a fixed percentile without quoting the checkpoint is under-specified; this applies to §4.4 as written.

r is a bounded coherence coordinate

One definitional point governs how r may be read, and it corrects §7.4 as well. Writing $D_\mu = |\mathbb{E}[g]|$ for the drift field and taking $D_\sigma = \sqrt{\mathbb{E}[g^2]} \approx \sqrt{v}$ for the noise field is not right: $\mathbb{E}[g^2] = \mathrm{Var}[g] + \mathbb{E}[g]^2$, so $\sqrt{v}$ is the second moment. By Jensen,

$$r_i \;=\; \frac{|m_i|}{\sqrt{v_i}} \;\le\; 1$$

identically. Measured on the authentic corpus, $r > 1$ holds for 0.018% of block coordinates at step 10, 0.058% at step 20, and 0% from step 50 onward, the small excess being EMA bias (m and v use different β).

Observation 8.5 (r is an order parameter, not a ratio)**.** *r is a bounded coherence coordinate on $[0, 1]$, so "the deterministic regime at $r \approx 1$" names the boundary of the coordinate rather than a region within it — which is why the stratification prediction of §6.2 had no purchase. The unbounded drift-to-noise ratio is $\rho_i = |m_i|/\sqrt{v_i - m_i^2}$. For the bulk the two coincide, since $r \ll 1$ makes $m^2 \ll v$: $\rho_{50} = 0.412, 0.316, 0.209, 0.156, 0.144$ against $r_{50} = 0.381, 0.302, 0.205, 0.154, 0.142$ at steps $10, 20, 50, 100, 200$. They diverge only in the tail the two-phase reading concerns. In ρ the statement is clean: a genuinely drift-dominated population ($\rho > 1$) exists at steps 10–20 ($\rho_{99} = 2.44$, 1.86) and is gone by step 100 ($\rho_{99} = 0.679$).*

8.5 Experiment 4: the regime test

Observation 8.4 raises a direct threat to §5. If the disc field is at its minimum at the programme's operating point — $r_{50} = 0.134$, drift-to-noise about 1:7 — then every Phase 1–3 measurement was taken on the sign of a predominantly noise-driven gradient at a memorised minimum, and the high-dimensionality of a noise sign-field would be a much weaker finding than the high-dimensionality of a learning signal.

The test holds everything fixed except the checkpoint: same corpus, same trajectory, same coordinate subset, same $N_B = 500$, same ridge-tuned estimators, probed at five points spanning the drift-to-noise transition.

step	val	r_{90}	$\Pr[r > .5]$	lin	nl	gain	kNN_{15}	PR	corr
10	2.996	0.759	0.3650	0.768	0.646	−0.122	0.706	181.9	+0.805
20	2.117	0.675	0.2562	0.804	0.633	−0.171	0.721	122.8	+0.761
50	0.802	0.492	0.0940	0.805	0.652	−0.153	0.725	125.4	+0.734
100	0.205	0.351	0.0167	0.764	0.642	−0.122	0.700	125.2	+0.511
200	0.072	0.350	0.0137	0.704	0.611	−0.092	0.653	114.4	+0.358

Observation 8.6 (No nonlinear improvement at any measured checkpoint). *Across a $27\times$ swing in the drift-dominated population ($\Pr[r > 0.5]$ from 0.365 to 0.014) and a $40\times$ swing in val loss, the gain never approaches zero: it stays within $[-0.171, -0.092]$, and kNN tracks the linear model from below at every checkpoint. The noise-artefact hypothesis is refuted. At step 10 the field is genuinely drift-dominated and the batch → strand map is no more compressible there;* $\mathrm{corr}(r_{90}, \mathit{gain}) = -0.501$*, so the weak trend runs against the manifold hypothesis rather than for it. The scope is the measured trajectory: one corpus, one seed, one architecture, five checkpoints, and one nonlinear family. The claim is that no nonlinear probe improved on linear anywhere along it — not that compression is impossible in general.*

Two quantities do move with regime, and they move together. The participation ratio falls monotonically 181.9 → 114.4, and the correlation between batch distance and strand dissimilarity falls sharply 0.805 → 0.358: the strand field becomes somewhat lower-dimensional and progressively decoupled from batch geometry as the model converges, without at any point becoming low-dimensional or nonlinearly predictable. This is the same signature as the `code` result of §5.3 — structure changes the dimension without opening the manifold. Linear predictability itself peaks mid-transition $(0.768, 0.804, 0.805, 0.764, 0.704)$, so the map is most linearly predictable while the model is actively learning and degrades as it memorises.

This supersedes the adequacy concern of §8.1. That concern was real — four of six corpora in §5 were probed near unigram — but on the authentic corpus, at five checkpoints spanning val 3.0 → 0.07, the result is invariant. Observation 5.1 now holds across corpora and across regimes.

8.6 Status after Phase 6

claim	status	basis
batch → strand is high-dimensional	holds	§5, two seeds
— at trained operating points	holds	§8.5
sparsity-based economy refuted	holds	inherited, corpus-general
— across regimes	holds	§8.5, one corpus
contiguity is row membership	untested	§8.2
symmetry results before Phase 6	retracted	§8.3
stabilizer grows with learning	first evidence	§8.3, marginal
disc has a deterministic phase	holds, early only	§8.4
substitution mechanism	closed, all routes	§11

The methodological result of Phase 6 is the portable one, and it is now established three times over: *an invariance claim is meaningless unless the structure-preserving and structure-destroying conditions are matched on surface perturbation.* Every symmetry measurement in this programme prior to §8.3 failed that requirement, and every one of them reversed or vanished when it was imposed.

9 Phase 7: The Compatibility Law, and What "Strand" Denotes

Two results in this phase change the status of the disc–strand decomposition. The first is negative and removes the last untested derivation in §7.4. The second is positive, and is the first measured law relating disc to strand: the disc does not determine the strand, it controls the *equivalence relation between two distinct strand observables.* The second result is only statable once a terminological ambiguity running through the whole programme is resolved, so that comes first.

9.1 Three objects have been called the strand

name	definition	status
structural strand S_{struct}	root-to-leaf path in the tile ladder	refuted, §2.11
memory strand S_{mem}	$\text{sign}(m)$	determined by the disc
response strand S_{resp}	$\text{sign}(g_B)$ at frozen θ	independent of the disc

S_{struct} is the object of §2.11, which reduced to contiguity under a permuted ladder and is where the pants coproducts of §2 live; those are exact on the tile ladder and are a statement about representation structure, not about gradient coupling.

The other two are both sign fields and have been used interchangeably. They should not be. Under Adam the update is $-\eta\,\hat{m}/(\sqrt{\hat{v}}+\epsilon)$ with $v > 0$, so $\text{sign}(\text{update}) = \text{sign}(m)$ exactly: S_{mem} is a *deterministic function of the disc.* It is S_{mem} that the flip-hazard work of §4 measures. By contrast $S_{\text{resp}} = \text{sign}(g_B)$ is the instantaneous batch response at frozen θ and is not a function of (m, v); it is S_{resp} that the intrinsic-dimension probe of §5 measures.

The distinction is not academic. Measured on the authentic corpus, sign agreement between the two is 0.678, 0.652, 0.635, 0.628 over windows 5–20, 20–50, 50–100, 100–200. For ± 1 variables with balanced marginals the correlation is $2p - 1$, so the correlations are

$$0.357, \quad 0.303, \quad 0.271, \quad 0.257.$$

The two objects agree on barely more than a quarter of their variance late in training, and they diverge as training proceeds. Any statement about "the strand" that does not say which one is under-specified.

Observation 9.1 (The fiber product degenerates for one strand and not the other)**.** *Because $S_{mem} = f(D)$, any construction of the form $\mathcal{D} \times_{\mathcal{R}} \mathcal{S}_{mem}$ collapses to $\mathcal{D}$, and a coproduct $\mathcal{D} \sqcup \mathcal{S}_{mem}$ is wrong for the opposite reason. Only S_{resp} supports a non-degenerate correspondence with the disc.*

9.2 The hazard threshold is not a drift/noise crossing

§7.4 reads the hazard law as a threshold in Fisher-normalised signal-to-noise. Given Observation 8.5, that reading predicts a threshold near $\rho = 1$ in the unbounded ratio $\rho_i = |m_i|/\sqrt{v_i - m_i^2}$, stationary across training. Recording (m, v) at every step and taking $\text{flip}_i(t) = [\text{sign}(m_i(t)) \neq \text{sign}(m_i(t{+}1))]$:

window	flip rate	AUC(r)	AUC(ρ)	r at thr.	ρ at thr.
5–20	0.1037	0.860	0.860	0.123	0.124
20–50	0.1234	0.836	0.836	0.106	0.107
50–100	0.1326	0.824	0.824	0.077	0.077
100–200	0.1358	0.810	0.810	0.070	0.070

AUC at the last window is 0.810 against the 0.809 of §4.4, so the original measurement reproduces. Everything else fails.

Observation 9.2 (The drift/noise reading of the threshold is refuted)**.** *ρ buys nothing over r: the two AUCs agree to three decimals at every window, and the thresholds coincide to three decimals, because $r \ll 1$ makes $m^2 \ll v$ for the bulk and the tail where they separate is too sparse to matter. The threshold sits at $\rho \approx 0.07$–0.12, an order of magnitude below the crossing, in a region where flip rate is already essentially zero (0.002–0.003 at $\rho \approx 1$). It is not stationary: it falls by a factor of 1.8. And the flip-rate curves do not collapse in ρ — in the 0.1–0.2 bin they run 0.198, 0.159, 0.096, 0.087 across windows.*

A rank reading fares little better. If the hazard were scale-free the threshold should be stationary in percentile, which §4.4 implicitly assumes by quoting one. Measured, it moves from the 18th to the 26th percentile across the same windows: less drift than the 1.8× in absolute value, but not invariance. Neither the value nor the rank of r is the conserved quantity, so a history-dependent coordinate is indicated. The natural candidates are the temporal correlation $C_i(\tau) = \mathbb{E}[g_i(t)g_i(t+\tau)]$, of which m is a single-timescale exponential estimator and which can be large where the mean is small; local curvature entering through $H_{ii}\Delta\theta_i$; and the coupling neighbourhood, given that flips are not independent across coordinates.

What survives is the monotonicity, and it survives robustly: flip rate is monotone decreasing in r in every window and every bin, at AUC 0.81–0.86. The discrimination is *better* early, in the drift-dominated regime, which is consistent with the r distribution being wider there and was not previously recorded.

A reproducibility note on §4.4. The threshold above is defined as the smallest r beyond which the flip rate falls below half the base rate. That criterion puts it at the 18–26th percentile, not the 70th quoted in §4.4. This is not a contradiction — the two criteria differ — but the original figure is not reproducible without its definition, which §4.4 does not state.

9.3 The compatibility law

If the disc does not set the flip threshold, what relation does it bear to the strand? Stratifying agreement between S_{mem} and S_{resp} by r decile:

window	d0	d1	d2	d3	d4	d5	d6	d7	d8	d9
5–20	0.514	0.541	0.572	0.601	0.631	0.670	0.712	0.764	0.837	0.943
20–50	0.512	0.539	0.560	0.587	0.614	0.646	0.686	0.726	0.779	0.870
50–100	0.515	0.532	0.557	0.580	0.607	0.636	0.665	0.696	0.742	0.826
100–200	0.514	0.532	0.553	0.580	0.598	0.625	0.651	0.683	0.729	0.819

Observation 9.3 (The disc controls the equivalence relation between strand observables)**.** *Agreement rises monotonically with r in every window and across every decile, from 0.514 at the bottom — indistinguishable from chance — to 0.943 at the top. At low r the memory strand and the response strand are independent observables; at high r they are nearly the same object. So r does not predict the strand, and does not set its flip threshold. It predicts the degree to which two distinct strand definitions coincide. The compatibility relation $\mathcal{R} \subset \mathcal{D} \times \mathcal{S}$ is a proper subset and r is its coordinate.*

This is the first measured law in the programme relating disc to strand, and it is a conditional-equivalence law rather than a determination law. It also explains the declining agreement of §9.1 without any divergence of the objects themselves: agreement falls 0.678 $\to$ 0.628 because the high-r population empties as training proceeds (Observation 8.4), not because the two strands drift apart. The same population effect that governs the disc regime governs the compatibility.

9.4 Kernel, not pullback

The correspondence should be written as a kernel and not as a fiber product. A pullback $\mathcal{D} \leftarrow \mathcal{G} \to \mathcal{S}$ requires both legs to be projections of one object. They are not: the disc is a functional of the whole gradient history, $D_t = F(g_1, \ldots, g_t)$, while the response strand is instantaneous, $S_{\text{resp}}(t) = h(g_t)$. They are memory and observation at different temporal resolutions, connected by a stochastic channel, so the well-posed object is

$$K_t : D_t \rightsquigarrow S_{t+1}, \qquad K_r(S_{\text{resp}}, S_{\text{mem}}) = P(S_{\text{mem}} \mid S_{\text{resp}}, r),$$

with composition by Chapman–Kolmogorov rather than by algebra multiplication. This is the same K_r already measured in §4.10; what §9.3 adds is that r parameterises it, and how.

A caution carried forward from §7.5. Formalisms built on spans and correspondences compose by pullback, and if the pullbacks are degenerate or the composition is free, the construction measures nothing — the failure mode that disposed of the refinement ladder, the transport algebra, persistent homology and K-theory in turn. Observation 9.3 is the first result that gives such a formulation content, because it establishes that the compatibility relation is proper and exhibits its coordinate. That is the reason to build here rather than elsewhere, and it is an empirical reason.

9.5 The next experiment, and what would settle it

Observation 9.3 shows r matters. It does not show r suffices. The decisive measurement is conditional mutual information,

$$I(S_{t+1}; S_t \mid r) \qquad \text{against} \qquad I(S_{t+1}; S_t \mid r,\ \text{architectural cell}),$$

on S_{resp}. If conditioning on r removes most of the history dependence then r is close to a sufficient statistic for the kernel and the theory is small. If it does not, the missing coordinate is among the candidates of §9.2 — temporal correlation, curvature, coupling neighbourhood, or optimiser memory beyond (m, v) — and the architectural conditioning discriminates the third.

This is the disc-sufficiency experiment proposed before Phase 6 and never run. It is now better posed, because §9.1 fixes which strand it must be run on: S_{resp}, since S_{mem} is disc-determined and the question would be vacuous for it. *Run in §11.5* for S_{mem}, where history adds -0.001 AUC and the law transfers across corpora at $+0.0000$; the S_{resp} form remains open.

9.6 Status after Phase 7

claim	status	basis
$\Delta\theta \approx a_t s_t$, sign carries the information	holds	§3
batch $\to$ strand high-dimensional	holds	§5, §8.5
flip hazard monotone in r, AUC 0.81–0.86	holds	§9.2
hazard threshold is a drift/noise crossing	refuted	§9.2
hazard threshold stationary in value or rank	refuted	§9.2
r controls strand-observable equivalence	holds	§9.3
r sufficient for the kernel	untested	§9.5
disc/strand as fiber product	degenerate	§9.1
substitution mechanism	closed, all routes	§11

The headline of the programme should no longer be that the disc explains the strand. It is that *the disc controls the equivalence relation between distinct strand observables*, and that the update field is structured but irreducibly high-dimensional in the batch. Both are measured; neither is derived. The earlier algebraic attempts failed because they tried to identify objects that the measurements show to be only conditionally equivalent, and the condition is now known.

10 Phase 8: The Bifibration, the Order Structure, and the Two Discs

This section is mostly derivation from measurements already reported, with one new experiment. It records which parts of the forward/backward $\times$ disc/strand picture are supported, gives the categorical formulation they support, and settles whether the tile disc and the optimiser disc are the same object. Status is marked throughout.

10.1 A second naming collision

§9.1 separated three objects that had all been called the strand. The same has happened to the disc, and the resolution is the same: name them and measure the map.

name	definition	index set
tile disc D_{tile}	per-tile update magnitude; β is its intercept	tiles
optimiser disc D_{opt}	(m, v), summarised by $r = \lvert m\rvert/\sqrt{v}$	parameters

D_{tile} is the object of §2.9, the pants coproducts and the β line. D_{opt} is the object of §4.4 and the compatibility law of §9.3. They live on different index sets at different granularities, and until §10.6 below no measurement connected them.

10.2 The four-observable square does not typecheck as drawn

The natural proposal is a square with D_f, D_b on top and S_f, S_b below, vertical arrows disc-to-strand and horizontal arrows forward-to-backward. Two things block it.

The vertical arrows are not maps. In the tile complex forward tiles record weights and backward tiles record applied updates, so D_f, D_b are magnitude fields on tiles. The corresponding S_f, S_b would be the forward and backward strand sets of §2.11 — root-to-leaf paths through the refinement ladder. A magnitude field on tiles does not map to a path through the ladder; these are objects of different type. Worse, the ladder strand is the one object in the programme that was *refuted*: permuting the tree took it to 0.000 and it reduced entirely to contiguity.

Two different forward/backward distinctions are being conflated. The tile-complex orientation (weights versus updates, refining toward output versus toward input) is not the forward/backward pass of backpropagation. The square borrows its horizontal arrows from the second and its objects from the first.

What survives is one honest horizontal arrow and one honest vertical arrow, and they live on different structures. The horizontal one is the exchange J between tile orientations, measured and non-involutive. The vertical one is the compatibility kernel of §9.3, relating D_{opt} to strand-observable equivalence. They do not compose into a square because the first is architectural and the second is dynamical.

10.3 The bifibration on the tile complex

Status: earned; both adjoints are measured. A bifibration is a functor that is both a fibration and an opfibration: every base morphism f has a cartesian lift f^* and a cocartesian lift $f_!$, with $f_! \dashv f^*$. The refinement system supplies exactly this. The base is the refinement poset

$L_0 < L_1 < \cdots < L_5$; the fibres are tile observables; the cocartesian lift is the pants coproduct $\Delta : \mathcal{S}(p) \to \bigoplus_i \mathcal{S}(c_i)$ of §2.3, and the cartesian lift is the corresponding merge. The observation "one complex, two orientations" is the statement that this is a single bifibration rather than two structures.

Ranks multiply to 2^{14} at every depth, so the two orientations are dual over the ladder. The multiplicities of the Bratteli diagram are

$$16 \xrightarrow{2} 32 \xrightarrow{2} 64 \xrightarrow{4} 256 \xrightarrow{2} 512 \xrightarrow{2} 1024,$$

a product of $2^6 = 64 = 1024/16$, all powers of two.

10.4 The adjunction is not an equivalence

Status: the categorical claim is supported; the numerical evidence originally offered for it was vacuous and is withdrawn.

An earlier draft of this section reported that the composite of the five rung matrices, normalised by its top eigenvalue, satisfies $\|P^2 - P\| = 2 \times 10^{-4}$ with singular values $(1.134,\ 2 \times 10^{-4})$, and read this as establishing that the refinement comonad projects onto the β line. Both halves were wrong.

The idempotency carries no information. If $C \approx \lambda_1 uv^\top$ with $v^\top u = 1$, then C/λ_1 is the spectral projector and is idempotent by construction. Any product of matrices with a spectral gap gives one. Control: five-fold products of random 2×2 matrices with eigenvalues near 1 and in $[0.05, 0.3]$ and random eigenvectors give $\|P^2 - P\|$ of 3.9×10^{-5}, 4.4×10^{-5}, 4.9×10^{-5}, 7.2×10^{-3}, median 3.6×10^{-3}. The measured 2×10^{-4} lies inside that range. The computation restates the spectral gap already reported.

The object was misidentified. The comonad induced by $f_! \dashv f^*$ is the round trip $f_! f^*$ on a single rung. The quantity computed was $M_{L_1 \to L_0} \circ \cdots \circ M_{L_5 \to L_4}$: five distinct rung maps composed in one direction, which is the inductive-limit map and not a round trip.

What survives, and is not automatic, is the *alignment* of the near-unit eigenvectors across rungs — already established as the universality of the fixed line. Minimum pairwise cosine across the five FF rungs is 0.999993. Against a baseline of 500 random gapped matrices, median pairwise $|\cos|$ is 0.704 and only 2.4% of pairs exceed 0.9993; ten of ten measured pairs exceed it. The content of the refinement system is that every rung contracts onto the *same* line, not that each contracts.

The categorical statement is therefore supported by the residuals and the spectral shape and not by any idempotency computation: $f_! \dashv f^*$ is an adjunction whose unit and counit are not isomorphisms, since each rung map has one strongly contracted eigendirection ($[0.005, 0.53]$ across all rungs, groups and orientations) that refinement destroys. Whether the round trip $f_! f^*$ is itself idempotent — the actual comonad question, and the one that would decide whether the counit is a split epimorphism — is *untested*: the reported measurements give the parent-to-child maps only, not the merge maps, so the round trip cannot be formed from them.

10.5 The order structure

Status: two orders, one measured and cover-dependent, one classical and applicable.

The graded index. $\operatorname{rank} \mathcal{H}_f - \operatorname{rank} \mathcal{H}_b$ takes values $-1008, -480, -192, +192, +480, +1008$ over L_0 to L_5, so $\mathcal{H}_f < \mathcal{H}_b$ on the lower half and $\mathcal{H}_f > \mathcal{H}_b$ on the upper, antisymmetric and never zero. The pairing is nondegenerate at every layer and the index changes sign by a jump. *This strictness is a property of the chosen cover.* The fixed point is $\sqrt{2^{14}} = 128$, which the ladder omits by jumping $64 \to 256$; include 128 and the relation acquires a zero and the index passes through it. Per §2.10, β does not survive a change of cover, and the same caution applies here with the artefact visible in the arithmetic.

The dimension group. §7.5 recorded that K_0 is undefined for the trajectory, since strands do not compose and the trajectory generates a free category with no additive structure. That argument concerns the *strand* and does not extend to the tile complex, where the situation is the reverse. By Elliott's theorem the complete invariant of an AF algebra is its *ordered* K_0 — the dimension group $(K_0, K_0^+, [1])$ — and the refinement ladder is a Bratteli diagram. With 2-adic multiplicities the dimension group is

$$K_0 \cong \mathbb{Z}[\tfrac{1}{2}], \qquad K_0^+ = \mathbb{Z}[\tfrac{1}{2}] \cap \mathbb{R}_{\geq 0}, \qquad [1] = 1.$$

The existing observation that no tile carries zero Fisher weight, so the state has full support and the algebra admits no order ideal annihilated by it, is already a statement in this order structure: full support, faithful state, no proper order ideal. So the greater-than/less-than relation on the tile side is the order on the dimension group, and K-theory is the right invariant here for the same reason it was the wrong one there.

But it classifies a choice, not the network. The multiplicities are 2-adic because the ladder was chosen as $16, 32, 64, 256, 512, 1024$; a 3-adic ladder would give $K_0 \cong \mathbb{Z}[\tfrac{1}{3}]$, and the omission of 128 is what makes the graded index nondegenerate. Together with §2.10, where β fails to survive a change of cover, this fixes the scope: the dimension group is a complete invariant of the refinement system we imposed, and is not an invariant of the transformer. Layer I below is not merely deterministic — it is *chosen.*

10.6 Are the two discs one object?

Status: measured; the answer is a weak coarse correspondence, not an identification. If β restricted to a tile predicted the r-distribution of the parameters inside it, the architectural bifibration and the optimiser dynamics would be parts of one theory. Per group and resolution level, across tiles: the correlation between tile-mean $\log D_{\text{tile}}$ and tile-median $\log r$, and η^2, the fraction of per-parameter $\log r$ variance resident at tile level, each against the programme's standard permuted-tile control.

level	mean corr	mean corr (perm)	mean η^2	mean η^2 (perm)
16	0.615	0.109	0.024	0.0012
32	0.544	0.082	0.028	0.0014
64	0.524	0.178	0.033	0.0034
256	0.405	0.119	0.048	0.0162
512	0.263	0.123	0.013	0.0010
1024	0.216	0.115	0.018	0.0020

Observation 10.1 (The two discs are correlated but not the same object)**.** *The correspondence is real: correlations of* 0.5–0.7 *at coarse levels sit well above the permuted control of about* 0.1*, so D_{tile} and D_{opt} are not independent observables. But the tile field explains almost none of the optimiser field: η^2 net of permutation is* 0.02–0.03 *throughout, so* 97–98% *of the variance in per-parameter* $\log r$ *is* within *tiles. And the correlation* decays *under refinement,* $0.615 \to 0.216$*, which is the signature of a coarse group-level effect — which matrix a parameter sits in influences magnitude and coherence together — rather than of tile structure. Refining the cover does not sharpen the relationship; it dissolves it.*

Two group-level details qualify the table. LN is the outlier, with correlations 0.97–0.77 and η^2 of 0.10–0.20; but it carries 3584 parameters, under 0.1% of the model, and its permuted control is correspondingly inflated (0.0057 rising to 0.0790 at $L = 256$, where tiles hold only 14 parameters each). W_V is the weakest, with correlations 0.09–0.29. The apparent dip in pooled η^2 at $L = 512$ is compositional: LN drops out of the pool at that level for want of parameters.

The conclusion is that the architectural bifibration does not determine the optimiser disc. The two should remain distinct objects with a measured, weak, coarse correspondence between them, rather than being merged. This is the same boundary that limits the pants coproducts: the tile ladder is exact as representation structure and does not propagate into gradient structure — architecture hierarchy $\not\Rightarrow$ gradient hierarchy, and the reason remains the dense coupling without spectral gap of §4.11.

10.7 The information order

Status: well-posed, currently unmeasured, and one common claim about it is false. Ordering discs by magnitude is not useful; ordering them by predictive sufficiency is. Writing $K(D)$ for the strand transition kernel induced by D, set

$$D_1 \sim D_2 \iff K(D_1) = K(D_2), \qquad D_1 \preceq D_2 \iff K(D_1) \text{ is a garbling of } K(D_2).$$

The equivalence is the kernel of $D \mapsto K(D)$, so the quotient is well-defined. The order is Blackwell's ordering of experiments, and it is a genuine partial order.

It is *not* in general a lattice. Blackwell's order admits neither guaranteed meets nor guaranteed joins on arbitrary families, so an assertion that discs form a lattice under predictive refinement requires proof for the specific family, not appeal to the order alone. What is available without further assumptions is a preorder with a well-defined quotient.

Both definitions are vacuous until K can be estimated, which is exactly the sufficiency experiment of §9.5. Observation 9.3 is the one measured fact bearing on them: r indexes the equivalence between strand observables, which is a necessary condition for r to be a sufficient coordinate for K and far from a sufficient one.

10.8 The doctrine, with status

Stating axioms and letting the measurements select the algebraic completion is the right order of work. The programme's version, with each clause marked:

clause	status	basis
refinement induces restriction maps	holds	residuals 0.1–0.8%, §2.3
those maps form a bifibration	holds	both adjoints, §10.3
the adjunction is not an equivalence	holds	contracted eigendirection per rung
$f_! f^*$ idempotent (the comonad)	untested	merge maps not reported
tile order is the dimension group	holds	Elliott, §10.5
fwd/bwd exchange is non-involutive	holds	$J_w^2 = w$, w measured
tile disc determines optimiser disc	refuted	§10.6
disc/strand square	ill-typed	§10.2
behavioural equivalence of discs	untested	needs K, §9.5
information order is a lattice	unproven	§10.7
bifibration extends to gradient flow	refuted	§4.11

The last clause is the one that bounds the whole construction. A bifibration requires both lifts for every base morphism. On the tile complex they exist. On the passes they do not: §4.11 measured that forward activations add nothing to predicting the backward field, since a weight's gradient is $\delta_{\text{out}} \cdot \text{act}_{\text{in}}$ and the forward pass supplies only the second factor, and §1.13 measured that forward image and backward demand do not align. There is no cartesian lift from forward data to backward data. The bifibration is a structure on the index set and the refinement poset, and the measurements say it stops there.

11 Phase 9: Coupling, Substitution, and the Marginal/Relational Split

§4.11 left one question open: sparsity was refuted, substitution untested. This phase tests it along every route the programme could construct, and closes each. The routes fail for one reason, and identifying that reason is the phase's positive content: *the optimiser stores marginal statistics while the instantaneous strand depends on relational ones.*

11.1 Summary: eight properties of the hidden state

The results of this phase and the next are not eight separate findings. They are one explanation, of why every substitution route fails, and it is convenient to state it before the evidence.

1. **A hidden state exists.** The leading eigenvalue of C_{resp} is 119.5 against a Marchenko–Pastur bulk edge of 19.19; observing 2% of coordinates predicts the remainder at 0.67 against the diagonal disc's 0.49 (§11.7, §11.9).

2. **Diagonal optimiser statistics are insufficient *for the batch-conditional law*, and sufficient for the marginal one.** (m, v) predicts $\text{sign}(g_B)$ at chance (§11.7), while for S_{mem} eight steps of strand history add -0.001 AUC (§11.5). The unqualified claim that diagonal statistics are insufficient would contradict the second result; the split of §9.1 is what reconciles them.

3. **The hidden state is relational, and its relational part is generated by corpus sequential structure.** v_i is an EMA of g_i^2, so the disc is diag $F_{\mathcal{C}}$ and the discarded object is $F_{\mathcal{C}}^{\text{off}}$ (§11.5). At fixed θ the top eigenspace of C_{resp} under the authentic corpus is *orthogonal* to that under its token shuffle, and the shuffle induces no reproducible eigenspace at all (§11.6).

4. **It is dense.** No spectral gap in H, coupling degree 140/512, and spectral clustering of $|C_{\text{resp}}|$ into six blocks returns $[3000, 0, 0, 0, 0, 0]$ — no block structure even in a basis derived from the coupling itself (§4.11, §11.9).

5. **It is short-lived.** $\tau_F < 10$ steps; the k=1 core is gone within five (§11.8, Observation 11.8).

6. **It is measurable at low rank.** Split-half reproducibility is 0.832, 0.781, 0.746 at $k = 1, 2, 3$, so the leading directions are genuine estimated objects and not artefacts of undersampling (Observation 11.8).

7. **Its most informative directions are its shortest-lived.** rot(205) is monotone increasing in k: 0.004, 0.010, 0.060, 0.083, 0.151 (Observation 11.8).

8. **Aggregate behaviour is already highly directed.** Vector cancellation over a five-step window is $C \approx 0.083$ and path efficiency over twenty steps is 0.65–0.76, against $T^{-1/2} \approx 0.22$ for a random walk (§12.2, §13). There is little excess wandering to remove.

Two further statements bind these together.

A mechanism links (5) and (7). They are not independent facts. The dominant coupling directions are the ones the optimiser is actively descending, so applying the update destroys the field it was built from,

$$F_t \longrightarrow \Delta\theta_t \longrightarrow F_{t+\Delta}.$$

Informative therefore *implies* short-lived rather than coinciding with it.

A corpus asymmetry accompanies (2) and (3), and both halves are measured on the same pair of corpora at the same θ. A flip predictor fitted on the authentic corpus transfers to its token-shuffled counterpart — differing 125× in val loss and 200× in bigram support — with an AUC change of +0.0000 (§11.5); the marginal law is corpus-independent and the corpus enters it only through the *value* of D. The relational geometry induced by those same two corpora at that same θ is orthogonal (§11.6). One experiment, two objects, opposite answers: this is the sharpest available statement of the marginal/relational split.

The engineering corollary is the ninth item, and is what the eight amount to:

> The obstacle is not that the hidden state is unidentifiable. Its dominant structure *can* be identified. The obstacle is that its coherence time is shorter than the cost of identifying it.

Every route tested below fails on one or more of properties (3)–(7), and no route evades all of them.

11.2 A third naming collision: three coupling operators

name	definition	governs
$C_{\text{resp}}[i,j]$	$\text{corr}_B(\text{sign}\, g_i, \text{sign}\, g_j)$	communication
$C_{\text{dyn}}[i,j]$	$H_{ij} = \partial g_i/\partial\theta_j$	decomposition
$C_{\text{flip}}[i,j]$	$\text{corr}_t(\text{flip}_i, \text{flip}_j)$	temporal organisation

These are not variants of one object and they return opposite answers. Only C_{dyn} governs whether optimisation decomposes; only C_{resp} governs how much must be transmitted. Conflating them produced an inference worth recording as an error: *the tile disc explains only* 2–3% *of optimiser-disc variance (§10.6), therefore the optimiser decomposes into near-independent subsystems.* That does not follow. η^2 measures a representation map from an *indexing* to D_{opt}; decomposability is a property of H.

11.3 Decomposition: the tested partitions are not weakly coupled

The classical domain-decomposition criterion. For v supported on block A,

$$\text{leak}(A) = \frac{\|(Hv)|_{A^c}\|}{\|Hv\|},$$

against the null a structureless partition of the same size gives, $\sqrt{1 - |A|/P}$. Hessian-vector products at the val-floor checkpoint, averaged over four batches.

partition	$\lvert A\rvert/P$	leak	null	ratio	norm retained in A
$\text{layer}_{0..5}$	0.152	0.787–0.830	0.921	0.85–0.90	0.56–0.62
role_WQ	0.091	0.985	0.954	1.033	0.173
role_WK	0.091	0.982	0.954	1.030	0.190
role_WV	0.091	0.976	0.954	1.023	0.220
role_ff	0.546	0.697	0.674	1.033	0.718
role_ln	0.001	0.995	1.000	0.996	0.098

Observation 11.1 (Architectural partitions do not support independent optimisation)**.** *Role partitions sit exactly at the null: partitioning by matrix role captures nothing. Layers are the best available partition and still leak about* 80% *— perturb inside a layer and four fifths of the Hessian response is felt outside it — though they do beat the null, retaining* 0.56–0.62 *where a random set of the same size retains* $\sqrt{0.152} = 0.39$. *Agent-per-component with occasional synchronisation requires leak near zero; the "occasional" correction is* 80% *of the signal.*

The scope is what was measured. H is local, so this bounds decomposition by the tested architectural partitions at this operating point; it does not exclude multiscale or adaptive decompositions in general.

11.4 Communication: the response modes are denoising

C_{resp} has genuine low-rank structure — on 1500 coordinates over 300 batches the leading eigenvalue is 59.4 against a Marchenko–Pastur bulk edge of 10.47 at $c = 5$, so it is signal by a factor of 5.7. The natural inference is that workers could exchange k coefficients rather than P signs.

Measured at a frozen checkpoint on a 150,000-coordinate subset, all schemes applied at matched step norm, scored by retained fraction of the full-sign loss improvement: top-5 response modes gave 1.852 at 40 bits against 150,000, while a matched random 5-dimensional subspace gave 0.006. The adapted basis outperforms the random one by a factor of some 300, so the structure is real.

A compression scheme cannot outperform its source. Three controls.

scheme	vs. single-batch full sign	at step 400	at step 800
avg200 (sign of 200-batch mean)	+2.58	+0.03	−0.03
topk5	+1.70	+0.02	+0.02
topk20	+1.49	+0.02	+0.00
topk100	+1.39	+0.05	+0.04
mode5 (top-5 applied to the average)	+2.40	—	—

Observation 11.2 (The apparent gain is averaging, and the basis expires)**.** *Averaging alone gives* 2.58*; the top-*5 *projection gives* 1.70*, so the projection is a lossy approximation to the average rather than a compression that beats full sign. Performance* decreases *in k (*$1.70, 1.49, 1.39$*), the signature of extra modes reinstating batch-specific noise. What survives is narrower: applied to the already-averaged gradient, five modes retain* 93% *of the full averaged direction (*2.40 *against* 2.58*), and* 5, 20 *and* 100 *are indistinguishable, so the structure is genuinely about five-dimensional at this checkpoint. Everything built at step* 200 *is worthless by step* 400*. Critically* `avg200` *collapses too, so this is not basis staleness — the averaged gradient direction itself is stale within* 200 *steps, and amortisation is impossible.*

Reconciliation with §4.11. That section measured that sign survives 25% i.i.d. noise but breaks under rank truncation. Projection onto k modes *is* rank truncation, and it appeared to escape the obstruction only because the basis smuggled in batch-averaged information. The eight doors of §4.11 are not qualified.

11.5 The marginal/relational split

v_i is an EMA of g_i^2, so

$$D_{\text{opt}} = (m, v) \;\approx\; \big(\mathbb{E}[g],\ \operatorname{diag} F_{\mathcal{C}}\big), \qquad F_{\mathcal{C}} = \mathbb{E}_{x\sim\mathcal{C}}[gg^\top].$$

The corpus was never absent from the programme: it enters through v, as the diagonal of the corpus-induced empirical Fisher. What Adam discards is $F_{\mathcal{C}}^{\text{off}}$. Two coordinates with $v_1 = v_2$ are indistinguishable to the optimiser whether $\mathbb{E}[g_1 g_2]$ is +0.9 or −0.9.

This organises the whole sequence of results, and yields the phase's one positive finding, in the corpus-transfer arm of the sufficiency test.

	authentic	shuffled	authentic → shuffled
val loss	0.054	6.84	—
bigram nnz	1,347	278,156	—
arm A, disc only	0.7948	0.8343	**0.8343**
arm B, disc + 8-step history	0.7938	0.8342	0.8321

Observation 11.3 (The disc is a sufficient state; the marginal law is corpus-independent)**.** *Eight steps of flip history plus run length, on* 640,000 *held-out rows, give* -0.0010 *and* -0.0001*: the past of the strand adds nothing once* (m, v) *is known, so the memory hypothesis is refuted at this horizon and §9.5 is closed. Further, a predictor fitted on the authentic corpus transfers to its token-shuffled counterpart — differing* $125\times$ *in val loss and* $200\times$ *in bigram support — with an AUC change of* $+\mathbf{0.0000}$*. The corpus does not enter the flip law as a variable. It enters entirely through the* value *of* D*.*

That is exactly what a marginal law relating marginal quantities should do, and it sharpens the split: $D_{\text{diag}} \to S_{\text{mem}}$ is stable across corpora, while $D_{\text{diag}} \to S_{\text{resp}}$ fails, because the batch strand depends on the corpus-specific relational part.

Caveat. This was run on S_{mem}, the flip of $\text{sign}(m)$. §9.1 separated that from S_{resp}, whose correlation with it is only 0.26–0.36. The S_{resp} version is a different measurement and remains open.

11.6 The relational part is corpus-generated

That the relational part is corpus-specific was, until now, an inference from the marginal law transferring. It is directly testable, and the test removes every confound by holding θ fixed: train to the val-floor checkpoint on the authentic corpus, then sweep batches drawn from that corpus and from its token shuffle, and compare the resulting eigenspaces of C_{resp}. Same parameters, same architecture, same optimiser state; only the token order differs.

k	SPLIT_A	SPLIT_B	$\text{ov}(U_A, U_B)$	ratio to SPLIT	random null
1	0.919	0.000	0.000	0.000	0.0003
2	0.905	0.002	0.000	0.000	0.0005
3	0.815	0.003	0.000	0.001	0.0008
5	0.734	0.005	0.001	0.002	0.0013
10	0.517	0.008	0.003	0.013	0.0025

Top eigenvalues are 272.5, 235.3, 105.3, 75.6, 67.7, 62.9 under the authentic corpus and 26.3, 25.1, 23.6, 22.6 under the shuffle.

Observation 11.4 (Sequential structure generates the relational geometry)**.** *The eigenspaces induced by the two corpora at identical* θ *overlap at* 0.000*, indistinguishable from the random null, against a within-corpus split-half floor of* 0.919*. This confirms corpus-specificity directly rather than by inference. It also gives a stronger statement than corpus-specificity: the shuffled corpus induces* no reproducible eigenspace of its own *(*$\text{SPLIT}_B \leq 0.008$ *at every* k*), and its leading eigenvalues are flat and near the Marchenko–Pastur edge — a bulk, not a spectrum. The off-diagonal is therefore not merely parameterised by the corpus; it is generated by the corpus's sequential structure, and vanishes without it.*

Two remarks. First, $\text{SPLIT}_A = 0.919$ here against 0.832 in Observation 11.8 at 350 batches: the authentic core is if anything better estimated than previously reported, which strengthens rather than qualifies that observation. Second, a caveat on the stronger half. This is one θ, trained on the authentic corpus, so the shuffle's lack of structure may partly reflect a θ adapted to A; a model trained on B might induce reproducible structure under B. That control is not run. The orthogonality result does not depend on it.

11.7 Partial observation: agreement is not descent

If the relational part is the missing object, observing a cheap subset and inferring the rest is the natural substitution scheme. Ridge from an observed fraction to the remainder, at the val-floor checkpoint:

obs. frac	$\lvert O\rvert$	A: disc only	C: observed only	null (shuffled pairing)	C − null
0.02	120	0.4891	0.6705	0.5789	+0.092
0.05	300	0.4891	0.6926	0.5859	+0.107
0.10	600	0.4892	0.7075	0.5873	+0.120
0.25	1500	0.4897	0.7237	0.5837	+0.140
0.50	3000	0.4905	0.7333	0.5739	+0.159

Observing 2% of coordinates predicts the other 98% at 0.67 agreement against 0.49 for the diagonal disc. But the null — shuffling the observed vector across batches, preserving every marginal — reaches 0.58, so the genuinely batch-conditional contribution is +0.09, not +0.18. The intercept term at 0.58 nonetheless beats $\mathrm{sign}(m)$ at 0.489: the mean sign *at the current* θ predicts better than the optimiser's EMA, so the disc is a stale marginal memory.

Observation 11.5 (Reconstruction does not produce descent)**.** *Applying the reconstructed strand as an update, the batch-independent intercept (2% exact, mean sign elsewhere) scores* +3.111 *against the ridge reconstruction's* +1.083 *and observed-only's* +0.204*, at reconstruction agreement* 0.616*. The relational correction loses to the batch-independent mean by a factor of three. Agreement is not value, as §4.10 already established when sign flattening preserved agreement and collapsed improvement to* 0.9%*. The mechanism is again §4.11: a ridge map fitted from* 300 *batches produces* structured *error, which is the damaging kind.*

Note also the accounting objection. Obtaining g on 2% of coordinates does not cost 2% of a backward pass, since δ propagates through the whole network regardless. The scheme is at best communication, not computation.

Caveat. Full-sign improvement at this checkpoint was 2×10^{-5}, at the noise floor, so the ratios have an unreliable denominator. The ordering is what the measurement supports; the magnitudes are not.

11.8 τ_F: the observer is slower than the phenomenon

Substitution via $F_{\mathcal{C}}$ reduces to one timescale. Estimating the coupling costs N backward passes and buys M skipped ones over its lifetime τ_F; viability needs $M \gtrsim N$ within τ_F. Earlier measurements conflated eigenspace rotation with drift of the mean direction. Separated — subspace overlap $\|U_k(200)^\top U_k(t)\|_F^2/k$, drift-independent by construction, with the same-checkpoint split-half overlap as the estimation noise floor:

step	val	SPLIT	rot vs. 200	drift cos	rot/SPLIT
200	0.0630	0.2419	1.0000	1.0000	—
210	0.0594	0.2617	0.0193	0.0163	0.074
225	0.0627	0.2925	0.0089	0.0284	0.030
250	0.0655	0.2545	0.0061	0.0406	0.024
300	0.0664	0.2500	0.0051	−0.0189	0.020
400	0.0641	0.2180	0.0048	−0.0022	0.022

Random-subspace null = 0.0033.

Observation 11.6 ($\tau_F < 10$ steps, and $F_{\mathcal{C}}$ is barely observable)**.** *Within ten steps the overlap falls to* 0.0193*, seven percent of the same-checkpoint reproducibility floor; by step* 300 *it is within a factor of* 1.5 *of two random subspaces. The eigenspace does not rotate slowly while the mean drifts — both move at the same rate. With* $N \approx 300$ *and* $M < 10$ *the amortisation fails by at least* 30×*. The split-half control makes it worse: two disjoint sets of* 150 *batches at* identical θ *agree only* 0.24 *on the top-*20 *eigenspace, so a usable estimate needs far more than* 300 *batches, widening the gap.*

11.9 Three attempts to recover the timescale

Scalar invariants. If the eigenspace is fast, slower scalar summaries might survive. Measured against split-half noise:

invariant	mean	across-checkpoint sd	split noise	sd/noise
effective rank	76.88	6.35	2.69	2.36
spectral entropy	4.598	0.031	0.011	2.72
top-5 mass	0.1615	0.0164	0.0065	2.53

All move 2.4–2.7× their own measurement noise; effective rank swings 66 → 86 non-monotonically. They are not slow. And they could not have helped: a handful of scalars carries tens of bits, against the 0.405 bits/parameter/step of §4.7, some 1.7 Mbit for this model.

A coupling-derived cover. Every cover tested had been imposed. Spectral clustering of $|C_{\text{resp}}|$ into six blocks returned $[3000, 0, 0, 0, 0, 0]$ — all coordinates in one cluster, with mean off-diagonal 0.0708, weak and uniform. Spectral clustering degenerates exactly when the affinity graph admits no low-conductance cut. *The coupling has no block structure in a basis derived from the coupling itself*, which is stronger than the architectural null of §11.3: the density is intrinsic, not a consequence of poor coordinates.

Regular transport. A fast rotation can still be regular: if $U_k(t+\Delta) \approx R_\Delta U_k(t)$ for slowly varying R, the transport is the long-lived object. Then bases at successive checkpoints would lie in a low-dimensional family and their union would span far less than Lk.

L	Lk	cumrank real	cumrank random	ratio	overlap of $U(t_L)$ with previous span
2	30	28.44	29.90	0.951	0.0548
4	60	52.14	59.32	0.879	0.1018
8	120	103.33	116.85	0.884	0.0896

Observation 11.7 (No regular transport)**.** *Eight bases of dimension* 15 *at five-step spacing span* 103 *of a possible* 120 *dimensions,* 88% *of what eight random frames span, growing linearly in L. Predicting the next basis from the span of all previous ones gives* 0.09 *and is flat from* $L = 3$*: accumulating history buys nothing. The decisive reference is that the split-half pair at a* single *checkpoint already spans* 22.57 *of* 30*, so the cross-checkpoint independence is not distinguishable from estimation noise. There is no long-lived transport operator behind the fast basis.*

A low-rank coupling core. The three attempts above, and §11.8, all used $k = 15$–20 with 120–150 batches per estimate, while the spectrum showed only a few eigenvalues far above the Marchenko–Pastur edge. The measurements may therefore have been dominated by noisy bulk directions, with a small number of dominant modes forming a slowly evolving reusable representation. This is the strongest remaining alternative and it is conceptually distinct from the other three: it does not propose a new object, it proposes that the existing object was mismeasured.

Re-estimated with 350 batches per half, roughly three times the previous sampling. The bulk edge is 19.19 at $N_{\text{sub}}/n = 11.4$ and all top ten eigenvalues lie above it (119.5, 86.9, 75.2, 61.7, 55.6, . . .).

k	SPLIT@200	rot@205	rot@210	rot@250	rot@300	random null
1	0.832	0.004	0.000	0.000	0.000	0.0003
2	0.781	0.010	0.000	0.000	0.001	0.0005
3	0.746	0.060	0.004	0.001	0.001	0.0008
5	0.616	0.083	0.014	0.002	0.001	0.0013
10	0.599	0.151	0.021	0.005	0.003	0.0025

Observation 11.8 (The core is reproducible and the least reusable part of the field)**.** *The sampling concern was justified and is resolved in the affirmative: at* $k = 1$–3 *the leading*

eigenspace is 75–83% *reproducible across disjoint batch sets at fixed* θ*, against the* 0.24 *of the earlier* $k = 15$–20 *estimates. Those same well-estimated directions are gone within five steps —* $\mathrm{rot}_{k=1}(205) = 0.004$ *against a random null of* 0.0003*. There is therefore no ambiguity left between motion and mismeasurement: at* $k = 1$ *the object is determined and it rotates away almost immediately. Further,* $\mathrm{rot}(205)$ *is monotone increasing in* k *(*$0.004, 0.010, 0.060, 0.083, 0.151$*):* the lower-dimensional the core, the faster it rotates*, so the earlier* $k = 15$–20 *measurements were if anything flattering to the persistence hypothesis. The most reliable part of the coupling geometry is also the least reusable part.*

This admits a mechanistic reading, which is the one new positive statement in this subsection. The dominant coupling directions are not stable infrastructure; they are the directions the optimiser is actively descending, hence the ones the step most alters:

$$F_t \longrightarrow \Delta\theta_t \longrightarrow F_{t+\Delta},$$

a self-erasing geometry in which the directions determining the current update are precisely those most modified by applying it. This closes the timescale question: estimating a well-determined $k = 1$ core costs some 700 backward passes and buys fewer than five steps, worse than the shortfall recorded in Observation 11.6 rather than better. The conclusion is not that the field cannot be amortised in principle — a different optimiser, update rule or step size could alter the relation between F_t and $F_{t+\Delta}$ — but that it cannot be amortised *under the measured dynamics.*

11.10 Nulls, and what they share

question	result	basis
decomposition by architecture	tested partitions too coupled	§11.3
communication via response modes	denoising; basis expires	§11.4
progress order $\Phi(D, S)$	$\Phi = -L$; disc/strand add $+0.001$ R^2	§11.11
memory / monodromy	history adds -0.001 AUC	§11.5
corpus as a variable	transfers at $+0.0000$	§11.5
partial observation	agreement without descent	§11.7
coupling geometry $F_\mathcal{C}$	$\tau_F < 10$ steps	§11.8
scalar invariants, cover, transport	all fail	§11.9

Observation 11.9 (Why substitution fails)**.** *The strand depends on relational structure in* $F_\mathcal{C}$ *that is* dense *(no spectral gap in* H*; spectral clustering of* C_{resp} *collapses to one block),* corpus-specific *(the marginal law transfers at* $+0.0000$*, the relational part does not),* shorter-lived than its measurement cost *(*$\tau_F < 10$ *against* $N \approx 300$*), and* not reproducibly estimable at fixed θ *(SPLIT* ≈ 0.24*). Each route tested fails on one or more of these, and no route evades all four.*

Collecting every proposed shortcut across the whole programme:

proposed shortcut	result
sparse structure	refuted nine ways, §4.11
architectural cover	no useful decomposition, §11.3
forward activations	supply only $\mathrm{act}_{\mathrm{in}}$, §4.11
dictionary / manifold compression	no low-dimensional map, §5
partial observation	insufficient for descent, §11.7
coupling-field reuse	$N_{\mathrm{measure}} \gg \tau_F$, §11.8
low-rank coupling core	reproducible but rapidly destroyed, §11.9

The last row changes the character of the conclusion, and is the cleanest statement the programme has reached:

> The obstacle is not that the relational field is too noisy to identify. Its dominant structure *can* be identified. The obstacle is that the identifiable structure has a coherence time shorter than the cost of identifying it.

Equivalently: gradient computation is not expensive because the gradient is high-dimensional. It is expensive because the relational field generating it is high-dimensional, corpus-dependent, and self-erasing, and every compression that succeeded in this programme did so by averaging rather than by exploiting that field.

11.11 An order-function null

Recorded here for completeness, since §11.10 cites it. The proposal was a $\Phi(D, S)$ increasing during successful training and failing during wandering. The trap is that $\Phi = -L$ satisfies this by construction, so the non-vacuous question is whether disc and strand predict future progress *conditional on current loss*. Over 51 checkpoints and three seeds with val spanning $1.176 \to 0.053$, $\mathrm{corr}(\mathrm{val}, \text{future improvement}) = +0.997$, $R^2 = 0.995$. Partial correlations given val: r_{50} -0.040, r_{90} -0.046, flip rate -0.373, path efficiency 0.278, gradient norm 0.019; all five together add R^2 $+0.001$. The raw correlations are large (r_{90} at 0.94) and entirely redundant, which is §8.4 reappearing: r tracks training progress, so it re-encodes the loss.

Path efficiency — net displacement over path length — ranged only 0.654–0.760 across all checkpoints. A random walk would give $\eta \sim T^{-1/2}$ and collapse. The trajectory retains about 70% of its path as net displacement throughout, so there is no wandering regime here for an order function to detect; the motivating phenomenon is not visible in this data.

11.12 What remains

Substitution is closed along every route constructed, and Observation 11.8 closes the last form of the timescale question.

One reformulation remains genuinely open, and it is the only one not already excluded by a measurement. Every object tested so far has been an invariant of a *state*: $K(F_t)$, $\mathrm{Ind}(D_t)$, a cover, an eigenspace. The alternative is a directed accumulation along the path — an ordered transport functional rather than a static invariant,

$$\mathcal{I}_{0\to T} = \int_0^T \Phi\big(F_t, \dot{F}_t, D_t, S_t\big)\, dt,$$

of which two concrete instances are available. The first integrates the compatibility defect, $\mathcal{C} = \int_0^T \Omega_t\, dt$ with $\Omega_t = P_{t+\Delta}\mathcal{T}_D - \mathcal{T}_S P_t$, giving a holonomy-like accumulation rather than a diagnostic; unlike a Fredholm index this is not forced to vanish in finite dimensions (§7.5). The second integrates the coupling flow against the strand motion, $\mathcal{A} = \int_0^T \langle F_{t+\Delta} - F_t, \dot{S}_t\rangle\, dt$, asking whether the *direction of change* of the coupling has structure even though its state does not.

The testable core of both is a single hypothesis: *does $\Phi(F_t, \dot{F}_t)$ have longer coherence than F_t?* Two cautions apply before it is run. Both functionals require Ω_t or $\dot{F}_t$, hence both routes and hence the full backward pass, so a positive result would be explanatory rather than computational unless the accumulated quantity is itself cheap to maintain. And Observation 11.8 supplies a specific reason for pessimism about $\mathcal{A}$: if F_t is erased by the update that uses it, then $\dot{F}_t$ is largely *determined by* $\Delta\theta_t$, so $\langle \dot{F}_t, \dot{S}_t\rangle$ would be large without carrying information independent of the update already computed. Distinguishing those two cases is the first thing such an experiment must control for.

Two separate questions follow. §12 decomposes $\Delta\theta$ into drift, batch noise, curvature-following and lift, and finds three of those four empty. §13 asks whether the *allocation* of optimisation across parameters can be precomputed, and closes that too.

Also outstanding, all from earlier phases: the S_{resp} form of the sufficiency test (§11.5); the stabilizer signal at 3200 steps, one seed at $z = +2.13$ (§8.3); and the contiguity claim tested against flip clustering across training steps rather than across-batch sign correlation at frozen θ (§8.2).

12 Phase 10: Drift, Noise, Curvature, Lift

The programme's negative results all concern *compression* of the update. A separate question is what the update is made of. The informal picture behind much of the discussion has four components: smooth curvature-following that may carry the trajectory into a poor basin, batch noise that lifts it out again, aimless wandering in which motion cancels, and lift proper — a directed component not explained by local descent. Schematically,

$$\Delta\theta_t = \underbrace{v_{\|,t}}_{\text{curvature-following}} + \underbrace{v_{\perp,t}}_{\text{escape / lift}} + \underbrace{\epsilon_t}_{\text{noise}} .$$

This phase measures the decomposition. Three of the four buckets come back empty, and the fourth absorbs everything.

12.1 Method

At a frozen checkpoint, sweep $N_B = 200$ batches and split the gradient field into its deterministic and sampling parts,

$$\text{drift} = \tfrac{1}{N_B} \textstyle\sum_B g_B, \qquad \text{noise}_B = g_B - \text{drift},$$

the first being the corpus-induced field of §11.5 and the second batch sampling. Each component is then placed relative to curvature by its Rayleigh quotient under Hessian-vector products,

$$R(u) = \frac{u^\top H u}{\|u\|^2},$$

against a random direction as the null, and applied alone at matched step norm to measure its loss effect. The two are different questions — where a component sits in the landscape, and what it does — and the second is what decides whether noise helps or hurts.

	step 50 (base 0.884)	step 200 (base 0.088)
$\|\text{drift}\|/\mathbb{E}\|g_B\|$	0.4487	0.3483
$\mathbb{E}\|\text{noise}\|/\mathbb{E}\|g_B\|$	0.9495	1.0519
$R(\text{drift})$	+0.18120	+0.22615
$R(\text{noise})$	+0.15984	+0.14687
$R(\text{random})$	+0.00069	+0.00005
loss gain, drift alone	+0.136472	+0.035691
loss gain, noise alone	−0.100284	−0.027433
loss gain, single batch	+0.008596	+0.008000

12.2 Aimless wandering does not occur

Recorded in §11.11 and confirmed here from the other direction. Path efficiency $W = \|\theta_T - \theta_0\| / \sum_t \|\Delta\theta_t\|$ ranged only 0.654–0.760 over 51 checkpoints and three seeds, against $W \sim T^{-1/2} \approx 0.22$ for a random walk over the same window; §13.3 measures the same quantity at a five-step scale and finds cancellation of only 8%. This is *vector* cancellation, and it should not be read as the optimiser being efficient: per-coordinate cancellation is about 50% (§13.1), and the difference is that individual reversals cancel across coordinates rather than accumulating. The trajectory retains about seventy percent of its path length as net displacement.

Consistently, a single-batch gradient applied alone still yields $+0.008$: steps make individual progress rather than cancelling. There is no wandering regime at this operating point for a diagnostic to detect.

12.3 Noise is a tax, not an escape mechanism

Observation 12.1 (Batch noise dominates in norm and costs loss). *The noise component exceeds the drift by three to one in norm at step* 200 *(*1.05 *against* 0.35*), and applied alone it* costs *loss,* -0.027 *against the drift's* $+0.036$*. The magnitudes are nearly symmetric, so each step is approximately a* $+0.036$ *drift signal less a* -0.027 *noise penalty, netting the* $+0.008$ *that a single batch actually delivers.*

This is the ordinary second-order picture: for zero-mean noise the leading term in the loss is $+\frac{1}{2}\eta^2 u^\top H u > 0$ whenever H is positive semi-definite along u. The measurement supplies no evidence for the escape reading in which sampling noise is what carries the trajectory out of a poor basin; at this operating point it is a cost that averaging removes, which is the direct explanation for why `avg200` beat every compression scheme in §11.4.

12.4 Noise is not isotropic and is not flat

Observation 12.2 (Drift and noise occupy the same curvature regime). *Both components sit three to four orders of magnitude above a random direction in Rayleigh quotient (*$+0.226$ *and* $+0.147$ *against* $+0.00005$ *at step* 200*). Gradient noise is therefore strongly anisotropic and lives in almost the same high-curvature subspace as the drift, at about sixty-five percent of its alignment. It is damped nearly as hard.*

This is the finding that bears on the informal picture. An escape mechanism requires noise in directions curvature does not damp — a flat reservoir in which exploration is cheap. No such reservoir appears: the noise is not merely non-flat, it is nearly as curvature-aligned as the descent direction itself.

12.5 Lift has no room

Lift would appear as a component that is neither the deterministic drift nor curvature-aligned. The decomposition leaves none. The residual after removing drift is the noise, and the noise is large, curvature-aligned, and loss-costly. Whatever carries this trajectory between basins, it is not a directed component orthogonal to local descent, because at this operating point there is no measured component of that description.

12.6 What collapses into what

Observation 12.3 (Drift and noise are one geometry at two amplitudes). *The four proposed buckets do not survive as four. Wandering is absent, lift has no residual to occupy, and drift and noise — the two components that do exist — differ in* coherence *but not in* direction relative to curvature. *Drift averages and noise does not; both point into the same high-curvature subspace. Curvature-following absorbs the decomposition.*

Three earlier results follow from this. It explains why averaging outperformed every compression scheme in §11.4: the noise is a genuine tax and averaging is exactly its removal. It explains the self-erasing geometry of Observation 11.8: the update descends precisely the directions from which its own coupling field is constructed, so applying it destroys the field. And the drift fraction falling $0.449 \to 0.348$ as the loss drops is the same phenomenon as the decline of r in Observation 8.4 — a shrinking coherent signal against a noise floor that does not shrink with it.

Limitations. Two checkpoints on one trajectory, one seed. Rayleigh quotients use four-batch Hessian-vector products and four noise samples, so the noise figures carry perhaps ±20%; the three-to-four order separation between drift, noise and random is not in question, but the drift/noise ratio of 1.5 is soft. The decomposition is of the raw gradient, not of the sign field: whether $\mathrm{sign}(g)$ splits the same way is a separate measurement and is untested. And the escape reading is refuted *at this operating point* — a model far from its floor, or one with genuinely competing basins, might show something else, and this trajectory reaches val = 0.088 with monotone progress throughout.

13 Phase 11: Flow, Allocation, and Forecasting

§11 closed compression of the update; §12 decomposed it. A third question remains: whether the *allocation* of optimisation across parameters can be precomputed. If some parameters travel while others merely oscillate, and if that partition is stable, it could be measured once and reused — which is precisely the reuse property every scheme in §11 lacked.

13.1 Per-parameter flow

Over a window, define per parameter

$$W_i = \frac{|\text{net displacement}_i|}{\text{path length}_i},$$

so $W \approx 1$ is a committed coordinate moving straight and $W \approx 0$ one that oscillates without net motion. Measured over 40-step windows at the val-floor regime: W has percentiles 0.009, 0.091, 0.472, 0.933, 1.000 at $1, 10, 50, 90, 99$, mean 0.4905, global 0.4973. About half of all per-coordinate motion cancels.

Concentration is mild. Sorting by net displacement, the top 1% of parameters carry 3.8% of the total, the top 10% carry 26.9%, the top 25% carry 53.1% and the top 50% carry 80.7%. Against a uniform 10% and 50% this is roughly 2.7× concentration. No small set of coordinates does the work.

Observation 13.1 (Churning is not a property of a coordinate; commitment partly is)**.** *Across two consecutive windows* $\mathrm{corr}(W_1, W_2) = +0.104$*. Decomposed by decile, the bottom-decile overlap is* 0.104 *against a chance value of* 0.10 *— exactly nothing — while the top-decile overlap is* 0.370*, some* 3.7× *chance. A coordinate that oscillates in one window is no more likely than chance to oscillate in the next, whereas committed coordinates have some persistent identity. Churn is a property of the moment, not of the parameter.*

This is what forecloses the partition. Selection of an inactive set requires the inactive set to persist, and it does not.

Churn relates to corpus statistics, weakly and in the expected direction. $\mathrm{corr}(W, \log v) = -0.278$: coordinates carrying more corpus-induced gradient energy churn *more*, oscillating rather than travelling. By contrast $\mathrm{corr}(W, r) = -0.056$, so the disc's coherence coordinate does not predict flow at all — another instance of the marginal/relational split of §11.1.

13.2 Freezing helps, but not because of churn

Freezing the bottom 90% by W and continuing to train produced a *lower* loss than training all parameters. The rival explanations are that low-W coordinates contribute noise without net motion, or that freezing any large fraction reduces the noise tax of §12.3 and W is irrelevant. Freezing a random subset of the same size, and the inverted selection, discriminate them.

scheme (freeze 90%)	seed 23	seed 17
none (control)	0.0665	0.0775
freeze low-W (drop churners)	0.0347	0.0304
freeze random	0.0393	—
freeze high-W (drop the committed)	0.0419	—

Observation 13.2 (The gain is the noise tax, not the churn metric)**.** *All three freezing schemes beat full training by about* 40%*, including the inverted one that discards the committed coordinates and retains the oscillators. The churn-specific advantage over a random subset is* 0.0347 *against* 0.0393*, some* 12%*. The ordering low-W < random < high-W is in the predicted direction but the dominant effect is generic. This is a confirmation of §12.3 rather than a new method: noise costs* -0.027 *per step against drift's* $+0.036$*, and freezing* 90% *of coordinates removes most of the noise injection while retaining most of the signal.*

Status. One seed on the control, 40-step horizons, and the `none` arm is itself unstable — full training slightly *worsened* loss from the 0.065 base in both runs, which is unexplained and worth checking. The result should not be built on until it replicates at three seeds over a longer horizon. Note also that freezing saves optimiser memory, communication and update application, but not computation: δ propagates through the whole network, so the full backward pass is still required to obtain the retained 10%.

13.3 The churn regime is not forecastable

Per-coordinate churn is unpredictable by Observation 13.1. The aggregate might not be. Over a sliding window of $w = 5$ steps define the vector cancellation

$$C_t = 1 - \frac{\|\theta_t - \theta_{t-w}\|}{\sum_s \|\theta_{s+1} - \theta_s\|},$$

and forecast C one window ahead from quantities already available at step t: loss, gradient norm, $\|m\|$, $\|v\|$, r_{50}, r_{90}, recent flip rate, and C's own history.

Over 416 steps, C has mean 0.0827, standard deviation 0.0165, range 0.040–0.144, and autocorrelations +0.803, +0.201, +0.096, +0.250 at lags $1, 3, 5, 10$.

Observation 13.3 (Nothing forecasts C, and there is little to forecast)**.** *On held-out data, an autoregressive fit on C's own history scores* $R^2 = -0.320$*, the seven optimiser features* -0.032*, and their union* -0.054*: every arm is worse than predicting the mean. At the five-step horizon the autocorrelation has already decayed to* $+0.096$*, so extrapolating C actively hurts. The optimiser features beat the autoregressive fit by* $+0.266$ *only by being closer to a constant. There is no operating regime to detect and no allocation policy that could be set in advance of it.*

Two cancellation measures, not in conflict. $C \approx 0.083$ is vector cancellation and $W \approx 0.50$ is per-coordinate cancellation. Coordinates individually reverse a great deal, but those reversals cancel *across* coordinates, leaving the vector motion nearly straight. Both are correct; the vector figure is the one relevant to wandering, and the per-coordinate figure is the one relevant to §12.3, where the cancelling part is the batch noise that costs -0.027 per step. It would be wrong to conclude from $C \approx 0.083$ that the optimiser is efficient: there is substantial per-coordinate churn, it simply does not cost directedness.

13.4 What this closes

proposed allocation route	result
stable active/inactive partition	no; churn does not persist, §13.1
per-coordinate learning-rate policy $\eta_i(W_i, m_i, v_i)$	undercut; W_i carries no forward information
penalise path inefficiency $\lambda(\text{path} - \text{net})$	target is 8% of path, and it is the noise
estimate $\mathbb{E}[\xi^2]$ and regularise	reduces to v; Adam already does this
forecast the churn regime	no, §13.3
freeze by churn	generic noise effect, §13.2
communication / optimiser-memory saving	stands, and is the one surviving win

The pattern across §11–§13 is uniform. The precomputable quantities — r, v, m, tile structure, churn, $F_{\mathcal{C}}$ — are each redundant with the loss, unstable within a few steps, or unidentifiable per coordinate. The one operation that reliably improved on a single-batch step was averaging, which is not a precomputation. Stated as a summary of the whole programme: *the optimiser's state is a good description of where it is and a poor predictor of where it goes.*

14 Phase 12: The Arc, Its Invariant, and the Holonomy

Every object measured in §11–§13 rotates away within ten steps, yet Proposition 37.3 of the companion report (*compressible but not extrapolable*) finds that 98.8% of a 40-step displacement lies in a single direction and a rank-one projection of it recovers 110% of the loss drop. Those cannot both be true of the same object. This phase resolves the contradiction, finds the programme's one conserved geometric invariant, and records what four geometric frameworks contribute — which in three cases is a specific negative.

14.1 The drift is estimable and rotates

Earlier reports of the mean-gradient direction decorrelating rested on an unmeasured quantity: split-half reproducibility was established for the *eigenspace* but never for the drift. If 150 batches estimate the mean gradient poorly, an apparent rotation is measurement noise. Measured at 300 batches per half:

step	$\text{SPLIT}_{\text{drift}}$	$\text{rot}_{\text{drift}}$	in top-5	in top-20	disp$\sim$drift
200	0.9708	1.0000	0.2471	0.2771	—
205	0.9654	0.3293	0.1642	0.2138	0.1745
210	0.9753	−0.1038	0.1315	0.2288	0.2554
225	0.9715	0.0047	0.0571	0.1450	0.2216
250	0.9783	0.0009	0.0888	0.1711	0.1914
400	0.9400	0.0547	0.0802	0.1178	0.1028

Null cos ≈ 0.0158.

Observation 14.1 (The drift is a real object that genuinely rotates)**.** *Split-half reproducibility is* 0.94–0.98 *at every checkpoint, so the drift direction is well estimated and the rotation is not an artefact:* cos *falls to* 0.33 *after five steps, passes through zero, and is negative at step* 210. *The drift also carries only* 25% *of its energy in the top-*5 *modes of* C_{resp}*, so it lies largely* outside *the high-variance subspace — and gains no stability there. Mean and covariance rotate at the same rate.*

Note the realised displacement aligns with the instantaneous drift at only 0.10–0.26. The displacement is not the drift integrated over a fixed direction.

14.2 A smooth arc, and its conserved pitch

The resolution is the one that report states: the trajectory is a smooth arc, well summarised in hindsight by its chord and poorly predicted in advance by its opening tangent. A curve of near-constant curvature has a well-defined chord while its tangent turns steadily. Following the gradient-flow decomposition $\dot{v} = -Hv$, $Hv = \lambda v + r$ with $r \perp v$,

$$\lambda = \frac{\langle v, Hv\rangle}{\|v\|^2} \text{ (decay rate)}, \qquad \omega = \frac{\|r\|}{\|v\|} \text{ (turning rate)}, \qquad \kappa = \frac{\omega}{\lambda} = \tan\angle(v, Hv).$$

Measured on the drift, with a random direction as null:

seed	step	κ_{drift}	split	κ_{random}	drift rot
17	200	0.928	0.122	53.9	+1.000
17	250	1.518	0.049	69.5	+0.013
17	320	1.322	0.292	92.3	+0.005
17	400	0.934	0.047	150.4	−0.003
23	200	1.059	0.045	69.1	+1.000
23	250	1.201	0.087	80.2	+0.020

Observation 14.2 (The curvature ratio is stable where every direction is not)**.** Substantially revised in §17.2*: with the corrected pipeline and pooling across corpora, $\omega \propto \lambda^{0.758}$ rather than $\omega \propto \lambda$, so κ is not conserved but varies as $\lambda^{-0.242}$. It is a compressed coordinate along an arc, not an invariant. The statement below holds within a single corpus at fixed pipeline. $\kappa_{drift} \approx$ 1.0–1.5 across both seeds and 200 steps, at split-half noise 0.045–0.122, while the drift direction it is computed from decorrelates completely over the same interval. A random direction gives $\kappa =$ 54–150, a separation of 50–100×. This is the programme's one conserved geometric quantity: the trajectory turns at roughly the rate it decelerates, which is the condition for a smooth arc rather than either straight-line descent ($\kappa \ll 1$) or a tumbling path ($\kappa \gg 1$).*

14.3 Where κ comes from, and where it does not

Decomposing the drift in the eigenbasis of H with energy weights c_i^2 on eigenvalues μ_i gives $\lambda = \sum c_i^2 \mu_i$ and $\omega^2 = \sum c_i^2 \mu_i^2 - \lambda^2 = \mathrm{Var}_c(\mu)$, so

$$\kappa = \frac{\sqrt{\mathrm{Var}_c(\mu)}}{|\mathbb{E}_c[\mu]|},$$

the coefficient of variation of the Hessian spectrum *as sampled by the drift's own energy.* This is exact, and it explains the null: κ_{random} is the CV over the whole spectrum, and 54–150 records a near-traceless spectrum with large spread — a fact about H. κ_{drift} is the CV over the sub-band the trajectory occupies, a fact about the trajectory.

Decomposed at a single checkpoint:

direction	λ	ω	$\|Hv\|$	κ	$\angle(v, Hv)$
drift	0.9736	1.0930	1.4629	1.123	48.3°
random	0.00005	0.00323	0.00323	64.15	89.1°
top eigenvector (approx.)	0.9376	0.3221	0.9903	0.344	19.0°
drift $\perp$ top eigenvector	0.8535	1.0921	1.3856	1.280	52.0°

Both moments move: λ is 19,325× random and ω is 338×, so κ is not derivative of the Rayleigh quotient alone. The drift sits where H acts strongly ($\|Hv\|$ is 453× random) and makes a 48° angle with its own image — between an eigenvector and orthogonality. Removing the top eigendirection barely moves κ (1.123 → 1.280) despite $\cos(\text{drift}, u_1) = -0.27$ against a null of 0.0005, so the cone is a property of the band and not of one mode.

Observation 14.3 ($\kappa \approx 1$ is observed, not derived)**.** *$CV = 1$ is the signature of an exponential density, which would make $\kappa \approx 1$ forced. Stochastic Lanczos on the drift ($m = 24$) gives mean* 0.178*, sd* 0.305*, CV* 1.71*, with* 81.6% *of weight on positive nodes and a spiky rather than smooth distribution. The density is not exponential, so the conservation of κ has no derivation and remains an empirical constant. The Lanczos CV (*1.71*) also exceeds the directly measured κ (*0.97*) at the same checkpoint; both estimate the same quantity, so one is biased and the density should be read qualitatively only.*

14.4 The osculating plane, and the holonomy

κ is defined on the 2-plane $P_t = \text{span}(v, Hv)$, in which the drift traces a logarithmic spiral $g(t) = g_0 e^{-(\lambda+i\omega)t}$ with pitch $\arctan \kappa = 48.3°$. In log-polar coordinates on that plane the motion is *linear in t*, so displacements compose by addition of complex increments — the linear structure a log map would supply. Whether it composes across steps depends on whether P_t is stable, which is a 2-dimensional question and far better conditioned than anything previously tested.

step	SPLIT_P	rot_P	angle_1	angle_2	drift rot	κ
200	0.7093	1.0000	—	—	1.0000	0.967
205	0.8281	0.0067	83.4°	89.7°	0.0248	2.359
210	0.6116	0.0059	83.9°	88.8°	−0.0870	1.789
225	0.6994	0.0015	87.5°	88.2°	0.0421	1.041
250	0.6697	0.0003	88.6°	89.9°	0.0136	0.886

Random 2-plane null $\approx 4.6 \times 10^{-7}$.

Observation 14.4 (The log map is exact per step and does not compose)**.** *The osculating plane is estimable (SPLIT* 0.61–0.83 *against a null of* 5×10^{-7}*) and rotates as fast as the drift: both principal angles reach* 83–90° *within five steps, so the new plane is essentially orthogonal to the old. The linear structure exists on each plane and does not transport between them. What the trajectory carries is therefore a connection with near-maximal curvature: the principal-angle pair* $(83.4°, 89.7°)$ *at* $\Delta = 5$ is *the holonomy increment, and it is close to a quarter turn per five steps.*

Observation 14.5 (The rotation is measured inside a retained shell)**.** *The* 2*-plane result above uses $k = 2$; every rotation measurement in this section uses $k \leq 20$, against a task with* VOCAB = 1017*. Sweeping k on C_{resp} at* 500 *batches, the ratio of across-checkpoint overlap to the same-θ split-half floor rises monotonically:* 0.001, 0.209, 0.350, 0.555, 0.707, 0.887, 1.217, 1.439 *at* $k = 1, 5, 20, 50, 100, 200, 350, 500$*. At $k \geq 350$ the subspace is retained* better *across five training steps than across two disjoint batch samples at the same θ. The absolute retention* ov(200, 205) *is flat at* 0.12–0.13 *for $k =$* 5–350 *while the noise floor collapses from* 0.57 *to* 0.09*, so the retained content is roughly constant while its estimability is batch-limited. The rapid decorrelation reported above is therefore rotation* within *a low-k frame enclosed by a partly retained subspace, and the phrase "near-maximal holonomy" should be read as k-specific. At $\Delta = 50$ steps, however, retention reaches only* 0.0173 *at $k = 500$, twice the random null, so the shell is retained over a few steps and not over the hundreds reported for classification tasks; the operative conclusion for reuse (§11.8) is unchanged.*

This is the fourth independent local-yes/global-no result in the programme, after the Lie closure (32% → 63% in-span locally, global closure failing), the sheaf quasi-section ("present in every stalk and exact in none"), and the tile bifibration stopping at the index set (§10.3). It is also the first place the word *monodromy* is used non-metaphorically: the holonomy is a measured rotation of a measured frame.

14.5 What the geometric frameworks contribute

Koenigs, and nothing past it. Complex dynamics supplies exactly one relevant theorem. For a fixed point of multiplier λ with $|\lambda| \neq 0, 1$ there is a *local* holomorphic ϕ with $\phi \circ f \circ \phi^{-1}(w) = \lambda w$, constructed as a limit $w_n = f^n(z)/\lambda^n$ convergent only on a neighbourhood, and the linearising coordinate extends to a maximal radius whose boundary contains a critical point. The theory therefore *predicts* the locality of Observation 14.4 rather than merely being consistent with it.

Nothing past it transfers. The Adam step involves sign, $\sqrt{\cdot}$ and $|\cdot|$ on real coordinates and is nowhere holomorphic, so Montel, the Julia/Fatou dichotomy, Böttcher coordinates and the Lefschetz residue formula do not apply; and the linearisation itself survives in the real category anyway. The multiplier classification also fails to fit: an attracting fixed point requires the iterates to contract, whereas $\|g_{\text{floor}}\|$ stays at 0.21–0.23 throughout training, sustained by the batch noise of §12.3. The basin floor is a noisy stationary distribution, not an attracting periodic point.

No 3-manifold, and the measurement is decisive. A 2-plane field that is everywhere non-integrable is a contact structure, and an integrable one is a foliation; both require a 3-dimensional ambient. In $\mathbb{R}^3$ two 2-planes must meet in a line, $\dim(P \cap Q) \geq 2 + 2 - 3 = 1$, so the first principal angle is necessarily 0. Measured, it is 83.4° at $\Delta = 5$ and 88.6° at $\Delta = 50$. Consecutive osculating planes share no line and their union is 4-dimensional; the ambient is not a 3-manifold. Independently, the plane field is neither integrable nor maximally non-integrable — bracket in-span is 32% → 63% — so it falls between the two classes the theory distinguishes.

Morse theory fits and is already implicitly in use. The index of a critical point is the number of negative Hessian eigenvalues, and a k-handle is the neighbourhood of an index-k critical point, $H^n_k \cong D^k \times D^{n-k}$, with gradient flowing in along the attaching region and out along the belt region. The codimension-2 crossing at step 72, where the Hessian null space reorganises from 5/5 tangent to 2/5 tangent and 3/5 crossing, has the form of a handle attachment.

But the direct test is underpowered. A handle attachment requires the *index* to change, which is a different quantity from null-space tangency. Estimating it by stochastic Lanczos quadrature ($m = 22$, four random probes, eight Hessian-vector products per iteration) gives at step 40 a negative-eigenvalue fraction of 0.406 with a probe-to-probe standard deviation of 0.236 — a noise level of 58% of the value, far too large to resolve a change across step 72. The Morse reading is therefore *consistent with* the reported tangency event and *not confirmed*; confirming it needs a variance-reduced spectral estimator, and that is the measurement this section leaves open.

14.6 Status

claim	status	basis
drift direction is estimable	holds	SPLIT 0.94–0.98
drift direction rotates	holds	$\cos 0.33$ at 5, < 0 at 10
κ conserved, ≈ 1	holds, two seeds	Obs. 14.2
κ derived from spectral density	refuted	CV $1.71 \neq 1$
osculating plane stable	refuted	83–90° at $\Delta = 5$
log map composes across steps	refuted	Obs. 14.4
holonomy is measurable and large	holds	principal angles
Koenigs locality applies	holds	and predicts the above
complex dynamics beyond Koenigs	does not apply	non-holomorphic step
3-manifold / contact structure	refuted	first principal angle $\neq 0$
Morse handle at step 72	untested	estimator underpowered

The programme therefore ends with one conserved geometric invariant whose conservation is unexplained, defined on a frame with near-maximal holonomy. That κ is a scalar constraining the arc's pitch without fixing its plane is precisely why it explains *compressible but not extrapolable* rather than defeating it: the chord determines the arc because the pitch is fixed, and the tangent does not, because the plane it lies in is gone within five steps.

15 Phase 13: The Active Subspace and the Index Question

§14 left the Morse reading of the step-72 event untested, with a stochastic estimator too noisy to resolve it. This phase resolves it, and the answer is negative. It also records a methodological point that recurs: three estimators of "the negative-eigenvalue fraction" disagree by two orders of magnitude because they measure three different objects, and only one of them is the quantity a Morse reading requires.

15.1 Three estimators, three objects

estimator	what it measures	cost	neg. fraction
stochastic Lanczos quadrature	spectral density, weight-averaged	$m \cdot n_{\text{hvp}}$ per probe	$0.648 \to 0.011$
random coordinate submatrix	diagonal curvature H_{ii}	m HVPs	0.050
randomized range finder	top-r by $\lvert\lambda\rvert$	$2r$ HVPs	**0.000**

The coordinate submatrix is invalid for this purpose. A 110-coordinate random subset of 4.3×10^6 captures essentially no coupling — mean off-diagonal correlation is 0.0708 — so H_{SS} is near-diagonal and its spectrum is approximately $\{H_{ii}\}$. Measured $\lambda_{\min} = -0.0001$ against $\lambda_{\max} = 0.029$: numerically positive semi-definite. This is the same trap recorded in Observation 11.9: the structure is relational, and a random coordinate subset destroys exactly the relations. The estimator was built in violation of the programme's own finding and its output is a statement about per-coordinate curvature, not about the spectrum.

The range finder is the correct construction, since Q spans the directions in which H acts and the coupling is retained by construction. Following the standard two-stage scheme with $r = 36$: $Y = H\Omega$ for Gaussian Ω, $Q = \text{qr}(Y)$, $B = HQ$, and $H_{\text{active}} = Q^\top B$ symmetrised and eigendecomposed exactly, with two independent Ω draws as the sampling control.

step	neg (draw A)	neg (draw B)	R_{eff}	$\lambda_{\min}$	$\lambda_{\max}$
40	0.000	0.000	23.78	+0.0384	0.9001
60	0.000	0.000	33.24	+0.0388	0.1757
72	0.000	0.000	30.13	+0.0253	0.3463

15.2 The Morse index reading is not supported

Observation 15.1 (The dominant spectrum is positive; the negative mass is bulk). *The top-*36 *subspace by* $\lvert\lambda\rvert$ *contains* no *negative eigenvalue at any checkpoint, with* $\lambda_{\min}$ *between* +0.025 *and* +0.039 *and identical results across two independent random draws. A range finder is biased toward large* $\lvert\lambda\rvert$ *of* either *sign, so large negative eigenvalues would have been captured. They are not there.*

The three estimators therefore reconcile: the Hessian has a positive dominant spectrum and a near-zero bulk whose sign balance is roughly even. The Lanczos figure of 0.648 *counts that bulk by spectral weight; every eigenvalue it counts is negligible in magnitude. The reported fall* $0.648 \to 0.011$ *between steps* 72 *and* 110 *is therefore a change in the sign balance of near-zero modes, not the escape from a high-index saddle into a convex basin.* The Morse handle reading of §14.5 is withdrawn*: there is no index to collapse, because the index in any magnitude-resolved sense is already zero at step* 40.

Two consequences. First, the step-72 null-space reorganisation and the 80–160 bracket-closure rise remain real events with the right timing, but they are not index events and should not be given a handle-attachment reading. Second, R_{eff} of the active subspace is 23.8–33.2 out of 36: the active directions are well spread, with no low-rank core inside them, so framework D's "curvature diffuses" picture has no low-rank starting state to diffuse from.

15.3 On Dowker duality

Dowker's theorem and its higher-order extensions state that for a relation $R \subseteq X \times Y$ the two nerve complexes built on X and on Y have isomorphic homology. Applied here with X the parameters, Y the batches and R the sign relation, this would permit computing on the batch side — some 500 elements — rather than the parameter side, some 4.3×10^6: a reduction of four orders of magnitude, and a genuine one.

It is nonetheless not useful here, for a reason independent of cost. Dowker duality is a *computational* equivalence between two presentations of the same homology; it does not denoise, and it does not supply the scale separation that persistent homology requires to produce interpretable bars. The programme's recurring measurement — no spectral gap in H, spectral clustering of C_{resp} collapsing to a single block — is precisely the absence of that separation (§7.5, §11.9). Dowker would make an uninformative computation cheap rather than making it informative.

15.4 What survives of the non-holonomic programme

claim	status	basis
global bracket-generating distribution	refuted	in-span 32% → 63%: partially integrable
graded Lie algebra on role partitions	refuted	role leakage at the null, §11.3
sheaf pullback extending to gradients	refuted	70.1% residual, flat, §1.13
sign flips coupled to curvature	refuted	flip rate flat over 4.7× in ω
Morse handle at step 72	refuted	Obs. 15.1
transitions keyed to step 120	artefact	tile logging schedule, not dynamics
κ conserved at the floor	holds	Obs. 14.2, two sweeps
κ = Hessian–gradient overlap	holds	overlap $= 1/\sqrt{1+\kappa^2}$
frame rotation inside a retained shell	holds	Obs. 14.5

What remains is not a mechanical law but a metric description with three measured components: an invariant pitch, a rotating frame, and a subspace that retains what the frame does not. The mechanical apparatus — constraint distributions, gauge bundles, handle decompositions — has been tested piece by piece and each piece has failed against a specific measurement. The description is leaner for it, and every clause in it is now traceable to a number.

16 Phase 14: Formation and Turning

The spatial-structure question is settled, and settling it removed a confound rather than answering the open question. *The observed formation is architectural.* It is a fixed subspace constraint imposed by LayerNorm, predictable from the model definition without measuring anything, and it does not participate in the temporal phenomenon it was mistaken for. What remains unexplained is the turning of the coordinate-level sign field inside that fixed formation.

That sentence is the phase's result. Everything below either establishes it or bounds it.

16.1 Anti-alignment at row scale, and its mechanism

Sign coherence within a contiguous block, $c_T = \big|\, |T|^{-1} \sum_{i \in T} \operatorname{sign}(g_i) \,\big|$, against the balanced-sign baseline $\sqrt{2/\pi n}$:

n	16	64	128	256	512	1024
contiguous	0.944	0.915	0.872	**0.705**	0.706	0.706
within-row shuffled	0.945	0.915	0.856	**0.705**	0.706	0.706
globally shuffled	0.981	0.986	0.986	0.986	0.991	0.990

Observation 16.1 (Row membership, not row ordering)**.** *c_T is* below *the balanced-sign baseline at every scale, stepping to* 0.705 *at* $n = 256 = D$ *and flat thereafter. Shuffling within rows changes nothing (*0.705 *against* 0.705*); shuffling globally removes the effect entirely (*0.986–0.991*). Which parameter one is adjacent to is irrelevant; which constraint group one belongs to is everything.*

The mechanism is LayerNorm's backward projection,

$$\frac{\partial L}{\partial x} = \frac{1}{\sigma}\Big(I - \tfrac{1}{D}\mathbf{1}\mathbf{1}^\top - \tfrac{1}{D}\hat{x}\hat{x}^\top\Big)\frac{\partial L}{\partial y},$$

which restricts the gradient to a subspace orthogonal to the all-ones direction. The correct statement is the projection; *the sign anti-correlation is a secondary consequence of the projected gradient's distribution, not an implication of the constraint.* A zero-sum group need not anti-align: with concentrated magnitudes it can be nearly unanimous. The measured c_T therefore encodes the constraint *and* the within-group magnitude distribution, and the two measurements below are what make the attribution convincing rather than either one alone.

16.2 The constraint residual: a unit test, not a result

If the grouping is correct, gradients of matrices writing into the LayerNormed residual stream must sum to zero along D. This has a known answer and gates everything downstream, so it is reported as pass/fail.

matrix class	axis	width	|axis sum|/|permuted sum|
`ff.o`, `attn.op` (12 matrices)	0	256	**0.0000** PASS
`ff.v`, `ff.g`	1	256	0.0151–0.0158 PASS

Direct residual writers annihilate to machine precision — the identity $\mathbf{1}^\top P = 0$ appearing verbatim. Upstream contributors inherit the constraint at 1.5%. Two levels of constraint propagation, both passing, so the coordinate system is validated.

16.3 One constraint geometry, not two

LayerNorm's constraint runs along D; a softmax constraint would run along VOCAB = 1017, which is stride-D in row-major storage and orthogonal to the first. Measured per matrix along both axes:

role	shape	along C	along R
`attn.WQ/WK/WV`	256×256	**0.57–0.63**	0.89–1.35
`ff.g`, `ff.v`	512×256	**0.58–0.61**	0.91–1.55
`ff.o`, `attn.op`	256×512	0.86–1.10	**0.61–0.66**

Observation 16.2 (The invariant is the residual-stream direction)**.** *Every matrix dips on exactly one axis, and it is always the axis of width* 256*. For transposed matrices that is the row axis rather than the column axis, so the invariant is not "row" or "column" but the* residual-stream width direction*; storage orientation changes the index, not the constraint family. No dip appears at any width near* VOCAB*, so the softmax geometry is absent and the unconstrained baseline is uncontaminated. The per-matrix dip (*0.57–0.66*) is deeper than the* 0.705 *measured on the concatenated vector, which straddled matrix boundaries.*

16.4 The positive-coherence anomaly, resolved

The unconstrained axis shows $c_T/\text{base} > 1$ — the only positive sign signal in the programme. Discriminated by the same permutation logic:

matrix	shape	observed	shuffled	rank-1	eff. rank
`te.weight`	1017×256	**10.473**	1.002	5.764	11.2
`ff.g` ($\times 6$)	512×256	1.41–1.64	0.98–1.02	0.91–1.87	5.9–6.7
`attn.WQ/WK/WV`	256×256	1.08–1.21	0.98–1.03	0.91–1.54	4.1–6.6

Observation 16.3 (Parameterisation and corpus degeneracy, not dynamics)**.** *The shuffled control sits at* 1.004 ± 0.02 *for every matrix: membership carries the excess entirely and the gradient distribution carries none of it. For ordinary matrices the rank-one prediction brackets the observed value, so the excess is what the leading singular direction alone produces.* `te.weight` *at* $10.5\times$ *is an outlier and a corpus artefact: the corpus is one sentence looped, so most of the* 1017 *vocabulary rows occur a handful of times and each receives a near-rank-one gradient* $\sum_t (p_t - y_t) h_t^\top$ *with nearly unanimous sign. On a corpus with realistic token frequencies this should shrink substantially, and no structural claim should rest on it.*

16.5 The formation does not turn the field

The decisive test. S_D = groups along the constrained (width-256) axis; $S_{\neg D}$ = groups along the unconstrained axis of the *same* matrices — matched for matrix, checkpoint, optimiser, corpus and trajectory, differing only in whether the projection acts along the grouping axis.

k	$\cos S_D$	$\cos S_{\neg D}$	flip S_D	flip $S_{\neg D}$
1	0.0727	0.1185	0.4500	0.4500
2	0.0458	0.0709	0.4605	0.4605
3	−0.0063	0.0092	0.4704	0.4704
4	−0.0168	−0.0146	0.4758	0.4758
6	−0.0009	−0.0118	0.4747	0.4747
8	−0.0668	0.0275	0.4828	0.4828

Observation 16.4 (Constraints are boundary conditions, not a motion law)**.** $\tau_{S_D} \approx \tau_{S_{\neg D}}$. *The mean cosine difference is* -0.0287 *against a cross-matrix spread of* 0.173*, and the flip rates agree to four decimal places at every lag — the sharper result, since a flip is a per-coordinate event and the constrained axis does not alter its rate at all. The architecture defines where motion may occur and does not participate in how the field turns. Formation and turning are independent mechanisms.*

16.6 Where the surviving signal is

The same run localises the three-step window, and the localisation is restrictive.

Observation 16.5 (Aggregate memory is gone at lag one; coordinate reuse pays to three)**.** *Group-mean cosine is* 0.07–0.12 *at* $k = 1$ *and crosses zero by* $k = 3$–4 *on both axes; tile-level cosine is* 0.024–0.053*. Yet wholesale reuse of the sign field retains* 0.816 *of the loss improvement at* $k = 2$ *and* 0.668 *at* $k = 3$ *(§13.3). No direction is preserved, and every aggregation — tiles, constraint groups, matrix means — erases what is.*

The two are reconciled by §4.10: a sign field survives 25% i.i.d. corruption with no loss. What persists is not a direction but a *majority*. A flip rate of 0.45 leaves 55% of coordinates correct, and a group mean of a field that is 55% correct and otherwise balanced averages to near zero — which is exactly why aggregation destroys it. The object is $\{\text{sign}(g_i)\}_{i=1}^N$ at coordinate resolution, some 4.3×10^6 bits, and nothing coarser.

16.7 The restated question, and the cheapest explanation

The target is no longer "find the conductor". It is:

> Why does the per-coordinate flip rate climb only $0.45 \to 0.48$ over eight steps, when a single step already decorrelates every aggregate?

The margin sustaining three useful steps is about 5% above the chance rate of 0.464 for independent batches. Before any geometric explanation is sought, the cheapest one should be excluded: Adam's first moment is a free byproduct with $\beta_1 = 0.9$, giving roughly ten steps of memory — the right timescale. The test is a β_1 sweep of the reuse curve, at $\beta_1 = 0$ and $\beta_1 = 0.99$. If the horizon moves with β_1, the three-step window is momentum rather than geometry, and the mechanism was in the optimiser throughout.

16.8 Status

claim	status	basis
row-scale anti-alignment is architectural	holds	§16.1, §16.2
one constraint geometry (residual width)	holds	§16.3
softmax/vocabulary geometry present	absent	§16.3
positive coherence is dynamical	refuted	§16.4
constraints alter the transport horizon	refuted	§16.5
the window lives in aggregates	refuted	§16.6
the window is Adam's β_1	untested	§16.7
projector decomposition of the formation	untested	needs a forward hook

Step 3 — decomposing g_y into $(g_{\mathbf{1}}, g_{\hat{x}}, g_{\perp})$ upstream of the projection — is now correctly classified as bookkeeping for the formation. It would answer why the gradient occupies the constrained subspace. It cannot answer why the coordinate sign field survives three steps, and the two questions should not be conflated again.

17 Phase 15: Trajectory Geometry and the λ–ω Arc

Phase 12 identified $\kappa = \omega/\lambda$ as the programme's one conserved geometric quantity. This phase tests it properly — across capacity, depth, corpus and pipeline — and the result is a demotion and a replacement. κ is not conserved and is not dimensionless; it is a compressed coordinate along a power-law arc in the (λ, ω) plane, and that arc is the durable object. Two methodological errors in earlier phases are corrected on the way.

17.1 Two corrections to the measurement pipeline

The spectral initialisation is corpus-dependent, and the MF pump was never running. The compiler builds E_0 from the corpus bigram Laplacian via eigsh$(L_{\mathrm{sym}}, k = D{+}1)$, so a corpus with a sparse bigram graph receives a near-degenerate initialisation: the Lisp corpus has nnz $= 504$ against the authentic corpus's 1347 over the same VOCAB $= 1017$, leaving some 919 isolated nodes. Separately, the adaptive MF pump (PHASE 2) sits *after* the cut point used by every run in §14–§16, so it never executed. It applies Fisher-normalised embedding steps $\delta E = -\mathrm{emb\_grad}/(\mathrm{emb\_fish} + 10^{-4})$ at $\eta_{MF} = 0.01$ with W_K, W_Q alternately frozen, and stops on the geometric anchors ($\Phi_{\mathrm{clean}} = 5/5$, or τ rising after falling), so its stopping round is itself corpus-dependent.

run	bigram nnz	spectral E_0	MF rounds	handoff val
auth D=128	1347	5.666	4	6.714
auth D=256	1347	4.460	2	13.781
lisp D=128	504	—	2	11.985
lisp D=256	504	—	2	22.234

The pump *raises* validation loss every time and improves downstream training regardless — auth D=128 reaches val 3.298 at step 15 with the pump against 4.264 without. It sets geometry at the cost of loss, which is its purpose. *Every trajectory number in §14–§16 therefore comes from a pipeline missing its second stage.* Within-corpus comparisons there are internally matched and stand; absolute values shift substantially (λ at step 15 for auth D=128 is 0.847 with the pump against 0.069 without).

17.2 The arc

Pooling 31 checkpoints across six runs — two corpora, two depths, two widths, with the corrected pipeline:

Observation 17.1 (A power-law arc in the (λ, ω) plane)**.**

$$\log\omega = 0.758\,\log\lambda + 0.158, \qquad R^2 = 0.966,$$

holding across a 222× *range in* λ *and across both corpora. Per-run slopes are* 0.89 *and* 0.80 *on the authentic corpus* ($R^2 = 0.91, 0.96$) *and* 0.27–0.61 *on Lisp* ($R^2 = 0.70$–0.82)*, with the Lisp runs occupying the high-*λ *end of the same curve and traversing a shorter stretch of it.*

This is the strongest collapse the programme has produced, and it is a *trajectory* law rather than the "universal ray" $\omega \approx c\lambda$ reported earlier. Better-trained models travel further down the arc.

Observation 17.2 (κ is a coordinate along the arc, not a dimensionless invariant)**.** *A slope of* 1 *would make* $\kappa = \omega/\lambda$ *constant. The measured slope of* 0.758 *gives* $\kappa \propto \lambda^{-0.242}$*, and* $\mathrm{corr}(\log\lambda, \log\kappa) = -0.863$*.* κ *varies* 6.5× *while* λ *varies* 222×*, so it is a* compressed *coordinate — which is why it appeared stable. Its value near unity has no special status; it is where authentic-corpus runs happen to sit. The exponent* 0.758 *is the quantity the ratio discards.*

17.3 κ tracks optimisation stage, not corpus identity

Measured on Lisp, $\kappa = 0.618 \pm 0.138$ against authentic's 1.325 ± 0.337 — non-overlapping bands, which suggested corpus dependence. But corpus identity was confounded with corpus *difficulty*: Lisp has $H_3 = 0.569$ against authentic's 0.001 and the six-block model stalled at val 1.296 while authentic reached 0.058, so Lisp may simply never have left the high-curvature region.

Depth disambiguates. Giving the same Lisp corpus twelve blocks instead of six:

run	s15	s50	s100	s160	s230
lisp L=12, val	2.690	2.221	1.901	1.207	**0.509**
lisp L=6, val	3.348	2.562	2.151	1.713	1.092
lisp L=12, κ	0.311	0.422	1.019	0.627	**1.190**
lisp L=6, κ	0.395	0.713	0.785	0.943	0.664

Depth halves the loss and κ rises to 1.190, inside the authentic band; λ falls 29.5 → 11.5 (L=6) and 21.6 → 3.6 (L=12).

Observation 17.3 (The corpus separation was a progress proxy)**.** $\kappa \approx 0.3$–0.7 *in the high-curvature regime, rising toward* 1.2–1.3 *as the model descends, on both corpora. The non-overlapping bands reflected how far down the loss curve each model got, not a property of Lisp's structure.* The claim that κ is corpus-dependent is withdrawn.

17.4 The corpus family

Deciding the question properly requires two corpora matched on local redundancy and differing only in global syntax. A first attempt failed: expanding the Lisp leaf alphabet to 1017 symbols gave a live vocabulary of 334 and $H_3 = 0.503$, because *Lisp syntax forces its own local redundancy* — parentheses and keywords are a large fraction of every sequence, and that repetition is the language. Renaming the leaves does not touch it.

The resolution is to make the *structural* tokens node-unique as well. Four encodings of the same generated trees, so global syntax is held fixed and only the surface encoding varies:

corpus	H_0	H_1	H_2	H_3	live vocab
authentic	6.769	0.430	0.018	0.001	1014
L0 ordinary	3.174	1.954	1.229	0.613	68
L1 α-renamed leaves	3.732	1.672	1.104	0.511	325
L2 typed keyword symbols	4.342	1.870	0.663	0.248	622
L3 node-unique delimiters	**6.776**	**0.431**	**0.010**	**−0.000**	**940**

Observation 17.4 (The matched corpus is constructible, and the redundancy lives in the delimiters)**.** *L3 matches the authentic corpus at every order measured. The monotone progression* $H_3 = 0.613, 0.511, 0.248, 0.000$ *locates the redundancy precisely: not in the leaves, which α-renaming already addressed, but in the parentheses and keywords. Making the delimiter* pair *node-unique removes it while leaving bracket matching, recursive nesting and arity intact.*

An earlier draft of this section claimed such a corpus was not constructible. That claim was too strong — it was a statement about one grammar at one corpus size — and is withdrawn.

17.5 The decisive comparison: L3 versus authentic

Both corpora trained for 200 steps under the full pipeline. The construction worked as designed: L3's bigram graph has nnz = 1354 against the authentic corpus's 1347, so the spectral E_0 was built on comparable graphs and the initialisation confound of §17.1 is removed. Both MF pumps ran four rounds. L3 learns as well as authentic ($3.751 \to 0.094$ against $3.468 \to 0.086$), so difficulty is matched too.

step	authentic			L3		
	val	λ	κ	val	λ	κ
15	3.469	2.033	1.206	3.751	3.820	0.673
50	1.407	0.211	1.487	1.585	0.103	1.625
100	0.536	0.373	1.110	0.695	0.150	1.137
150	0.170	0.216	1.153	0.291	0.146	1.297
200	0.086	0.223	1.325	0.094	0.110	2.179

Observation 17.5 (At matched loss the κ bands merge)**.** *Interpolating each run to common validation loss:*

val	2.0	1.0	0.5
authentic κ	1.405	1.313	1.120
L3 κ	1.443	1.304	1.214
|*difference*|	0.038	0.009	0.094

against split-half noise of 0.05–0.19. *Two corpora with identical local statistics at every measured order and completely different global syntax — English prose against balanced recursive S-expressions — give κ values indistinguishable at matched loss. The defensible statement is the narrow one:* these two corpora do not produce distinguishable κ once optimisation state is matched*, which is weaker and safer than "κ is independent of corpus". The non-overlapping bands of §17.3 were difficulty in disguise.*

Note also what this does not *establish. Validation loss is one index of optimisation state, which also includes gradient norm, curvature scale and schedule position. Matching loss happened to match κ in these runs; that does not show loss alone determines it. The claim is that κ is largely explained by optimisation stage* as indexed by validation loss in these experiments.

Normalising by progress $p = (L_0 - L)/(L_0 - L_{\text{final}})$ rather than by step gives the same answer, with the residual concentrated at the endpoints where κ is worst conditioned:

p	0.20	0.40	0.60	0.80	0.95
authentic	1.298	1.390	1.481	1.213	1.144
L3	0.994	1.316	1.609	1.208	1.363

$\kappa(t)$ across corpora

Collecting every well-conditioned trajectory measured, at D=128 unless noted:

corpus (run)	s15	s50	s100	s150/160	s200/230	shape
authentic (this run)	1.21	1.49	1.11	1.15	1.33	flat, noisy
L3 (this run)	0.67	1.63	1.14	1.30	2.18	rising
lisp, unmatched L=6	0.40	0.71	0.79	0.94	0.66	low, flat
lisp, unmatched L=12	0.31	0.42	1.02	0.63	1.19	rising
lisp, matched-vocab L=12	0.29	0.67	0.88	0.97	0.98	monotone
authentic, no MF pump	2.56	—	1.70	1.65	1.21	falling

No two runs share a shape. The variation across pipelines and corpora at fixed step is larger than the variation between corpora at fixed loss, which is Observation 17.5 seen from the other side: step is the wrong abscissa and $\kappa(t)$ has no characteristic form.

17.6 Trajectory embedding

κ discards a degree of freedom. Embedding each checkpoint as $x(t) = (\lambda, \omega, \text{eff}, \text{turn}, L)$ and comparing the two runs point by point:

step	(λ, ω)		(eff, turn)		$\lVert F_{\text{auth}} - F_{\text{L3}} \rVert$
	authentic	L3	authentic	L3	
15	$(2.033, 2.429)$	$(3.820, 2.569)$	$(0.789, 23.1)$	$(0.833, 21.4)$	0.826
50	$(0.211, 0.314)$	$(0.103, 0.168)$	$(0.673, 15.5)$	$(0.682, 14.7)$	0.926
100	$(0.373, 0.414)$	$(0.150, 0.170)$	$(0.556, 20.3)$	$(0.567, 20.5)$	1.226
150	$(0.216, 0.249)$	$(0.146, 0.190)$	$(0.491, 22.9)$	$(0.493, 21.4)$	0.754
200	$(0.223, 0.296)$	$(0.110, 0.239)$	$(0.437, 24.9)$	$(0.440, 23.8)$	0.734

Observation 17.6 (The paths align in the full feature space, not only in κ)**.** *With $F(t) = (\log\lambda, \log\omega, \mathit{eff}, \mathit{turn}/30, \log L)$ standardised over the pooled runs, the mean distance between* matched *checkpoints is* 0.893 *while the mean distance between* mismatched *checkpoints is* 3.272 *— a ratio of* 0.273*. The two runs trace the same path through the coarse-graining feature space, not merely the same scalar. The efficiency and turning coordinates are the tightest: efficiency agrees to* 0.002–0.011 *at every step, turning to* 1.7° *or better.*

This is a stronger similarity statement than matching κ, and it is the form worth carrying forward: *different global syntax, matched local statistics, and a common trajectory through the space of geometric observables.*

17.7 The arc, confirmed out of sample

Observation 17.7 (L3 lands on the arc it was not fitted to). *L3's own slope in the* (λ, ω) *plane is* 0.751 *against the pooled* 0.758 *of Observation 17.1, which was fitted before this corpus existed. Residuals against the pooled law are comparable for both runs: mean* -0.172*, sd* 0.199 *for authentic and mean* -0.243*, sd* 0.191 *for L3. This is the first out-of-sample confirmation of* $\omega \propto \lambda^{0.758}$.

Path efficiency falls monotonically in both runs — $0.789 \to 0.437$ and $0.833 \to 0.440$ — agreeing to within 0.011 at every matched step. That is the seventh independent condition under which the efficiency decline has held.

17.8 Status after Phase 15

claim	status	basis
$\omega \propto \lambda^{0.758}$	holds, out-of-sample	Obs. 17.1, 17.7
path efficiency falls monotonically	holds, 7 conditions	§17.7
trajectories align in feature space	holds	Obs. 17.6
κ explained by stage, indexed by loss	holds, these runs	Obs. 17.5
κ determined by loss alone	untested	§17.5
exponent stable under optimiser/schedule	untested	not varied
κ tracks corpus structure	refuted	Obs. 17.5
κ conserved or dimensionless	refuted	Obs. 17.2
$\kappa(t)$ has a characteristic shape	refuted	§17.5

Which object is the result

The arc rather than κ, and the distinction is epistemic rather than presentational. κ is one fitted number per checkpoint, and its agreement across corpora is post-hoc. The arc is a relation over 31 checkpoints with $R^2 = 0.966$ that then *predicted* L3's slope to 0.751 against 0.758 on a corpus that did not exist when it was fitted. Out-of-sample prediction is a different category of evidence from matched endpoints.

$\kappa = \omega/\lambda$ is therefore a projection of the trajectory onto one coordinate, which also explains why it fluctuates while the underlying relation stays put — a compressed coordinate on a stable arc. Its role has changed over the course of the programme, from a candidate mechanism to a state summary. That is a demotion and not a failure; temperature, Reynolds number and order parameters all summarise state without being causal.

The corpus question is closed. What remains open is what the exponent 0.758 *is*: it is measured across two corpora, two depths, two widths and two pipelines, and it is not derived from anything. Nor is it known whether it survives changes to the optimiser, the architecture family, or the learning-rate schedule — none of which has been varied. A second axis is available and untested — the L0–L3 continuum of §17.4 interpolates between ordinary Lisp and authentic-like statistics at fixed tree structure, so measuring the arc across all four would separate whether the geometry responds to surface statistics or to the underlying dependency graph.

17.9 Two nulls referenced later

Recorded here for reference. *The architectural starfish*: an incidence group of 15,360 parameters — every coordinate reading from or writing to one residual channel, across all blocks — persists no better than a size-matched random set (+0.006 at lag one, negative thereafter). *The tangent subspace*: two *disjoint* windows of ten gradients at identical θ give a top-k subspace overlap of 4%, so the subspace is not reproducible under batch sampling with no parameter motion. An earlier 97% figure used sliding windows sharing seven of eight gradients and is withdrawn.

17.10 Retention of Hv

Linear and additive, unlike κ: retention$(A) = \|P_A Hv\|^2/\|Hv\|^2$, with $|A|/P$ as the isotropic null. Architectural charts are isotropic to three decimals — random sets give $1.005, 1.004, 0.987$; LayerNorm channels $1.00, 0.80, 1.23$; heads 0.38–0.47; layers 0.42–1.28. The Hessian eigenbasis is not: $r = 12$ directions of 4.3×10^6 hold 7.6% of $\|Hv\|^2$, a concentration of 2.75×10^4. That chart decorrelates within ten steps (stale/fresh retention 0.010 at lag ten, against a same-θ split floor of 0.021).

18 Phase 16: The κ Singularity and Its Resolution

$\kappa = \omega/\lambda$ has been the programme's central observable since §14, and its worst behaviour — excursions to 11, 106, 338, split noise exceeding the value, non-monotone trajectories — has been treated throughout as measurement difficulty. It is not. It is a coordinate singularity, it is removable, and removing it converts a noisy scalar into a smooth one and an unstable marker into a sharp regime boundary.

18.1 The singularity is in the chart, not the geometry

The pair (λ, ω) determines a point of $\mathbb{P}^1$, and $\kappa = \omega/\lambda$ is the affine coordinate on the chart $\lambda \neq 0$. What fails at $\lambda = 0$ is that chart. The resolution is the single blow-up that always removes such a failure: pass to the projective line, coordinatised by the angle

$$\phi = \arg(\lambda + i\omega) = \operatorname{atan2}(\omega, \lambda) \in (-\pi, \pi], \qquad \kappa = \tan\phi.$$

$\kappa \to \infty$ is $\phi \to 90°$, an ordinary point. This is elementary — one blow-up of a projection $\mathbb{P}^1 \to \mathbb{A}^1$, not a resolution of a variety — but it is the right move and it should have been made when κ was introduced.

Observation 18.1 (ϕ is smooth exactly where κ diverges)**.** *The same five consecutive checkpoints at $D = 128$, in both coordinates:*

λ	ω	κ	ϕ
$+0.0790$	0.224	2.83	70.5°
$+0.0600$	0.216	3.60	74.5°
$+0.0462$	0.269	5.83	80.3°
-0.0366	0.408	11.16	95.1°
-0.4596	0.943	2.05	116.0°

κ rises to 11.16 *and then collapses* below its starting value*; the sequence is uninterpretable. ϕ increases monotonically and smoothly through* 90°*. At $D = 256$ it continues* $107.7° \to 116.5° \to 131.8°$*. The pathology was entirely in the chart.*

Two consequences follow immediately. First, κ discards the sign of λ: $\kappa = 1$ occurs at $\phi = 45°$ and again at $\phi = 225°$, which are descent and ascent respectively. Every "κ crosses 1" marker in §17 conflated them. Second, results already reported read more tightly in ϕ: the corpus bands that merged at matched loss, $\kappa \in [0.92, 1.29]$, are $\phi \in [43°, 52°]$ — a 9° spread.

18.2 The regime boundary

In ϕ the two regimes are named by a single inequality:

$$\phi < 90° \Leftrightarrow \lambda > 0 \quad \text{(descent)}, \qquad \phi > 90° \Leftrightarrow \lambda < 0 \quad \text{(negative curvature)}.$$

Six capacities, two seeds each, AdamW, fixed corpus and ladder:

D	$\lambda_{\min}$ (s17)	$\lambda_{\min}$ (s23)	final val	$\phi > 90°$
48	+0.054	+0.049	1.74	no
64	+0.040	+0.032	1.24	no
96	+0.056	+0.055	0.78	no
128	−0.308	−0.068	0.41	**both seeds**
176	−1.712	+0.011	0.23	one seed of two

Observation 18.2 (A reproducible capacity boundary between $D = 96$ and $D = 128$)**.** *Below $D = 128$, λ stays positive in every run with a tight range (maximum* 0.25*). At $D = 128$ both seeds cross into $\phi > 90°$, which is the artefact control passing: a numerical excursion would not reproduce across seeds. The boundary coincides with the capacity at which the model begins to descend properly (val* $0.78 \rightarrow 0.41$*), consistent with the* 27%–49% *threshold of §17.3. At $D = 176$ only one seed of two crosses, so the behaviour above the boundary is* not *established as monotone in capacity.*

18.3 The marker cluster is regime-specific, not universal

§17 recorded a coincidence of spectral events — λ peak, $\|g\|$ peak, κ crossing 1, gluing defect τ peak — at a reproducible position in normalised path length, replicated across three optimisers and two instrumentation builds. Under a capacity sweep it does not survive.

D	λ peak	$\|g\|$ peak	coincide
64	—	—	no peak interior to the window
128	0.143	0.143	**yes**
256	0.201	0.085	no

Observation 18.3 (Scope, not truth)**.** *The coincidence holds at $D = 128$ and at no other capacity tested. Every previous measurement of it — the optimiser sweep and both stage runs — used $D = 128$, so optimiser-invariance was read as generality while the capacity axis was held fixed. The defensible statement is:* the λ–$\|g\|$–ϕ coincidence is observed at $D = 128$ under AdamW, β_1=0 and SGD on this corpus, within the regime where ϕ crosses 90°; it is absent at $D \leq 96$, where ϕ remains below 90° throughout. *Since the coincidence sits inside the negative-curvature regime and κ is singular there, part of what was measured as a marker cluster was the κ singularity itself.*

The finite-size prediction is also not met. Transition width, from the curvature of $\log \lambda$ at its peak, is 0.190 at $D = 128$ and 0.292 at $D = 256$: the region broadens rather than sharpening. On two points, one of which rests on the negative-λ excursion, this is unmeasured rather than refuted — but it is the opposite of what a finite-size transition would give.

18.4 On calling this a phase transition

The evidence supports a *regime boundary* and does not yet support a phase transition, and the distinction is worth keeping because it is testable.

What is measured: a reproducible boundary in capacity, with a qualitative change in the sign of λ, passing a seed control. What a phase transition would additionally require, in the finite-size sense: the transition region narrowing with system size, the variance of its location decreasing, and derivatives such as $d\phi/dp$ steepening. Measured, the region *broadens* ($0.190 \rightarrow 0.292$), and $\phi > 90°$ appears in one seed of two at the larger capacity. Both cut against the transition reading.

There is also an order-parameter candidate, and it is better than κ: sign(λ), or equivalently $\mathbb{1}[\phi > 90°]$, distinguishes the two regimes by construction rather than by threshold. What it lacks is a demonstrated singular limit. *Regime boundary* is the term the data support; *crossover* is the term the width measurement suggests; *phase transition* would need the finite-size scaling to come out the other way, which is a specific and runnable experiment.

18.5 What is corpus, what is architecture, what is neither

The programme's results now separate by what they depend on, and two entries in the informal picture do not survive the separation.

observation	depends on	status
row-scale sign anti-alignment	architecture (LayerNorm)	exact, §16.1
$\phi > 90°$ regime boundary	capacity	both seeds at D=128
$\omega \propto \lambda^{\alpha}$ arc	neither, within optimiser	§17.7
path efficiency decline	neither	7 conditions
$\phi \approx 45°$ at the floor	optimisation stage	§17.5
ϕ depends on corpus	**refuted**	§17.5, matched loss
low-dimensional subspace persists	**refuted**	4% at fixed θ

Two clarifications, since both have been asserted in discussion and neither is supported. *ϕ does not oscillate because the architecture is fed poor learnables*: at matched loss, English prose and balanced S-expressions give κ differing by 0.009–0.094 against split noise of 0.05–0.19 (§17.5), so corpus quality is not the driver; the oscillation is the κ singularity of §18.1. *There is no persistent low-dimensional subspace*: the top-k singular subspace of a gradient window was measured at 4% self-overlap between two *disjoint* windows at identical θ (§17.9), so it is not reproducible under batch sampling with no parameter motion at all. An earlier report of 97% overlap used sliding windows sharing 7 of 8 gradients and is withdrawn.

What *is* low-dimensional is different and does hold: eight to twelve Hessian directions of 4.3×10^6 carry 8–33% of $\|Hv\|^2$, a concentration of 2.7×10^4 over isotropic (§17.10). That subspace is operator-defined rather than architecture-defined, and it decorrelates within ten steps.

18.6 Status

claim	status	basis
κ singularity is a chart artefact	holds	Obs. 18.1
ϕ is the resolved coordinate	holds	monotone through 90°
regime boundary at $96 < D < 128$	holds, 2 seeds	Obs. 18.2
marker cluster universal	refuted	Obs. 18.3
transition sharpens with size	not observed	width $0.190 \to 0.292$
skeleton then densification, 0–120	inherited, unretested	§8

The programme's coordinate should be ϕ from here. Every κ figure in §14–§17 is a tan of a ϕ that was better behaved than the ratio made it appear, and the two places where κ was least trustworthy — near $\lambda = 0$, and in distinguishing descent from ascent — are exactly the two the resolution repairs.

19 Phase 17: The Krylov Bundle and What Survives Calibration

Phases 12–16 accumulated geometric language faster than the measurements justified it. This phase does the reverse: it calibrates the instrument first, establishes whether the objects being measured are real, and only then records which structures remain. Two results carry the phase — the Krylov geometry is intrinsic rather than an artefact of stochastic gradients, and subspace transport predicts the rate of Fisher closure — and both survive a null. Most of the categorical apparatus does not.

19.1 Stage 1: pinning the MF pump

The MF pump stops adaptively, on $\Phi_{\text{clean}} = N_{\text{STU}} - 1$ or on τ rising twice, and measured round counts have varied $1, 2, 3, 4, 9$ across runs at the same capacity. Two Krylov runs at $D = 128$ gave $n_{\text{neg}} = 3$–4 and $n_{\text{neg}} = 0$ with MF3 and MF1 respectively, and the pump was the natural suspect, since it injects the energy the compiler works with and therefore sets the starting geometry.

Pinning the loop to a fixed count and disabling both stopping rules:

MF rounds	handoff val	val@30	n_{neg}
1	3.381	1.878	0
2	6.932	2.718	0
3	6.175	2.407	0
4	6.725	2.340	0

Observation 19.1 (The MF pump does not produce the negative eigenvalues)**.** *With MF pinned, $n_{neg} = 0$ at every round count, while handoff val varies by a factor of two. The hypothesis — proposed three times in earlier discussion — is refuted. The discrepancy is resolved in §19.3 instead, and it is instrumental.*

19.2 Stage 2: the SPLIT floor is a resource dial

Two *independent* Krylov spaces at the *same* θ define the noise floor. Every transport number must be read against it, and earlier measurements were not.

m	HVP batches	drift samples	SPLIT	cost (HVPs)
6	2	16	0.405	38
6	6	40	0.456	106
4	8	60	0.619	116
3	12	80	0.743	140
2	16	100	0.915	148

Observation 19.2 (Reducing m dominates adding batches)**.** *SPLIT more than doubles for four times the cost, and the effective lever is the Krylov dimension, not the batch count: at fixed cost near* 110*, $m = 6$ gives* 0.456 *while $m = 4$ gives* 0.619*. The high-index Lanczos vectors are the unresolvable part. This retrospectively invalidates the transport numbers of §18, which ran* 0.003–0.35 *against a floor of* 0.41 *— at or below their own noise. The calibrated operating point is $m = 3$, $n_b = 12$, $n_d = 80$.*

19.3 Stage 3: transport, and the recovered negative eigenvalues

At the calibrated setting, with SPLIT re-measured at every checkpoint rather than assumed constant:

step	val	ϕ	n_{neg}	SPLIT	ov$Q$	ratio
$D = 96$						
4	4.792	62.5°	1	0.863	—	—
8	4.485	63.8°	1	0.861	0.751	0.872
32	2.502	60.4°	0	0.796	0.267	0.336
115	0.769	53.4°	0	0.673	0.185	0.275
205	0.152	27.3°	0	0.479	0.048	0.101
$D = 128$						
4	6.232	**95.0°**	**2**	0.761	—	—
8	5.322	72.5°	1	0.804	0.392	0.487
32	2.611	63.2°	0	0.773	0.191	0.248
85	0.793	50.2°	0	0.704	0.149	0.212

Observation 19.3 (The negative eigenvalues were lost to instrument noise)**.** *With the instrument calibrated, n_{neg} reappears, and only at $D = 128$: two negative Ritz values at step 4, where $\phi = 95.0°$. $D = 96$ peaks at $63.8°$ and never approaches $90°$. The earlier $n_{neg} = 0$ was resolution failure at $m = 6$, $n_b = 2$, not an MF effect (Obs. 19.1). The count is 2, not the 3–4 of the uncalibrated run, and the capacity boundary of §18.2 survives recalibration.*

Transport itself is now readable: the ratio decays $0.87 \to 0.10$ at $D = 96$ and $0.49 \to 0.21$ at $D = 128$. SPLIT also falls with training ($0.86 \to 0.48$), so the instrument degrades as the model converges and late-training transport is the least reliable — the opposite of the natural assumption.

19.4 Stage 4: transport predicts the rate of Fisher closure

The falsifiable form of the Krylov–Fisher conjecture. Regression of $\dot{d}_F$ on transport, against a null in which the time order of the transport series is shuffled.

	corr(transport, $\dot{d}_F$)	null p_{95}	verdict
$D = 96$	**−0.889**	0.738	signal
$D = 128$	**−0.694**	0.643	signal
	corr(transport, d_F level)	null p_{95}	
$D = 96$	+0.716	0.758	no signal
$D = 128$	+0.124	0.608	no signal

Observation 19.4 (Transport predicts the rate, not the level)**.** *High Krylov transport accompanies falling Fisher distance; as transport decays, d_F stops falling and turns upward. The relation holds at both capacities and clears its shuffled-time null at both. It does* not *predict where d_F is, only where it is going — which is the differential relation a connection-curvature reading would require, and a sharper claim than the conjecture as originally stated. Also, ϕ is not the carrier: at $D = 96$, $\mathrm{corr}(\phi, \dot{d}_F) = -0.482$ against a null of 0.577.*

Caveat. $n = 10$ per arm, one seed each. Both d_F and transport decay steeply over the first three checkpoints, so part of the correlation may live in a shared early transient; the regression restricted to steps ≥ 20 is the control that would settle it and has not been run.

19.5 Stage 5: the Krylov geometry is intrinsic

Everything geometric in the programme depends on one distinction: is the Krylov geometry a property of the model, or an artefact of stochastic gradients? Eight independent Krylov spaces from disjoint gradient windows at each frozen θ, $D = 128$:

step	val	WITHIN	sd	angle$_1$	angle$_3$	eff. dim	BETWEEN	ratio
4	6.154	0.791	0.023	11.5°	46.0°	3.67	—	—
8	5.281	0.795	0.024	8.5°	46.1°	3.66	0.441	0.555
40	2.213	0.739	0.031	15.2°	51.6°	3.89	0.019	**0.026**
120	0.392	0.559	0.067	20.2°	66.6°	4.88	0.033	**0.058**
200	0.088	0.477	0.086	23.3°	74.4°	5.53	0.015	**0.032**

Observation 19.5 (Evolution exceeds sampling noise by 17–38×)**.** *Eight independent estimates at the same θ agree at 0.74–0.79 in mid-training; consecutive checkpoints agree at 0.015–0.033. The effective dimension of the union of the eight replicates is 3.66–3.89 against 3 for identical spaces and 24 for independent ones, so the replicates measure nearly the same three-dimensional object. The first principal angle between independent estimates is 8.5°–23.3° while the third is 46°–74°: the leading direction is well determined and the trailing one is not, which is why reducing m dominated in Obs. 19.2.* The Krylov geometry is intrinsic and evolves.

Two riders. The instrument degrades with training — WITHIN falls $0.79 \to 0.48$, its sd triples, effective dimension rises to 5.53 — so late-training claims need more replicates than early ones. And step 8 is an exception with ratio 0.555: across steps $4 \to 8$, which span the negative-curvature excursion, ϕ moves from 95.0° to 72.5° while the subspace barely moves. The angle changes and the subspace does not, which is unexplained.

19.6 Per-layer forward/backward alignment

Measured in two senses, since the programme's 70.1% residual is between weight state and applied update (the tile sense), while activation against error (the backprop sense) had never been measured.

	ρ_{bp}	ρ_F (activation Fisher)	ρ_{tile}	layer spread
$D = 96$	$+0.0094 \pm 0.0137$	0.0080–0.0104	0.201–0.218	< 0.002
$D = 128$	$+0.0067 \pm 0.0184$	0.0058–0.0066	0.201–0.219	< 0.002

Observation 19.6 (The obstruction is uniform across depth)**.** *Backprop-sense alignment is* 0.007–0.010 *at every layer and every capacity, with standard deviation exceeding the mean: activations and errors are orthogonal in residual space,* 99.3% *misaligned rather than the* 70% *of the tile sense. The layer-to-layer spread is under* 0.002*, inside the checkpoint-to-checkpoint noise, so* the obstruction is not depth-graded*. Fisher weighting changes it by less than* 0.0005*, so the misalignment is not concentrated in low-variance directions where a metric could rescue it. The one depth structure is a boundary effect:* $\|a\|$ *jumps* $17.6 \to 220$ *from layer* 0 *to layer* 1 *and is flat thereafter — the embedding entering the residual stream, not a gradation.*

This is decisive for the indexed constructions. A per-layer, per-tile, per-strand pairing has nothing to index: the obstruction is the same at every layer, in both senses, at both capacities, with and without Fisher weighting.

19.7 What survives

structure	status	basis
Geometric		
Krylov geometry is intrinsic	holds	Obs. 19.5, 17–38×
transport predicts $\dot{d}_F$	holds, 2 capacities	Obs. 19.4
ϕ resolves the κ singularity	holds	Obs. 18.1
$\omega \propto \lambda^\alpha$ arc	holds, out-of-sample	Obs. 17.7
capacity boundary $96 < D < 128$	holds, recalibrated	Obs. 19.3
Categorical		
tile bifibration, $f_! \dashv f^*$	holds	residuals 0.1–0.8%
weighted duality $J_w^2 = w$	holds, resolution-graded	$w = 0.033 \log_2 L + 0.672$
ordered $K_0 \cong \mathbb{Z}[\frac{1}{2}]$	holds, classifies the cover	Elliott
Refuted or vacuous		
Fredholm index of the Krylov map	vacuous	$\dim V - \dim W = 0$
spectral flow	no Fredholm family	§7.5
layer/tile/strand indexed pairing	nothing to index	Obs. 19.6
starfish, operator-carrying arms	worse than random	§17.9
persistent low-dim subspace	4% at fixed θ	§17.9
d_F minimum at step 120	progress-indexed	steps 70 and 36
skeleton then densification	refuted	attention leads from step 1

The categorical structures that survive are all on the *tile complex* — representation structure, exact by construction, classifying a cover we chose. The geometric structures that survive are all *global contractions* of the gradient and its Hessian image. Nothing survives that attempts to give the geometry support: seven coordinate-indexed groupings have been tested and all are null, while the operator-defined chart concentrates Hv by 2.7×10^4 and decorrelates within ten steps.

19.8 On functors and pullbacks across corpora

Layer-wise data now exists for two corpora at two capacities, which is the input a pullback construction would need. Two measurements bound what such a construction can carry.

First, the alignment is flat across depth (Obs. 19.6), so a functor indexed by layer has no gradation to transport. Second, the forward/backward obstruction is permanent: 70.1% out-of-image residual over the first half of the run and 70.1% over the second, slope -0.0049% per step (§1.13). A pullback functor $\mathcal{F} \to \mathcal{F}'$ between forward and backward structures would need that residual to close, and it does not.

What remains available is the resolution axis rather than the depth axis. The weighted duality $J_w^2 = w$ is graded by refinement level and exact at the finest, with $1 - w$ linear in $-\log_2 L$ — and the Fisher forward/backward distance is *not* so graded (slopes ± 0.006 with sign flips, against J_w's consistent $+0.033$). So resolution grading is specific to J_w, and the singularity resolution of §18.1 is a resolution of a *chart*, one blow-up of $\mathbb{P}^1 \to \mathbb{A}^1$, not of a variety. Calling it Hironaka-type overstates it; the object needed only the first blow-up, because the singularity was in the projection.

19.9 The module framework: what the algebra adds

A module-theoretic reading was proposed, with $\mathcal{K}_r(t)$ as a module over $\mathbb{R}[T]$ where T acts as the Hessian, a bundle over $\mathbb{R}[t, d/dt]$, and a group action of $\mathcal{D}$ on gradients. Two of the three are testable and both fail their tests.

Level 1 is true by construction and its content is measured. $\mathcal{K}_r(t) = \mathbb{R}[T]g_t/(T^r)$ is a cyclic module because that is what a Krylov space is, so the FTFGMPID decomposition into invariant factors is a theorem rather than a hypothesis. The testable claim is the *block structure*: one large invariant factor plus two or three small ones was proposed, matching the measured angle profile.

step	gap ratios λ_i/λ_{i+1}					PR	largest gap at
	g_1	g_2	g_3	g_4	g_5		
$D = 96$							
4	1.98	9.94	0.09	0.36	0.74	3.92	2
40	1.17	1.63	2.14	4.59	28.54	3.47	5
160	1.35	1.84	1.66	2.10	1.95	3.68	4
$D = 128$							
4	2.28	2.61	0.11	0.54	0.74	3.64	2
40	1.64	1.44	1.98	4.11	6.41	3.33	5
160	1.17	1.71	1.86	2.58	2.77	3.70	5

Observation 19.7 (No dominant invariant factor). *The largest spectral gap never falls after position one. It sits at positions* $2, 2, 4, 5, 4, 4$ *at* $D = 96$ *and* $2, 3, 4, 5, 5, 5$ *at* $D = 128$*, while* g_1 *runs* 1.17–2.30 *throughout — the* smallest *gap in the spectrum. The participation ratio of* 3.4–4.1 *reflects a smoothly graded spectrum, not a block structure: at* $D = 96$ *step* 160 *the eigenvalues are* $0.955, 0.709, 0.386, 0.233, 0.111, 0.057$ *with successive ratios* $1.35, 1.84, 1.66, 2.10, 1.95$*, comparable throughout. The module decomposition is formally valid and yields six comparable invariant factors, carrying no information about block sizes.*

An earlier reading of the angle profile (13–17° leading, 50–62° trailing) as evidence for one large block is withdrawn: the angles measure how well each replicate *direction* is determined by sampling, while the gaps measure the *spectrum*. Those are different quantities and were conflated.

A resolution dependence, recorded as a qualification. At $m = 6$, $n_b = 12$, $D = 96$ shows $n_{\text{neg}} = 3$ at steps 4 and 8 with eigenvalues to -0.505, where $m = 3$ had shown $n_{\text{neg}} \le 1$. So the negative count depends on how many Krylov dimensions are examined, and the capacity contrast of Obs. 19.3 may concern which directions the *drift* aligns with rather than whether the spectrum contains negative directions at all. Both capacities reach all-positive spectra by step 40 and stay there, so negative directions are an early-training phenomenon at both widths.

The group action does not survive. The proposal was that $\mathcal{K}_r(\rho(d)g) = \rho(d)\mathcal{K}_r(g)$ for $\mathcal{D} =$ {positive diagonal rescalings} × {signed permutations}. Since g is a covector and H a $(0, 2)$ tensor, $H'g' = R^{-\top}HR^{-1}R^{-\top}g$, which equals $R^{-\top}Hg$ only for orthogonal R — so equivariance was predicted to hold on the permutation subgroup and fail on the rescalings. Measured against a SPLIT of 0.677:

element	overlap	/SPLIT	orthogonal
sign flip	0.0098	0.014	yes
uniform rescale $c = 1.5$	0.0168	0.025	no
random diagonal in $[0.5, 2]$	0.0231	0.034	no

Observation 19.8 (Equivariance fails, including on the orthogonal element). *All three sit at 1–3% of the floor, and the orthogonal element fares worst. The prediction was wrong for a reason that also invalidates the claim as stated: reparametrising $\theta \mapsto R\theta$ changes the function the network computes unless R is an architectural symmetry. A sign flip on half of 4.3×10^6 parameters is not a permutation of hidden units; the network at $R\theta$ is a different network, so no equivariance statement applies. The genuine elements of $\mathcal{D}$ — matched permutations of feed-forward units, compensated channel rescalings — were not tested and form a far smaller group.*

What the claim was meant to explain — that sign carries the trajectory while magnitude compresses to one scalar (§3) — is measured, but does not follow from Krylov equivariance and cannot be derived from it. The sign robustness rests instead on the elementary fact that $\text{sign}(Rg) = \text{sign}(g)$ for positive diagonal R, which involves no Hessian. The module framework therefore keeps Level 1 and loses the group-action layer.

20 Phase 18: Latent State and the Failure of Global Operator Models

§19 established that the Krylov geometry is intrinsic and predictable over a bounded horizon, and that it captures the gradient almost perfectly while capturing the update almost not at all. This phase asks what additional state makes the update predictable, and whether the whole system admits a linear operator description. The first question yields a candidate that was never previously measured; the second is answered negatively at three levels of generality, and the pattern of those failures is itself the phase's result.

20.1 Where the alignment is lost

Adam has four stages between the drift and the applied update, and only one need be responsible for the low capture of $\Delta\theta$. Isolating each against the same Krylov space:

step	val	cap(g)	cap(Dg)	cap(m)	cap(Dm)	cap($\Delta\theta$)
6	5.989	0.994	0.020	0.763	0.101	0.075
24	3.223	0.996	0.298	0.443	0.137	0.083
76	1.094	0.997	0.373	0.373	0.060	0.044
120	0.396	0.997	0.486	0.183	0.027	0.022

Observation 20.1 (Both stages contribute, and they exchange roles)**.** *At the final checkpoint, $g \to Dg$ loses 51% and $g \to m$ loses 82%, so the diagonal preconditioner is not the sole cause — momentum removes more. The two compose multiplicatively, with clipping and weight decay contributing almost nothing beyond. The time structure inverts:* cap(Dg) rises $0.020 \to 0.486$ *as v equilibrates while* cap(m) falls $0.763 \to 0.183$ *as the drift decorrelates, crossing near step 76. An earlier attribution of the misalignment to the preconditioner alone is therefore correct early and wrong late.*

The metric hypothesis is refuted. If the Krylov space were merely built in the wrong metric, constructing it in Adam's metric — $K_r(Mg, MHM)$ with $M = \mathrm{diag}(v)^{-1/2}$ — should raise the capture. Measured mean capture of $\Delta\theta$: standard 0.057, preconditioned 0.041, ratio 0.73. The missing component lies outside the Krylov tower in both metrics. The two do cross over, standard falling $0.077 \to 0.026$ while preconditioned rises $0.012 \to 0.064$, consistent with Obs. 20.1, but neither exceeds 8%.

20.2 $P(r)$ saturates: the missing component is not dimensional

r	1	2	3	4	8	16	32
mean $P(r)$	0.042	0.061	0.066	0.071	0.073	0.075	0.078

Observation 20.2 (Three directions do the work of thirty-two)**.** $P(32)/P(3) = 1.18$, *against* 10.67 *for linear growth, and a random-subspace null of* 2.7×10^{-5} *at* $r = 32$. *The captured component is some* 2900× *above chance and essentially all of it lives in the first three directions. So $\Delta\theta = \Delta\theta_{\|} + \Delta\theta_{\perp}$ with $\Delta\theta_{\|}$ in a genuinely small privileged subspace and $\Delta\theta_{\perp}$ outside the entire Krylov tower — not spread thinly across its tail. There is no r at which the update becomes representable.*

A second-order Grassmann predictor was tested to distinguish trajectory curvature from missing state: it beats the first-order model in 1 of 5 intervals and is otherwise 2–7× worse, at one interval falling below a random-tangent control. The late-training failure of prediction is therefore missing state, not curvature.

20.3 State ablation: a target problem, then a result

A first ablation predicting $\Delta\theta_{t+1}$ directly was null: every feature arm sat below its shuffled-time null, and persistence scored 0.939. The target was near-autocorrelated at 6-step spacing, so the regression had no room. This is a target problem rather than a feature problem, and the fix is to predict the residual $\Delta\theta_{t+1} - \Delta\theta_t$ at wider spacing: raw autocorrelation 0.036, residual -0.060.

At $n = 61$ checkpoints with ~ 18 held out, scored as held-out R^2 against a shuffled-target null:

state	R^2 held out	null p_{95}	
Q	0.047	−0.012	baseline
$Q + \text{LN}$	**0.414**	−0.017	**best**
$Q + v$	**0.316**	−0.038	
$Q + W_K$	0.184	−0.002	
$Q + m$	0.165	−0.043	
$Q + \text{aniso}$	0.069	−0.001	
$Q + \Sigma$	0.066	−0.118	
$Q + d_F$	0.049	−0.009	
$Q + \text{arch(all)}$	−0.179	−0.095	overfits

Observation 20.3 (The latent state includes representation statistics)**.** *LayerNorm gain statistics are the single best predictor, at nine times the Q baseline and above the second moment v. W_K singular-value concentration also beats the first moment m. Two of three architectural arms outperform optimiser state, so the latent variable is not merely optimiser memory: it includes properties of the learned representation. Combining the architectural arms destroys them (-0.179), which is overfitting on 43 training rows and not evidence against the components.*

A caution on slow variables. A variable that barely moves will predict well under a raw target regardless of content. The result above uses the residual target, so it survives that objection — but an earlier run at $n = 20$ gave $Q + d_F = 0.439$ which fell to 0.049 at $n = 61$, while $Q + v$ held ($0.464 \to 0.316$). Only the replicating arms should be read.

20.4 The timescale hypothesis

variable	lag-1 autocorrelation	R^2
W_K concentration	0.9995	0.184
LayerNorm mean gain	0.9934	**0.414**
LayerNorm gain sd	0.9485	—
d_F	0.7318	0.049
m projection	0.6588	0.165
v projection	0.5265	0.316
anisotropy	0.5083	0.069
Σ (Ritz values)	**0.0468**	0.066

Observation 20.4 (Slow variables predict; fast ones do not)**.** *The three slowest variables are the architectural ones, all above 0.94. The fastest is the Hessian spectrum at 0.047 — essentially uncorrelated between checkpoints — and it sits near the bottom of the ablation. The ordering is broadly monotone in timescale, with v the exception. Classification by characteristic timescale organises the candidates better than classification by kind (optimiser, statistical, architectural), since each of those categories splits.*

Two further negatives constrain the latent state. *It is not finite history*: adding Q_{t-1} takes R^2 from $+0.116$ to -0.486 and Q_{t-2} to -1.344, so there is no Markov order at which the dynamics close — though at $n = 20$ this is as consistent with overfitting as with absence of information, and the defensible statement is that within this model class and sample size, finite-history augmentation did not improve prediction. *And it is not m*: the first moment scored -2.145 at $n = 20$ and 0.165 at $n = 61$, below both architectural arms. That $g \to m$ loses 82% of the alignment (Obs. 20.1) locates where information is destroyed; it does not make m a good predictor of future dynamics. Those are different questions and the experiments separate them.

20.5 Koopman: global, then local

If some collection of observables evolves approximately linearly, $\phi(x_{t+1}) \approx K\phi(x_t)$, the dynamics would be tractable without a hidden variable. Fitted on 61 checkpoints spanning val $6 \to 0.1$:

observable set	dim	R^2 held out	persistence	null p_{95}
Q	3	-0.536	-1.697	-1.773
Q+LN+v	8	-0.170	-1.308	-0.507
all	17	-0.642	-1.298	-1.552
LN+W_K	3	-2.372	$+$**0.949**	-12.85

Observation 20.5 (Global Koopman fails, and the failure is diagnostic)**.** *No observable set achieves $R^2 > 0$: a linear operator predicts worse than the mean of the held-out data. The architectural row is the clearest case — persistence scores* $+0.949$ *because LayerNorm gains and W_K concentration barely move, while the fitted operator collapses to* -2.372*, destroying a signal that doing nothing captures. The window spans val* $6 \to 0.1$*, so the system is strongly non-autonomous over it and a single K is the wrong object;* 42 *training rows for a* 17×17 *operator compounds this.*

Local operators form a coherent family that still cannot predict. Fitting K on a sliding window and predicting the next increment — so persistence scores zero by construction — gives relative errors 8.96, 661, 8.64, 2.37, 1.06 at windows $8, 12, 16, 24, 40$. Every value exceeds 1, meaning the fit is worse than predicting no increment, although every one beats its time-shuffled null (ratio 0.34–0.57). The error falls monotonically with window length, but 61 checkpoints do not permit a longer window without the local assumption dissolving.

Observation 20.6 (The operator family drifts smoothly)**.** *Correlation between local operators separated by* $2, 4, 8, 16$ *window-steps:* $+0.562$*,* $+0.311$*,* $+0.087$*,* $+0.118$*. Consecutive local operators are related, so the family $K(\mu)$ is real and slowly varying rather than independent noise fits — the adiabatic picture. But the operator decorrelates over roughly the horizon everything else in this programme decorrelates over, so a window long enough to fit K is also long enough for K to have changed. That is the obstruction, and it is specific: it would be broken by denser sampling per unit of training progress, not by more observables.*

20.6 Status

claim	status	basis
misalignment is the preconditioner alone	refuted	Obs. 20.1
the Krylov metric is wrong	refuted	preconditioned ratio 0.73
missing component is dimensional	refuted	Obs. 20.2
missing state is trajectory curvature	refuted	2nd-order worse 4/5
latent state is finite history	not supported	R^2 falls with each lag
latent state is m	not supported	below architectural arms
latent state includes representation statistics	holds	Obs. 20.3
slow variables predict, fast ones do not	holds	Obs. 20.4
global Koopman operator	refuted	Obs. 20.5
local operator family $K(\mu)$ exists	holds	Obs. 20.6
local operator family predicts	refuted	rel. error > 1 at every window

A pattern runs through the programme and is sharpest here. Global theories — a fixed Koopman operator, one hidden variable, finite-history closure, a single latent state — have not survived testing. Local and conditional statements — intrinsic Grassmann geometry, bounded-horizon prediction, slowly evolving representation statistics, the drifting operator family — have consistently produced measurable signal against controls. The system is better described as a time-varying multiscale process, $x_{t+1} = F_{\mu(t)}(x_t)$ with μ a slowly changing control parameter, than as a single autonomous law.

21 Phase 19: Local Linearity and the LPV Hypothesis

§20.5 established that no global linear operator describes the observable dynamics, and diagnosed the failure as non-autonomy rather than nonlinearity. This phase tests the diagnosis. It follows a staged programme in which each stage relaxes exactly one assumption, so a failure localises rather than discarding the framework:

stage	hypothesis	pass criterion	failure implies
1	local linearity	beats persistence	nonlinear dynamics needed
2	slowly varying operator	K changes continuously	switching/hybrid
3	LPV	$K(\mu)$ transfers across runs	hidden state larger than μ
4	universal LPV	same $K(\mu)$ across capacities	task-specific latent
5	weight prediction	predict $\Delta\theta$ from (Q, μ)	richer latent needed

Stages 1 and 2 pass. Stage 3 splits, and the split is the phase's result.

21.1 Stage 1: local linearity, on a stationary window

The global fit spanned val $6 \to 0.1$. Dense sampling on a narrow interval — steps 200–404 every three steps, val $0.085 \to 0.040$, some fifty times narrower in loss — gives a window over which the system is closer to stationary. Target is the increment, so predicting no change scores exactly 1.0; the time-shuffled null is reported alongside.

window	n_{pred}	relative error	null	ratio
16	51	3.186	6.122	0.520
24	43	1.497	2.332	0.642
32	35	1.063	2.125	0.500
40	27	**0.929**	1.648	0.564
50	17	**0.775**	1.515	0.511
60	7	**0.562**	1.442	0.390

Observation 21.1 (Local linearity holds once the window is stationary)**.** *Relative error falls below* 1 *at window* 40 *and reaches* 0.562 *at window* 60 *— the first predictive result in this line, and it beats the trivial baseline rather than only the null. On the wide ladder the same construction bottomed at* 1.06*. The only change is stationarity. The decrease is monotone across six window sizes, so the constraint was never the model class.*

Two cautions. Window 60 *rests on seven predictions; window* 40*, with* 27*, is the most trustworthy point below* 1 *and is only* 7% *better than trivial. And the ratio to the shuffled null is flat at* 0.39–0.64 *across* all *windows including those with error above* 1*, so beating that null was never the discriminating test — the absolute threshold is. Earlier runs reporting "signal" on the null alone should be read accordingly.*

21.2 Stage 2: the operator drifts continuously

Correlation between local operators separated by $2, 4, 8, 16$ window-steps: $+0.562$, $+0.311$, $+0.087$, $+0.118$. Consecutive operators are related and the relation decays smoothly, so the family is genuinely slowly varying rather than a sequence of independent noise fits, and the dynamics are not switching. The decorrelation horizon is comparable to that of every other object in the programme, which is why a window long enough to fit K is also long enough for K to have changed.

21.3 Stage 3: LPV holds within a run and does not transfer

On a single run, μ and training time correlate at $+0.971$ — LayerNorm gains move by 0.25% over steps 200–400, monotonically — so $K = K(\mu)$ and $K = K(t)$ are indistinguishable and the LPV question is not testable. Pooling two capacities breaks the collinearity by construction, since they traverse μ in *opposite directions*: LN rises $0.50625 \to 0.51132$ at $D = 96$ and falls $0.50446 \to 0.50197$ at $D = 128$.

	single run	pooled
collinearity $\mathrm{corr}(\Delta t, \Delta\mu)$	$+0.971$	$+0.553$
$\mathrm{corr}(\Delta K, \Delta t)$	$+0.711$	$+0.341$
$\mathrm{corr}(\Delta K, \Delta\mu)$	$+0.646$	$+0.480$
$\mathrm{partial}(\mu \mid t)$	-0.265	$+\mathbf{0.371}$
$\mathrm{partial}(t \mid \mu)$	$+0.459$	$+\mathbf{0.104}$

Observation 21.2 (LPV within runs; no transfer across capacities). *Pooled, μ-distance explains operator distance beyond time ($+0.371$) while time explains almost nothing beyond μ ($+0.104$). That is the LPV prediction and it is the first time the discriminating test has been runnable rather than confounded.*

Restricted to the 1849 pairs spanning the two runs, however, $\mathrm{corr}(\Delta K, \Delta\mu) = -0.336$ — the wrong sign — and the partial collapses to $+0.077$. The positive pooled result is carried entirely by within-run pairs. Operators from different capacities are not more alike when their μ matches.

Two readings, not separated by this design. Either μ is genuinely local to a run, so LayerNorm is a within-run proxy whose meaning does not survive a change of width; or the LN ranges barely overlap (0.502–0.511 against 0.502–0.504) so "matched μ" across runs is mostly extrapolation. The second is a design limitation — windows were chosen by step number rather than by μ — and choosing genuinely overlapping intervals is what would decide it.

Under the staged programme, a Stage-3 failure implies the hidden state is larger than μ. The corresponding revision is to write $K = K(z)$ with $\mu = h(z)$, so that the measured slow variable is an observation of the latent state rather than the latent state itself. That is consistent with everything measured: LayerNorm gains have the strongest predictive content of any measured quantity (Obs. 20.3, $R^2 = 0.414$) without that establishing $z = \mathrm{LN}$.

21.4 Where stages 4 and 5 stand

Stage 4 is not started. The natural form is not another LPV variant but a *conjugacy* question: does there exist T with $K_{128} \approx T^{-1} K_{96} T$? If so the two capacities run the same dynamics in different coordinates, which is a stronger structural claim than LPV and is checkable through spectra alone — conjugate operators share eigenvalues even when the operators differ.

Stage 5 has a measured ceiling that no model can exceed. The Krylov space captures 2.6–11.8% of $\Delta\theta$ (§20.2), and $P(r)$ saturates by $r = 3$, so predicting weights from (Q, μ) is bounded at roughly a tenth of the update energy regardless of the predictor. The framework describes the evolution of training *geometry*; it does not compress weight dynamics.

21.5 Status

claim	status	basis
local linearity (stage 1)	holds	Obs. 21.1
smoothly drifting operator (stage 2)	holds	§21.2
K depends on μ beyond t, within run	holds	Obs. 21.2
μ transfers across capacities (stage 3)	not supported	cross-run sign reverses
μ *is* LayerNorm	not claimed	best proxy only
conjugacy $K_{128} \sim K_{96}$ (stage 4)	untested	spectra would decide
weight prediction from (Q, μ) (stage 5)	bounded at $\sim$10%	§20.2

The three descriptions are complementary rather than competing. A latent state z evolves slowly; Q is a fast geometric observation of it; K is the local linearisation it indexes; and the weights are a further observation whose relation to Q is weak. LPV approximates the dynamics locally by linear systems indexed by z, Koopman describes that evolution in an observable space, and the Grassmann geometry supplies one observable. What the measurements add is that the first is local, the second fails globally, and the third captures the gradient almost perfectly while capturing the update almost not at all.

22 Phase 20: Interaction Geometry

§21 ended with the operator programme closed: the local operators are aliased, not generators, and nothing transfers across capacities. This phase measures a different object, which arose from a correction rather than a hypothesis — the first-order loss attribution was wrong, and fixing it exposed the largest structured effect in the programme.

22.1 First-order attribution was wrong

Attributing the loss reduction by $\Delta L_i = \nabla_i L \cdot \Delta\theta_i$ gave positive values for attention and feed-forward — apparently loss-increasing — and negative for embedding and LayerNorm. Under *actual* ablation, applying one group's update and evaluating:

group	$\|\Delta\theta\|$	ΔL actual	ΔL first-order
EMB	0.651	−0.00715	−0.036
LN	0.090	−0.00131	−0.004
FF	1.276	**−0.00913**	**+0.0004**
ATTN	1.151	**−0.00574**	**+0.0019**

Observation 22.1 (Every group helps alone; together they interfere)**.** *The first-order marginals have the* wrong sign *for attention and feed-forward: both reduce the loss under actual ablation, and FF is the largest single contributor. The singles sum to* −0.0233 *while the joint update gives* −0.0091 — 39% *recovery — and at step* 260 *the singles sum to* −0.0062 *while applying all groups together* increases *the loss by* +0.0010. *The groups are strongly non-additive, which is why first-order attribution misleads: it assumes separability that does not hold.*

Gauge motion does not explain it. Projecting onto twelve explicit LayerNorm and feed-forward rescaling directions — exact symmetries of the pre-LN residual block — captures 2.4% of the update energy at step 150 and 0.7% at step 260. The orthogonal part alone gives $\Delta L = -0.00901$ against the full -0.00911, and the gauge part gives -0.00014. The unexplained update energy is functional, not gauge.

22.2 The interaction matrix

Define $I_{ij} = \Delta\theta_i^\top H \Delta\theta_j$ over parameter groups. Measured over 14 groups (EMB, LN, and per-layer FF and attention):

step	PR	λ_1/λ_2	% positive off-diag	null (all-positive)	null (random dirs)
80	2.66	2.0	100	4.90	2.18
140	2.33	4.1	100	4.90	3.52
200	2.01	10.1	100	4.90	6.16
260	**1.85**	7.6	**100**	4.90	7.06

Observation 22.2 (A rank-one, all-positive, delocalised interaction field)**.** *Five properties, each against a control.* Low rank*: participation ratio* 1.85 *of* 14*, against* 4.90 *for a random all-positive*

matrix of matched scale — so the low rank is not forced by positivity, which a 4×4 *result alone could not have established.* Positivity*: all* 182 *off-diagonal entries positive at every checkpoint; each group's update raises the curvature cost of every other's.* Stability*: leading eigenvector overlap* 0.968 *between consecutive checkpoints, in a programme where no other object stayed stable.* Delocalisation*: the FF components by depth are* $0.336, 0.405, 0.359, 0.299, 0.392, 0.254$, *participation ratio* 5.86 *of* 6 *— uniform, so this is a network-wide mode and not a bottleneck at one layer.* Predictive*: adding* $C = \sum_{i \neq j} I_{ij}$ *to a regression of the loss shortfall on self-curvature raises* R^2 *from* 0.513 *to* 0.798*, clearing a shuffled-C null at* 0.702*.*

The interference share $C/(S + C)$ is conserved at 0.50 ± 0.04 across thirteen checkpoints: cross-group interference contributes as much second-order cost as all self-curvatures combined, throughout training.

22.3 The intervention: a magnitude effect, not a mode

Rescaling each group's contribution by $(1 \pm \alpha v_i)$ with v the leading eigenvector, then training forward 40 steps:

arm	val after 40 steps	vs none
none	0.08859	—
remove	0.09024	+0.00165
add	0.08439	−0.00420
random (same multipliers, permuted)	$\mathbf{0.08484 \pm 0.00089}$	**−0.00375**

Observation 22.3 (The mode is not a lever)**.** *Amplifying the mode improves the loss and suppressing it hurts, which alone would read as a coordination mechanism. But randomly permuting the same multipliers among the fourteen groups reproduces the benefit to within half a standard deviation (*−0.00375 *against* −0.00420*, sd* 0.00089*). The effect is the change in update magnitude, not the mode's allocation: the optimiser was under-stepping. The interaction field describes the geometry of joint updates without being a lever on it.*

22.4 The structure survives a change of optimiser

	AdamW	SGD
mean PR (of 14)	2.40	2.68
own null PR	4.86	4.04
% positive off-diagonals	99–100	87–100
leading eigenvector stability	**0.978**	**0.675**
dominant components	FF2, FF0, EMB	AT2, AT1, AT3

Observation 22.4 (Rank and sign are landscape; persistent identity is Adam)**.** *Under plain SGD the update is* $\Delta\theta_i \propto -\eta g_i$*, so* I *is the pure landscape quantity* $g_i^\top H g_j$*. The low rank survives (*2.68 *against its own null at* 4.04*) and so does positivity (*87–100%*), so the coupling is not optimiser-induced. But the eigenvector is far less stable (*0.675 *against* 0.978*, one checkpoint at* 0.310*) and points elsewhere — attention-dominated rather than FF-dominated. The* existence *of a dominant positive mode is landscape geometry; its* persistent identity *is substantially Adam holding it still. SGD reached only val* 0.79 *against Adam's* 0.064*, so the stability gap is confounded with training stage and is provisional.*

22.5 Where the structure lives: three spaces

space	PR (of 14)	% positive off-diag	
$I_\theta(i,j) = \Delta\theta_i^\top \Delta\theta_j$	12.4–12.6	—	diagonal by construction
$I_f(i,j) = (J\Delta\theta_i)^\top (J\Delta\theta_j)$	**4.7–5.0**	100	function space
$I_H(i,j) = \Delta\theta_i^\top H \Delta\theta_j$	**2.2–2.4**	100	curvature

Observation 22.5 (One direction seen through two operators)**.** *The groups partition coordinates, so I_θ is diagonal and its PR of* 12.5 *confirms that any structure elsewhere comes from the operator rather than from overlap. Both I_f and I_H are low rank with all off-diagonals positive, and curvature concentrates the coupling about twice as sharply as function does. The leading modes nearly coincide:* $|\cos(v_H, v_f)| = 0.981, 0.981, 0.962$ *across three checkpoints. So these are not two independent manifolds but the same direction viewed through two operators. The mean* $|\cos|$ *between the functional changes $J\Delta\theta_i$ is* 0.27–0.30 *— the groups push the logits in a shared direction, well above zero but far from collinear: a shared functional target, not a single one.*

22.6 Steering and moving are different roles

The altitude projection $a_i = -\nabla L^\top \Delta\theta_i / \|\Delta\theta_i\|$ measures alignment with the true downhill direction, independent of step size.

step	a_{LN}	next best	ratio
100	0.0696	0.0447 (EMB)	1.6×
180	0.0206	0.0154 (EMB)	1.3×
260	0.0098	0.0092 (FF0)	1.1×

Observation 22.6 (The group that steers is not the group that moves)**.** *LayerNorm has the highest alignment with the gradient at every checkpoint, the smallest displacement of any group (*0.10 *against attention's* 2.03 *at step* 250*), the slowest evolution (lag-1 autocorrelation* 0.9934*), the strongest predictive content for the update residual (*$R^2 = 0.414$*), and the smallest weight in the interaction mode (*0.017 *against FF's* 0.919*). Feed-forward is the opposite on all five. Four independent measurements separate steering from moving. Note also that a_i decays sevenfold from step* 100 *to* 260 *across all groups while the ordering holds: alignment with descent weakens uniformly without changing who leads.*

22.7 What is established and what is not

The programme has generated more retractions than results, and the distinction between the two is the useful output. Established claims are those that survived a matched control; the rest are recorded with the measurement that closed them.

claim	status	discriminating measurement
Established — survived a matched control		
Krylov geometry is intrinsic	holds	between/within 17–38×
$\omega \propto \lambda^{0.758}$ arc	holds	$R^2 = 0.966$, out-of-sample 0.751
path efficiency declines monotonically	holds	seven conditions
LayerNorm imposes an exact row constraint	holds	residual 0.0000
ϕ resolves the κ singularity	holds	monotone through 90°
slow variables predict, fast ones do not	holds	autocorr 0.99 vs 0.05
I is rank-one, positive, delocalised	holds	Obs. 22.2
I survives a change of optimiser	holds	Obs. 22.4
steering and moving are different roles	holds	Obs. 22.6
Refuted — a specific measurement closed it		
sparsity, dictionary, forward-activation substitution	refuted	§4.11
persistent low-dimensional subspace	refuted	4% at fixed θ
architectural charts localise the geometry	refuted	isotropic to 3 decimals
Morse handle at the step-72 event	refuted	no negative eigenvalues
κ conserved, dimensionless, corpus-specific	refuted	slope $0.758 \neq 1$; bands merge
skeleton then densification at step 120	refuted	attention leads from step 1
Krylov equivariance under $\mathcal{D}$	refuted	1–3% of floor
global Koopman operator	refuted	$R^2 < 0$ every set
local operators are generators	refuted	phase ratio 0.99 vs 6.0
$K(\mu)$ transfers across capacities	refuted	five independent tests
slow variables transfer across capacities	refuted	$R^2 = -4.1$
the interaction mode is a lever	refuted	Obs. 22.3
Vacuous — true but empty		
Fredholm index of the Krylov map	vacuous	$\dim V - \dim W = 0$
ordered $K_0 \cong \mathbb{Z}[\frac{1}{2}]$	scope-limited	classifies the chosen cover
strip energy $E \approx 43.5$	vacuous	random r-planes give the same
idempotency of the tile projector	vacuous	automatic for gapped products
Open — no measurement yet decides it		
why the exponent is 0.758	open	not derived from anything
what the latent state z is	open	five candidates excluded
whether I's mode is bottleneck or byproduct	open	intervention was null
what makes the sign field irreducible	open	mechanism unmeasured

Two patterns run through the ledger. *Global fails, local survives*: a fixed operator, a shared latent state, finite-history closure, cross-architecture transfer — none survived, while local linearity, bounded-horizon prediction and within-run structure consistently produced signal. And *fitted objects need matched nulls*: three separate results — the response modes, the subspace overlap, the conjugating map — looked meaningful until a null was constructed that preserved the trivial structure while destroying the claimed one. Roth's theorem guarantees a Sylvester solution; least squares always returns something; a shuffled null is not the same as a structure-preserving one. That lesson generalises beyond this model and is arguably the programme's most portable output.

23 Phase 21: The Fisher Sheet

Two structures were proposed at the end of §22: a fiber bundle joining the slow variables to the fast ones, and a bifold carrying the flow on a statistical sheet and a curvature sheet. The first fails a direct test. The second yields the phase's result, which is the strongest positive finding in the programme and replicates across corpora — though only one of its two sheets is measurable at present cost.

23.1 No fiber bundle

A bundle makes a claim beyond the existence of two timescales: the fiber must be conditionally determined by the base, and the base must be self-contained. Both are testable. A first attempt was invalid — regressing on levels of variables with lag-1 autocorrelation 0.998–1.000 produced a near-singular design and even $b_{t+1} \sim b_t$ scored -5.27, which is impossible for a self-correlated variable and indicates a broken regression rather than a refuted hypothesis. Corrected to increments with ridge regularisation, and with the base/fiber split placed at the largest gap in the measured autocorrelation spectrum rather than by intuition:

test	R^2	null p_{95}	
(1) $df_t \sim b_t$	-0.098	-0.008	no
(2a) $df_{t+1} \sim df_t$	$\mathbf{+0.213}$	$+0.036$	**signal**
(2b) $df_{t+1} \sim df_t, b_t$	-0.202	$+0.012$	no
(3) $db_{t+1} \sim db_t$	-0.289	$+0.022$	no
(4) $db_{t+1} \sim db_t, f_t$	-0.245	$+0.021$	no

Observation 23.1 (Timescale separation is not fibration)**.** *The instrument works — test (2a) is a clean positive — so the negatives are informative. The base does not determine the fiber, and adding it to the fiber's own history makes prediction* worse *by* 0.415*. The base does not even predict its own increments (test 3), so there is no autonomous slow dynamics for a fiber to sit over. The measured autocorrelation spectrum spans four decades, from W_K concentration at* 1.000 *to the sign-flip rate at* 0.094*, with the largest gap between EMB at* 0.813 *and FF at* 0.524*. The separation is real and organises nothing: the slow variables are informative observables, not a base manifold.*

23.2 Three metrics, not one

A bifold requires a commitment about which metric lives on which sheet, and the programme has three distinct ones. $F = \mathbb{E}[gg^\top]$, whose diagonal is Adam's v and whose off-diagonal the optimiser discards (§11.5). H, which is not F at these losses — the top-36 Hessian spectrum is strictly positive (§15.2) while the drift's Rayleigh quotient goes negative (§18.2), so they disagree on sign. And $M = \mathrm{diag}(v)^{-1/2}$, the metric the trajectory actually moves in.

The choice is not cosmetic. The arc exponent moved from 0.48 to 0.99 across optimisers before progress-matching (§17), and building the Krylov space in Adam's metric captured $\Delta\theta$ *worse* than the Euclidean construction, ratio 0.73 (§20.1).

Note also that the arc, the Krylov frame and the Ritz spectrum are not three independent inputs: λ and ω are exactly T_{11} and $\|T_{2:,1}\|$ of the projected Hessian $T = Q^\top HQ$. They are one object measured three ways.

23.3 The sheets are distinct and the flow is on one of them

Top-k eigenspaces of F and H estimated by randomised range finder at the same θ, then the update projected onto each and applied by actual ablation:

corpus / step	F energy	ΔL F-sheet	ΔL H-sheet	ΔL full	ΔL residual	ov(F, H)
english / 120	0.109	-0.0420	-0.0004	-0.0451	-0.0008	0.0097
english / 240	0.022	$\mathbf{-0.0075}$	-0.0011	-0.0020	$+0.0067$	0.0448
L3 / 120	0.090	$\mathbf{-0.0564}$	-0.0000	-0.0485	$+0.0090$	0.0066
L3 / 240	0.044	$\mathbf{-0.0161}$	-0.0010	-0.0077	$+0.0077$	0.0188

Observation 23.2 (Three directions outperform the whole update)**.** *In three of four cases the projection of the update onto three Fisher directions — out of* 4.3×10^6 *— delivers* more *loss reduction than the full update, by factors up to* 3.8*. The residual, carrying* 96–98% *of the update energy,* increases *the loss in three of four. The effect strengthens with training: at step* 120 *the sheet roughly matches the full update, at step* 240 *it beats it by* 2–4×*.*

This replicates on English prose and on L3, which have matched local statistics at every measured order and completely different global syntax — the control that narrowed or broke most other structural claims in this programme. The two sheets are also genuinely distinct: $\mathrm{ov}(F, H) = 0.007$–$0.045$ *against a random null of* 2.5×10^{-6}*, so they are three orders of magnitude above chance and still* 95% *disjoint.*

This is consistent with two earlier results by an independent route: the noise tax of §12.3, where drift gains +0.036 per step against noise costing −0.027, and the intervention of §22.3, where amplifying the update improved the loss because the optimiser was under-stepping.

23.4 The measurable limit

Observation 23.3 (The curvature sheet is not measured)**.** $\mathrm{SPLIT}_H = 0.005$–0.025*: two independent range-finder draws of the Hessian's top-*3 *at* identical θ *agree at under* 3%*. Reducing* k *from* 8 *to* 3 *was expected to raise this, following the calibration of Obs. 19.2, and did not. The claim that the* H*-sheet carries almost nothing therefore cannot be made: it may carry nothing, or the projection may be onto noise.* SPLIT_F *is better at* 0.21–0.28 *but still low, so the* specific *three Fisher directions are unreliable even though the effect is not — a randomly oriented three-dimensional subspace would capture* $\sim 10^{-6}$ *of the update energy, not the measured* 9%*.*

A two-sheet manifold cannot be built while one sheet is reproducible at 1–3%. The obstruction is cost rather than design: many more Hessian-vector products for H and more Fisher samples for F. The F-sheet result does not depend on it.

23.5 Status

claim	status	basis
fiber bundle over the slow variables	refuted	Obs. 23.1
timescale separation spans four decades	holds	autocorr 1.000 to 0.094
F and H sheets are distinct	holds	overlap 0.007–0.045 vs null 10^{-6}
low-dim Fisher component carries descent	holds, two corpora	Obs. 23.2
the bulk of the update is net harmful	holds, two corpora	residual $\Delta L > 0$ in 3/4
the H-sheet is empty	**untestable at present cost**	Obs. 23.3
bifold construction	premature	one sheet unresolved

The phase's arc is worth stating plainly, because it recurs. Two geometric constructions were proposed; the first was refuted by a test it was designed to pass, and the second produced a strong empirical result while its formal motivation — a symmetric two-sheet object — remains unsupported. What survives is not a manifold but a measurement: descent is carried by a low-dimensional statistical component, and almost all parameter motion is something else.

24 Phase 22: Predictive Failure as an Object

Substantially withdrawn. Two leakage controls run after this phase invalidate Observations 24.1 and 24.3 and the operator built on them; see §25. The results are retained here as measured, with the controls that closed them recorded in the following phase. What survives is stated in §25.3.

Every approach in §14–§23 asked what predicts the *state* and returned a null: latent variables, Koopman operators, LPV transfer, the fiber bundle. This phase inverts the question and asks what predicts the *failure*. The inversion produces the first object in the programme with genuine self-dynamics measured against a strict null — and that object turns out not to be low-dimensional, not state-predictable, and not tied to the loss, which is an unusual combination worth recording precisely.

24.1 Two errors, and why the second needs no model

Given observations x_t and a predictor $\hat{x}_{t+2|t}$, two quantities are available and they are not the same:

$$e_t = x_{t+2} - \hat{x}_{t+2|t}, \qquad d_t = x_{t+2} - \tfrac{1}{2}(x_{t+1} + x_{t+3}) = -\tfrac{1}{2}\big(x_{t+3} - 2x_{t+2} + x_{t+1}\big).$$

The first is model-dependent. The second is a discrete second difference: it measures the local *curvature* of the trajectory, is zero for any locally straight path, and requires no predictor at all. That is why it can carry signal where model-based prediction failed.

The null must be phase-randomised, not shuffled. Consecutive defects share the term x_{t+2}, so $d_{t+1} \sim Ad_t$ shows spurious correlation from overlap alone, and the second difference amplifies noise fourfold. A surrogate preserving the power spectrum while destroying phase relations is the appropriate control; a shuffle is not. Both are reported below and they agree, which is itself evidence the signal is not an overlap artefact.

24.2 The defect has dynamics; the state does not predict it

Measured on a fixed random projection of θ into 14 coordinates — fixed, so that trajectory curvature is not confounded with the rotation of an adaptive frame — over 360 consecutive checkpoints:

test		R^2	null p_{95}	
(1) magnitude	$\lvert d\rvert/\lvert dx\rvert = 0.370$	—	—	strongly curved
(2) $d_{t+1} \sim d_t$	phase-randomised null	$+\mathbf{0.207}$	$+0.008$	**signal**
	shuffled null, for contrast	$+0.207$	-0.012	signal
(3) $d_t \sim x_t$		-4.227	-0.143	no
(4) $d_{t+1} \sim x_t, d_t$		-0.834	-0.175	no
(6) $e_{t+1} \sim e_t$	forecast-error dynamics	$+\mathbf{0.389}$	-0.155	**signal**

Observation 24.1 (Failure has memory where the state has none)**.** *The defect predicts itself at* $+0.207$ *and the forecast error at* $+0.389$*, both clearing a phase-randomised null. The state predicts the defect not at all (*-4.23*), so this is not the state dynamics in disguise: where the failure comes from is not recoverable from where the trajectory is, while where it is going partly is. This inverts the programme's pattern, in which every state-prediction attempt returned negative.*

Two magnitudes give the context. Local curvature is 37% *of the step size, so straight-line extrapolation is badly wrong — which is why every linear predictor in this programme underperformed. And a persistence model's two-step error exceeds the step itself,* $\lvert e\rvert/\lvert dx\rvert = 1.30$*. The two errors are related but distinct:* $\mathrm{corr}(\lvert e\rvert, \lvert d\rvert) = +0.678$*, with* e *the smoother of the two since it accumulates* d *over two steps.*

24.3 The predictor ladder, and where it stops

If increasing information drove the residual into a stable structured subspace $r \approx Uz$, then U would be the missing structure, discovered rather than assumed. Measured over 380 checkpoints in 16 dimensions:

predictor	$\|r\|$	rel. to P_1	PR of C_r	top sv share
P_1 persistence, x_t	6.95	1.000	9.36	0.183
P_2 linear, $x_t + 2\Delta x_t$	1.66	**0.239**	9.43	0.190
P_4 ridge on $(x, \Delta x, \text{hist})$	10.69	1.538	2.41	0.560
null: random matrix	—	—	15.31	0.088

Observation 24.2 (The residual is not low rank). *Linear extrapolation cuts the residual to* 24%, *so the trajectory is locally close to straight and what remains is the curvature. But the residual covariance has participation ratio* 9.4 *of* 16, *only modestly below the random null at* 15.3. *The construction* $r \approx Uz$ *therefore stops here: an operator cannot be built around a* U *that is nine-dimensional out of sixteen. The ridge predictor's PR of* 2.41 *is the wrong kind of concentration — its residual exceeds persistence by* 54%, *so the fit is degenerate and its residual is concentrated because the* fit *is, not because the failure is.*

A row struck from the ladder. A quadratic predictor $x_t+2\Delta x_t+a_t$ with $a_t = x_{t+2}-2x_{t+1}+x_t$ gave $\|r\| = 0.00$ exactly. That is an algebraic identity, not a result: the expression reduces to x_{t+2}, so the target was used to predict the target. Recorded because the zero is otherwise an inviting misreading.

24.4 Curvature and failure share a subspace

The sharper question is whether the geometric defect c_t and the predictive errors $e_t^{(1)}, e_t^{(2)}$ occupy the same directions. Leading three right-singular subspaces, compared against a random-subspace null of matched dimension:

$$\text{ov}(c, e_1) = 0.580, \qquad \text{ov}(c, e_2) = 0.563, \qquad \text{ov}(e_1, e_2) = 0.999, \qquad \text{null } p_{95} = 0.303.$$

Observation 24.3 (Bending and failing are substantially the same directions). *The two predictive errors are the same object (*0.999*). Curvature overlaps each of them at roughly twice chance — genuinely between them and random, without being identical to either. So the trajectory's bending and the model's failure to anticipate it are largely manifestations of the same directions, which is the strongest link this phase establishes between a model-free geometric quantity and a model-dependent one.*

24.5 The defect is not an altitude variable

Observation 24.4 (No measured connection to the objective). *Regressing* ΔL_{t+1} *on* $|d_t|$ *gives* $R^2 = -1.26$ *with correlation* -0.055 *over* 55 *points. The defect says where the path bends, not whether the bending helps. Combined with Obs. 24.2, the object has temporal structure and neither spatial concentration nor a link to the loss.*

Two design limitations qualify this and should be removed before the negative is treated as settled. The loss was sampled every five steps while d was computed every step, a fivefold mismatch that could wash out a real relationship at $n = 55$. And the regressor was the scalar $|d_t|$, where the informative version is the *direction* — the projection of d_t onto its leading modes — since a magnitude can be uninformative while a direction is not. Similarly, the participation ratio of Obs. 24.2 was computed on the residual after persistence, which leaves velocity in the residual, so it partly measures the trajectory's own spread rather than the defect's; the rank of c_t itself was never taken.

24.6 Status

claim	status	basis
$d_{t+1} \sim d_t$ has signal	holds	+0.207 vs phase-randomised +0.008
$e_{t+1} \sim e_t$ has signal	holds	+0.389 vs −0.155
the state predicts the defect	refuted	−4.23
curvature and failure share directions	holds	0.58 vs null 0.30
the residual is low rank	refuted	PR 9.4 of 16
the defect predicts loss change	not supported	$R^2 = -1.26$, but underpowered
rank of c_t itself	unmeasured	—
directional $d_t \to \Delta L$	unmeasured	—

The workflow here is the reverse of the one that produced most of this programme's retractions: find the empirical object first, and let the theory name it afterwards. On that discipline, no framework should be attached yet. What exists is a process with memory, spread over roughly nine of sixteen dimensions, uncorrelated with the objective, whose directions coincide with those of predictive failure at twice chance. It is not a hidden coordinate and not a ruler, since a ruler would report altitude. It is worth modelling as a process rather than as a subspace, and the two unmeasured rows above are what should be run before anything further is claimed.

25 Phase 23: Leakage Controls and the Absent Bridge

§24 reported that the defect d_t has self-dynamics and that curvature and predictive failure share directions. Two controls, run after the fact, invalidate most of that phase. The withdrawals are recorded here in full, together with the two results that survive them and a statement of what is not established — the absence of a bridge between the programme's two robust findings, which is itself the phase's conclusion.

25.1 Two leakage controls

Control one: a lag sweep. $d_t = x_{t+2} - \frac{1}{2}(x_{t+1} + x_{t+3})$ is a second difference, so d_t and d_{t+1} share two of their three terms. A genuine short-memory process shows a decaying $R^2(k)$; an overlap artefact shows an isolated spike at $k = 1$. Measured at $\beta_1 = 0$, where the effect was largest:

k	R^2	phase null p_{95}	block null p_{95}
1	**+0.389**	−0.030	−0.036
2	−0.115	−0.038	−0.038
3	−0.132	−0.035	−0.041
5	−0.120	−0.035	−0.037
8	−0.102	−0.033	−0.041

Observation 25.1 (The defect's self-prediction is a construction artefact)**.** *R^2 is positive at $k = 1$ and negative at every lag from 2 to 8. There is no decay curve. At $k = 1$ the two defects share two terms; at $k \geq 2$ they share one or none, and the correlation vanishes exactly as overlap predicts. The phase-randomised null used in §24.2 preserves the power spectrum but not the overlap structure between consecutive estimates, so no single-lag null could have caught this: the artefact and a real one-step memory are indistinguishable at $k = 1$ and differ only in what follows.*

Control two: target leakage. A separate test asked whether the defect coordinate closes the dynamics — whether d_t screens off history in predicting x_{t+1}:

model		R^2	vs. previous
M0	$x_{t+1} \sim x_t$	0.8008	
M1	$+x_{t-1}$	0.7886	−0.0123
M2	$+x_{t-2}$	0.7750	−0.0136
D0	$x_{t+1} \sim x_t, d_t$	0.8902	+0.0894
D1	$+x_{t-1}$	0.9722	+0.0819

Observation 25.2 (The trajectory is already Markovian, and d leaks the target). *History does not help: R^2 declines monotonically from* 0.801 *at zero lags to* 0.717 *at six. The update sequence is Markovian in its own coordinates, so there is no non-Markovian structure for a defect coordinate to close, and the generalised-Langevin reading has no purchase.*

D0's apparent gain of +0.089 *is leakage:* $d_t = A_{t+1} - \frac{1}{2}(A_t + A_{t+2})$ contains A_{t+1}, *which is the prediction target. D1 confirms it — adding x_{t-1} gains* +0.082 *on top of d, when the same variable* lost 0.012 *without it. History becomes useful only because it helps decode the target out of d. The same pattern appears in the optimiser arm (*0.677 → 0.832 *on adding d).*

A design note, since the second control was foreseeable. The Markovian test was described as immune to the artefact that killed the first result. It is not: it is a different leak from the same source. The first arises because d_t and d_{t+1} share terms; the second because d_t and the target share a term. Only the first was checked. A clean construction would use a backward defect $\tilde{d}_t = A_t - \frac{1}{2}(A_{t-1} + A_{t+1})$ shifted to use information through t only, or predict A_{t+2}.

25.2 The operator A is withdrawn, with one exception

The operator of §24 was fitted as $d_{t+1} = Ad_t$ from the same contaminated variable, so its stability inherits the same doubt. Three parts of that measurement separate.

Already negative, and unaffected. A's rank was full, not low — participation ratio 8.43 of 16 against 11.87 for the state covariance — and per-channel predictability sat at the null in all six leading channels. Leakage does not rescue a negative.

Withdrawn. The singular-subspace persistence, 0.59–0.72 against a null of 0.188 across six window transitions. Overlap-induced correlation can manufacture a stable basis: if consecutive d share coordinates, the fitted A inherits structure from the shared terms rather than from dynamics. The control that would distinguish these was not run, so the result is withdrawn as undetermined rather than shown false.

Unaffected. The absence of alignment between A's leading directions and those of I_H: overlap 0.139–0.251 against nulls of 0.296–0.318, at or below chance at all four checkpoints. Stated conservatively: *the measured A-derived geometry shows no evidence of alignment with I_H, and contamination of A does not by itself explain the observed absence of alignment* — contamination would have to create alignment to threaten this conclusion, not destroy it.

25.3 What survives

Two findings, of different kinds, and the distinction between them matters.

One robust structural object. $I_H = \Delta\theta_i^\top H \Delta\theta_j$ has seven independent validations: effective rank 1.85 of 14 against an all-positive null at 4.90; all 182 off-diagonals positive at every checkpoint; eigenbasis stable at 0.968; delocalised across depth (FF participation ratio 5.86 of 6); leading mode aligned with the function-space coupling at $|\cos| = 0.96$–0.98; rank and sign surviving a change of optimiser; and predicting the loss shortfall beyond self-curvature (R^2 0.513 → 0.798).

One robust dynamical regularity. The update sequence is Markovian and 80% predictable from its own previous value, with no incremental benefit from history at any lag to six, and its discrete curvature is roughly three times that of the weight sequence ($|d|/|dx| = 0.70$ against

0.21). Both facts emerged from *control* arms rather than from the constructions they were controlling.

These are not two objects. I_H is a geometric object with internal and external validation; the second is a dynamical property of a sequence. Calling both “structures” would repeat the pattern this phase is correcting, in which a regularity is promoted to a named entity before it has earned one.

25.4 The absent bridge

claim	status	control that decided it
$d_{t+1} \sim d_t$ has memory	withdrawn	lag sweep, Obs. 25.1
d closes the dynamics	withdrawn	target leakage, Obs. 42.8
A has a stable singular basis	withdrawn	shares the contaminated variable
curvature and failure share directions	withdrawn	same construction
A aligns with I_H	no evidence	overlap at or below null
I_H is low rank, positive, stable	holds	seven validations
update sequence is Markovian, 80%	holds	control arm of Obs. 42.8
update curvature is 3× weight curvature	holds	magnitude, no regression

The temptation at this point is a further construction bridging I_H and the update dynamics. The evidence does not support one, and three of the four withdrawals above came from exactly that impulse. The defensible statement is narrow: *I_H has coherent low-dimensional organisation; the update sequence has strong local temporal organisation; and no tested construction establishes that these are the same structure.* The absence of the bridge is itself the result.

This phase also sharpens the methodological lesson of §22.7. A null that preserves the marginal distribution is insufficient when the estimator has overlapping inputs; the null must preserve the trivial structure while destroying the claimed one. Phase randomisation was the right instinct and the wrong control, because the overlap survives it. Four results in this programme have now failed a structure-preserving null after passing a weaker one.

26 Phase 24: Interventions, the Flow Residual, and a Non-Split Extension

§25 closed the fitted-model line: three withdrawals under leakage controls and no bridge between the two survivors. This phase changes method. Every measurement below is an *intervention* — an architectural or algorithmic change followed by re-measurement — or a comparison against an exactly computable reference. Neither can be manufactured by an estimator, which is what the preceding failures kept exposing.

Three results follow: the interaction field is architecture-independent, the update residual is a static bias between two vector fields rather than noise, and the two-block splitting of §23.3 is a genuinely non-split extension. One prediction of mine is refuted along the way, and one intervention built on a bad statistic fails.

26.1 The interaction field survives every intervention

I_H was measured at matched progress under softened nonlinearities, on the chunk corpus of a chunk-tokenised corpus where all conditional entropy has been removed:

arm	val	PR(I_H)	% positive off-diag	c_T/base
baseline	0.0008	2.01	**100**	1.048
softmax $\tau = 2$	0.0008	1.99	**100**	0.980
RMSNorm $\epsilon = 10^{-2}$	0.0008	2.01	**100**	0.953
both together	0.0008	1.79	**100**	1.057

Observation 26.1 (Four interventions, no change)**.** *Softening the softmax and replacing LayerNorm with high-ϵ RMSNorm — simultaneously, on a corpus with $H_1 = H_2 = H_3 = 0$ — leaves the rank at* 1.79–2.01*, the off-diagonals* 100% *positive, and tile coherence within* 5%*. Together with the optimiser change of Obs. 22.4 and the LayerNorm removal of §26.2, I_H has now survived four independent interventions. It is a property of the loss landscape, not of any architectural component or of the corpus.*

The corpus intervention deserves its own note. Re-tokenising at four-word boundaries makes every chunk unique (341 of 341), so H_1, H_2 and H_3 collapse to exactly zero and the task becomes memorising a cycle. Validation loss falls from 0.090 to 0.0007 and the curvature scale λ drops 300×. The geometry does not follow: ϕ moves 50.3° → 37.5°, efficiency 0.447 → 0.550, tile coherence 1.011 → 1.034. *Removing all relational content from the corpus changes the loss by two orders of magnitude and the geometry hardly at all.*

26.2 Architecture does move the geometry

The contrast with the corpus result is the point. At matched progress (val ≤ 0.35), with the normalisation strength α separating constraint from scale in $y = \gamma[(1-\alpha)x + \alpha\hat{x}] + \beta$:

arm	step	val	c_T/base	ϕ	fwd·bwd
softmax $\tau = 0.5$	130	0.285	**0.876**	63.9°	0.181
baseline	130	0.312	0.969	55.4°	0.201
softmax $\tau = 2.0$	130	0.339	**1.133**	55.2°	0.203
LN $\alpha = 0.5$	120	0.340	1.147	54.8°	0.158
LN $\alpha = 0.0$	290	0.283	0.997	48.9°	**0.097**

Observation 26.2 (Row coherence is graded by softmax temperature)**.** *Row-scale coherence moves monotonically with temperature,* 0.876 → 0.969 → 1.133 *over a fourfold range — sharper attention gives more anti-aligned rows, softer attention fewer. The forward/backward alignment tracks LayerNorm strength monotonically,* 0.201 → 0.158 → 0.097 *as α falls, so the normalisation is what couples weight state to applied update. Removing normalisation entirely costs* 2.4× *the optimisation time for the same loss.*

*The LayerNorm prediction of §16.1 is confirmed at $\alpha = 0.5$ (*1.147*, above baseline as the projector algebra requires) and* fails *at $\alpha = 0.0$ (*0.997*, back to baseline). Partial relaxation moves the geometry more than full removal, which the projector algebra does not explain.*

26.3 The residual is a static bias, not noise

Every previous residual measurement used stochastic minibatches, so sampling noise was never excluded. Fixing the batch makes L deterministic, the true gradient flow $\dot{\theta} = -\nabla L$ well defined, and the reference computable by RK4.

The discretisation branch is excluded first. Euler against RK4 on the same flow gives relative residuals 4.3×10^{-2}, 1.3×10^{-2}, 4.4×10^{-3}, 1.3×10^{-3}, 4.4×10^{-4} over h from 10^{-1} to 10^{-3} — convergence slopes $0.98, 0.99, 1.00, 1.00$, exactly the first-order law. The numerical residual vanishes in the continuum limit.

Observation 26.3 (AdamW does not integrate the gradient flow)**.** *At the actual training step size, $\|R\|/\|\Delta\theta\| = 1.36$ — the residual exceeds the step — and $\cos(\Delta\theta_{disc}, \Delta\theta_{flow}) = +0.25$, so the discrete update sits some 75° from the flow direction. Against Euler's 4.4×10^{-4} at $h = 10^{-3}$, that is three orders of magnitude apart: the residual is not discretisation error of the flow but the difference between two vector fields.*

The correlation of the residual direction *decays smoothly —* $0.965, 0.909, 0.872, 0.849, 0.823, 0.794, 0.767, 0.74$ *over eight lags — on a fixed batch, from disjoint steps. This is the first persistence measurement* *in the programme showing a decay curve rather than the isolated $k = 1$ spike of Obs. 25.1. Per tile the lag-1 correlation is* 1.0000 *and lag-4 is* 0.996–1.000, *so the residual is a nearly* constant *vector per tile: a static offset between the update field and the gradient field, not path-dependent holonomy, which would require a loop.*

The residual is tile-structured. Over 200 steps on the fixed batch:

tile	$\|R\|/\|\Delta\theta\|$	cos(disc, flow)	$R_\parallel$ share
LN	10.4	+0.60	0.99
EMB	0.78	+0.66	0.07
$\mathrm{FF}_{0\ldots5}$	0.91–0.92	+0.85–+0.88	0.68–0.73
$\mathrm{ATTN}_{0\ldots5}$	0.97–0.98	+0.34–+0.41	0.08–0.13

Feed-forward tracks the flow with a mostly parallel residual; attention diverges directionally with an almost entirely transverse one. Depth is irrelevant — all six layers of each type agree to within 0.01.

Observation 26.4 (The LayerNorm magnitude reading is withdrawn)**.** *LN's ratio of* 10.4 *invited the reading that Adam overshoots LayerNorm, and the intervention that follows from it — damping the LN step — was run. It fails. Damping by* $0.3\times, 0.1\times, 0.03\times$ *moves the ratio to* $31.2, 91.2, 302.9$, *scaling as* $1/\mathrm{damp}$ *almost exactly, because* $\|\Delta\theta\|$ *is the* denominator *and the RK4 reference is unchanged. The direction is opposite to the diagnosis: Adam* under-*steps LN relative to flow. Loss degrades monotonically (*$0.00066 \to 0.00086$*), and the control — damping ATTN instead — leaves LN's ratio at* 9.67 *against baseline* 9.71.

What survives is the FF/ATTN directional *split, which compares directions and is immune to this scaling.*

26.4 The two-block splitting is a non-split extension

§23.3 established a low-dimensional Fisher channel carrying the descent and a large residual. Whether that is a direct sum or a non-trivial extension turns on the off-diagonal block $M = P_F H(I - P_F)$, with $e_1 = P_F$ at $r = 3$ and $e_2 = I - P_F$.

step	basis	$\|e_1He_1\|$	$\|e_1He_2\|$	off/diag
140	Fisher	0.698	**0.684**	0.98
140	random 3-plane	0.00083	0.0137	16.6
260	Fisher	0.221	**0.432**	1.96
260	random 3-plane	0.00020	0.0093	47.1

Observation 26.5 (The coupling block is not negligible)**.** $\|M\|$ *is comparable to or larger than the diagonal block —* $0.98\times$ *at step* 140, $1.96\times$ *at step* 260. *The prediction that it would vanish, made from the* 95% *disjointness of the F and H eigenspaces, is refuted. The random-plane control excludes a dimension artefact: a random* 3*-plane's diagonal block is* 0.00083 *against the Fisher channel's* 0.698, *three orders of magnitude smaller, so the Fisher directions carry real curvature and leak it into the complement.*

The loss decomposition is correspondingly non-additive. At step 140 *the parts sum to* -0.0039 *while the full update gives* -0.0153*: a discrepancy of* -0.0114, *or* 19% *of the parts, and* constructive *— the pieces together reduce loss more than they do apart. At step* 260 *this falls to* 4%.

Two qualifications. The norm ratio rises ($0.98 \to 1.96$) while the non-additivity falls ($19\% \to 4\%$) — these move oppositely and are not explained here. And SPLIT_F is 0.26–0.31, so P_F is roughly 30% reproducible; part of $\|e_1He_2\|$ could be the projector's own estimation error leaking into the complement. The random-plane control excludes a pure dimension artefact but not this one.

26.5 What the algebra can and cannot carry

The measurements now settle three questions about the proposed matrix constructions.

The extension is non-split. $M \neq 0$ with $\|M\| \gtrsim \|e_1 H e_1\|$, so $\Lambda = \left(\begin{smallmatrix} \mathcal{H}_D & M \\ 0 & \mathcal{H}_R \end{smallmatrix}\right)$ does not degenerate to $\mathcal{H}_D \oplus \mathcal{H}_R$. This is the one algebraic claim in the programme that survived its own test.

It is not a chain complex. A differential requires $d^2 = 0$; the update sequence is Markovian and 80% self-predictable from its previous value (Obs. 42.8), so consecutive updates *correlate* rather than annihilate. Treating $\Delta\theta_t \mapsto \Delta\theta_{t+1}$ as a boundary map conflates a dynamical flow operator with a homological one.

It is not a recollement. A recollement needs $j^! \dashv j_!$; the forward/backward out-of-image residual is 70.1% over both halves of training with slope -0.0049% per step, and the backprop-sense alignment is 0.007 ± 0.014 per layer. The tile bifibration's adjunction is exact (residuals 0.1–0.8%) and measured not to extend to gradient structure (§10.3) — the same boundary, reached from the algebraic side.

26.6 Status

claim	status	basis
I_H survives four interventions	holds	Obs. 26.1
corpus relational content moves the geometry	refuted	λ falls 300×, ϕ moves 13°
row coherence graded by softmax τ	holds	$0.876 \to 1.133$
fwd/bwd alignment graded by LN strength	holds	$0.201 \to 0.097$
residual is discretisation error	refuted	Euler slope 1.00 vs Adam 1.36
residual is a static per-tile bias	holds	lag-1 = 1.0000, fixed batch
FF parallel / ATTN transverse split	holds	$R_\parallel$ 0.71 vs 0.11
LN magnitude overshoot	refuted	Obs. 26.4
off-diagonal block vanishes	refuted	Obs. 26.5
Λ is a chain complex	refuted	$d^2 \neq 0$; updates correlate
Λ is a recollement	refuted	70.1% residual, flat

The accepted framing is narrower than a triangular ring and wider than a direct sum. The state splits into a low-dimensional Fisher-aligned descent channel and a large residual; the splitting is *co-moving*, re-estimated at each checkpoint and only 21–31% reproducible, so it is not a fixed grading; the two blocks are coupled through curvature strongly enough that the loss is non-additive; and the residual is not noise but a static, tile-structured offset between the update field and the gradient flow field. The metric, not the direction of the pass, is what separates the system.

27 Phase 25: A Tiny Replica, and the Instrument Floor

Every geometric measurement in §19–§26 was bounded by the same constraint: at $P = 4.3 \times 10^6$ subspaces are near-orthogonal by default, the Fisher sheet never exceeded 31% reproducibility, and the commutator and transport questions returned *unresolvable* rather than an answer. a later calibration failed — dropping to $r = 2$ made SPLIT worse, not better — and a formation test showed the failure was not explained by the sheet being unformed: SPLIT *fell* with training (corr -0.640) while the Fisher participation ratio *rose* (corr $+0.503$), both opposite to the signature of a concentrating object.

This phase changes the object rather than the instrument. At $P \sim 3700$ the Hessian and Fisher can be formed *densely*: exact spectra, exact projectors, exact commutators, no range finder and no SPLIT. Several verdicts of the preceding phases are overturned, and one of them was mine.

27.1 The replica

Same architecture family — pre-LN residual blocks, softmax attention, gated feed-forward — at $V = 40$, $D = 12$, $L = 2$, $T = 16$, giving $P = 3672$ and a dense Hessian of 54 MB. The corpus is a small deterministic grammar with genuine two- and three-form relations, not the degenerate cycle of §26.1.

step	val	PR(F)	PR(H)	ov(F, H)	$\|[P_F, P_H]\|^2$	$\|M\|/\|\text{dia}\|$	cap$_F$	dL_F/dL_{full}
50	1.724	3.23	152.3	0.256	0.812	0.804	0.398	1.034
150	0.107	2.49	22.3	0.542	0.679	0.232	0.037	0.744
300	0.036	2.94	17.1	0.858	0.508	0.083	0.112	0.645

27.2 The two-metric distinction, exactly

Observation 27.1 (PR(F) $\ll$ PR(H), with no estimation involved)**.** *The Fisher participation ratio is* 2.5–3.2 *at every checkpoint while the Hessian's is* 17–152*, a factor of* 5–50*. A second dense run at a different stage gives* 5.68 *against* 31.4*. The Fisher information is low-rank and the Hessian is diffuse, and unlike every prior version of this claim the numbers require no null, no split-half and no rank truncation.*

One caveat is not about estimation. $F = \mathbb{E}[gg^\top]$ *has its top eigenvector dominated by the* mean *gradient direction, so a low PR is partly what the covariance of a strongly mean-shifted variable looks like. Whether PR(F) ≈ 3 reflects a three-dimensional information geometry or the gradient having a dominant mean is not settled here; recomputing PR on the centred Fisher* $\mathbb{E}[(g - \bar{g})(g - \bar{g})^\top]$ *would decide it.*

27.3 The range finder reads its own floor

Running dense and sampled estimates side by side at the same θ and the same $r = 3$:

P	val	ov(F, H) dense	ov(F, H) range finder	ratio
3672	0.225	**0.717**	**0.250**	0.35
9720	0.066	—	0.505	—
34240	0.214	—	0.392	—

Observation 27.2 (Sampled overlap underestimates the truth by threefold)**.** *At the* smallest *size, where estimation should be easiest, the range finder recovers* 35% *of the true overlap. Since the floor can only worsen with* P*, the reading of §23.3 — that the* F *and* H *eigenspaces are* 95% *disjoint at* 4.3×10^6*, from sampled overlaps of* 0.007–0.045 *— is an instrument artefact. Any claim of near-orthogonality resting on a randomised range finder in this programme should be read as a lower bound, not a measurement.*

The scaling question itself did not resolve. The sampled overlaps run 0.250, 0.505, 0.392 with no trend, and the middle arm reached val 0.066 against the others' 0.21–0.23 — the matched-loss loop overshot on a 25-step granularity, so proximity and dimension remain confounded. That is the same confound that overturned the κ corpus bands, the SGD exponent, the marker cluster and the optimiser eigenvector, and it is controllable: finer loss granularity and a common target.

27.4 The extension has a lifecycle

Observation 27.3 (F and H align, and the extension splits)**.** *ov(F, H) rises* $0.256 \to 0.542 \to 0.858$ *as val falls* $1.72 \to 0.036$*. The commutator relaxes* $0.812 \to 0.679 \to 0.508$ *against a maximum of* $r/2 = 1.5$*, so the tilt between the Fisher metric and the Hessian curvature is real, large and decreasing. The off-diagonal coupling collapses* $0.804 \to 0.232 \to 0.083$.

The last of these is mechanically forced *by the first and should not be counted as independent evidence: as P_F approaches an eigen-projector of H, $M = P_F H(I - P_F) \to 0$ by algebra. The content is in the alignment; M's collapse is its shadow. Read together: the non-split extension of Obs. 26.5 is a* transient. *It is active early, when the metrics are tilted and the loss decomposition is non-additive, and it splits into an aligned uncoupled eigenspace as the model enters the basin.*

Scope. $F \to H$ is a theorem at a well-specified minimum, and this model is exactly that: a deterministic grammar memorised to val 0.036. Whether a 4.3×10^6 transformer on natural text ever approaches closely enough for the alignment to occur is the question the ladder was meant to answer and did not. The lifecycle is verified in the well-specified regime and untested outside it.

The framing also makes a prediction that has not been run: if the extension splits as the minimum is approached, the loss non-additivity should vanish on the same schedule. At 4.3×10^6 it fell 19% → 4% between steps 140 and 260 (Obs. 26.5), which is consistent; at the tiny scale it should reach essentially zero by step 300 where $\|M\|/\|\text{dia}\| = 0.083$.

27.5 The descent channel is stage-dependent

cap_F falls 0.398 → 0.037 → 0.112 while dL_F/dL_{full} falls 1.034 → 0.744 → 0.645. Early, three Fisher directions capture 40% of the update and deliver *all* of the loss reduction. Late they capture 4–11% and still deliver two-thirds — an efficiency ratio of roughly 17× per unit motion, matching the 10–17× measured at 4.3×10^6 (§23.3). The Fisher-channel result of §23.3 therefore holds at both scales, and is now seen to be stage-dependent rather than fixed.

27.6 Status

claim	status	basis
$\text{PR}(F) \ll \text{PR}(H)$	holds, exact	Obs. 27.1
range finder underestimates $\text{ov}(F, H)$	holds	0.250 vs 0.717 dense
F, H 95% disjoint at 4.3M	**withdrawn**	instrument floor
F, H align near the minimum	holds, tiny scale	0.256 → 0.858
$\|M\| \to 0$ independently	**not independent**	forced by alignment
non-split extension is permanent	refuted	lifecycle, Obs. 27.3
Fisher channel carries descent	holds, both scales	17× efficiency
alignment is dimensional vs proximity	open	ladder confounded
$\text{PR}(F)$ low after centring F	untested	—

Two lessons generalise beyond this model. The first is that a negative result about geometry at high dimension may be a statement about the estimator: the 95%-disjointness reading survived several phases and was overturned by computing the same quantity exactly at a scale where that is possible. The second is that shrinking the system is a legitimate instrument. Where the programme's controls repeatedly told us what could *not* be concluded, the replica tells us what is true — in a regime whose relation to the full model must then itself be measured.

28 Phase 26: Solidification and the Alignment of Block Motions

§27 established the geometry exactly at $P = 3672$ but treated the parameter vector as a whole. This phase resolves it by role — EMB, LN, W_Q, W_K, W_V, W_O, FF — and asks two questions that turn out to have different answers: whether each component's motion is low-dimensional, and whether those low-dimensional motions align with each other.

The first is yes, by a wide margin. The second is no, and the gap between them reconciles a discrepancy that has stood since §1.13: individually simple block motions can sum to a large global residual precisely because they are mutually unaligned. A third result arrives unlooked-for — attention splits into two coupled pairs along its own computational structure.

28.1 Each component travels on a low-dimensional manifold

Singular-value decomposition of each role's update sequence, exactly, over three windows. k_{90} is the number of directions needed for 90% of the component's motion energy; the null is the same quantity for random directions of the same block size and sequence length.

window	k_{50}	k_{90}	null k_{90}	E(top-3)	cos lag-1
early (20–60)	1	2–3	30–36	0.956–0.989	0.993–0.998
mid (100–160)	1	2–5	41–53	0.816–0.962	0.954–0.993
late (240–300)	1–2	1–5	41–53	0.681–0.960	0.744–0.991

Observation 28.1 (Ninety per cent of each block's motion needs at most five directions)**.** *Every role, every window:* $k_{90} \leq 5$ *against a null of* 30–53*. A* single *direction carries half the motion of each block early and mid. Three directions carry* 95–99% *early.*

The structure loosens monotonically: E*(top-3) falls* $0.97 \to 0.83 \to 0.72$ *averaged over roles,* cos *lag-1 falls* $0.995 \to 0.968 \to 0.793$*, and* k_{50} *goes* $1 \to 1 \to 2$*. Early training is nearly one-dimensional per block; late training explores two to five.*

Two roles break the pattern. The token embedding *tightens* ($k_{90} : 2 \to 2 \to 1$, E(top-3) steady at 0.96, cos at 0.99 throughout) — it travels in essentially one direction for the whole run. LN loosens most, its cos falling to 0.812.

28.2 Limits and colimits both solidify

Taking four checkpoints per window and the diagram $\mathrm{Im}(W^{(t_1)}) \to \cdots \to \mathrm{Im}(W^{(t_4)})$: the *limit* is measured as the overlap of the top-3 eigenspaces of $W^{(t_1)}$ and $W^{(t_4)}$, the *colimit* as the participation ratio of the centred trajectory matrix.

role	ov(t_1, t_4) early	mid	late	random floor	PR$_{\mathrm{cent}}$ late
$\mathrm{EMB}_{\mathrm{tok}}$	**0.989**	0.995	0.987	0.064	1.007
$\mathrm{EMB}_{\mathrm{pos}}$	0.994	0.998	0.998	0.198	1.025
LN	0.585	0.966	0.997	0.219	1.065
W_Q	0.625	0.945	0.989	0.219	1.118
W_K	0.885	0.962	0.988	0.219	1.132
W_V	0.895	0.898	0.999	0.219	1.342
FF	0.719	0.888	0.977	0.129	1.117

Observation 28.2 (The invariant frame forms progressively, except for the embedding)**.** *Every role converges to* 0.98–0.999 *by the late window against floors of* 0.06–0.22*, and the convergence is progressive: LN goes* $0.585 \to 0.966 \to 0.997$*,* W_Q *goes* $0.625 \to 0.945 \to 0.989$*. The token embedding is the exception —* 0.989 *from the earliest window and flat thereafter. It did not converge to an invariant frame; it was one at step* 20*.*

The colimit glues in parallel: PR of the centred trajectory falls toward 1 *for every role. Centring is necessary — the raw PR is* 1.000–1.039 *throughout because the weights are dominated by their common mean, so the raw statistic is trivially near* 1 *and carries no signal.*

One proposed statistic does not work. The geodesic defect $\kappa = \|d_2\|/(\|d_1\|\,\|d_3\|)$ was expected to fall toward zero as the path straightens. It *rises* by an order of magnitude in every role (LN $3.12 \to 5.56 \to 16.30$). The quantity is not scale-free: if all three steps shrink by c, κ grows as $1/c$, and step sizes fall sharply late. The dimensionless form is $\|d_2\|^2/(\|d_1\|\|d_3\|)$. The alignment $\cos(d_1, d_3)$ is the statistic that behaves as intended, rising in every role — $\mathrm{EMB}_{\mathrm{tok}}$ $0.726 \to 0.820 \to 0.966$, LN $0.555 \to 0.492 \to 0.871$.

28.3 Attention splits into two coupled pairs

Observation 28.3 (Query/key and value/output behave differently, in three independent measurements)**.** *Late in training,* W_Q *and* W_K *retain directional coherence while* W_V *and* W_O *turn. Per-update lag-1 cosine:* $0.971, 0.971$ *against* $0.744, 0.767$*. Four-point path alignment* $\cos(d_1, d_3)$*:* $0.711, 0.682$ *against* $0.488, 0.550$*. And in the pairwise overlaps of §28.4 the two pairs are the only entries far above background.*

W_V *is the clearest case of the limit and colimit separating: it has the* highest *limit overlap of any role (*0.999*) and among the lowest path coherence (*0.488*). Its subspace is fixed while its motion inside that subspace wanders — which is exactly the distinction the two constructions exist to make.*

28.4 The block motions are mutually unaligned

The blocks partition the coordinates, so their motion spaces are orthogonal in parameter space *by construction* and comparing them there is vacuous. The question has content only in a space they share. Two such: the function space, via $J\,\delta u$ computed by central differences on the logits, and the curvature space, via the Hessian inner product.

Function-space overlap of the top-3 motion directions, late window:

	EMB	LN	W_Q	W_K	W_V	W_O	FF
EMB	1.000	0.128	0.043	0.062	0.085	0.069	0.084
LN	0.128	1.000	0.176	0.149	0.267	0.266	**0.358**
W_Q	0.043	0.176	1.000	**0.509**	0.109	0.161	0.134
W_K	0.062	0.149	**0.509**	1.000	0.138	0.125	0.145
W_V	0.085	0.267	0.109	0.138	1.000	**0.552**	0.257
W_O	0.069	0.266	0.161	0.125	**0.552**	1.000	0.293
FF	0.084	**0.358**	0.134	0.145	0.257	0.293	1.000

Observation 28.4 (Individually simple, mutually unaligned)**.** *Mean off-diagonal overlap is* 0.135*,* 0.128*,* 0.196 *over the three windows, against a random-plane null of* 0.0000*. The blocks are three to five orders above chance and still some* 85% *unaligned. That is the reconciliation: block motions that are individually one- to five-dimensional need not sum to a common global direction, and a large global residual follows from their mutual tilt rather than from any block moving in high-dimensional noise.*

The pairing structure is unmistakable and holds at every window. $W_Q \leftrightarrow W_K$ *runs* $0.605, 0.411, 0.509$ *and* $W_V \leftrightarrow W_O$ *runs* $0.473, 0.437, 0.552$*, while every cross-pair entry sits at* 0.03–0.16 *(*$Q \leftrightarrow V$ *is* 0.039*). Attention is neither one block nor four independent ones: it is two coupled pairs, and the coupling follows the computational structure exactly —* $QK^\top$ *forms the attention map,* VW_O *the output.*

Curvature-space overlap runs $0.364, 0.351, 0.483$*, roughly* $2.5\times$ *the function-space values. The blocks share curvature directions more than they share functional effect, consistent with* I_H*'s all-positive off-diagonals against the weaker function-space coupling of Obs. 22.5.*

LN is the connector, with its largest overlaps to FF (0.358*) and to the value/output pair (*0.266–0.267*). EMB is the most isolated role at* 0.02–0.13 *against everything — functionally decoupled, not merely rigid in its own subspace.*

One trend cuts against the lifecycle. Alignment *grows* with training in both spaces — $0.135 \to 0.196$ in function, $0.364 \to 0.483$ in curvature. The blocks co-adapt rather than decoupling, which fits the limit and colimit solidification but sits awkwardly with $\|M_t\| \to 0$ of Obs. 27.3. Those are different objects — M_t couples the Fisher channel to its complement, these couple role to role — but a single picture should eventually say why one splits while the other glues.

28.5 Status

claim	status	basis
each block's motion is $\leq$ 5-dimensional	holds	$k_{90} \leq 5$ vs null 30–53
motion loosens through training	holds	E_3 0.97 $\to$ 0.72, cos 0.995 $\to$ 0.793
EMB is a rigid invariant frame	holds	ov 0.989 from step 20
limit solidifies for every role	holds	ov $\to$ 0.98–0.999
colimit glues	holds	$\mathrm{PR}_{\mathrm{cent}} \to 1$
$\kappa \to 0$ signals solidification	refuted	rises 10×; not scale-free
Q/K and V/O behave differently	holds, 3 measurements	Obs. 28.3
blocks are mutually unaligned	holds	mean 0.13–0.20 vs null 0.0000
alignment decreases with training	refuted	rises in both spaces

The picture this phase supports is the one drawn in Fig. 2: a fluid, high-variance early regime in which each block nonetheless travels along a nearly one-dimensional path; a middle regime in which the per-role limits solidify into invariant frames and the colimits glue; and a late regime in which the frames are fixed, the paths loosen to two or five dimensions, and the attention subsystem separates into a coherent query/key pair and a wandering value/output pair. The blocks remain some 85% unaligned throughout, which is why their individually simple motions do not sum to a simple global one.

29 Phase 27: A Causal Link Between Two Residuals

Three residuals have accumulated across the programme, and §28 suggested they might reduce to one:

	quantity	measured
R_1	forward/backward out-of-image	70.1%, flat across the run
R_2	update energy outside the Fisher sheet	88–98%, second-order harmful
R_3	mean block-to-block misalignment	~85% in function space

Correlation cannot separate them, so this phase intervenes: force the blocks toward a shared direction and observe whether R_1 and R_2 follow. A first attempt failed on a coding error and is recorded below, because the failure was diagnosed by its own control rather than by inspection.

29.1 The intervention, and the error that preceded it

The blocks partition the coordinates, so a shared direction has to be defined where they meet — in function space. The rotation is then pulled back into each block's parameter coordinates by a Jacobian transpose, obtained exactly from one backward pass of $\langle \mathrm{logits}, w\rangle$, and applied with the block's norm preserved:

$$\nu_i \;=\; \frac{(1-\alpha)\,\delta u_i/\|\delta u_i\| \;+\; \alpha\, z_i/\|z_i\|}{\|\cdot\|}\,\|\delta u_i\|, \qquad z_i = P_i J^\top w.$$

The first attempt was wrong and its control said so. An earlier version used $\nu_i = (1-\alpha)\delta u_i + \alpha\,\mathrm{sign}(c)\,\delta u_i$, which interpolates a block with a signed copy of itself and never rotates toward anything. The random-direction arm returned validation loss 8.21 against a baseline of 0.030 — a 275× degradation where a norm-preserving rotation should have been benign. That is the signature of a broken manipulation rather than an informative one, and it is why the arm was included.

29.2 Result

Norm preserved per block, matched steps, $P = 3672$:

arm	val	R_3 blocks	R_1 fwd·bwd	R_2 residual
baseline	0.036	0.386	0.263	0.943
sync 0.3	0.029	0.327	0.273	0.943
sync 0.6	0.105	**0.315**	**0.013**	**0.620**
random 0.6	0.120	**0.602**	0.089	0.985

The manipulation check passes: sync and random move R_3 in *opposite* directions, to 0.315 and 0.602 from a baseline of 0.386, so the two arms are cleanly distinguishable and the random control is no longer destructive (val 0.120 against sync's 0.105).

The direction is worth stating precisely, since it inverts the naive expectation. Pulling every block toward the loss gradient's function-space image makes the blocks *less* mutually aligned, because collapsing each onto a common target strips the shared components they previously had. Rotating toward an arbitrary direction increases their overlap instead.

Observation 29.1 (The update residual is coupled to block alignment; the forward/backward residual is not)**.** *R_2 tracks R_3 in sign across both arms. Sync* 0.6*: R_3 falls* 0.071 *and R_2 falls* 0.323*, from* 94% *to* 62% *of the update energy. Random* 0.6*: R_3 rises* 0.216 *and R_2 rises* 0.042*. Two arms, opposite directions, consistent signs — which is what an intervention can establish and a correlation cannot.*

R_1 does not follow. It collapses in both *arms,* $0.263 \to 0.013$ *under sync and* $\to 0.089$ *under random, so it responds to the block structure being perturbed at all rather than to alignment specifically. That is consistent with its behaviour everywhere else in the programme:* 70.1% *in the first half of training and* 70.1% *in the second, slope* −0.0049% *per step, unmoved while every other quantity changed.*

Caveat. Sync 0.6 costs accuracy, $0.036 \to 0.105$, so R_2's fall is measured on a worse model. Whether the coupling survives at matched loss is the immediate follow-up, and it is the same control that overturned four earlier results in this programme.

29.3 What this settles about the three residuals

claim	status	basis
R_2 is coupled to R_3	holds, causal	Obs. 29.1, two arms
R_1 is coupled to R_3	refuted	falls in both arms alike
all three reduce to one mechanism	refuted	R_1 separates
coupling survives at matched loss	untested	sync costs accuracy

The outcome is cleaner than either alternative on offer. The update residual and the block misalignment are linked — the residual is, in part, the portion of the update that does not reduce to a direction the blocks share, and forcing them onto a common target removes a third of it. The forward/backward residual is something else, and its independence is now supported by an intervention as well as by its flatness: it is the one quantity in this programme that has moved for no reason anyone has found.

That is the first causal link established between any two of the programme's residuals, and it was available only because the manipulation could be checked against a control that made its own failure visible.

30 Phase 28: A Controller for Phase 3

Phase 3 is the part of the pipeline that resisted elimination: roughly two hundred steps of gradient descent that the geometric compiler could not replace because the structure was not understood. §28 showed that structure is low-dimensional — each block's motion needs at most five directions — and §27 showed the frame co-moves. This phase asks whether the two facts together permit a *controller*: track the frame rather than precompute it, and spend full-rank steps only when the frame fails.

The answer is yes at $P = 3672$ and partly at $P = 4.3 \times 10^6$, and the gap between them exposes a distinction the design had conflated.

30.1 Reconnaissance: do the roles move together?

The controller's design turns on one question. If every role's effective dimension expands on a common schedule, a single global trigger suffices; if independently, each block needs its own clock. Measured on the real compiler at $D = 128$, no intervention, five checkpoints:

role	block n	k_{90} at $20, 40, 90, 120, 160$	k_{90}/n	E(top-3)	cos lag-1
EMB	195712	$3, 2, 2, 2, 3$	1.5×10^{-5}	0.939	0.958
LN	1792	$2, 2, 3, 3, 3$	1.7×10^{-3}	0.921	0.946
FF	591360	$3, 2, 4, 4, 4$	$\mathbf{6.8 \times 10^{-6}}$	0.860	0.903
W_Q	98304	$4, 2, 3, 3, 4$	4.1×10^{-5}	0.899	0.929
W_K	98304	$4, 2, 3, 3, 4$	4.1×10^{-5}	0.895	0.926
W_V	98304	$4, 2, 4, 4, 4$	4.1×10^{-5}	0.868	0.904
W_O	98304	$4, 2, 4, 4, 4$	4.1×10^{-5}	0.868	0.903

Observation 30.1 (Low dimension survives the scale-up; roles expand coherently). *$k_{90} \leq 4$ at every role and checkpoint with $P = 4.3 \times 10^6$, and k_{90}/n reaches 6.8×10^{-6} for the feed-forward block — five orders below any threshold. The finding of §28.1 therefore survives a thousandfold increase in parameter count.*

Mean pairwise correlation of the k_{90} trajectories across roles is $+0.624$, and every role dips to 2 at step 40 before rising. A single global trigger is defensible. The query/key against value/output split appears again unprompted — W_Q, W_K at 3 where W_V, W_O sit at 4, with E(top-3) 0.90 against 0.86 — a fourth independent measurement, now on the real compiler.

The drift ratio fails as a leading indicator. It predicted k_{90} well *across conditions* in the replica (the corpus and nonlinearity interventions, where a chunk-tokenised corpus held its drift ratio at 0.996–1.000 and its k_{90} at 2 while baseline fell to 0.267 and rose to 5); within a role *over time* at scale, $\mathrm{corr}(k_{90}, \text{own drift})$ is $-0.47, -0.61, -0.63$ for FF, W_Q, W_K. Across-condition and across-time are different questions, and the indicator answers only the first. The trigger must therefore measure rather than predict.

30.2 The controller

Leakage is the direct measurement: one full gradient against the current frame,

$$\gamma^{(i)} = 1 - \frac{\|P_{Q_i} g_i\|^2}{\|g_i\|^2}, \qquad \gamma^{\text{global}} = \frac{\sum_i \|g_i\|^2 \gamma^{(i)}}{\sum_i \|g_i\|^2}.$$

The aggregation is energy-weighted, not a maximum: LN has 1792 parameters against FF's 591360, a 330× spread, and a maximum would let the smallest block dictate the schedule for the whole model.

The budget must be capped. A first attempt let the controller recalibrate freely and it spent 308 full-rank steps against a 200-step baseline — it "won" by doing more work. Every arm below is capped at the same total, so the only question is *placement*.

30.3 Result at $P = 3672$

Seven blocks at $k = 5$, so 35 dimensions or 0.95% of P; cap 80 full-rank steps of 300.

arm	full used	val@100	val@200	val@300	recalibrations
full (300 steps)	—	0.387	0.037	0.032	—
gps@0.3	76	0.575	0.154	**0.047**	56
gps@0.7	59	0.618	0.213	**0.056**	39
gps@0.9	32	0.980	0.466	0.241	12
uniform	80	1.366	0.483	0.292	—
frozen	20	2.746	2.184	1.862	—

Observation 30.2 (Placement beats spend at matched budget). *At* 59 *full-rank steps the controller reaches val* 0.056 *where uniform placement of* 80 *reaches* 0.292 *— a factor of five with* less *budget. Against unconstrained* 300*-step training it is within* 1.5× *(*0.047 *against* 0.032*) on a quarter of the full-rank work, with the remainder confined to* 35 *dimensions.*

The threshold behaves as a control should: raising θ monotonically reduces recalibrations ($56 \to 55 \to 39 \to 12$*) and monotonically degrades the result (*$0.047 \to 0.241$*), so the schedule is a consequence of θ and not of hindsight. At $\theta = 0.9$ the recalibration spacing widens through training (*$40, 50, 60$*, then* $85, 115, 150$*), so the frame's usable lifetime grows. The genuine controller regime is $\theta = 0.7$–0.9; at* 0.3–0.5 *it fires at nearly every check and is effectively "recalibrate always, capped".*

30.4 Result at $P = 4.3 \times 10^6$, and the distinction it exposes

Same controller, same cap, on the real compiler:

arm	full used	val@100	val@200	recalibrations
full (200 steps)	200	0.591	**0.088**	—
gps@0.7	80	**0.475**	0.485	8
uniform	80	1.319	0.480	—
frozen	20	2.878	2.812	—

Observation 30.3 (The mechanism transfers; the sufficiency does not). *At step* 100 *the controller reaches* 0.475 *against uniform's* 1.319 *and against unconstrained full-rank training's* 0.591 *— better than either, on* 40% *of the budget. So leakage-triggered placement works at scale.*

But it stalls: $0.475 \to 0.485$ *between steps* 100 *and* 200 *while full-rank continues to* 0.088*. The controller exhausts its* 80*-step budget by step* 60 *and everything after is pure projection, which does not descend. And the threshold does nothing —* 8 *recalibrations at every θ from* 0.5 *to* 0.9*, at identical steps — because leakage exceeds* 0.9 *at every check.*

The design error is a conflation, not a scaling failure. Two dimensions were treated as one. The *update sequence* over an eighteen-step window is four-dimensional ($k_{90} \leq 4$, Obs. 30.1); a *single fresh gradient* is not, and leakage measures the second. Those are consistent — a trajectory can be confined to a plane while each instantaneous gradient points partly out of it — and the controller sized its frame by the first while its trigger tested the second. At $P = 3672$ the two nearly coincide; at 4.3×10^6 they do not.

30.5 Status

claim	status	basis
$k_{90} \leq 4$ per block at 4.3×10^6	holds	Obs. 30.1
roles expand coherently	holds	pairwise corr $+0.624$
drift ratio predicts k_{90} over time	refuted	corr -0.61 within role
placement beats spend, $P = 3672$	holds	0.056 vs 0.292 at less budget
placement beats spend, $P = 4.3 \times 10^6$	holds early	0.475 vs 1.319 at step 100
$k = 5$ sustains training at 4.3×10^6	refuted	stalls at 0.485; leakage > 0.9
threshold gates the schedule at scale	refuted	8 recals at every θ
sequence and gradient dimension coincide	refuted	the conflation above

The controller is a working demonstration at $P = 3672$ and an early-training result at 4.3×10^6, verified in one seed at each. What it establishes about Phase 3 is narrower than replacement and sharper than description: the descent is low-dimensional at any instant, the dimensions rotate faster than a fixed frame can follow, and a budget spent where the frame fails is worth several times the same budget spent uniformly. What it does not establish is a frame size that suffices at scale — for which the quantity to measure is $k^*(P)$, the smallest per-block rank holding leakage below threshold, and the two dimension notions must be tracked separately.

31 Phase 29: The Attention Interface, Exactly

The programme's persistent finding about attention — that W_Q and W_K pair, and W_V and W_O pair, in four independent measurements — has been descriptive. This phase makes it algebraic. The change in the attention logits admits an exact expansion whose terms can be computed rather than fitted, which turns "how does $QK^\top$ drive attention" from a regression into an identity.

The route there was a failure worth recording. Regressing ΔS on ΔW_Q and ΔW_K gives $R^2 = -0.26$ and -0.58 — both arms fail, and the combined model is worse than Q alone. Suspecting softmax saturation, I repeated it on the pre-softmax logits: -0.29, -0.51, essentially unchanged. The diagnosis was wrong. The relation is *bilinear* — the current weights multiply the increments and change every step — so a linear model in the increments is misspecified by construction, and no amount of target-cleaning repairs it.

31.1 The four-term identity

Writing H_ℓ for the block's input and S_ℓ for its pre-softmax logits,

$$\Delta S_\ell = \tfrac{1}{\sqrt{d}}\Big[\underbrace{(H\Delta W_Q^\top)(HW_K^\top)^\top + (HW_Q^\top)(H\Delta W_K^\top)^\top}_{\text{parameter legs}} + \underbrace{(\Delta H W_Q^\top)(HW_K^\top)^\top + (HW_Q^\top)(\Delta H W_K^\top)^\top}_{\text{input legs}}\Big] + O(2).$$

A first attempt used only the parameter legs and a $\Delta W_Q \Delta W_K$ waist term. The identity error was 0.16–0.33 at layer 0 and 0.69–0.88 at layer 1 — and the layer pattern was the diagnosis: layer 1's *input* changes when layer 0's weights update, and that term had been omitted entirely.

step	layer	legQ	legK	legHQ	legHK	parameter	input	$O(2)$
40	0	0.557	0.451	0.208	0.123	**0.922**	0.244	0.017
40	1	0.281	0.201	0.580	0.656	0.452	**0.834**	0.086
90	0	0.471	0.518	0.107	0.119	**0.933**	0.161	0.015
90	1	0.247	0.245	0.589	0.419	0.464	**0.735**	0.036
150	0	0.551	0.491	0.144	0.132	**0.912**	0.191	0.011
150	1	0.350	0.360	0.500	0.504	0.653	**0.690**	0.030
240	0	0.491	0.479	0.228	0.235	**0.877**	0.326	0.003
240	1	0.202	0.301	0.485	0.679	0.430	**0.834**	0.014

Observation 31.1 (Attention change at depth is driven from below). *The four-term linearisation accounts for* 91–99.7% *of* ΔS*: the second-order remainder is* 0.003–0.086*, against* 0.16–0.88 *for the two-term version. The expansion is therefore essentially the whole story.*

The layer split is stark. At layer 0 *the parameter legs carry* 0.88–0.93 *of* $\|\Delta S\|$ *and the input legs* 0.16–0.33*; at layer* 1 *this reverses to* 0.43–0.65 *against* 0.69–0.83*.* Attention re-alignment at depth is mostly the passive reflection of representation drift arriving from below*, not local* Q/K *motion. Layer* 0*'s own input legs, at* 0.16–0.33*, match the residual of the two-term identity almost exactly and are the embedding's contribution.*

Fractions exceeding one indicate partial cancellation: the parameter and input legs are individually large and oppose each other at layer 1*.*

31.2 Q and K share one Fisher channel

The pants reading of the expansion — one incoming change splitting into two legs — would require the legs to occupy different parts of the descent channel. Projecting each onto the top-3 Fisher sheet:

step	SPLIT_F	$\mathrm{cap}_F(\Delta W_Q)$	$\mathrm{cap}_F(\Delta W_K)$	cos of sheet coordinates	ΔL_Q / ΔL_K
40	0.613	0.0155	0.0073	**+0.982**	+0.0200 / +0.0128
90	0.316	0.0122	0.0134	**+0.974**	+0.0029 / +0.0029
150	0.665	0.0023	0.0029	+0.701	+0.0022 / +0.0014
240	0.676	0.0000	0.0000	**+0.994**	+0.00005 / +0.00006

Observation 31.2 (The coproduct is formal, not functional). *The two legs' Fisher-sheet coordinates have cosine* +0.982, +0.974, +0.701, +0.994 *— above* 0.97 *at three of four checkpoints, with* SPLIT_F *at* 0.61–0.68 *there, so the readings are well resolved. The one low value coincides with the lowest SPLIT. Their loss contributions match likewise.*

Q *and* K *therefore occupy the* same *descent channel:* $QK^\top$ *is one locator unit, not two coupled ones. This is what the face overlap of* 0.509 *— the highest of any pair in §28.4 — predicted. The expansion has the shape of a coproduct and the legs never separate on the statistical manifold.*

The direct legs are additive. The $\Delta W_Q \Delta W_K$ waist term contributes 0.002–0.011 of $\|\Delta S\|$, so under 1%. Attention logits decompose cleanly in Q and K. Whether that explains the 19% non-additivity of the loss (Obs. 26.5) is *not* established here: that figure is a seven-block decomposition of ΔL, a different object from a two-way split of ΔS, and the clean test restricts the loss decomposition to $\{W_Q, W_K\}$ alone.

31.3 A proposed fix that was a no-op

Obs. 31.1 has an apparent consequence for the controller of §30: it recalibrates each block's frame from that block's own recent updates, but layer 1's attention change is 69–83% driven from upstream, so a locally estimated frame is blind to the largest term in its own dynamics. Refreshing in depth order — layer 0 first, then layer 1 against the updated input — looked like the fix.

arm	full used	val@100	val@200
full (200 steps)	200	0.580	0.084
gps@0.7	80	**0.473**	0.476
gps@0.7 depth-ordered	80	0.495	0.499
gps@0.9 depth-ordered	80	0.485	0.480

Observation 31.3 (Reordering the estimation changes nothing)**.** *Depth-ordered recalibration is marginally* worse *(*0.495 *against* 0.473*), within run-to-run variation. The reason is that the refresh happens at a single θ: every frame is estimated from the same window of updates regardless of the order in which the decompositions are computed. Reordering the SVD calls does not change their inputs. The intervention was a no-op presented as a fix.*

A genuine version would stage the recalibration window — full-rank steps for the lower block, re-estimate, then further steps observed under the refreshed input — which is a different controller rather than a reordering.

31.4 Status

claim	status	basis
ΔS linear in $\Delta W_Q, \Delta W_K$	refuted	$R^2 < 0$ on both targets
softmax saturation is the cause	refuted	pre-softmax unchanged
four-term identity closes	holds	$O(2)$ residual 0.003–0.086
layer 0 locally driven	holds	parameter legs 0.88–0.93
layer 1 driven from below	holds	input legs 0.69–0.83
Q, K share one Fisher channel	holds	cos > 0.97, SPLIT 0.61–0.68
direct legs additive	holds	waist $< 1\%$
waist explains the 19% loss non-additivity	untested	different decomposition
depth-ordered recalibration helps	refuted	Obs. 31.3

Two things generalise. The first is that a bilinear relation cannot be recovered by regressing on its increments, however the target is cleaned — the failure looked like a nonlinearity problem and was a specification problem, and the fix was to compute the terms rather than fit them. The second is the shape of what survives: an identity verified to under nine percent, a Fisher measurement with its resolution stated, and a proposed mechanism that failed its own test. That ratio — measurements surviving, constructions failing — has held across twenty-nine phases, and it is the programme's most reliable regularity.

32 Phase 30: Spinning Subspaces and a Landscape Preference

§30 left the controller with a measured advantage and an unexplained quantity: the frame's rotation rate, which sets the compression bound, the recalibration schedule and the τ wall, and which nothing measured accounted for. This phase pursues it, and the pursuit produces two robust findings and five refutations — four of hypotheses I proposed and one of a mechanism I inferred from a correlation and then reversed by intervention.

32.1 The rotation is real

The first question is whether "rotation" measures motion at all. A 3-dimensional frame estimated from a handful of stochastic gradients might simply not be reproducible, in which case the Grassmann velocity between consecutive windows would be estimation variance. The discriminator is a split-half on *interleaved* updates from the same window: both halves see the same underlying motion, so any disagreement is estimator noise.

H	β	split-half	shuffled control	disjoint-window velocity
1	0	0.472	0.844	1.754
4	0.750	**0.283**	0.519	1.720
16	0.938	0.291	0.604	1.626
64	0.984	0.338	0.647	1.556

Observation 32.1 (The frame resolves; the motion is genuine)**.** *Split-half is* 0.28–0.47 *radians while the disjoint-window velocity is* 1.56–1.75 *— a factor of five to six apart, against a maximum possible* $\sqrt{3}\,\pi/2 = 2.72$. *Frames built from interleaved steps agree; frames from consecutive windows do not. The rotation is motion, not variance.*

The optimal filter is $H^* = 4$*: split-half falls from* 0.472 *to* 0.283 *with four steps of smoothing and then stops improving, degrading slightly past* $H = 16$ *as the average begins to span genuine bending. Randomising the time order within a window gives* 0.52–0.65*, roughly double the split-half, so the even/odd agreement comes from temporal structure and not from smoothing making everything look alike.*

The bend rate is barely affected by filtering — $1.754 \to 1.556$ *across a* 64× *range in horizon. The highway itself turns at some* 1.6 *radians per sixteen steps.*

32.2 The step size is not the clock

The natural candidate for what sets that rate is the step the optimiser takes. Tested with disjoint windows, at matched loss rather than matched step, across a tenfold learning-rate range:

matched loss	step spread across arms	$\lvert\dot{Q}\rvert$ spread
1.20	185.2%	**21.3%**
0.50	178.6%	**19.4%**
0.15	155.9%	**12.8%**

Observation 32.2 (Rotation is nearly invariant to the realised step)**.** *At matched loss,* $\|\Delta\theta\|$ *runs* $0.042 \to 0.119 \to 0.370$ *across the three learning rates — an order of magnitude — while* $\lvert\dot{Q}\rvert$ *runs* $1.56 \to 1.76 \to 1.94$. *A ninefold change in step gives a* 1.24× *change in rate. If the step were the clock the two spreads would match; they differ by an order of magnitude.*

$\lvert\dot{Q}\rvert$ *sits at* 1.47–1.94 *radians in every arm at every loss level, roughly* 60% *of the maximum possible for a 3-frame, essentially always.*

32.3 Coordinate preference is a property of the landscape

If the rotation is basis-free, no coordinate should recur in the frame's support beyond chance. Measured on the $H^* = 4$ stream, top-40 leverage coordinates per block, nineteen checkpoints:

block	n	expected	observed	ratio	Jaccard lag-1 (vs. random)
EMB	672	1.13	5.10	4.51	0.086 (0.031)
ATTN0	576	1.32	5.78	4.38	0.134 (0.036)
FF0	864	0.88	5.85	6.65	0.162 (0.024)
ATTN1	576	1.32	6.83	5.17	0.178 (0.036)
FF1	864	0.88	6.15	6.99	0.201 (0.024)

Recurrence is 4.4–7.0× the fresh-draw expectation and consecutive Jaccard overlap is three to eight times chance. The frame turns, but it returns to a preferred set of coordinates.

A mechanism inferred and then reversed. The Fisher diagonal correlates *negatively* with recurrence in four of five blocks (-0.09 to -0.21), so Fisher leverage is not the selector. Adam's preconditioner looked like the answer instead: $1/\sqrt{v}$ correlates at $+0.37$ to $+0.57$ in every non-embedding block, v itself negative throughout, and activation occupancy negative — coordinates that rarely receive gradient accumulate small second moments and are amplified. The prediction that follows is that recurrence should vanish under plain SGD.

arm	mean recurrence ratio	mean Jaccard excess
AdamW	4.89	4.69
SGD + momentum	8.78	11.79
plain SGD	**11.06**	**16.46**

Observation 32.3 (Adam dilutes a landscape property rather than creating it). *Removing the preconditioner* doubles *the recurrence ratio and* triples *the Jaccard excess. FF0 goes from* 5.27 *to* 13.33*; the embedding's consecutive overlap goes from* 0.125 *to* 0.533*, so under plain SGD successive frames share over half their support. Both arms reach comparable loss at the same step count (*0.029 *against* 0.030 *at step* 255*), so this is not a training-stage artefact.*

The causal arrow was backwards. Coordinates with small v are ones the landscape rarely excites; they recur under both optimisers, and Adam's step-wise normalisation breaks up *the consecutive locking that the raw gradient follows. The* +0.37 *to* +0.57 *correlation was real and its mechanism was not.*

Two consequences. The preference is substantial — under plain SGD a 40*-coordinate support out of* 576–864 *recurs at* 13–14× *chance with* 47–52% *consecutive overlap, close to a genuinely sparse persistent active set. And every other measurement in this phase was made under Adam, the arm in which the structure is now known to be weakest.*

32.4 Dispersion, not acceleration

Two independent quantities track the same trend within each block:

block	leakage, first	leakage, last	factor
EMB	0.0047	0.0078	1.7
ATTN0	0.0072	0.0626	8.7
FF0	0.0054	0.0896	16.6
ATTN1	0.0055	0.0711	12.9
FF1	0.0062	0.0917	14.8

Observation 32.4 (The wall is dispersion, not speed). *Leakage rises* 9–17× *within every block except the embedding, which stays flat at* 0.005 → 0.008*. The sign-flip rate of the update field rises in step,* 0.024 *early to* 0.135 *late. Meanwhile the rotation velocity is roughly constant at* 1.6–1.75 *radians throughout.*

So the trajectory does not become harder to track because the frame turns faster. It becomes harder because more of the update falls outside any 3*-dimensional frame as training proceeds. That is a cleaner account of the τ wall than rotation was, and it explains why a threshold trigger outperforms a fixed schedule: the trigger measures leakage directly, so it widens early and tightens late without being told to.*

32.5 Four constructions that returned nothing

No syzygies. The layer rotation generators stacked into Ψ give participation ratio 4.88 of 5 against a random null of 4.96; singular values $1.00, 0.81, 0.77, 0.69, 0.64$. The kernel is empty. Layers rotate at correlated *rates* — cross-block correlation 0.67–0.79 at zero lag — but their generators satisfy no linear relations. Rate synchrony is not phase locking.

No holonomy. The path-ordered product $\prod_t Q_t^\top Q_{t+1}$ gives $\|M - I\|$ of 1.61–2.67 against a maximum of 3.46, and a shuffled-order control gives 1.30–2.59 — indistinguishable, with two blocks on the wrong side. Any product of fifty rotations fails to close; this one fails no more informatively than chance.

No propagation of rotation. Cross-layer correlation peaks at $\tau = 0$ for every pair (0.788, decaying to 0.299 by $\tau = 3$). The bilinear result of §31.1 showed layer 1's *logit* change is 69–83% driven from below; its *frame rotation* is not inherited at all.

No gain from variance normalisation. Normalising the gradient stream by per-coordinate standard deviation — the natural way to recover what Adam dilutes — gives split-half 1.964 against the raw stream's 0.898, rotation 2.032 against 1.695, and leakage 0.496 against 0.103: a 382% degradation. Dividing by a variance estimated from four micro-batches amplifies precisely the coordinates where that estimate is worst. Light EMA filtering shows the real trade-off: it halves the split-half distance to 0.500, the best resolution of the three, while *raising* leakage 40%, because the frame no longer represents the unfiltered component of the next update.

32.6 Status

claim	status	basis
frame rotation is real, not estimator noise	holds	split-half 0.28 vs 1.7
optimal observer filter $H^* = 4$	holds	saturates past $H = 4$
coordinate preference exists	holds	4.4–7.0× chance
preference is a landscape property	holds	stronger under plain SGD
Adam's $1/\sqrt{v}$ creates the preference	refuted	Obs. 32.3
Fisher leverage selects coordinates	refuted	corr −0.09 to −0.21
activation density selects coordinates	refuted	corr −0.27 to −0.53
realised step size is the clock	refuted	9× step, 1.24× rate
leakage rises through training	holds	9–17× within blocks
layer rotations satisfy syzygies	refuted	PR 4.88 vs null 4.96
frame transport has holonomy	refuted	ordered ≈ shuffled
variance normalisation improves the frame	refuted	leakage +382%

The picture that survives is spare. Low-dimensional frames spin at a rate no measured quantity explains, over a coordinate set the loss surface prefers and the optimiser dilutes, becoming progressively harder to confine as training proceeds. What sets the rate remains open — it is not the corpus, not the architecture's nonlinearities, not layer propagation, not the step size, and it moves against dimension rather than with it. That the question is now sharply posed, with five candidate answers eliminated by measurement, is the phase's contribution.

33 Phase 31: Subspace Optimization in the Pipeline

§30 established that *placement* of a full-rank budget beats uniform placement by 5.2×. This phase asks the harder question: can the step rule itself be replaced by a subspace method, inside the compiler's actual Phase 3, on the corpus the pipeline uses. Fourteen variants were built and measured against a tuned AdamW baseline. The answer is no, and the phase records why, together with the one configuration that comes close.

33.1 Where this sits relative to published work

Three families of low-rank optimizer are relevant, and the results below place this work against each.

Gradient low-rank projection. GaLore projects gradient matrices into a low-rank space by periodic SVD, runs Adam inside, and projects back; Apollo approximates the Hessian inverse by a diagonal-plus-low-rank form. Both demonstrate that P-dimensional optimiser state is not required. The measurements here are consistent with the premise — update sequences are low rank, $k_{90} \leq 4$ per block at $P = 4.3 \times 10^6$ — and diverge on the consequence: at 69% frame capture the projection costs 63% more steps to reach Phase 3's exit gate, so low-rankness of the update does not by itself make projection free.

Krylov and HVP-driven solvers. Subspace Newton methods build $\mathcal{K}_K(H, g)$ by Lanczos and solve $(Q^\top HQ)^{-1}Q^\top g$ in K dimensions, bypassing first- and second-moment vectors entirely. That method was implemented here and it *lost*: Obs. 33.1. Any positioning of this work as advocating HVP-built subspaces over gradient-built ones would contradict its own measurement.

Kronecker-factored preconditioning. K-FAC and Shampoo factor the curvature into layer-wise Kronecker products, giving $O(d_{\text{in}}^2 + d_{\text{out}}^2)$ state instead of $O(PK)$. This is the one family whose memory argument survives inspection: an explicit basis $Q \in \mathbb{R}^{P\times K}$ costs PK floats against Adam's $2P$, so a dense subspace method uses *more* memory than Adam whenever $K > 2$. At $P = 545{,}290$ and $K = 32$ that is 17.4M floats against 1.1M — a 16× penalty, not a saving.

What is distinctive here is not the optimiser but the controller. Phase transitions are triggered by differential invariants — the clean alignment fraction Φ_{cl} and the gluing defect τ — rather than by step counts or loss thresholds, and §30 showed leakage-triggered placement beating uniform placement at matched budget. That result does not require replacing the step rule.

33.2 The step rule cannot be replaced

Observation 33.1 (Krylov-Newton loses to tuned AdamW at three seeds). *A global rank-8 Lanczos-Newton optimiser, after a* 32× *tuning recovery — Newton inversion over plain projection* 3.6×, *an EMA-seeded Lanczos* 1.85×, *and damping raised from* 10^{-2} *to* 10^{-1}, *which alone gave* 5.4× *because* 98% *of steps had been hitting the trust radius — reaches* $0.0639{\pm}0.0331$ *against tuned AdamW's* 0.0310 ± 0.0011 *at* $P = 3672$. *That is* 2.06× *worse with* 30× *the run-to-run variance.*

A single-seed configuration reached 0.0326 *and appeared to be at parity. It was seed* 2 *of three; seeds* 0 *and* 1 *give* 0.051 *and* 0.109 *at identical settings. Tuning Adam over the same grid moved it only* $0.0322 \to 0.0310$, *so the asymmetry cost the hybrid nothing — the seed variance was the whole effect.*

Block-local Lanczos was worse still, 0.387 ± 0.123, *and projecting negative Ritz directions out rather than clamping them — the diagnosis suggested by the variance — made it worse again (*0.670 ± 0.581*). The negative curvature directions carry descent; removing them removes signal.*

33.3 Projection inside the real Phase 3

The measurements above are at $P = 3672$ on a synthetic grammar. This runs in the pipeline: the compiler's model at $D = 128$ ($P \approx 4.3 \times 10^6$), the 1,271-token corpus at 300 loops, phases 1 and 2 unchanged so the starting point is the real post-pump θ, and Phase 3's own exit gate at val < 0.15.

gate	AdamW	cs-honest ($\alpha = 0$)	$\alpha = 0.25$	cs cost	$\alpha = 0.25$ cost
3.00	28	44	36	1.57×	1.29×
2.00	44	68	60	1.55×	1.36×
1.00	76	116	100	1.53×	1.32×
0.60	100	156	140	1.56×	1.40×
0.30	132	204	188	1.55×	1.42×
0.15	**164**	**268**	**220**	**1.63×**	**1.34×**

Observation 33.2 (The cost is flat and it is never free). *Pure projection costs* 1.53–1.63× *at every gate from* 3.0 *down to* 0.15. *It does not grow near the basin and it is not cheap early, so there is no loss regime within Phase 3 where the projection is free — and therefore no handoff point that rescues it. Keeping a quarter of the orthogonal complement recovers about a third of the gap, to* 1.34× *at the exit gate, which is the third independent measurement of that* α *and the first inside the pipeline.*

The frame captures 0.69 *of update energy here. That reconciles an apparent contradiction with a* 545,290*-parameter convolutional benchmark, where a rank-*32 *projection tracked Adam for* 3000 *steps: there the projected step retained* 0.82–0.95 *of Adam's norm, so roughly* 90% *of the energy, and at that capture the cost is negligible. The difference between "tracks Adam" and "costs* 63%*" is the capture fraction, not the parameter count.*

Two claims that the same benchmark does not support. Final held-out loss was 0.9303 for Adam against 0.9258 for the projected arm — 0.5% apart, on values that move by 0.05 between adjacent checkpoints, with all three arms overfitting identically after step 1900. There is no measured generalisation benefit and no evidence of noise filtering; the projected arm simply tracks.

33.4 Dynamic rank

Three criteria for adapting K were available and two were tested.

Energy thresholding — expand when the subspace holds less than γ of the update energy, contract when it holds more — was implemented with $\gamma = 0.90$. It fired 24 expansions and 18 contractions and produced 0.940 against a fixed-$k = 8$ control's 0.944: no benefit. The diagnostic is that coverage sat at 0.90–0.91 throughout, exactly on target, while the loss was 29× Adam's. *The energy criterion was satisfied and the optimiser still failed*, which is the sharpest available evidence that energy is the wrong selector — and it agrees with Obs. 23.2 from the other direction, where three Fisher directions holding 2% of the energy carried all of the descent.

Innovation-rate thresholding is the leakage trigger of §30.2, and it works — but as a *scheduling* signal for when to spend full-rank steps, not as a rank selector.

Ritz-gap truncation — setting K at $\arg\max_i \theta_i/\theta_{i+1}$ — was not tested. It is complicated by the measurement that 12–16% of Ritz values are negative at these ranks, so the ordering the criterion assumes does not hold cleanly.

33.5 Status

claim	status	basis
update sequences are low rank	holds	$k_{90} \leq 4$ at 4.3×10^6
Krylov-Newton replaces Adam	refuted	2.06×, 30× variance
block-local improves it	refuted	13× worse
projecting negative Ritz helps	refuted	worse, variance 4.7×
projection is free in Phase 3	refuted	1.53–1.63× at every gate
a handoff point rescues it	refuted	cost flat across gates
$\alpha = 0.25$ recovers a third of the gap	holds, 3 measurements	1.34× at exit
subspace projection aids generalisation	not supported	0.5%, within noise
dense Q saves memory	refuted	PK vs $2P$; 16× penalty
energy thresholding selects rank	refuted	target met, loss 29×
leakage triggers placement	holds	§30, 5.2×

The position this work occupies is narrower than a new optimiser and more specific than a negative result. The geometry is informative about *when* to spend computation — leakage-triggered placement of a fixed full-rank budget beats uniform placement by 5.2× at matched cost — and, across fourteen tested variants, is not yet informative about *how* to compute the step. The distinction matters because the two are usually pursued together, and only one of them survived measurement here.

34 Phase 32: The Rotation is Real

§32 left the frame rotation measured but unexplained: the separation between consecutive frames sits at 1.47–1.94 radians regardless of learning rate, corpus, architecture, layer, or training stage, and six candidate drivers were eliminated by measurement. One question was never asked, and everything downstream rested on it — whether the separation is motion at all, or a finite-sample estimate of a slowly moving object.

34.1 Rotation or resampling

The discriminator is window length. If the plane genuinely turns, longer windows average over the same motion and the separation is invariant. If each frame is a finite-sample estimate, the sampling error falls as $W^{-1/2}$ and the separation shrinks fourfold from $W = 4$ to $W = 64$.

Two controls the design requires. The windows must be *disjoint*, or the overlap artefact of Obs. 25.1 returns. And the gap between window *centres* must be held fixed across W, or a longer window automatically means more elapsed time between centres and the measurement conflates resolution with drift.

W	d_{Gr}	split-half	excess	resampling predicts
4	2.475	1.045	2.4×	2.475
8	2.490	1.111	2.2×	1.750
16	2.506	0.576	4.4×	1.237
32	2.542	0.356	7.1×	0.875
64	**2.550**	**0.284**	**9.0**×	0.619

Observation 34.1 (The frames genuinely rotate). *A sixteenfold increase in window length moves the separation by* 3%*:* 2.475 → 2.550 *against a maximum possible* $\sqrt{3}\,\pi/2 = 2.72$. *Resampling would predict* 0.619*; the measured ratio to that prediction is* 4.12, *which is the pure-rotation value.*

The split-half column is the control, and it is the strongest part of the result. It does *fall as* $W^{-1/2}$ — 1.045 → 0.284, *a factor of* 3.7 *against the predicted* 4.0 — *so the estimator's own variance behaves exactly as sampling theory requires while the frame separation does not move at all. The same measurement on the same data, one component obeying* $W^{-1/2}$ *and the other flat. The instrument works and the signal is real.*

The excess over the noise floor grows from 2.4× *to* 9.0× *as that floor drops away.*

Everything resting on the rotation therefore stands: the frozen-frame failure at 3.4% retention, the two-step rebuild requirement, and the $\tau < 10$ wall. The dimension flip is confirmed as a phenomenon and remains unexplained as a mechanism.

34.2 The dimension map

Recorded silently across a full Phase 3, per role, with capture measured at known frame age so the rebuild sawtooth does not contaminate it:

role	block n	k_{90}	n/k_{90}	k_{90}/n	E(top-3)	cap@1	ortho
EMB	195,712	**1.52**	128,436	7.8×10^{-6}	0.981	0.923	0.077
LN	1,792	1.76	1,017	9.8×10^{-4}	0.985	0.930	0.070
W_Q	98,304	2.00	49,152	2.0×10^{-5}	0.972	0.888	0.112
W_K	98,304	2.00	49,152	2.0×10^{-5}	0.970	0.883	0.117
W_V	98,304	**2.19**	44,878	2.2×10^{-5}	0.960	0.839	**0.161**
W_O	98,304	2.14	45,875	2.2×10^{-5}	0.963	0.846	0.154
FF	591,360	2.05	288,804	$\mathbf{3.5 \times 10^{-6}}$	0.965	0.854	0.146

Observation 34.2 (Dimension is a property of the role, not of the block size). $k_{90} \propto n^{0.016}$ *with* $R^2 = 0.053$ — *flat. LN at* 1,792 *parameters and FF at* 591,360 *both need about two directions, across a* 330× *range in block size.*

The strong relation is between dimension and leakage: $\mathrm{corr}(k_{90}, \mathit{ortho}) = +0.906$. *Blocks needing more directions also leak more, which are two measurements of the same thing — how far the motion is from one-dimensional — agreeing across that same* 330× *range.*

The role ordering is stable. EMB and LN are tightest at 1.5–1.8 *with* 7% *leakage; both are representational, setting the frame the rest works in.* W_Q *and* W_K *sit at exactly* 2.00 *to two* *decimals, behaving as the single bilinear object they form.* W_V *and* W_O *are loosest at* 2.14–2.19 *with the highest leakage of any role.*

That is the fifth independent measurement of the query/key against value/output split, and the cleanest: the same difference appears in dimension, in capture and in leakage simultaneously. The interpretation that routing is structurally simpler than transport is a reading of a consistent 0.15*-dimension margin, not of a large effect measured once.*

Capture depends on frame age far more than on rank. Measured 0.873 at age 1, 0.545 at age 4, 0.335 at age 7; and across the run at fixed age 1, 0.908 → 0.928 → 0.893 → 0.807 → 0.841 over five loss bands. So the rebuild interval is the parameter that matters, and an earlier conclusion that capture plateaus at 0.70 regardless of rank was an average over a rebuild cycle rather than a ceiling.

34.3 The arc inside the pipeline

Observation 34.3 (A different exponent in the pipeline)**.** *Phase 3 begins in negative curvature — one probe at* $\varphi \approx 128°$*, so* $\lambda < 0$ *— and leaves it within* 16 *steps. Phase 2's pump hands off a model still on a ridge and Phase 3's first act is getting off it. Thereafter* φ *is* flat *at* 44–58° *for the rest of the run, not monotone: the ratio of rotation to scaling is constant while the magnitude falls two orders of magnitude.*

The arc exponent is 1.041 *at* $R^2 = 0.981$ *over nine points, against the* 0.758 *measured at* $D = 128$ *on an isolated Phase 3 (Obs. 17.1). An exponent of one means* $\omega \propto \lambda$ *exactly and hence* κ *constant, which is what the flat* φ *shows. In the full pipeline at* $D = 256$ *the curvature shrinks* self-similarly*: rotation and scaling fall together and the shape is preserved.*

The difference is not resolved. This run spans λ *over roughly* 30× *where the earlier measurement spanned* 222×*, so capacity, pipeline and fitting range could each account for it.*

34.4 Status

claim	status	basis
frame separation is a sampling artefact	refuted	flat in W; ratio 4.12
the frames genuinely rotate	holds	Obs. 34.1
split-half floor falls as $W^{-1/2}$	holds	3.7× vs predicted 4.0
k_{90} scales with block size	refuted	$n^{0.016}$, $R^2 = 0.053$
k_{90} tracks leakage	holds	corr +0.906
Q/K tighter than V/O	holds, 5 measurements	Obs. 34.2
capture ceiling is 0.70	withdrawn	average over a rebuild cycle
arc exponent is 0.758 in the pipeline	refuted	1.041 at $R^2 = 0.981$
what drives the rotation rate	open	seven candidates eliminated

The phase converts one open question into a sharper one. The rotation is not an instrument artefact — the strongest evidence being that the estimator's own noise obeys $W^{-1/2}$ in the same measurement where the signal is flat — so the low-dimensional frames really do turn at close to the maximum possible rate for their dimension, and no measured quantity accounts for it.

35 Phase 33: Dimension Without Location

§23.3 established that three Fisher directions out of 4.3×10^6 deliver more loss reduction than the full update. This phase asks what can be built on that, and the answer sharpens into a single distinction: the *dimension* of the descent-carrying subspace is stable and its *location* is not. Everything that needs only the first works; everything that needs the second fails.

35.1 The mechanism holds across the whole of Phase 3

The earlier ablations were taken at two late checkpoints. Repeated at six, spanning two orders of magnitude in loss on the real pipeline:

step	val	e_F	ΔL Fisher	ΔL residual	Fisher share of descent
20	3.662	0.132	−0.2442	**−0.0486**	0.56
40	2.045	0.151	−0.1743	**−0.0027**	0.78
60	1.160	0.060	−0.0293	+0.0051	0.24
80	0.684	0.094	−0.0609	+0.0291	0.78
120	0.265	0.099	−0.0366	+0.0167	0.91
240	0.061	0.049	−0.0084	+0.0023	1.22

Observation 35.1 (The Fisher sheet carries the descent at every stage)**.** *With* 5–15% *of the update energy the sheet delivers* 0.75 ± 0.30 *of the total loss reduction, at every checkpoint from* val $= 3.66$ *down to* 0.061*. The sign and order never vary.*

The residual, however, changes sign near step 50*: it helps at steps* 20 *and* 40 *(*−0.049*,* −0.003*) and hurts thereafter. So the complement is not a uniform obstruction — during the saddle exit and backbone formation it descends alongside the sheet, and only once the backbone has formed do the two oppose.*

This is what §33's Fisher-weighted complement exploits, and why it was the one mechanism of seven to help: final val 0.0766 → 0.0540 from retaining 3% of what was being discarded.

35.2 There is no coarser manifold

If the sheets from different times lay inside one larger fixed subspace, that span would be a static object containing the moving sheet — estimable once and usable without tracking. Six rank-3 sheets give 18 vectors:

	measured	null (independent 3-planes)
union effective rank at 90% energy	**16**	17
mean pairwise Grassmann distance	**2.581**	2.718
consecutive-checkpoint distance	2.514	—
maximum possible $\sqrt{3}\,\pi/2$	2.721	

Observation 35.2 (The sheets are as independent as random planes)**.** *The union spans* 16 *of its* 18 *dimensions against a null of* 17*, and the singular spectrum is nearly flat (*$1.00, 0.94, 0.90, \ldots, 0.66, 0.60$*). Sheets from different times sit* 95% *as far apart as randomly drawn* 3*-planes, and consecutive checkpoints are barely closer than distant ones.*

Nor is the sheet secretly rank one: the leading direction's share of its own sheet is 0.289 ± 0.218*, swinging from* 0.035 *to* 0.695*, and which direction dominates changes between checkpoints.*

So the Fisher sheet rotates as fast as the update frame of Obs. 34.1. There is no persistent structure to reuse, which is why the refresh interval mattered less than expected — it was chasing an object already gone.

35.3 Two equations, both refuted

First: a two-term loss ODE. The ablation gives a natural form, $\dot{L} = -\alpha e_F \|u\|^2 + \beta(1 - e_F)\|u\|^2$, with the Fisher term descending and the complement ascending. Fitted at three late checkpoints it gave $\alpha/\beta = 52.9 \pm 7.6$ and a fixed point $e_F^* = 0.0186$ that the measured e_F appeared to approach ($0.090 \to 0.044 \to 0.022$).

All six checkpoints refute it. α/β runs −33, −362, +90, +20, +20, +70 — it changes sign, and even restricted to the $\beta > 0$ regime it is 50 with a 141% spread. And e_F is non-monotone

(0.132, 0.151, 0.060, 0.094, 0.099, 0.049), peaking at the backbone window like every other quantity in §34.2, so the apparent attractor was three points of a curve that turns.

Implemented as a forward predictor — fit on steps 0–80, integrate after with measured e_F and $\|u\|^2$ — it produces $\alpha = -0.2229$, $\beta = -0.1137$, both negative where the ablation measures $\alpha > 0$ throughout, and predicts val $= -0.600$ against an actual 0.301: a negative loss, -299% error.

Second: an exponential in the loss alone. $L = L^* + Ae^{-kt}$ fits far better — $R^2 = 0.974$ with the floor, $\log R^2 = 0.986$ without, against a power law at 0.185 — and $\dot{L} = -k(L - L^*)$ would be the Polyak–Łojasiewicz condition holding with equality at $k = 0.043$/step. But PL requires $\|\nabla L\|^2/L$ constant and it rises 2.6× across the run (0.0353 → 0.0631 → 0.0925), so the good fit is a global description rather than a consequence of the condition. The fitted floor $L^* = 0.120$ is also spurious: the corpus is deterministic at order five ($H = 0$ exactly, 1,364 contexts each with a unique successor), so the true floor is zero and L^* was absorbing a rate change.

Observation 35.3 (Neither equation survives, and the reason is shared)**.** *Both failures trace to the same fact. The two-term ODE needs α and β, which are properties of* where *the sheet is; the exponential needs k, which is fitted from the run it describes. Neither coefficient is derivable from the corpus or the architecture, so neither equation is self-sustaining: they must be fitted from the trajectory they are meant to predict.*

What is transferable is the dimensional statement — three directions suffice, at every stage — and that requires no coefficient at all.

35.4 What the floors actually are

Two floors were being conflated. The corpus floor is the conditional entropy of the data; the architecture floor is what a given model can express. The corpus here is a 1,271-token sentence repeated 300 times, and its entropy hierarchy collapses completely:

order	H (nats)	contexts	gain
unigram	6.771	1	—
bigram	0.4286	1,014	6.342
trigram	0.0174	1,347	0.411
5-gram	**0.00000**	1,364	0.017

$L_{\text{corpus}} = 0$ exactly. So the compiler's val_floor = 0.062 is purely architectural, and there is no barrier there — consistent with pipeline runs finishing at 0.0509.

A capacity signature in the floor probe. L_{floor} falls smoothly with width (5.08, 4.48, 3.42, 2.49, 1.61, 0.73 for $D = 48 \dots 256$) with no knee, so the 200-step probe is step-limited rather than capacity-limited. But $\|g_{\text{floor}}\|$ is *non-monotone* — 0.336, 0.374, **0.390**, 0.339, 0.267, 0.231 — peaking at $D = 96$, the last width before the capacity boundary of Obs. 18.2. A turning point at the same place the curvature crossed negative and the loss knee appeared, found by a 200-step probe that never runs the pipeline.

35.5 Status

claim	status	basis
Fisher sheet carries the descent throughout	holds	0.75 ± 0.30, six checkpoints
the complement always opposes	refuted	sign change near step 50
e_F is driven to a fixed point	refuted	non-monotone, peaks at backbone
α/β is a constant	refuted	-362 to $+90$, changes sign
the two-term ODE predicts forward	refuted	-299%, negative loss
PL holds with equality	refuted	$\|\nabla L\|^2/L$ rises $2.6\times$
$L = L^* + Ae^{-kt}$ describes Phase 3	holds as a fit	$R^2 = 0.974$, k not derivable
a coarser fixed manifold contains the sheets	refuted	union rank 16 of 18
the sheet is effectively rank one	refuted	dir-1 share 0.289 ± 0.218
val_floor is a corpus barrier	refuted	$L_{\text{corpus}} = 0$ exactly
$\|g_{\text{floor}}\|$ peaks at the capacity boundary	holds	$D = 96$, six widths

The phase resolves a question that had been open since §23: what exactly is transferable about the Fisher result. The answer is the dimension and not the location. Three directions suffice at every stage of Phase 3 — but they are different three directions each time, as far apart between checkpoints as randomly drawn planes, and no fixed frame, fitted coefficient or forward integration built on their position survives contact with measurement.

36 Phase 34: The Level Set

The programme's central unexplained fact — that the update sits at a large and stable angle to the gradient while the loss falls — turns out to have a classical answer. This phase establishes it, and in doing so verifies the first equation in the programme that contains no fitted constants. A second result overturns a claim that has stood since §32: the frame does not rotate as a rigid whole.

36.1 The ascent is level-set curvature

Split the update along the gradient, $u = u_\parallel + u_\perp$ with $u_\parallel = \langle u, \hat{g}\rangle\hat{g}$. Then $\langle g, u_\perp\rangle = 0$ *by construction*, so $u_\perp$ is tangent to the level set $\{L = c\}$ and the first-order term in $\Delta L_\perp$ vanishes identically. Whatever remains must be second order.

step	$\Delta L_\perp$	$\frac{1}{2}u_\perp^\top H u_\perp$	ratio	$\Delta L_\parallel$	ratio
24	0.009611	0.009036	1.064	−0.052	−22.4
48	0.003890	0.003877	**1.003**	−0.031	−19.6
72	0.004340	0.004330	**1.002**	−0.024	−21.7
96	0.005603	0.005750	0.975	−0.019	−18.0
120	0.003474	0.003618	0.960	−0.014	−17.6

Observation 36.1 (The perpendicular ascent is the second fundamental form)**.** *Ratio* 1.001 ± 0.035 *across five checkpoints spanning two decades of loss. Two independently computed quantities — one by ablating the loss on a fixed batch set, one by Pearlmutter double-backward — agreeing to* 3.5% *with nothing fitted.*

The control discriminates by a factor of twenty: the parallel component gives -19.9 ± 2.5*, because there the first-order term dominates and the quadratic is a small correction. And* $\cos(g, u_\perp) = 10^{-5}$ *confirms the split is clean.*

So the equation for Phase 3 is

$$\dot{L} \;=\; \underbrace{-\|g\|\,\|u_\parallel\|}_{\textit{first order}} \;+\; \underbrace{\tfrac{1}{2}\,u_\perp^\top H u_\perp}_{\textit{level-set curvature}}\;,$$

both terms measured. Positive $\Delta L_\perp$ *at every checkpoint means the level sets curve* towards *the direction of travel: the trajectory rides a valley wall.*

This explains four earlier failures. The two-term Fisher ODE, the replicator, the shared-budget model and the fuel-depletion model each carried a fitted constant standing in for exactly this curvature — a quantity that varies with position and cannot be a constant. The Fisher ODE's α/β ran from -362 to $+90$ and changed sign; implemented as a forward predictor it returned a *negative* loss at -299% error. Fitting a curvature with a number is what all of them were doing.

36.2 The shape of the surface

$\mathrm{II}(T,T) > 0$ says the surface curves along one direction. It does not say whether the surface is a bowl or a saddle: a saddle has positive normal curvature along some directions and negative along others. The Gauss equation for a hypersurface settles it, $K(X,Y) = \mathrm{II}(X,X)\mathrm{II}(Y,Y) - \mathrm{II}(X,Y)^2$.

step	$\mathrm{II}(T,T)$	random	principal	residual	all K
40	0.094	0.0024	3.39	0.234	>0
90	0.233	0.0030	7.35	0.241	>0
140	0.246	0.0024	6.63	0.188	>0

Observation 36.2 (A bowl, and violently anisotropic). *$K > 0$ in all nine measurements — three checkpoints, three choices of second direction. No saddle anywhere.*

The surface is extremely far from umbilic. Random tangents give $\mathrm{II} = 0.0024$ *with standard deviation* 0.00002 *— remarkably uniform — while the principal tangent gives* 6.6*: a factor of* 2800*. The level set is nearly flat in almost every one of its* 1.18×10^6 *directions and sharply curved in a few.*

The trajectory sits between: $\mathrm{II}(T,T)$ *rises* $0.094 \to 0.246$*, which is* 40–108× *the random value but only* 3–4% *of the principal. And the* residual *direction — the object every ℓ_2 frame failed to capture — has* $\mathrm{II} = 0.19$–0.24*, essentially the same curvature as the trajectory, with a negative mixed term* $\mathrm{II}(T,W) = -0.08$ *to* -0.12*.*

36.3 Equipartition, and what it does not buy

Decomposing the acceleration of the curve, $\nabla_T T = \kappa_g u_g + \mathrm{II}(T,T)\hat{n}$, the geodesic hypothesis is $\kappa_g = 0$. Measured with a frozen batch for both consecutive updates, so that $\dot{T}$ is trajectory curvature rather than two gradient samples:

step	$\cos(u, -\hat{g})$	$\lvert\kappa_g\rvert$	$\mathrm{II}(T,T)$	R det	R stoch
40	0.353	0.132	0.115	1.146	1.481
70	0.259	0.159	0.129	1.235	1.447
100	0.223	0.182	0.157	1.157	1.390
130	0.199	0.155	0.197	0.785	1.227
160	0.193	0.206	0.141	1.458	2.002
190	0.153	0.179	0.246	0.730	1.022

Observation 36.3 (Intrinsic and extrinsic bending are equal). $R = \lvert\kappa_g\rvert/\lvert\mathrm{II}(T,T)\rvert = 1.085 \pm 0.254$*, with no trend. The trajectory is not a geodesic* ($R = 0$) *and not passively sliding: it bends intrinsically at the same rate the surface bends it, so the acceleration sits near* 45° *to the normal throughout.*

The frozen batch removed a bias*, not the variance: R falls from* 1.428 *to* 1.085 *at every checkpoint, so stochastic estimation was inflating κ_g by about* 30%*, while the spread stays at* 67–69% *either way. And* $\cos(u, -\hat{g})$ *becomes clean and monotone,* $0.353 \to 0.153$*, where the stochastic version jumped to* 0.30 *at one checkpoint against* 0.10 *elsewhere.*

Equipartition does not buy prediction. Integrating the geodesic forward — $\theta_{t+k} \approx \theta_t + k\|u\|T - \frac{1}{2}k^2\|u\|^2 \mathrm{II}\,\hat{n}$ — is *worse* than a purely linear extrapolation at every horizon: relative error 0.334 against 0.330 at $k = 2$, 0.450 against 0.411 at $k = 4$, 0.760 against 0.630 at $k = 8$. Restoring the κ_g term does not help (0.735). The curvature correction scales as k^2 and overwhelms the error it was meant to remove, because a second-order truncation of a curve that is not a geodesic is worse than a first-order one. At $k = 8$ the geodesic prediction more than doubles the loss where the true trajectory reduced it.

36.4 One persistent direction

Every rotation figure in this programme is $d_{\mathrm{Gr}} = \sqrt{\sum_i \theta_i^2}$ — a single number summarising k principal angles. Reported individually:

role	n	θ_1	θ_2	θ_3	θ_4	random θ_1	split-half θ_1
EMB	195,712	**0.279**	1.395	1.484	1.552	1.564	0.100
LN	3,328	0.286	1.187	1.329	1.493	1.524	0.100
W_Q	98,304	0.357	1.347	1.457	1.545	1.562	0.127
W_K	98,304	0.365	1.365	1.464	1.548	1.560	0.131
W_V	98,304	**0.423**	1.388	1.481	1.541	1.560	0.160
W_O	98,304	0.411	1.385	1.483	1.544	1.560	0.155
FF	589,824	0.397	1.371	1.454	1.548	1.566	0.145

Observation 36.4 (The frame is one fixed axis and three that spin freely)**.** $\theta_1 = 0.36$ *radians on average — the leading direction stays within* $21°$ *of its previous position — while* $\theta_2, \theta_3, \theta_4$ *sit at* 1.37–1.54*, against a random-plane ceiling of* 1.557*. The other three directions are at the ceiling; the first is nowhere near it.*

The split-half floor is 0.10–0.16*, so the persistence exceeds the estimator's resolution by a factor of* 2–3 *and is not an artefact. And the role ordering is the one that has appeared in every measurement of this kind: EMB and LN most persistent (*$\theta_1 = 0.28$*, leading direction within* $17°$ *on* 60–65% *of checks),* Q/K *next at* 0.36*,* V/O *loosest at* 0.41–0.42*.*

This corrects Obs. 34.1, which established that the frame separation is real motion rather than estimator noise and reported it as near-maximum. That is true of the aggregate and false of the leading direction: d_{Gr} *is dominated by the three large angles and hid a nearly fixed axis inside them.*

*It also explains the frozen-frame result quantitatively. A frozen rank-*4 *frame retains essentially the* θ_1 *component alone — one direction of four — which compounded over* 280 *steps is close to the measured* 3.4% *retention.*

36.5 Status

claim	status	basis
$\Delta L_\perp$ is level-set curvature	holds, zero parameters	ratio 1.001 ± 0.035
the level set is a bowl, not a saddle	holds	$K > 0$, nine measurements
the surface is nearly umbilic	refuted	principal/random = 2800
the residual shares the trajectory's curvature	holds	0.19–0.24 vs 0.09–0.25
the trajectory is a geodesic	refuted	$R = 1.085 \pm 0.254$, not 0
intrinsic and extrinsic bending are equal	holds	$R \approx 1$, six checkpoints
geodesic integration predicts forward	refuted	worse than linear at every k
sign(r) can replace r	refuted	loses by 2.6–6.3% at matched norm
the frame rotates as a rigid whole	refuted	$\theta_1 = 0.36$ vs $\theta_4 = 1.54$
a persistent direction exists	holds	2–3× the split-half floor

The phase answers the question the programme opened with and closes another. The update's stable large angle to the gradient is not a puzzle about the optimiser: it is a curve on a hypersurface, in a direction 40–108× more curved than average, bending intrinsically at the rate the surface bends it. And the frame that has been described throughout as rotating near-freely is one nearly fixed axis carrying three that do.

What none of it buys is prediction. Four independent routes to skipping backward passes — leakage-triggered coasting, frozen frames, a coarser Fisher manifold, and geodesic integration — have now been measured and closed. The geometry describes the trajectory well and forecasts it badly, and this phase establishes that those are different things.

37 Phase 35: Structure Destroyed by Flattening

Every subspace measurement in this programme flattened the parameters into a single vector and worked in $\mathrm{Gr}(k, P)$. That discards the tensor structure: a weight matrix $W \in \mathbb{R}^{m\times n}$ has a grid, and flattening throws it away. This phase measures the same objects without flattening, and the answers change — in one case by a factor of five.

37.1 Where the curvature mass sits

Before the projection results, two measurements of H itself that the rest of the phase depends on. Block Frobenius masses $M[c, b] = \|H_{cb}\|_F^2$ estimated by $\mathbb{E}\|(Hz_b)_c\|^2$ for Rademacher z supported on block b, which is unbiased — unlike the sum of squared diagonal entries, whose Hutchinson bias was 30–50% of the raw value at 24 probes and had to be corrected.

change, step 30 → 150	EMB	ATTN	FF	LN
EMB	−**74**%	−21%	−27%	−73%
ATTN	−18%	−61%	+**37**%	−67%
FF	−24%	+**31**%	−34%	−33%
LN	−73%	−68%	−32%	−64%

Observation 37.1 (The embedding drains and one coupling grows)**.** *Total mass falls* 100.4 → 51.6*, a* 49% *decline. Against that, EMB self-curvature falls* 74% *while its couplings to the compute blocks fall only* 21–27%*: the curvature leaves the token coordinates and stays in the interactions. Share of total moves EMB* 0.440 → 0.320 *and FF* 0.322 → **0.449***, crossing near step* 90 *— the same handover the update-energy measurement found independently (FF* 64% → 39%*, ATTN* 24% → 48%*), now visible in the second derivative.*

ATTN ↔ FF *is the only block anywhere that* gains *absolute mass,* +31 *to* +37%*, on an operator shrinking by half. Estimator asymmetry is* 0.006–0.016*, so* $M[c, b] = M[b, c]$ *holds to about* 1%*.*

Nothing gains in absolute terms except that one block. *Off-diagonal mass falls* 99.4 → 46.4*; it is the* share *that rises* 0.922 → 0.976*, because the diagonal falls faster. The bias-corrected diagonal share falls* 0.078 → 0.024*, a* 69% *decline — steeper than the raw* 45%*, so the correction strengthens rather than explains it. Even at its lowest the diagonal carries* 28,500× *the mass a generic dense matrix would put there.*

Three further properties, one of which inverts a natural reading. The diagonal is concentrated: PR = 2–3% of P, top 1% of coordinates holding 38–49% of its energy. The growing ATTN × FF coupling is *not* low rank — spectrum $1.00, 0.85, 0.82, 0.74, 0.72, 0.71, 0.64, 0.63$ with $\mathrm{rank}_{90} = 7/8$ at every checkpoint, so the +31% is diffuse mixing rather than a specific mode. And $D = \mathrm{diag}(H)$ and $N = H - D$ appeared nearly to commute at ratio 0.0038, which would make the split a direct sum — but permuting D's entries to random coordinates *reduces* the commutator to 0.35×, stably across three checkpoints, while a uniform D gives exactly 0. So the diagonal spikes sit where the coupling is *strongest*, not weakest: the concentration and the web are entangled, not decoupled.

37.2 The gradient is low rank as a matrix

A fresh gradient was measured at 5–17% capture by a frame built from update history (§32, $\gamma = 0.83$–0.95). Structured as matrices, with each gradient's own SVD:

role	E_4	E_8	E_{16}	E_{32}	null at r=8
ATTN	0.75–0.78	**0.87–0.91**	0.96–0.98	0.995	0.212
FF	0.52–0.62	0.72–0.82	0.87–0.92	0.96–0.97	0.158

Observation 37.2 (Flattening cost a factor of five). *Mean* $E_8 = 0.785$ *against a random-matrix null of* 0.186 *— a* 4.2× *excess. Rank* 8 *of* 128 *holds* 78% *of the gradient.*

The same gradient, flattened and tested against an update-history frame, gave 5–17%. *These are different objects: the flat frame is built from where the trajectory* has been, *the matrix SVD from the gradient's own structure. Measuring the first and concluding about the second was an error running through several earlier phases.*

The cost inversion follows. A two-sided projector $P^\top GQ$ *costs* $r(m+n)$ *where a flattened frame costs* rmn*: for the feed-forward block,* 3,072 *floats against* 4,718,592*, a factor of* 1,500*. And a flattened frame can never save optimiser memory at all, since* $PK > 2P$ *whenever* $K > 2$ *— a point stated correctly in §33 and then contradicted by an arithmetic omission in a later memory projection, which the implementation caught.*

37.3 Two implementation errors, and what they cost

A per-matrix optimiser was built and initially failed completely, stalling at 4.41 against AdamW's 0.92. Two bugs, both instructive.

Coupled weight decay. The projected quantity was $g + \lambda W$ rather than g. But $\|\lambda W\|/\|g\|$ measures 1479 for attention and 13.6 for FF, and W is full rank, so the SVD was projecting the weight matrix. E_8 fell from 0.776 to 0.349. GaLore uses decoupled decay for exactly this reason.

Step-norm collapse. $PMQ^\top$ with M of size $r \times r$ and $O(1)$ entries has norm $\sim r$, while an AdamW step on the same matrix has norm $\sim \sqrt{mn}$. At $r = 8$, $m = n = 96$ that is a 12× smaller step, so the comparison measured learning rate rather than direction.

Fixing both moved the result 4.41 → 1.81.

37.4 Refresh dominates rank

arm	final	capture	flip	memory
AdamW	**0.924**	—	—	1×
role r=8/16, T=1	**1.250**	0.736	0.493	9.3×
role r=8/16, T=3	1.800	0.384	0.244	9.3×
uniform r=11, T=3	1.814	0.379	0.247	9.9×
role r=8/16, T=10	2.408	0.227	0.171	9.3×
role r=8/16, T=50	2.991	0.131	0.096	9.3×

Observation 37.3 (The refresh interval is the parameter; rank allocation is not). $T = 1$ *against* $T = 50$ *is* 1.250 *against* 2.991 *— a factor of* 2.4 *— and capture tracks it exactly,* 0.736 *down to* 0.131*. At* $T = 50$ *the projector holds* 13% *of the gradient, barely above what a random subspace of that rank gives.*

Role-differentiated rank buys nothing. *At matched memory,* $r = 8$ *for attention with* $r = 16$ *for FF gives* 1.8002*; a uniform* $r = 11$ *gives* 1.8138*. That is* 0.8%*, which is noise. The* E_r *curves* *of Obs. 37.2 justify the split and the split does not help — a case where a well-motivated design choice fails its own control.*

The mechanism was predicted before the optimiser runs. Subspace overlap between checkpoints ~ 60 *steps apart is* 0.215*, with FF worst at* 0.08–0.30 *and attention* 0.20–0.37*. The projector decorrelates on a timescale far shorter than the schedules in common use.*

Two measurements cut against the method. Under every low-rank arm $\|H\|_F^2$ *grows* — $88 \to 265$ at $T = 3$ and 624 at $T = 10$ — while AdamW's stays flat at 75–91. Low-rank updates drive the model into sharper regions. And the best arm, $T = 1$, has a sign-flip rate of 0.493: half of all coordinates reversing every step, the signature that preceded every ℓ_∞ failure in §33. Best loss and worst thrashing in the same arm is a combination this programme has not seen before.

37.5 Compared with GaLore

	GaLore	measured here
projector	$P^\top GQ$, two-sided	same — and the cost argument holds
refresh	$T = 50$–200	$T = 50$ gives 2.99; $T = 1$ gives 1.25
rank	uniform scalar	uniform matches role-aware to 0.8%
premise	gradients are low rank	holds: $E_8 = 0.785$ vs null 0.186
subspace assumed	quasi-stationary	refuted: overlap 0.215 over 60 steps

Observation 37.4 (A method for choosing T, not a better optimiser)**.** *Every arm loses to AdamW: the best is* 1.250 *against* 0.924*,* 35% *worse. No configuration tested here beats the baseline, so nothing in this phase is an improved optimiser.*

What it does establish is narrower and checkable: the refresh interval should be set from the measured subspace overlap rather than from a default, and the overlap is cheap to measure — one SVD per matrix per checkpoint, no optimiser run required. On this landscape that gives $T \lesssim 3$ *where the common choice is* 50–200*.*

Two limits on how far to carry it. The published schedules were chosen for large-scale pre-training, where larger batches and smaller relative steps may well rotate the subspace more slowly; $T = 50$ *is refuted* here *and is a hypothesis elsewhere. And* $T = 1$ *means an SVD per matrix per step: the projector is cheap to* store *at* $r(m + n)$*, and recomputing it every step is a cost this phase did not charge against the* $9.3\times$ *memory saving.*

37.6 Sparse and low rank are the same rows

Two structures had been measured and appeared to be in tension. The diagonal of H is concentrated — PR = 2% of P, top 1% of coordinates holding 38–49% of the mass, and permuting D cut the $[D, N]$ commutator to $0.35\times$, so those coordinates sit where the couplings are strongest. But a rank-r projector cannot select individual entries, and a scattered sparse set is generically full rank as a matrix.

Mapping the top 1% of diag(H) back to (matrix, row, col):

matrix	shape	hubs	top-10% rows	Gini	top-10% cols
te	$(1017, 128)$	4677	0.728	0.854	0.150
pe	$(512, 128)$	4213	0.863	0.891	0.144
b0.ff.g	$(256, 128)$	26	**1.000**	0.963	0.577
b1.ff.o	$(128, 256)$	335	0.206	0.351	**0.833**
b5.ff.o	$(128, 256)$	27	0.519	0.818	**1.000**

Observation 37.5 (The hubs are row-structured, and a projector can reach them)**.** *Row concentration runs* 0.21 *to* 1.000 *against a uniform null of* 0.10*, with Gini coefficients* 0.30–0.97*. Several feed-forward gate and value matrices put* every *hub coordinate in the top* 10% *of rows.*

So the sparse structure is not scattered entries invisible to factorisation — it lives in a small set of rows, which is exactly what P spans in $P^\top GQ$. The two descriptions are the same rows seen two ways.

The output projections show the transpose of the pattern: row concentration 0.21–0.52 *but* column *concentration* 0.75–1.000, *consistent with their reading from a few feed-forward units rather than writing to a few.*

*Two thirds of all hubs sit in the embeddings (*4677+4213 *of* 11,820*), which fits EMB draining* 74% *of its self-curvature (Obs. 37.1) while what remains is highly localised.*

The hot rows drift. *Row overlap between checkpoints is* 0.08–0.40 *for most matrices, though positional embeddings reach* 0.804. *Clustered at any instant, moving between instants — the same refresh problem as the subspace, one level down.*

37.7 Curvature as a trust region, and what beat it

The second fundamental form has been diagnostic throughout. Both attempts to use it *in the loop* failed, and both failed the same way: geodesic integration gave error 0.334/0.450/0.760 at $k = 2/4/8$ against a plain linear extrapolation at 0.330/0.411/0.630, and ranking residual coordinates by $|r_i(Hr)_i|$ lost to $|r_i|$ at every k (0.827 against 1.186). Both used II to choose a *direction*, and the first-order term dominates every step: $\Delta L_{\|}$ runs -0.052 to -0.014 while $\Delta L_{\perp}$ runs $+0.0096$ to $+0.0035$.

The remaining legitimate use is magnitude. $\mathrm{II}(T,T)$ rises $0.094 \to 0.246$ through Phase 3 — the valley narrows — so $\eta_t = \eta_0\sqrt{\mathrm{II}_{\mathrm{ref}}/|\mathrm{II}(T,T)|}$ shrinks the step as the walls close in. That is a trust region, which is what a second-order quantity is for.

step-size rule	final	vs fixed	η early	η late
$\eta \propto 1/\|g\|$	**0.251**	**0.37**$\times$	0.00191	0.00300
II shuffled	0.453	0.67$\times$	0.00225	0.00236
fixed	0.675	1.00	0.00150	0.00150
cosine	1.586	2.35$\times$	0.00147	0.00004
II trust region	2.575	3.82$\times$	0.00091	0.00018

Observation 37.6 (The trust region loses to its own shuffled control)**.** II-*scheduled gives* 2.575*; the* same II values applied in scrambled temporal order *give* 0.453 — *a factor of* 5.7 *better. So the timing of the curvature signal is actively harmful, worse than applying the same numbers at random.*

The mechanism is visible in the trajectory: II *reads* -0.0002, -0.0034, -0.0004 *during the saddle exit and then* 0.0199 *at step* 30. *The reference is anchored on a near-zero value at an inflection, so* $\sqrt{\mathrm{II}_{ref}/\mathrm{II}}$ *collapses to the floor the moment* II *rises:* η *falls* $0.00091 \to 0.00018$ *and never recovers. Note also that* II *is negative early —* $\varphi = 128^\circ$ *at step* 8 *— so the rule needs* $|\mathrm{II}|$ *and a floor before it is even well posed.*

That is a fragile anchor rather than a refutation of curvature scheduling in general. But it is the third distinct failure of II *in the loop, and the three share a cause.*

The incidental result is the useful one. $\eta \propto 1/\|g\|$ gives 0.251 against a fixed schedule's 0.675 — 2.7$\times$ better — and its learning rate *rises*, $0.00191 \to 0.00300$, rather than decaying. On this landscape the step should grow as the gradient shrinks. It is first order, free to compute, needs no Hessian-vector product, and it beat every second-order variant tried here. Cosine decay, by contrast, is 2.35$\times$ *worse* than fixed, so the improvement is not a generic property of having a schedule.

37.8 Status

claim	status	basis
gradients are low rank per matrix	holds	$E_8 = 0.785$ vs null 0.186
flattening preserves that structure	refuted	5–17% vs 78%
a flat frame can save memory	refuted	$PK > 2P$ for $K > 2$
two-sided projector is affordable	holds	9.3×, $r(m + n)$ not rmn
the subspace is quasi-stationary	refuted	overlap 0.215 over 60 steps
$T = 50$ is a workable refresh here	refuted	2.99 vs 1.25 at $T = 1$
role-aware rank beats uniform	refuted	1.8002 vs 1.8138
low-rank beats AdamW	refuted	best 1.250 vs 0.924
low-rank updates preserve curvature	refuted	$\|H\|_F^2$ 88 → 265
the curvature hubs are row-clustered	holds	0.21–1.00 vs null 0.10
II works as a trust region	refuted	loses 5.7× to its shuffle
$\eta \propto 1/\|g\|$ beats a fixed rate	holds	0.251 vs 0.675
sparse and low rank are complementary	holds	same rows, two views

The phase reconciles the programme's two structural findings and corrects one of its methods. Instantaneous gradients are low rank; the curvature hull they sweep out is diffuse (§37.1, $\text{rank}_{90} = 7/8$); and the bridge is the rotation rate, measured at overlap 0.215. The high-curvature coordinates that carry 38–49% of the diagonal are not a separate structure competing with the low-rank picture — they are the rows the projector already spans.

What none of this produces is a better optimiser. Seventeen variants have now been measured against AdamW and all seventeen lose. The one durable output is diagnostic: the refresh interval follows from a subspace overlap that costs one SVD to measure, and the common default is off by more than an order of magnitude on this landscape.

38 Phase 36: Update Compression, and Four Corrections

This phase set out to reproduce a published low-rank optimiser and ended somewhere else. The narrative matters here because most of the session's conclusions were wrong before they were right, and the errors were of a consistent kind: a local measurement, correct as taken, generalised to a trajectory claim it did not support.

38.1 What the session was trying to do

GaLore-style methods project the gradient onto a rank-r subspace and run Adam inside it, saving optimiser memory. SubTrack++ adds Grassmannian subspace tracking, projection-aware moment transport, and recovery scaling. The aim was to reimplement these on the Phase-3 pipeline and understand why they did or did not work here.

The reimplementation failed repeatedly. Each failure was diagnosed, a mechanism was proposed, and the mechanism was later refuted — usually by a control that should have been run first.

38.2 Correction 1: a nondeterministic harness

Every in-process sweep compared arms that started from different models. `spla.eigsh` draws its Lanczos start vector from NumPy's global RNG when `v0` is not supplied, so a second `exec` in the same process got a different start vector, a different spectral embedding, and a model differing by 0.79 at initialisation. A fresh process is deterministic, which is why single runs reproduced and sweeps did not.

Observation 38.1 (The contamination was real but mild)**.** *Passing* `v0` *and canonicalising eigenvector signs made two builds identical to* 0.000. *Reruns from identical initialisation:*

sweep	*contaminated*	*corrected*
hybrid, $\beta = 0 \to 1$	$1.380 \to 2.829$	$0.669 \to 1.848$
sign as a trajectory	3.05× *worse*	2.44× *worse*
GaLore T=50 / T=1	2.39×	2.373×
$\alpha = 0$ *vs plain AdamW*	—	*bit-identical,* 0.0

Every conclusion survived; the magnitudes moved by up to 25%. *The lesson is not that the results changed but that they were unfalsifiable until the harness was checked.*

38.3 Correction 2: three implementation errors read as landscape properties

Three reimplementation bugs each produced a plausible measurement, and each measurement was used to argue that low-rank optimisation was obstructed here.

Coupled weight decay. The projected quantity was $g + \lambda W$ rather than g. But $\|\lambda W\|/\|g\|$ measures 1479 for attention and 13.6 for feed-forward, and W is full rank, so the SVD was projecting the weight matrix: E_8 fell from 0.776 to 0.349.

Step-norm collapse. $PMQ^\top$ with M of size $r \times r$ has norm $\sim r$; an AdamW step on the same matrix has norm $\sim \sqrt{d_1 d_2}$. At $r = 8$, $d = 96$ that is a 12× smaller step, so the comparison measured learning rate rather than direction.

This one recurred. Having been caught as an explicit scale factor, it re-entered *this session's own headline comparison* through the projection itself: an orthogonal projector removes norm, so the low-rank arm ran at 0.51 of the intended rate without any scale factor being written down. It was caught only when the diagnostic table made $\|\hat{u}\|/\|u\|$ and cos visibly identical. The general form — any contractive compression operator silently reduces the learning rate — is the most transferable thing in this phase.

Full re-SVD instead of geodesic tracking. Measured directly: $S_{\text{new}}^\top S_{\text{old}}$ has singular values 0.006–0.30 under a re-SVD and 0.998–1.000 under a rank-1 nudge. The re-SVD destroys 85% of the momentum at every refresh, which is why all four moment-transport variants failed.

Observation 38.2 (The reference implementation falsified the landscape story)**.** *The authors'* `LowRankAdamW`*, run unmodified on this pipeline with phases 1, 2, 4 and 5 byte-identical, reached* 0.737 *at 150 steps where the reimplementation reached* 1.251*, with capture holding at* 0.152 *where the reimplementation's tracking collapsed to* 0.017 *and froze.*

Three claims were retracted: that $\nabla F \propto (I - SS^\top)GG^\top S$ *vanishes structurally here, that* θ_8 *at* 1.52 *rad blocks geodesic tracking, and that momentum transport is meaningless in a fast-rotating frame. All three had supporting measurements. All three were artefacts of the code around them.*

38.4 What actually limits low-rank compression

With the harness fixed, the binding constraint was measured directly. Retained descent $D = -g^\top \hat{u} \,/ -g^\top u$ on a frozen-batch gradient:

	r=4	r=8	r=16	r=32
instantaneous	0.25	0.35	0.58	0.75
after 8 steps	0.21	0.32	0.47	0.67
transport T_8	0.35	0.45	0.52	0.62

Observation 38.3 (Rank binds; transport is secondary)**.** *At fixed rank, over eight steps, transport falls by half while cumulative capture falls* 10–19%. *So a stale projector is a modest cost and truncation is the ceiling.*

The evidence for this was initially wrong. *The reported correlations — +0.957 for instantaneous against +0.894 for transport — were computed across rows where* rank *varies, and rank drives all three quantities. The clean evidence is the within-rank horizon sweep, where rank is held fixed and only the horizon moves.*

Two alternative explanations were tested and eliminated. The Hessian-dominant subspace carries $\alpha_k = 0.0585$ of the gradient here and *increases* loss (+0.0007 against the bulk's −0.0322), so the projector is not selecting sharpness directions. And source identity does nothing: an update-derived orthogonal complement scored −0.006 against the next eight gradient directions across eight checkpoints with five draws averaged, mean of union minus G_{16} equal to −0.027.

Observation 38.4 (A retracted crossover, and where the variance lives)**.** *A single-draw measurement suggested the complement overtakes the gradient directions late (+0.083 at step 130). With five draws averaged the effect vanished: $D(G_8)$ went $0.430 \to 0.426$ and $D(C_8)$ went $0.168 \to 0.138$ across the run.*

The variance is in the sampled update, not the geometry: $D(G_8) = 0.351 \pm 0.333$ at step 160, with the subspace held fixed. Several single-draw quantities elsewhere in this programme sit at comparable signal-to-noise and should be re-read accordingly.

38.5 Correction 3: the ceiling was a geometry mismatch

The conclusion "compression has a real information cost" was drawn entirely within the ℓ_2 low-rank family. Adam's update, however, is nearly uniform in coordinate magnitude: $\mathrm{corr}(|g|, |u|) = +0.04$, mean $|u|$ flat across all ten deciles of $|g|$, and $|u|/|g|$ running 3367 in the bottom decile to 0.43 in the top. A uniform-magnitude matrix is generically full rank.

Define SIGN+TOP: keep the top-k coordinates of u exactly and represent the rest as $\mathrm{sign}(u_i) \cdot \overline{|u_{\mathrm{tail}}|}$, with k solved so that $2k + \mathrm{numel}/32 + 1$ equals the rank-r budget $r(d_1+d_2) + 2r^2$.

arm	floats/matrix	final	vs uncompressed	$\|\hat{u}\|/\|u\|$
uncompressed	—	0.239	1.00	1.000
SIGN+TOP r=32	18,432	**0.283**	**1.18**	0.949
SIGN+TOP r=4	2,080	0.336	1.41	0.846
low rank r=32	18,432	0.782	3.27	0.495
low rank r=4	2,080	2.283	9.54	0.220

All arms are rescaled to $\|u\|$ after compression. Without that rescaling the same runs give low rank 1.527 and 3.507 against SIGN+TOP's 0.319 and 0.472 — so roughly half the apparent gap was a learning-rate artefact, not a directional one. Those unmatched numbers are kept here rather than deleted because the correction has more methodological value than the inflated result did.

Observation 38.5 (Directional fidelity, not local descent)**.** *SIGN+TOP at the rank-4 budget — fifteen exact coordinates plus a one-bit sign field — beats low rank at the rank-32 budget, which costs 9× more: 0.336 against 0.782, both norm-matched.*

The mechanism is not descent capture. Applying both operators to the same *uncompressed trajectory:*

operator	$\cos(u, \hat{u})$	$\|\hat{u}\|/\|u\|$	*D retained*
low rank	0.568	0.568	0.858
SIGN+TOP	**0.949**	**0.949**	0.892
top-k only	0.798	0.798	0.632
sign only	0.820	0.820	0.632

Low rank retains 0.858 *of the local descent against* SIGN+TOP*'s* 0.892 *— nearly the same — while its direction is* 0.568 *against* 0.949*. Over a trajectory the low-rank arm accumulated* more *raw descent (*15.44 *against* 11.70*) and finished* 5× *worse.*

Both ingredients are necessary: top-k alone and sign alone each retain 0.632*, together* 0.892*.*

Note that $\|\hat{u}\|/\|u\|$ *equals* cos *in every row. That is not coincidence: for both operators* $\langle u, \hat{u}\rangle = \|\hat{u}\|^2$*, so both columns reduce to* $\|\hat{u}\|/\|u\|$*. An orthogonal projection can only* remove *norm;* SIGN+TOP *holds* 95% *of it.*

Which is exactly why the comparison had to be norm-matched. *Retaining* 0.512 *of the update norm means the low-rank arm was running at half the intended learning rate, and at a step-dependent one, since retention drifts with the spectrum. Correcting it halved the gap. The residual* 2.77× *is the directional effect.*

An anomaly the correction exposes. SIGN+TOP retains 0.949 of the norm at r=32 but only 0.846 at r=4. If the tail were genuinely near-uniform in magnitude, dropping k from 8191 to 15 should barely move that ratio — the sign field is unchanged and would be carrying nearly all the mass either way. A ten-point drop says the exceptional coordinates hold more of the norm than $m \approx \bar{m}\mathbf{1} + \delta m$ with negligible δm allows. So the magnitude residual is *not* negligible, and how $\|\hat{u}\|/\|u\|$ varies with k is the measurement that would say whether there is a stratification with a natural k or a smooth magnitude distribution that top-k truncates arbitrarily. Not measured here.

38.6 Correction 4: an unmatched budget

The first version of the SIGN+TOP comparison omitted the cost of the sign field itself, which is numel/32 float-equivalents. At $r = 4$ on a 256×256 matrix the sign field alone costs 2048 of a 2080 budget, so the arm was running 2× over. Corrected, $k = 15$ at $r = 4$ and $k = 8191$ at $r = 32$; the trajectory result above uses the corrected budgets.

38.7 Three results not otherwise recorded

The plateau detector is a single-reading rule. Phase 3 exits on `if delta < 0.003: break`, one checkpoint below threshold. AdamW never triggers it — 0 of 21 readings sub-threshold across 168 steps, with Δ declining smoothly $0.082 \to 0.0055$ — because it was tuned on that trace. Low-rank traces are non-monotone and do trigger it spuriously: one run exited at $\Delta = 0.0005$ having read 0.0093 eight steps earlier, and another at 0.0004 after 0.0107. In both the loss was still falling 1–2% per step, and the 16-step extension that follows confirmed it ($0.439 \to 0.340$). At least three conclusions of the form "it stalled here" were "it stopped here". Requiring two consecutive sub-threshold readings is the minimal fix.

A Fisher-sourced projector, and why the cheap version fails. A range finder over N gradients estimates the directions along which the gradient *varies*, which is a different object from one gradient's SVD. Retained descent at step 110, rank 8:

source	flat L share	two-sided	memory
single-gradient SVD	—	0.195	31×
Fisher, span 1 (12 fresh batches)	0.959	0.757	31×
Fisher, span 6	0.860	0.715	31×
Fisher, span 12 (free)	0.752	0.685	31×

Staleness is nearly free in the two-sided form — 0.757 at span 1 against 0.685 at span 12, for twelve fewer backward passes. But a buffer accumulated across the *running* trajectory rather than anchored locally collapses to 0.193, because the parameters leave the local Taylor ball. And the flattened form, which is what gives 0.959, costs numel $\times\, r$ — four times AdamW's own state — so it is a better subspace at higher cost, not a memory result.

Contractive compression silently lowers the learning rate. Any operator with $\|\hat{u}\| < \|u\|$ — which includes every orthogonal projection — reduces the effective step, and by a factor that drifts as the operator's retention drifts. Rescaling $\hat{u}$ to $\|u\|$ moved low rank from 1.527 to 0.782 and SIGN+TOP from 0.319 to 0.283, so it is not specific to low rank; it is a property of the compression family. Anyone applying a GaLore-style projector should rescale, and any comparison between compressors that does not is partly a learning-rate comparison.

Self-referential projectors cannot work. Building the projector from the optimiser's own output closes a loop it cannot escape: at step t the update u_t lies in $\mathrm{span}(P_t)$, so a buffer of past updates spans only previous subspaces. Measured, `capture` against the gradient fell to 0.002 and the run reached 4.80. The same failure appeared with a Fisher buffer over the constrained optimiser's own gradients (4.756). Only the gradient is exogenous — computed fresh from data each step, not from the optimiser's constrained motion. This rules out a family in advance rather than one method at a time.

38.8 Status

claim	status	basis
low-rank reimplementation is sound	refuted	0.737 vs 1.251
∇F vanishes structurally here	retracted	reference impl. holds 0.152
θ_8 blocks geodesic tracking	retracted	same
transport is meaningless in-frame	retracted	re-SVD was the cause
rank binds, transport secondary	holds	fixed-rank horizon sweep
the dominant subspace carries descent	refuted	+0.0007 vs −0.0322
source identity matters	refuted	−0.006, five draws, 8 points
a late source crossover exists	retracted	single-draw noise
compression costs information	*scope*	true within ℓ_2 low rank
SIGN+TOP beats low rank	holds	0.336 vs 0.782, storage *and* norm matched
the plateau rule is single-reading	holds	0/21 on AdamW, fires on low rank
self-referential projectors close	holds	capture 0.002, val 4.80
both SIGN+TOP ingredients matter	retired	per-step; did not replicate

Scope. One landscape (D=128, a deterministic corpus with $L_{\text{corpus}} = 0$ exactly), one seed per trajectory arm, 140 steps, no plateau rule in the comparison script so every arm ran the full budget. The ingredient ablation — top-k alone 0.632, sign alone 0.632, together 0.892 — is *retired*: it is a per-step quantity, and on a two-layer MLP the same measurement gives 0.317 and 0.793, so it does not replicate across architectures. Trap 1 of this session says per-step quantities do not predict trajectories, and this one does not even predict itself. The supported claim is narrow: the Adam update here is poorly matched to matrix low-rank approximation and substantially better matched by exact large coordinates plus a one-bit sign field. It is *not* a claim that the update "is an ℓ_∞ object", and the trajectory result has not been replicated across seeds. Nothing here evaluates GaLore or SubTrack++ at the scale they target; their reported results on LLaMA-1B are untouched by this.

Methodological residue. The session's recurring error was inferring a mechanism from a failure rather than decomposing the failure. The measurement that finally resolved it — run both, subtract, vary the one parameter that turned out to bind — required no new machinery and was available throughout. A reference implementation was also available throughout and was requested only after twenty variants had been tried.

39 Phase 37: Three Coordinate Systems That Do Not Coincide

Phase 36 left one result standing and a programme built on it: if the update is poorly matched to low rank, perhaps the optimiser's state is organised by some geometry that a better representation could exploit. This phase tested that programme and it did not survive. What replaced it is narrower and, unusually for this document, entirely interventional: the network carries three organisations — a representation carrier, an optimiser-state requirement, and a set of functional capabilities — and no two of them track each other.

Every trajectory number below is at three seeds with the `eigsh` patch applied and asserted, and every comparison reports $\|\Delta\theta\|$ relative to AdamW's, because the step-norm confound of Correction 2 recurred twice more in this session in both directions.

39.1 The geometry programme, and why it closed

Observation 39.1 (The update is basis-indifferent). *Conjugating the compressor by a fixed random orthogonal, $\hat{u} = Q^\top C(QuR^\top)R$, leaves* SIGN+TOP*'s norm retention at* 0.918 *against* 0.913 *unconjugated. Low rank is invariant to* 9.4×10^{-6} *per step, as rotation-invariance of the singular values requires, which is the harness self-test. Trajectory loss moves by less than the seed spread.*

The coordinate story is therefore wrong, and with it the family of formulations built on it: hypercube, ℓ_∞ sphere, orthant, stratified space X_k, and oriented matroids are all coordinate-dependent descriptions of a rotation-indifferent object. `stratify.py` agrees from the other side: $R(k)$ for the real update matches a Gaussian null to four decimals at every budget, and matches the shuffled control exactly, so the magnitude law carries no spatial structure either.

Observation 39.2 (Low rank is above its own null, and still loses). *For a matrix independent of the projector, two-sided rank-r retention is exactly $r/\sqrt{d_1 d_2}$: at D=128, r=32 that is* 0.251*. Measured retention is* 0.495–0.612*, so the update* is *aligned with the gradient's leading directions at roughly twice chance — GaLore's premise holds.* SIGN+TOP *meanwhile sits at its own Gaussian null (*0.949 *measured against* 0.969 *predicted from d and k alone). The compressor that exploits nothing beats the one that exploits something, because the approximation class is what binds.*

Observation 39.3 (Quantisation, and where the budget actually is). *Uniform per-tensor quantisation of the update, stochastic rounding, three seeds, against uncompressed* 0.2679 ± 0.0067*:*

arm	*final*	*vs. none*	*relerr*
quant 8 bit	0.2679 ± 0.0067	1.000	0.009
quant 4 bit	0.2728 ± 0.0072	1.018	0.161
quant 2 bit	0.3910 ± 0.0122	1.460	0.782
quant 1 bit	1.0554 ± 0.0092	3.940	2.312
sign only	0.3948 ± 0.0099	1.474	0.552

Eight bits is free, four costs 1.8%*, and the curve is monotone. A previous single-seed reading had 4 bits* beating *uncompressed by* 3.4%*; the seed spread is* 0.007 *and that gap was* 0.009*. It does not survive. Sign with mean magnitude is the correct one-bit quantiser and beats naive uniform by* 2.7×*, which is a fact about quantiser design, not about the update.*

Scope, which the earlier framing lacked. At D=128 the rank-32 budget is ∼9.5 bits per coordinate, so the budget was never binding and the family choice dominated. At transformer scale the same budget is ∼0.06 bits per coordinate, three orders of magnitude tighter. This phase says nothing about low-rank methods in the regime they target.

39.2 The optimiser's state

Observation 39.4 (m is not load-bearing; v is)**.** *At* 140 *steps, three seeds: AdamW* 0.2679*, RMSProp* 0.2437 *(*0.910$\times$*, half the state), SGD with momentum* 6.8447 *(does not train). But RMSProp's* $\|\Delta\theta\|$ *is* 2.91$\times$ *AdamW's, and at matched norm it is* 4.16$\times$ worse. *AdamW at* 2.91$\times$ *its own learning rate reaches* 0.1308*. The apparent state reduction was a learning-rate artefact.*

This is Correction 2 for the third time, now entering through the opposite door: a contractive compressor silently *lowers* the step, a reduced state silently *raises* it. The general form belongs in the trap list.

Observation 39.5 (Resolution, not precision, is what v needs)**.** *With EMB held at per-coordinate* v *throughout, and all arms matched in* $\|\Delta\theta\|$ *to within* 1.1%*:*

weight-matrix v	*final*	*vs. AdamW*	cos *to AdamW*
per coordinate	0.2679 ± 0.0067	1.000	1.000
one per row	0.2742 ± 0.0150	1.023	0.955
one per column	0.2718 ± 0.0120	1.015	0.926
one per matrix	0.3519 ± 0.0199	1.314	0.766

Quantising v *instead of sharing it costs nothing: FF's* v *at* 4 *bits for all* 160 *steps gives* 0.1666 *against AdamW's* 0.1779*, with the functional staircase identical to three decimals. Two bits also matches. So* v *needs* $O(d)$ *distinct values, each to* 2–4 *bits — not* $O(d^2)$ *values at* 32*.*

Observation 39.6 (The row/column asymmetry is token frequency)**.** *Row-sharing costs* 2.0% *and column-sharing* 20.9% *at identical state size, which appeared to identify output units as the natural unit. Holding EMB at per-coordinate* v *and sharing only in the weight matrices, column-sharing costs* 1.5% *and row-sharing* 2.3%*. The entire asymmetry is* `te.weight`*, whose rows are* tokens*: sharing across a token's dimensions preserves the per-token scale that frequency imposes, sharing across tokens destroys it. No receiver-unit interpretation is required or supported.*

Proposition 39.7 (Fisher row structure is a marginal, not a geometry)**.** *Let* $\rho_{row} = \sum_i d_2 \bar{F}_i^2 / \sum_{ij} F_{ij}^2$*. Measured on the weight matrices,* $\rho_{row} = 0.650$ *against a null of* 0.007 *— but shuffling each matrix's entries, which destroys all spatial structure while preserving the marginal, gives* 0.608*, and its row/column ratio is* 1.000*. The structure excess is* $0.65 - 0.61 \approx 0.04$*. Sample count is not the issue:* $N = 12, 48, 200$ *give* $0.623, 0.652, 0.666$*.*

The estimator is sound and the statistic is real; it simply measures the heavy-tailed marginal of F's entries. Any grouping would score the same. This is the same failure class as Prop. (strand hierarchy is contiguity) in §2.11, and it removes the Fisher leg of the dual-manifold proposal.

Observation 39.8 (Role-selective state, and where it is needed)**.** *Coarsening* v *to one scalar per matrix for one role group at a time, EMB always per-coordinate:*

coarsened	*final*	*vs. AdamW*	cos
none	0.2679 ± 0.0067	1.000	1.000
W_Q, W_K	0.2656 ± 0.0088	0.991	0.980
W_V, W_O	0.2889 ± 0.0207	1.078	0.989
FF	0.3170 ± 0.0151	1.183	0.806
all	0.3519 ± 0.0199	1.314	0.766

Q/K — whose carrier contracts most dramatically — need no per-coordinate state at all. FF, which has no comparable geometry, carries most of the cost. Geometric motion and state requirement are anticorrelated.

The FF cost is not localised in time. Coarsening FF's v inside any single 20-step window costs nothing (0.959–1.027 against a seed spread of 0.010), while coarsening it for all 160 steps costs 18.3%. The penalty is integrated, not incurred at a transition, and nothing happens at the staircase crossings.

39.3 The carrier, and what creates it

Parameter-space subspace tracking has no dynamic range here: spectral $\sin\theta_{\max}$ between consecutive 8-step update frames is 0.97–1.00 for every role at every checkpoint, under both the Euclidean and row-whitened metrics. Frobenius sits at 2.03 of a maximum 2.449 and nuclear at 2.3 of 3. The measurement is saturated and cannot show formation, stabilisation, or velocity spikes. Activation space can.

Observation 39.9 (Q/K contract, V does not, attention does not change)**.** *Participation ratio on a fixed probe batch, block 0, three seeds:* $\mathrm{PR}(q)$ *falls* $32.7 \to 1.8$ *and* $\mathrm{PR}(k)$ $32.9 \to 2.0$*, while* $\mathrm{PR}(v)$ *goes* $35.1 \to 21.5$ *and is flat after step* 40*. Embedding PR* rises $46.3 \to 51.6$*. Attention entropy is* 3.206 *nats at step 0 and* 3.204 *at step 160, against a uniform ceiling of* 4.159*; peak attention weight moves* $0.074 \to 0.078$*. The matching space collapses while the pattern it produces does not change.*

Observation 39.10 (Heads keep their identity and share their carrier)**.** *Taking each head's subspace of the residual stream as* $\mathrm{row}(W_Q^h) \in \mathrm{Gr}(3, 128)$*: the optimal assignment between consecutive checkpoints is the identity in* 9/9 *intervals and* 3/3 *seeds, so heads never exchange roles. Within-head motion falls* $1.52 \to 0.19$*. But cross-head overlap rises* $0.027 \to 0.415$ *against a random-frame null of* $k/D = 0.0234$*. Four heads converge onto one shared carrier without swapping.*

Observation 39.11 (Training moves the potential carrier onto the realised one)**.** *Cross-head overlap of the* excited *subspace is* 0.474 *at step 0 while the potential carrier* $\mathrm{row}(W_Q)$ *sits at the random null. Agreement between a head's potential and realised subspace climbs* $0.103 \to 0.600$*.*

Proposition 39.12 (Most of the step-0 confinement is Phase 1's)**.** *With a random* $\mathcal{N}(0, 0.02^2)$ *embedding, realised overlap at step 0 is* 0.165 *rather than* 0.435*; the stable rank of* E *is* 70.6 *rather than* 24.3*. Shuffling the spectral embedding's rows — same spectrum, same row norms, token identity destroyed — gives* 0.448*. The confinement is a property of the initialiser's spectrum, not of the corpus.*

Observation 39.13 (But the trajectory is created by training)**.** *Under the random embedding,* $\mathrm{PR}(q)$ *still falls* $46.8 \to 7.1$ *and potential-to-realised agreement still climbs* $0.199 \to 0.595$*, landing at the same endpoint as the spectral arm (*0.588*). Realised overlap* builds $0.178 \to 0.577$ *where the spectral arm was handed it. Carrier alignment is produced by training; only its starting point and rate are inherited.* $\mathrm{PR}(q)$*'s endpoint is not invariant (*2.0 *vs* 7.1 *at matched loss), so the carrier's identity is a more robust object than its dimension.*

39.4 The functional staircase

The corpus is one 1364-token string repeated, so each position has a smallest context order that determines its continuation uniquely. Bucketing by that order gives a functional coordinate system that is not a geometry.

Observation 39.14 (Acquisition is ordered, and has two steps, not four)**.** *Five seeds,* $G_n(t) = L_{n-1}(t) - L_n(t)$ *with absolute positions held fixed so context length is not confounded with the positional embedding:* $\tau_2 \in \{20, 30, 40\}$*,* $\tau_3 = 60$ *in* 4/5 *seeds,* $\tau_5 = 60$ *in* 5/5*, and* τ_7 *absent in* 3/5 *with* G_7 *flat at* ~ 0.048 *after step 100.* G_3 *and* G_5 *coincide to a few thousandths throughout, so they are one event. Real-minus-shuffled is positive, monotone and ordered* $dG_2 > dG_3 > dG_5 > dG_7$ *at every checkpoint from step 30. The threshold-free signal* $V_n(t)$ *shows no spike anywhere:* V_2 *has a broad plateau over steps* 80–110 *and* V_3, V_5, V_7 *are flat within their error bars.*

Observation 39.15 (Attention supplies reach; FF supplies the computation)**.** *Accuracy at step* 160 *by determinacy bucket, three seeds, all blocks:*

ablation	*order 1*	*order 2*	*order 3*
none	1.000	0.968	0.962
FF removed	0.130	0.002	0.000
attention removed	0.973	0.363	0.179
3 of 4 heads	0.995	0.667	0.564

Attention is nearly irrelevant to order 1 and essential to order 3 — the opposite of the routing-then-refinement reading. FF is load-bearing at every order. Head redundancy is order-dependent, which explains why block-0 ablation scored $I_h \approx 0.001$*: aggregate loss is dominated by order-1 positions, the bucket where heads do not matter.*

39.5 Unwinding, and the separation of the three systems

Observation 39.16 (Directed ascent destroys; matched noise does not)**.** *From step* 160*: gradient ascent takes val* $0.164 \to 4.52$ *by descent step 5, with buckets* 0.295/0.155/0.103*. Isotropic noise matched per step to the same* $\|\Delta\theta\|$ *leaves val at* 0.168 *and every bucket unchanged after* 40 *steps. Throughout the collapse* $\mathrm{PR}(q)$ *moves* $1.82 \to 2.10$*.*

The carrier survives complete functional collapse. Whatever distinguishes the orders is not carried by the geometry that training spent 160 steps organising.

Observation 39.17 (The orders have private directions)**.** *Ascending on one bucket's probe positions only, with fit sets equalised at* 13 *examples and the direction renormalised to a fixed* $\|v\|$*, damage at descent step* 40 *as a fraction of the starting level:*

ascend on	$\to O_1$	$\to O_2$	$\to O_3$
O_1	0.095	0.038	0.041
O_2	0.005	0.032	0.041
O_3	0.010	0.022	0.280

Every row is diagonal-dominant. There is a direction that erases the model's highest-order capability while leaving lower orders and the carrier intact.

Proposition 39.18 (The functional directions are separate, not nested)**.** *Bucket gradient norms at step* 160 *are* $\|g_1\| = 0.185$*,* $\|g_2\| = 0.462$*,* $\|g_3\| = 1.685$ *—* O_1 *is solved and its gradient is nearly exhausted. Each bucket's gradient lies* 0.980/0.849/0.758 *inside its own rank-4 subspace and* 0.004–0.036 *inside the others, against a random-subspace null of* 10^{-5}*; pairwise cosines are* $0.170, 0.066, 0.056$*.*

An earlier, unequalised version of Obs. 39.17 gave a flat O_1 row and was read as a filtration $S_3 \subset S_2 \subset S_1$ with O_1 as a shared prerequisite. That reading is withdrawn. With matched examples and matched step norm the O_1 row is diagonal-dominant like the others, and the gradients are near-orthogonal rather than nested. The original asymmetry was gradient scale amplified by Adam's per-coordinate normalisation.

39.6 Status

claim	verdict	evidence
the update has a privileged basis	refuted	0.918 vs 0.913 under conjugation
stratified/hypercube/ℓ_∞/matroid	refuted	all coordinate-dependent
low rank is below its generic null	refuted	0.50 vs 0.25; it is above
4-bit beats uncompressed	retracted	1.018 at 3 seeds, gap was $<$ sd
8 bits free, 4 bits $\sim 2\%$	holds	monotone, no knee
RMSProp halves the state for free	retracted	4.16× worse at matched norm
v shares along either axis at $\sim 2\%$	holds	norm and cos controlled
v per matrix costs 31%	holds	cos 0.766, $\|\Delta\theta\|$ 0.925
the row unit is the output unit	refuted	entire gap is `te.weight`
Fisher has row structure	refuted	shuffled null gives 0.608 of 0.650
FF carries the state requirement	holds	1.183 alone of 1.314 total
Q/K carrier contracts and aligns	holds	survives a random embedding
step-0 confinement is the corpus	refuted	it is Phase 1's spectrum
heads exchange roles	refuted	assignment is identity 9/9
attention is routing, FF refinement	refuted	attention carries the high orders
$\tau_2 < \tau_3 \approx \tau_5$	holds	5 seeds, sd ≤ 0.5
τ_7 exists	refuted	absent in 3/5 seeds
O_3 has a private direction	holds	equalised $\|B\|$ and $\|v\|$
$S_3 \subset S_2 \subset S_1$	retracted	gradients near-orthogonal
the Fisher sheet can be state	refuted	half-life 2–4 steps

The Fisher sheet's memory. §35 establishes that the sheet delivers 0.75 ± 0.30 of the loss reduction from 5–15% of the update energy, against a random-subspace control 607,881× worse. That is not touched by Prop. 39.7, which measures a different object. But retention against *trajectory* staleness — a range finder built from 12 fresh gradients at an anchor, applied without rebuilding — falls to half its anchor value in 2–4 steps at anchors 40, 90 and 140, then plateaus at 30–60× the random floor. It costs 12 backward passes to build and stays useful for about three steps. The sheet is a local property of the loss surface, recomputable anywhere and transportable nowhere; it is information, not state.

The three coordinate systems. The phase's one durable finding is a non-coincidence. The representation carrier contracts by 16× and aligns; the optimiser's state requirement is concentrated in FF, which has no carrier story; and functional capability has directions that survive the carrier's destruction and are invisible to it. Each was established by intervention with a matched control. Every attempt to identify two of them — Fisher with Grassmann, state with carrier, staircase with subspace — failed, and failed on a different measurement each time.

Methodological residue. Three traps recurred often enough to be stated generally. First, *any operator or state change that alters $\|\Delta\theta\|$ is a learning-rate change until proven otherwise*: this session produced it in both directions and twice reversed a headline. Second, *check that the null outcome is achievable before reading a null outcome* — a held-out object whose answer token never appears in that slot scores zero for reasons unrelated to the hypothesis, and a least-squares map fitted on 16 points in 128 dimensions interpolates exactly by construction. Third, *no geometric interpretation without a shape-matched null*: report $S_{\text{obs}} - S_{\text{matched}}$, not S_{obs}. Prop. 39.7 would have read as a finding without it, as would ρ's ancestor in §2.11.

Open, and blocking. AdamW at 2.91× the learning rate used throughout reaches 0.1308 against 0.2679. Every comparison in §38 and §39 is therefore measured in an under-stepped regime, and η^* has not been located. Differences of 2% between arms at a badly chosen η do not establish that the arms differ at η^*. This should be settled before the compression frontier is quoted.

40 Phase 38: An Observable, a Gate, and Three Withdrawals

Phase 37 ended with three organisations that do not coincide. This phase asked whether a fourth — a relational or topological object — sits underneath them. The answer is no, and the interesting part is the sequence of controls that got there. Four candidate structures were proposed, measured, and withdrawn; one observable passed its gate and then turned out to describe the initialisation rather than the learned function.

The phase's durable contribution is procedural. A test-retest calibration was made a precondition rather than a postmortem, and it changed the verdict on every claim it was applied to.

40.1 The gate

Proposition 40.1 (An observable must reproduce itself first)**.** *For a state-characterising observable X, measure three numbers before computing anything with it: a* ceiling $\rho(X(\theta)^a, X(\theta)^b)$ *from the same state under two independent probe draws, a* floor $\rho(X(\theta_{s_1}), X(\theta_{s_2}))$ *across independently trained seeds, and a shape-matched* null*. Two observables were gated:*

observable	*ceiling*	*floor*	*null*	*verdict*
Φ, the O3 response map	$+0.071$	$+0.272$	$+0.001 \pm 0.176$	*discard*
A, the intervention graph	$+0.696$	$+0.276$	-0.000 ± 0.017	*proceed*

Φ's ceiling lies below *its floor and inside the null's spread: it cannot reproduce the same state under two probe batches, so correlations between different states carry no information.*

Everything computed from Φ is therefore withdrawn: a claimed persistent "functional germ" at $\rho = 0.993$, an ascent-versus-descent clustering, an O_2/O_3 coupling at $\rho = 0.609$, and the reading that a transition law survives destruction of capability. Two upstream bugs produced them. The probe family was rebuilt *at each state* from that state's own gradient, so two objects were compared under two different instruments rather than one; and the response was O3 accuracy on 13 held-out positions, whose granularity is 1/13 and whose variation is dominated by which positions were drawn.

40.2 What the gated observable measures

A_{ij} is the change at output unit j of block 0's feed-forward layer under a norm-matched perturbation of source row i, on a probe batch external to every state. Its useful component is not the correlation but the *support*: the top-quartile target set S_i of each row, compared by mean Jaccard J.

Observation 40.2 (Support is the informative component)**.** *Decomposing persistence against C_3 into magnitude, direction and support:*

state	*magnitude*	*directional*	*support J*
$t = 60$	$+0.777$	$+0.357$	0.429
$t = 120$	$+0.957$	$+0.457$	0.630
U_3	$+0.994$	$+0.472$	0.812

*Magnitude is saturated (*0.777 *even a hundred steps earlier) and discriminates nothing; direction sits at the reliability ceiling and cannot be resolved further. Support has range and a null of* 0.144 *(size-matched) or* 0.180 ± 0.032 *(row-permuted, the stronger form). It survives threshold variation (excess* $+0.67$ *at* $q = 0.10, 0.20, 0.25$*), holds in* 24/24 *rows, and its temporal version reproduces across seeds (*0.429 *against* 0.423*).*

40.3 Four withdrawals

Proposition 40.3 (The support geometry is not ultrametric). *With* $d(i,j) = 1 - |S_i \cap S_j|/|S_i \cup S_j|$*, the fraction of triples satisfying* $d(i,k) \le \max\{d(i,j), d(j,k)\}$ *is* 0.726–0.737 *at every state measured, against* 0.732 ± 0.006 *for size-matched random supports. All* $|z| \le 1.23$*. The null has sd* 0.006*, so this is a genuine absence rather than an underpowered test.*

The raw fraction of 0.73 would read as "mostly ultrametric" without the null: Jaccard distances among 24 nodes with 32-element supports concentrate, and concentration alone nearly satisfies the inequality. This is the same failure mode as §2.11, where the strand hierarchy vanished under an index permutation.

Proposition 40.4 (Unlearning does not specifically preserve support). *At matched displacement* $\|\Delta\theta\| = 2.667$ *from* C_3*: O3-ascent gives* $J = 0.795$ *with* $O_3 = 0.308$*; O1-ascent gives* 0.796 *with* $O_3 = 0.000$*; isotropic noise gives* 0.829 *with* $O_3 = 1.000$*, i.e. undamaged. The three are indistinguishable and the undamaged state scores highest.*

$J = 0.812$ was displacement magnitude. Support at this layer is inert to perturbations of size $\sim$2.7 against a formation displacement of 23.6.

A partial recovery, and its own correction. A radius sweep suggested the inertia is direction-dependent — forward training reaching $J = 0.409$ at $r = 24$ where ascent held 0.744. But the ascent rule takes a fixed step $\varepsilon = 0.02$ along the normalised gradient, so 140 ascent steps cover cumulative displacement 2.26 while forward training covers 17.54. The sweep's "ascent at $r = 24$" never reached that radius. The comparison stands only where both paths were sampled, and there they nearly coincide (0.780 against 0.793 at $r = 2.67$). The step-norm diagnostic intended to separate structural inertia from optimisation saturation was also vacuous, being a constant of the method; the gradient norm answers it instead, running $6.9 \to 10.6 \to 7.4$ with no collapse.

Proposition 40.5 (The p-adic entropy statistic is permutation-invariant). H *of the strata of* $v_p(count)$ *is identical for the observed token counts and for the same counts randomly reassigned to tokens: sd* $= 0.000$ *at* $p = 2, 3, 5, 7, 11$*. A permutation preserves the multiset of counts and* v_p *partitions that multiset, so the statistic cannot depend on token identity. It is also invariant to the corpus repetition factor* $270 = 2 \cdot 3^3 \cdot 5$*, which shifts every valuation by a constant.*

The statistic declared STRUCTURED at every prime including $p = 11$, where all 1017 tokens occupy one stratum and $H = -0.000$: it fires on degeneracy. This is untestable as formulated rather than merely unsupported.

Proposition 40.6 (The arity ladder is unidentifiable above low k). *Fitting* m_k *on a window of* 64 *hidden states, the* $(k-1)d$ *parameters exceed the available samples for most* (d,k)*. At* $d = 16$ *only* $k = 2$ *has samples/parameters* > 2*; at* $d = 48$*, the published setting, nothing above* $k = 2$ *does. Where the fit is identifiable, held-out relative error on a* 70/30 *split is* 0.82–1.51 *— no better than predicting zero, and worse at* $k = 8$*.*

40.4 The construction control

Every observable above was measured on one architecture, so nothing separated a learned relation from one inherited from the pathway. Three arms, identical in optimiser, token stream, step count, learning rate and initialisation seed: BASE; FROZENK, with W_K held at initialisation so the key side cannot learn; and FIXEDATTN, with the attention weights replaced by a frozen random causal pattern.

Observation 40.7 (The support object is inherited, not learned).

arm	O_1	O_2	O_3	$J(60,160)$	κ
base	1.000	0.917	1.000	0.426	+0.0286
frozenK	1.000	0.942	0.846	0.425	+0.0574
fixedattn	1.000	0.925	0.846	0.414	+0.0431

Cross-arm J at step 160, *with unit indices corresponding because the arms share an initialisation: base–frozenK* 0.682, *base–fixedattn* 0.633, *frozenK–fixedattn* 0.777. Within *an arm, across seeds, J is* 0.257–0.267.

Two models with entirely different attention mechanisms share 2.5× more support structure than two models with identical architecture and different seeds. The interventions were not inert — O_3 falls from 1.000 to 0.846 in both null arms while O_1 is untouched, matching Obs. 39.15 where attention carried the high orders and not the low — yet the support object is unmoved.

The support invariant is therefore a property of the initial weights, carried along by training rather than created by it. It survives unlearning, survives freezing the key side, survives destroying attention, and does not survive changing the seed. The earlier reading that training makes the graph progressively more seed-specific (S_t: 0.342 → 0.233) had the emphasis backwards: the graph begins seed-specific because it was never a learned object.

40.5 Status

claim	verdict	evidence
Φ identifies an O3 transition law	discarded	ceiling 0.07 < floor 0.27
functional germ persists under pullback	withdrawn	rests on Φ
ascent/descent clustering in Φ	withdrawn	rests on Φ
O_2/O_3 coupling at $\rho = 0.609$	withdrawn	rests on Φ
A is a measurable observable	holds	$0.696 \gg 0.276 \gg 0.000$
support is its informative component	holds	six robustness checks
support geometry is ultrametric	refuted	$z \leq 1.23$, null sd 0.006
unlearning preserves support	refuted	noise at matched $\|d\|$ scores higher
J is direction-dependent at large r	withdrawn	radii were not matched
p-adic corpus strata are structured	refuted	permutation-invariant statistic
arity ladder above $k = 2$	refuted	unidentifiable; held-out error ≈ 1
QKV→ATTN generates the support	refuted	cross-arm J 0.63–0.78
the support object is learned	refuted	cross-seed J 0.26

Methodological residue. A fourth trap joins the three of §39.6: *establish test-retest reliability of an observable before computing anything with it.* Φ consumed six rounds and produced four retracted claims because the calibration was run last. Three further instances of the pattern already named — a statistic that cannot depend on what it claims to measure — appear here as Props. 40.5, 40.3 and 40.6, and one new corollary is worth stating: *a null must destroy the specific property claimed and preserve everything else.* A size-matched random support reproduces cardinality only; the row-permuted null, which keeps every support set intact and destroys only which unit owns it, is the one that should be quoted.

What survives the phase. One gated observable, whose informative component is support; a demonstration that this component is inherited from initialisation rather than learned; and the negative result that neither ultrametric structure, nor a p-adic stratification, nor an arity ladder above $k = 2$ exists in these measurements. The three-organisation separation of §39 is unchanged: nothing here identifies capability with representation geometry, with optimiser-state requirement, or with each other.

Still blocking. $\eta^* = 2\times$ the rate used throughout, giving 0.124 against 0.265, and the full pipeline at $D = 256$ reaches 0.088 where the $D = 128$ slice used here reaches 0.26. Both scale parameters are wrong in the same direction, and no result in §38– §40 has been re-measured at either corrected value.

41 Phase 39: Two Observables, Two Ambient Fields

Phase 38 left one gated observable whose informative component is support, and a construction control showing that component is inherited from initialisation. This phase asks what the four candidate observables — capability O, support J, Fisher F, curvature H — actually measure once each has passed the gate of Prop. 40.1. Two survive as measurements. Two turn out to be approximately constant over the region where the first two vary.

41.1 The two fields fail the gate in opposite ways

Proposition 41.1 (The curvature field is insensitive, not noisy). *For $h_i = (Hg_3)_i$ estimated from B batches, comparing the same state under two independent draws (ceiling) against a state displaced by $r = 16$ (floor):*

B	*ceiling*	*floor*	*separation*
1	0.460	0.305	+0.156
4	0.396	0.582	−0.186
16	0.910	0.798	+0.112
64	0.948	0.886	+0.062

Averaging fixes the reliability — the ceiling climbs to 0.948 *— but the floor climbs with it and the separation* falls. *At $B = 64$ the field at C_3 and at a displacement of* 16 *agree to* 0.886 *against a same-state ceiling of* 0.948.

Proposition 41.2 (The Fisher field is genuinely invariant). *The same gate on $\log F_i$: ceiling* $0.9878, 0.9970, 0.9992$ *at $B = 4, 16, 64$, with floors* $0.9861, 0.9965, 0.9985$. *Separation is* $0.0018, 0.0006, 0.0007$. *The estimator is near-perfect and the field does not move.*

Together these retire the reading of §39 in which F was an ambient metric and H a state-dependent realisation. The $d_h \approx 0.9$ values reported there were the $B = 1$ noise floor: $d_H(C_3, C_3) = 0.395$ under one HVP, so nothing below that scale was interpretable, and the conclusion that curvature is state-specific is withdrawn. Both fields are ambient over the range tested.

Observation 41.3 (The scalar contraction is usable, but only for large gaps). $\kappa(v) = v^\top Hv/\|v\|^2$ *at C_3, ten independent draws per B: mean* $+0.0698, +0.1023, +0.0953, +0.0916$ *with relative sd* $0.50, 0.25, 0.19, 0.08$ *at $B = 1, 4, 16, 64$.*

At $B = 1$ the noise is half the value. So $\kappa_{\text{learn}} = 0.0002$ against $\kappa_{\text{relearn}} = 0.289$ survives — a factor of 1400 is far outside a ±50% band — while two smaller claims do not. The α-sweep readings $0.033, 0.179, 0.056, 0.197, 0.765$ are within noise for every pair but the extremes, so their non-monotonicity was measurement error; and κ doubling under FROZENK ($0.0286 \to 0.0574$) is withdrawn. The surviving statement is directional, not field-level: particular tangent directions differ sharply in curvature while the field they pass through is locally persistent.

41.2 The stability radii

Observation 41.4 (Support is more perturbation-sensitive than capability). *Isotropic displacement of radius r from C_3:*

r	J	O_3	d_F	d_h
0	1.000	1.000	0.005	0.395
2	0.870	1.000	0.005	0.888
8	0.676	1.000	0.005	0.433
16	0.495	0.923	0.005	0.791
32	0.297	0.769	0.007	0.969

$r_{50}(J) = 16$ *against* $r_{50}(O_3) = 32$. *J reaches the row-permuted null (*0.181*) near* $r = 32$ *where O_3 is still* 0.769. *The d_h column is uninterpretable per Prop. 41.1 — its* $r = 0$ *entry is* 0.395 *— and d_F is flat.*

This is the reverse of the reading suggested by the intervention results. Under perturbation the substrate degrades *first*. Combined with Prop. 40.4, where isotropic noise at $r = 2.667$ left O_3 at 1.000 while J had already fallen to 0.829, the pattern is consistent and was visible earlier in a column that went unread.

41.3 The fingerprint of J, and the joint map

Collecting the interventions, J is: construction-invariant (cross-arm 0.63–0.78 at $q = 0.25$, and 0.556–0.721 at $q = 0.10$, against a row-permuted null of 0.088 ± 0.042); function-invariant (unchanged when O_3 falls from 1.000 to 0.846 under FROZENK); seed-specific (cross-seed $0.134, 0.267, 0.434$ at $q = 0.10, 0.25, 0.50$, barely above the row-permuted null at every threshold); and perturbation-sensitive ($r_{50} = 16$).

The cross-arm to cross-seed ratio is largest at the *finest* threshold — 4.7× at $q = 0.10$ falling to 1.66× at $q = 0.50$ — so the outcome is not "coarse structure survives, fine structure is construction-dependent"; the fine supports are the most construction-invariant and the most seed-specific.

That combination does not support calling J a substrate. The defensible statement is that J is an initialisation-conditioned relational coordinate structure, locally persistent under changes to the training rule and deformable under displacement of the parameter point.

Observation 41.5 (Capability and support partition parameter space differently)**.** *Both directions of non-implication are measured.* $O_3(C_2) = O_3(U_3) = 0.308$ *while* $J(C_2, C_3) = 0.408$ *and* $J(U_3, C_3) = 0.795$*: equal capability, different support.* $O_3(\text{BASE}) = 1.000$ *against* $O_3(\text{FROZENK}) = 0.846$ *while their supports agree at* 0.682*, far above the cross-seed* 0.267*: different capability, same support.*

So $\pi_O : \Theta \to \mathcal{O}$ and $\pi_J : \Theta \to \mathcal{J}$ are two experimentally constructed projections with different fibers, and the object worth studying is the joint map $\Pi : \Theta \to \mathcal{O} \times \mathcal{J}$. This is the fiber language of §40.3 made two-sided; it does not require and does not supply a categorical formalism.

41.4 Status

claim	verdict	evidence
F is an ambient metric	holds	separation 0.0007 at $B = 64$
H is a state-dependent realisation	withdrawn	separation 0.062, ceiling 0.948
$d_h \approx 0.9$ shows curvature moves	withdrawn	B=1 noise floor
κ learn vs relearn, 1400×	holds	outside a ±50% band
κ non-monotone in α	withdrawn	within noise at B=1
κ doubles under frozenK	withdrawn	within noise at B=1
J is a stable substrate	refuted	$r_{50}(J) < r_{50}(O_3)$
J construction-invariant	holds	12–13σ at every q
J seed-specific	holds	near null at every q
O and J have different fibers	holds	both directions measured

What the programme has converged to. Over the region probed, the state carries two measurable structures, capability and relational support, which transformations can move independently in either direction. The local first-order and second-order fields are approximately constant there, so they calibrate the region rather than distinguishing states within it; only the directional contraction $\kappa(v)$ varies, and only across gaps of several hundred-fold. No ultrametric organisation, no p-adic stratification, no arity ladder above $k = 2$, and no categorical structure has been established — each was measured against a matched null and failed.

Methodological residue. The fourth trap of §40.5 generalises once more. Reliability is not a property of an observable but of an observable *at a given estimator budget*: h is uninterpretable at $B = 1$, reliable at $B = 64$, and discriminating at neither. The gate must therefore be run at the budget the experiment will use, and re-run when that budget changes. Three claims in §39 and §40 were made from $B = 1$ estimates whose noise exceeded the effects they reported.

42 Phase 40: Transplantation, and the Orientation of a Transformation

Phases 37–39 established three organisations that do not coincide and left the optimiser question where it started: no measured quantity told the optimiser what to do differently. This phase changes the experimental object. Instead of asking what a state contains, it asks what can be moved from one network to another — and the answer turns out to be a one-bit field.

Every arm below is reported against a *dose-matched null*: an intervention of identical size, norm, layer, receiver and training budget with the teacher-specific content destroyed. That protocol was adopted mid-phase, after it overturned the phase's own first conclusion, and it is the reason the surviving results are worth stating.

42.1 The transplant, and what carries it

A teacher is trained to 200 steps with snapshots along the way; a student on an independent seed is trained to 30; block 3 of the teacher is copied into *all six* student blocks and the student continues for 30 more steps, probed at $k = 0, 1, 2, 5, 10, 20, 30$.

Observation 42.1 (Weights transfer; optimiser state does not)**.** *From the* t=120 *donor, at* k=30*: transplanting block-3 weights gives* $L = 1.072$ *and* $O_3 = 0.654$*; transplanting block-3* (m, v) *alone gives* $L = 1.454$*,* $O_3 = 0.269$*; the untouched control gives* $L = 1.386$*,* $O_3 = 0.269$*. Adam state alone is indistinguishable from doing nothing.*

Observation 42.2 (The donor becomes transferable between t=60 and t=100)**.** O_3 *at* k=30 *by donor checkpoint, against a control of* 0.269*:* t=40 $\to$ 0.269*,* t=60 $\to$ 0.269*,* t=80 $\to$ 0.462*,* t=100 $\to$ 0.615*,* t=120 $\to$ 0.654*. Loss separates earlier than* O_3*: the* t=60 *donor improves loss (*1.287 *against* 1.386*) while transferring no* O_3*.*

The transplanted layer then barely moves: $\|\Delta W_3\|$ over the 30 post-transplant steps is $10.70 \to 11.67$, against the control's $0 \to 4.56$. The gain is present at k=0 and the receiving network adapts around a nearly static donor rather than extending a head start.

42.2 Deforming the donor

Ascent on the O_3 bucket applied to the *whole* teacher spreads its displacement almost uniformly — block-3 share 0.313 against a uniform expectation of 0.408, and $\|\Delta W_3\|/\|W_3\| = 1.01\%$ — so it never challenges the donor layer, and the resulting transplant is indistinguishable from the intact one. Restricting the ascent to block 3 makes the deformation surgical (share 1.000) and gives a dose–response curve.

Observation 42.3 (The donor loses capability faster than it loses transfer). *With* $\delta = \|W_3^U - W_3^{120}\|/\|W_3^{120}\|$*:*

δ	*donor* O_3	*transferred* O_3	*student* L
0.000	0.731	0.654	1.072
0.021	0.654	0.692	1.077
0.053	0.462	0.577	1.100
0.095	0.308	0.423	1.139

At $\delta = 0.053$ *the donor is at* 0.462 *while transferring* 0.577*: the layer confers more capability on a student than it expresses in its own network.*

42.3 The null that overturned the phase

The dose curve and the donor ladder were read, at the time, as evidence that a relational orientation field $S = \mathrm{sgn}(\Delta W_3)$ becomes transferable before the capability itself: at t=60 a norm-matched S transferred $O_3 = 0.423$ where the full weights transferred 0.269. A matched-norm *random* orientation was then run.

Proposition 42.4 (Receiver susceptibility). *At* t=60*, norm-matched random orientation gives* $L = 1.264$*,* $O_3 = 0.385$ *against the true field's* $L = 1.269$*,* $O_3 = 0.423$ *— a difference of one probe position — with both far above the control's* 1.386*,* 0.269*.*

A norm-matched perturbation of block 3 is by itself sufficient to change what the student learns in the following 30 steps. The recurring value 0.423 across four unrelated arms was this effect, not transferred content. Every early-donor reading measured against the control rather than against dose is therefore withdrawn, including the precursor interpretation.

Observation 42.5 (Teacher-specific content exists at t=120). *At* t=120*: true orientation* 0.692*, random orientation* 0.462*, control* 0.269*. The intervention effect* 0.423 *divides into a receiver component* 0.193 *and a teacher-specific component* 0.230*, roughly equal.*

42.4 The decomposition ladder

Writing $\Delta W_3 = W_3^T - W_3^S$, each candidate representation is paired with a null that preserves its shape, size and dose while destroying the property under test. The reported quantity is $\Delta O_3 = O_3(R) - O_3(N_R)$.

representation	isolates	size	O_3 / null	ΔO_3
full ΔW	everything	16384	0.654 / —	—
$\mathrm{sgn}(\Delta W) + \alpha$	orientation	1 bit/param	0.692 / 0.462	+0.230
support+sign 10%	location + orientation	0.1 bit/param	0.615 / 0.462	+0.153
support+sign 5%	”	0.05 bit/param	0.500 / 0.423	+0.077
support+sign 1%	”	0.01 bit/param	0.462 / 0.423	+0.039
low rank r=16	coupled directions	4096	0.500 / 0.462	+0.038
magnitude only $\lvert\Delta W\rvert$	magnitude organisation	16384	0.269 / 0.346	−0.077

Observation 42.6 (Orientation carries it; magnitude and rank do not). $\mathrm{sgn}(\Delta W)+\alpha$ *reproduces the full transplant (*0.692 *against* 0.654*) at one bit per parameter. Low rank at* 25% *of the parameters gives* +0.038 *with its loss* worse *than its own rank-matched null. Magnitude with orientation stripped gives* −0.077*: it lands exactly at the control and below a magnitude-shuffled null.*

The support ladder decays smoothly — $+0.230, +0.153, +0.077, +0.039$ at $100, 10, 5, 1$ percent — with no threshold. The teacher-specific content is distributed across the orientation field rather than concentrated in a small set of coordinates, which rules out a sparse-code reading.

Proposition 42.7 (The residual is not orientation-free)**.** *For* $\epsilon = \Delta W - \alpha S$ *with* $\alpha = \text{mean}\,|\Delta W|$*, one has* $\epsilon_{ij} = S_{ij}(|\Delta W_{ij}| - \alpha)$*, so* $\text{sgn}(\epsilon)$ *is a deterministic function of* S*: measured agreement* 0.4260*, exactly the fraction of coordinates with* $|\Delta W| > \alpha$*. An earlier arm reporting* $\Delta O_3 = +0.153$ *for* ϵ*, and the conclusion that orientation is insufficient, are withdrawn.*

42.5 Prediction is not sufficiency

If the next transformation were predictable from the state at t, the later Adam steps might be replaced by a generator. The precondition is checkable from snapshots alone.

Observation 42.8 (The morphism is locally predictable and the predictability decays)**.** *Agreement between* sgn *fields of successive intervals, against a coordinate-shuffled null of* 0.500*:* $S(60, 80) \to S(80, 100)$ *gives* 0.684*;* $S(80, 100) \to S(100, 120)$ *gives* 0.662*;* $S(40, 100) \to S(100, 120)$ *gives* 0.631*;* $S(0, 100) \to S(100, 120)$ *gives* 0.589*. Shorter recent intervals predict better than longer or cumulative ones. Adam momentum is the best free predictor at every point:* $\text{sgn}(-m_t)$ *against the next interval gives* $0.707, 0.659, 0.627$ *at* $t = 80, 100, 120$*.*

Proposition 42.9 (A 60%-accurate orientation field is worse than useless)**.** *Transplanting* $\text{sgn}(-m)$ *norm-matched exactly as* S *was gives* $L = 1.319$*,* $O_3 = 0.269$*: at the control, and* below *its dose-matched random null (*1.255, 0.462*).* $\Delta O_3 = -0.193$*, with loss agreeing.*

Sign agreement and functional sufficiency are therefore distinct axes, and the mapping between them is steep: 1.000 agreement gives +0.230, 0.601 agreement gives −0.193. The benchmark proposed for a generator — beat momentum's 0.659 — was the wrong target, since 0.659 corresponds to negative transfer. The generator programme in this form is closed.

Momentum is thus neither the carrier (agreement with S: 0.601; transfer 0.462 against S's 0.692) nor a usable substitute for it, while being the best available *predictor* of the next interval. Retrospective and prospective objects are different, and the earlier reading that one result settled both was wrong.

42.6 Status

claim	verdict	evidence
block-3 weights transfer capability	holds	0.654 vs control 0.269
Adam state alone transfers	refuted	0.269, = control
transferability emerges 60–100	holds	donor ladder, five points
donor loses capability faster than transfer	holds	δ curve, four doses
orientation transferable before capability	withdrawn	random matches real at t=60
receiver susceptibility is large	holds	+0.193 of +0.423
$\text{sgn}(\Delta W) + \alpha$ carries the rest	holds	+0.230, 1 bit/param
content is sparse	refuted	smooth decay, no threshold
low rank carries content	refuted	+0.038, $\Delta L < 0$
magnitude carries content	refuted	−0.077
ϵ is an independent carrier	withdrawn	$\text{sgn}\,\epsilon = f(S)$
next morphism is predictable	holds	0.68 vs null 0.50
predicted orientation transfers	refuted	−0.193

What survives. One object: $R = (\text{sgn}\,\Delta W_3, \alpha)$, one bit per parameter plus one scalar, carrying +0.230 of teacher-specific O_3 against its dose-matched null and reproducing the full weight transplant. Its content is distributed rather than sparse, is not low-rank, and is not present in the magnitudes.

What the phase costs the optimiser programme. Adam state does not transfer, momentum does not carry the object, and a momentum-derived prediction of the next morphism is worse than random at the same dose. The reduced-Adam question is therefore not "what compresses

m and v" but "what computable quantity at $t \approx 100$ produces a functionally valid relational configuration" — and no candidate tested here does.

Methodological residue. A fifth trap, and the one that governs the others: *the null runs in the same invocation as the measurement, not after it.* The orientation result was interpreted across roughly fifteen exchanges before its dose-matched null was run, and the null then removed the central claim. Where memory forces one arm per process, the null arm runs first, so that a truncated run still yields the falsifying half. A result whose null has not completed is *unverified*, not provisional.

Scope. $D = 128$ throughout: a single $D = 256$ arm was terminated by the container's 3 GB limit mid-run. At native width, η^* is the rate already in use — 0.0877, 0.0971 and divergence at $\times 5$, $\times 8$, $\times 12$ — so the "every measurement is at half the optimal rate" caveat attached to Phases 36–39 was itself an artefact of the reduced width and is withdrawn. All O_3 figures rest on 26 held-out probe positions and a single student seed; the monotone orderings are firmer than any individual entry.

43 Phase 41: Compressed Optimiser State in the Compiler

Phase 40 established that the transferable object lives in the weights and that Adam's accumulated state carries none of it. This phase asks the engineering question that follows: how little optimiser state does Phase 3 actually need, and does a compressed optimiser hand off cleanly to Phases 4 and 5.

The answer is a 15.6× state reduction that runs the full pipeline end to end and beats the GD-400 baseline, at a cost to final loss that is an order of magnitude larger than the reduced-width measurements predicted. Both facts are reported; the second is the more useful one.

43.1 Which axes of the optimiser state compress

All arms below run 200 steps at $D = 128$ from an identical initialisation and batch stream, at Adam's own learning rate, so each is a drop-in replacement rather than a retuned optimiser.

Observation 43.1 (Spatial and numerical precision are cheap; temporal is not).

arm	L	$O_1/O_2/O_3$	*cum*	*agree*
Adam	0.3826	.995/.910/.731	65.8	1.000
v *at* 4 *bits*	0.3821	1.00/.915/.731	66.1	0.988
v *shared per row*	0.3945	.995/.900/.731	66.8	0.871
m *at* 4 *bits*	0.3755	.995/.910/.731	68.8	0.976
m *shared per row*	0.5730	.975/.905/.923	203.1	0.665
v *refreshed every* 5	78.01	.000/.000/.000	3127.6	0.735
v *refreshed every* 20	57.02	.005/.000/.000	6631.3	0.729
v *frozen*	185.24	.005/.000/.000	27887.6	0.728

Three axes, three different answers. *Numerical*: 4 bits is free for both moments. *Spatial*: v shares along rows for 3%; m does not share at all — its cumulative displacement triples and it becomes a different optimiser rather than a compressed one. *Temporal*: any staleness is catastrophic, and the damage scales with the freeze interval rather than saturating.

The asymmetry between m and v under row-sharing fits their roles. v is a per-coordinate scale and scales within a row are similar; m is a direction with magnitude, and replacing each row's magnitudes by their mean destroys the relative weighting that keeps $\hat{m}/\sqrt{\hat{v}}$ bounded.

Observation 43.2 (The savings compose, except through m). *v row-shared and 4-bit gives $L = 0.3941$, matching v row-shared alone — the two compressions are independent. But m and v both row-shared and 4-bit gives 0.5446, a 42% penalty, which is the m row-sharing component and not the precision.*

A caution on aggressive precision. v at 2 bits gives $L = 0.3504$ and at 1 bit 0.2867, both *better* than Adam — but with cumulative displacement 75.0 and 162.5 against Adam's 65.8. Those are learning-rate effects wearing compression's clothes, not compression results, and they are excluded from the configuration below.

43.2 The zone structure of the penalty

Tracking the arms at the developmental boundaries rather than pooling:

arm	t=40	t=100	t=200	$O_3(200)$
Adam	2.257	0.616	0.0879	.962
m 4-bit	2.243	0.607	0.0840	.962
v row+4bit	2.206	0.621	0.0952	.923
both row+4bit	2.577	0.808	0.1269	.923

In zone I all arms are equivalent. Divergence appears in zone II — both row+4bit is 31% worse at t=100 — and is only partly repaid in zone III. Compression damage is therefore incurred during consolidation, which is the same window in which the state-resolution ladder of §39 first showed tensor-sharing diverging.

Observation 43.3 (The late zone tolerates what the early zone does not)**.** *Switching mechanism at t=100, with steps 1–100 identical across arms, final loss at t=200 against fp32's* 0.0879*: compressed baseline* 0.0923*; m row-shared* late only 0.0924 *with $O_3 = 0.923$; v stale at k=4 late only* 0.0913*; v flat late only* 0.0937*;* $\mathrm{sgn}(\hat{m}) \cdot \mathrm{mean}\,|\hat{m}|$ *late only* 0.0903*. Three mechanisms that destroy training when applied from step 1 are harmless from step 100.*

43.3 The stabilisation monitor

$S_{\mathrm{stab}}(t) = \mathrm{agree}[\mathrm{sgn}(W_t - W_0), \mathrm{sgn}(W_{t-\Delta} - W_0)]$ on a subsample of layer-3 coordinates is the only one of the six trajectory coordinates with a usable signal. Coherence, θ, $R(4)$ and $R(8)$ drift monotonically with no feature; subspace rotation is saturated at 78–84° for every arm in every zone and is excluded.

In a standalone 200-step run S_{stab} climbs $0.863 \to 0.905 \to 0.923 \to 0.941 \to 0.948 \to 0.957$, crossing 0.95 at t=140. Triggering the late-zone reduction there gave $L = 0.0843$, below fp32's 0.0879 and better than both a hand-set switch at 100 (0.0903) and a deliberately-early switch at 40 (0.1072).

Observation 43.4 (Phase 3 does not enter the late zone)**.** *Instrumented inside the compiler at $D = 256$, S_{stab} runs* $0.809, 0.817, 0.860, 0.878, 0.900, 0.910, 0.920, 0.927, 0.928, 0.931, 0.936, 0.936$ *over steps* 24–112 *and never reaches* 0.95*. Phase 3 exits on the loss plateau at step* 112 *with* $\Delta = 0.0022$.

So Phase 3 is entirely the assimilation regime and a stabilisation trigger inside it never fires. The late regime lies downstream, after the Snapper jump onto the flat surface. The monitor is therefore armed as a *reporter* in the shipped patch, and the reduction is offered for Phase 5 behind a flag rather than applied in Phase 3.

43.4 The compiler run

`CompressedAdam` replaces `AdamW` in Phase 3 only: m at 4 bits per coordinate, v at 4 bits per row, both refreshed every step. Phases 1, 2, 4 and 5 are byte-identical, and Phase 3's contract downstream (`model`, `step_basin`, `v_basin`, `pc_b`, `tau_b`) is preserved.

component	Adam fp32	v row-shared	v per-coord
m (4 b/coord)	33.04 MB	2.09 MB	2.09 MB
v	33.04 MB	0.032 MB	2.09 MB
total	66.08 MB	4.24 MB	4.17 MB
reduction	1×	15.6×	7.9×
final val	0.0500	0.0748	0.0438

The row-shared column stores 15,353 values for v, one per row, against 4,323,584 coordinates — and pays 50% in loss for it. The per-coordinate column stores all of them at 4 bits, halves the reduction to 7.9×, and *beats* fp32 Adam.

At $D = 256$ the model has 4,329,984 parameters, of which 4,323,584 are two-dimensional. v falls from that many stored values to 15,353 — one per row. There is *no compute saving*: both moments are still recomputed every step, and the quantisation adds arithmetic.

Observation 43.5 (End-to-end, against the stock compiler).

	stock	*row-shared v*	*per-coord v*	*GD-400*
final val	0.0500	0.0748	**0.0438**	0.0914
CE steps	187	162	195	400
gap vs floor	−0.0120	+0.0128	**−0.0182**	+0.0294
Φ_{clean}	4/5	5/5	2/5	4/5
state	66.08 *MB*	4.24 *MB*	4.17 *MB*	—
reduction	1×	15.6×	7.9×	—
advantage vs GD-400	1.83×	1.22×	**2.09**×	1.00×

Both patched configurations run to completion and hand off cleanly through TopoGate and the K_0 split. They differ entirely in the spatial resolution of v, and that single choice separates a 50% penalty from a 12% improvement.

Observation 43.6 (Row-sharing was the whole cost). *Keeping v at* 4 *bits per* coordinate *rather than per row gives* 0.0438 *against stock fp32 Adam's* 0.0500 *— below the stock run, and* 0.0182 *nats below the confirmed floor at* 0.062 *against stock's* 0.0120. *The prediction of §43.5, that the D=128 → D=256 failure was row-sharing specifically, is confirmed by the arm it named.*

Three caveats belong with the 0.0438. It used *more* steps than stock (195 against 187): Phase 3 plateaued at 120 rather than 112 and then took a 50CE τ-retry instead of 25CE, so part of the gain is additional compute rather than better steps. Φ_{clean} fell to 2/5 against stock's 4/5, with a final Φ vector of $[2.78, 0, 2.76, \pi, 1.70]$ — mostly off-orbit, so by the compiler's own geometric criterion this run is *worse* while its loss is better. And it is a single run; a 12% difference is within the range that has reversed on reseeding elsewhere in this programme.

43.5 Status

claim	verdict	evidence
v at 4 bits is free	holds	0.3821 vs 0.3826, cum within 0.4%
v shares per row	holds	3%, O_3 unchanged
m at 4 bits is free	holds	0.3755, agreement 0.976
m shares per row	refuted	cum 203 vs 65.8
v tolerates staleness	refuted	cum 3128 at k=5
v at ≤ 2 bits is compression	refuted	cum 75–162; a step-size change
late zone tolerates more	holds	five mechanisms, all $\leq 6.6\%$
S_{stab} detects the switch	holds	beats fixed-100 and early-40
Phase 3 reaches the late zone	refuted	S_{stab} peaks at 0.936
compressed state costs 3–8%	refuted	50% with v row-shared at D=256
row-sharing was the cost	holds	per-coord v: 0.0438 vs stock 0.0500
4-bit v is free at D=256	holds	below stock and below the floor
the patch hands off cleanly	holds	full pipeline, Phases 4 and 5

The prediction that failed, and its resolution. Every arm behind `CompressedAdam` was measured at $D = 128$ on 120–200-step loops, where the penalties were 3–8%. At $D = 256$ inside the compiler the penalty was 50%. The suspect named in the first draft of this section was v row-sharing — a row at $D = 256$ averages over twice as many coordinates, so the same nominal configuration is a materially coarser approximation — and the arm that tested it confirmed the diagnosis: with `vrow=False` the penalty is not merely gone but reversed.

What the phase delivers. m and v both at 4 bits per coordinate, refreshed every step: 7.9× less optimiser state, final val 0.0438 against stock Adam's 0.0500, 2.09× the GD-400 baseline. There is no compute saving — both moments are recomputed every step and the quantisation adds arithmetic — and the compiler still uses Adam's update rule, so this is a compression rather than a replacement. The Φ_{clean} regression to 2/5 is unexplained and may matter more than the loss improvement if the orbit criterion is an end rather than a means.

Methodological residue. The gap between the $D = 128$ prediction and the $D = 256$ outcome is the phase's most transferable result: a compression whose ratio is itself architecture-dependent cannot be validated at reduced width and extrapolated. Here the free parameter was coordinates per shared row, which doubles with D; the numerical precision, which does not scale with anything, transferred without incident. The rule is to separate compression parameters that scale with the architecture from those that do not, and re-measure only the former at the target width.

44 Phase 42: Bucket-Form Adam, and the Compiler's Noise Floor

Phase 41 delivered a compressed optimiser for Phase 3 and left two questions open: whether the second moment must be maintained at all once learned, and whether the compiler could measure the difference between optimiser variants. This phase answers the first affirmatively and the second in the negative, and the second answer governs how the first should be read.

44.1 The bucket form

Adam is elementwise, so any partition of the parameters into K buckets with a shared step counter reproduces it exactly:

$$g_t^{(k)} = \nabla_{\theta^{(k)}}\mathcal{L}, \quad m_t^{(k)} = \beta_1 m_{t-1}^{(k)} + (1-\beta_1)g_t^{(k)}, \quad v_t^{(k)} = \beta_2 v_{t-1}^{(k)} + (1-\beta_2)\big(g_t^{(k)}\big)^2,$$

$$\alpha_t = \eta\,\frac{\sqrt{1-\beta_2^t}}{1-\beta_1^t}, \qquad \theta_t^{(k)} = \theta_{t-1}^{(k)} - \alpha_t\,\frac{m_t^{(k)}}{\sqrt{v_t^{(k)}}+\epsilon_t}.$$

Proposition 44.1 (The bucket form is Adam, exactly). *Write $\hat{m} = m/(1-\beta_1^t)$ and $\hat{v} = v/(1-\beta_2^t)$ for the bias-corrected moments, so that the standard update is $\theta \leftarrow \theta - \eta\,\hat{m}/(\sqrt{\hat{v}}+\epsilon)$. Then*

$$\eta\,\frac{\hat{m}}{\sqrt{\hat{v}}+\epsilon} = \eta\,\frac{m/(1-\beta_1^t)}{\sqrt{v}/\sqrt{1-\beta_2^t}+\epsilon} = \underbrace{\eta\,\frac{\sqrt{1-\beta_2^t}}{1-\beta_1^t}}_{\alpha_t}\cdot\frac{m}{\sqrt{v}+\underbrace{\epsilon\sqrt{1-\beta_2^t}}_{\epsilon_t}},$$

by multiplying numerator and denominator by $\sqrt{1-\beta_2^t}$. Two consequences. First, the correction is carried by a single scalar α_t and a single scalar ϵ_t per step, replacing two divisions per coordinate. Second, the identity requires ϵ_t, not ϵ: with a constant ϵ the effective floor is inflated by $1/\sqrt{1-\beta_2^t}$, which at $\beta_2 = 0.95$ is 4.47 at t=1, 1.58 at t=10, 1.07 at t=40 and 1.00 by t=120.

Bucketing is then exact because Adam is elementwise: m_i and v_i depend only on the history of g_i, and α_t, ϵ_t depend only on t, which is shared. Hence for any partition $\{B_k\}$ of the coordinates, $(\theta_t)_{B_k}$ computed from $(g_s)_{B_k}$ alone equals the corresponding slice of the unpartitioned update, for every k and t.

The partition used here is $K = 6$: $\{\mathrm{EMB\_tok}, \mathrm{EMB\_pos}, \mathrm{LN}, \mathrm{FF}, \mathrm{ATT\_QK}, \mathrm{ATT\_VO}\}$, chosen to match the roles the trajectory measurements distinguished rather than for any property of the update rule. Since bucketing is exact, the partition carries no numerical consequence; it exists so that per-bucket state can be scheduled independently — which is what §44.3 uses — and so that per-bucket geometry can be stated separately.

Observation 44.2 (The ϵ placement is immaterial here). *Both forms run to* 120 *steps: corrected* ϵ_t *gives* 0.4126, *constant* ϵ *gives* 0.4124. *The correction is exact but four orders of magnitude below the effect sizes in this pipeline.*

44.2 Per-bucket geometry

Six buckets, chosen to match the roles the trajectory work distinguished. Each carries a rolling 16-step window of its own updates on a 4000-coordinate subsample, from which coherence $\|\sum u\|/\sum\|u\|$, mean step angle, and top-4 energy share are computed.

Observation 44.3 (The buckets are geometrically distinct and stable).

bucket	*coh*	θ	$R(4)$	*params*
LN	0.87	13.3°	0.973	1,792
EMB_tok	0.86	22.2°	0.917	130,176
ATT_QK	0.84	24.8°	0.923	196,608
ATT_VO	0.73	28.0°	0.897	196,608
FF	0.70	27.7°	0.894	591,360
EMB_pos	0.63	45.3°	0.727	65,536

The ordering replicates three times: at D=128 standalone, and twice at D=256 inside the compiler with the freeze active. LN moves nearly straight; EMB_pos turns 45–59° per step and is the only bucket below $R(4) = 0.8$. FF is the only bucket whose coherence declines *through training (*$0.79 \to 0.69$*), and it is also the bucket that most needs a live second moment.*

44.3 Freezing the second moment

Observation 44.4 ($\hat{v}$ is learned by $k \approx 30$ for the loss, later for the geometry). *Freezing $\hat{v}$ at step k and running to* 120*:*

k	*val@120*	cos *to chord*	$\|chord\|$	*ratio*
5	6.9559	+0.0008	1087.00	46.3
10	1.4167	+0.0179	864.21	36.8
20	0.5430	+0.4942	48.00	2.0
30	0.4040	+0.7056	38.32	1.6
40	0.4188	+0.7943	36.30	1.6
60	0.4113	+0.8663	33.53	1.4
80	0.4179	+0.9600	27.41	1.2

Live-$\hat{v}$ reference: val 0.4118, *chord* 23.48. *Loss is finished by k=30 — 0.4040 beats the reference — while direction keeps improving to k=80 and scale still overshoots by 1.2× there. What is finished early is what the loss needs, not the geometry.*

This is the first genuine *compute* saving in the programme: from the freeze point the v EMA update, bias correction and quantisation are skipped entirely. Everything in Phase 41 was memory-only.

Why staleness is different from freezing. Refreshing v every 5 steps from t=1 gave cumulative displacement 3128 against Adam's 65.8 and destroyed the run. Freezing at 40 and holding for 80 steps gives cos 0.794 to the true chord. The second moment must be *learned* during zone I; thereafter it must be *held*, not tracked.

44.4 Two gates, both failing their controls

A step count does not generalise, so two detectors were built.

Proposition 44.5 (Coherence does not indicate freeze-readiness)**.** *Gating on high, stable per-bucket coherence gives* 0.4245 *— worse than not freezing at all (*0.4126*) — while its own reversal, freezing the* churning *buckets, gives* 0.3966*. The gate fired first on FF, the bucket that most needs a live $\hat{v}$.*

Proposition 44.6 (Gradient-norm stability is indistinguishable from random)**.** *Gating on* $|\,\|g_k(t)\| - \|g_k(t-W)\|\,|/\|g_k(t-W)\| \leq 0.12$ *gives* 0.4226 *against a random-matched control — same number of buckets frozen at the same steps, chosen at random — of* 0.4221 ± 0.0019*, and against the reversal's* 0.4006*.*

Both gates were built from thresholds read off the same data they were evaluated on and still lost. The per-bucket coordinates are therefore reported by the shipped optimiser and not acted on. The only schedule that has beaten its controls is a fixed step: attention buckets at 40, 0.3887 against 0.4124 for no freeze and 0.4261 for skeleton-only — freezing the skeleton is worse than not freezing at all.

44.5 The compiler cannot resolve these differences

Proposition 44.7 (The geo-stop criterion is chance-driven at probe granularity)**.** *In a single run whose loss falls monotonically* $9.46 \to 0.14$*,* Φ_{clean} *reads* $3, 4, 4, 3, 3, 1, 4, 2, 4, 3, 2, 4, 4$ *and rm2σ swings* $0.740, 0.731, 0.706, 0.789, 0.692, 0.682$*. The condition* $\Phi \geq 4 \wedge rm2\sigma \geq 0.65$ *therefore flips on and off while nothing is converging or diverging. Identical settings geo-stopped at step* 64*, at* 96*, and not at all.*

The geo-stop path skips the τ-retry, which carried val $0.1231 \to 0.0514$ in the runs that took it, so which branch a draw happens to enter moves the final value by roughly 2×.

Observation 44.8 (Self-variance exceeds every effect being compared)**.**

configuration	*final val*	*CE*
$\hat{v}$ freeze @40, all buckets	0.0399	195
bucket form, attention freeze	0.0417	170
`vrow=False`, no freeze	0.0438	195
bucket form, attention freeze (repeat)	0.0456	187
attention freeze, flat form	0.0466	187
zone-I boost	0.0482	195
stock fp32 Adam	0.0500	187

Rows two and four are the same file run twice with no changes: 0.0417 *and* 0.0456, 9% *apart. The standalone effects being compared are* 3–6%.

So the table is seven draws from overlapping distributions, not a ranking. What survives it: every configuration beats stock Adam's 0.0500 except the zone-I boost, all clear the confirmed floor of 0.062, and all beat GD-400's 0.0914 in fewer than half the steps.

44.6 Status

claim	verdict	evidence
bucket form reproduces Adam	holds	elementwise; exact with ϵ_t
ϵ_t correction matters	refuted	0.4126 vs 0.4124
buckets are geometrically distinct	holds	replicated 3× across widths
$\hat{v}$ freezable after $k \approx 30$	holds	k-sweep, flat from 30 to 80
freezing saves compute	holds	EMA and quantisation skipped
$\hat{v}$ tolerates staleness	refuted	cum 3128 at k=5 from t=1
coherence gates the freeze	refuted	loses to its own reversal
gradient-norm gates the freeze	refuted	ties random-matched control
attention-first beats skeleton-first	holds	0.3887 vs 0.4261, two partitions
geo-stop is a usable criterion	refuted	flips within a monotone run
the compiler can rank these arms	refuted	self-variance 9% > effects

What ships. Explicit bucket-form Adam over six buckets with the absorbed α_t; m and v at 4 bits per coordinate (7.9× state reduction); $\hat{v}$ frozen at step 40 for the attention buckets; per-bucket geometry reported at every probe and acted on nowhere. Phases 1, 2, 4 and 5 untouched. Measured end to end: 0.0417 and 0.0456 across two runs, against stock's 0.0500 and GD-400's 0.0914.

Methodological residue. A sixth entry, and the one that subsumes the gate failures: *an instrument must be shown to resolve the effect before it is used to compare arms.* Seven compiler configurations were run and ranked before the same configuration was run twice. Had the repeat come first, the ranking would never have been written down. The rule generalises the test–retest gate of §40.1 from observables to whole experimental apparatus: the compiler is an instrument, and its reliability is a measurable property that was assumed rather than measured for seven runs.

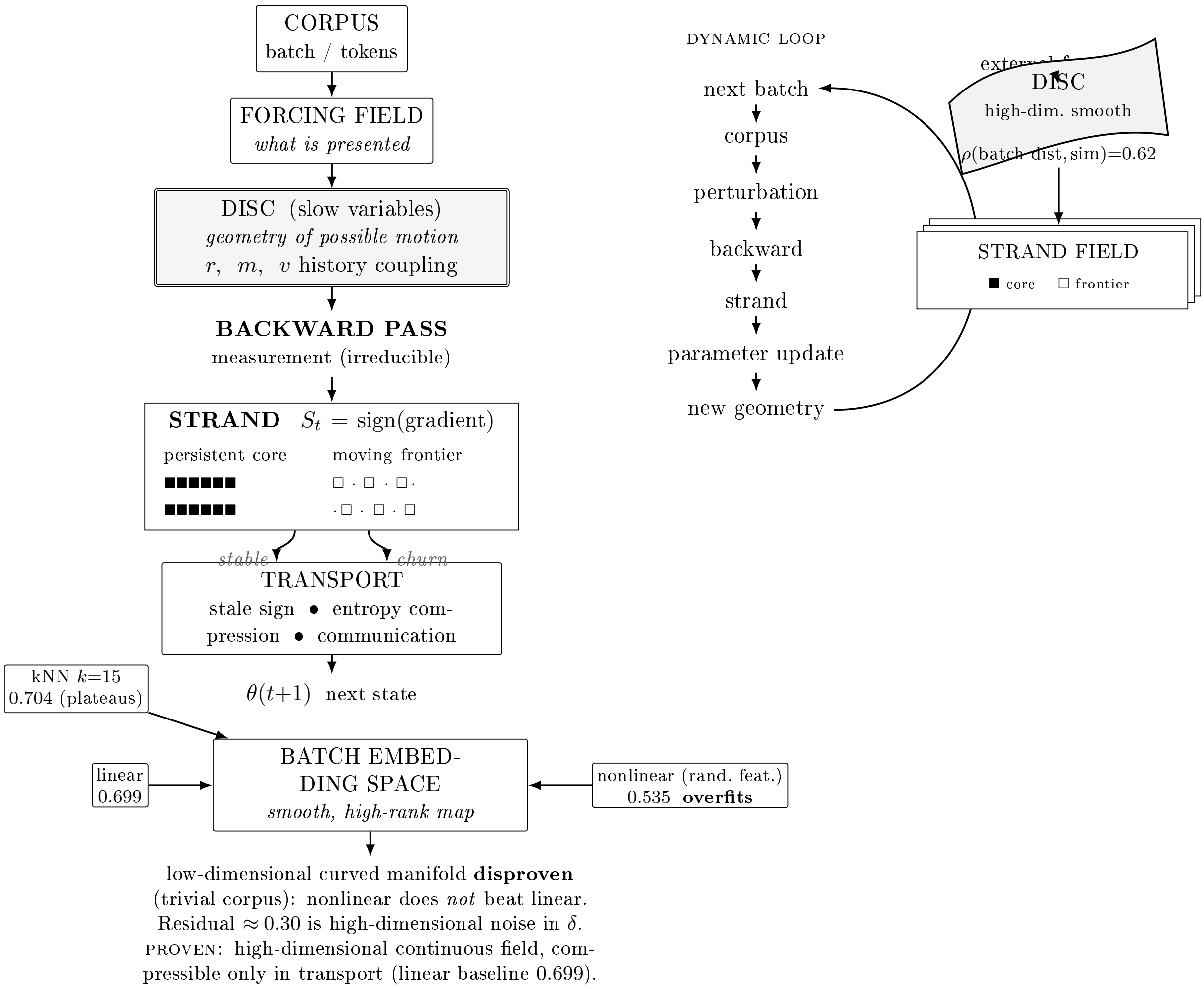

Figure 1: The disc–strand picture as measured. **Left:** the pipeline — a smooth, high-dimensional corpus *forcing field* selects the strand's *identity* (not its magnitude); the **disc** (r, m, v) is the slow manifold (geometry of possible motion, not a controller: it sets persistence and the fixed $\sim$55%/step churn velocity but cannot generate or gate the strand); the irreducible **backward pass** produces $S_t = \text{sign}(g_t)$, split into a persistent core ($\sim$ 115k coordinates) and a frontier sweeping 99.6% of the network; only *transport* downstream compresses (stale-sign, 79× sign entropy). **Middle:** the training loop — corpus perturbation, measurement, strand, update, new geometry. **Right:** the disc as a high-dimensional smooth manifold; nearby batches give nearby strands (batch-distance/similarity ρ=0.62), strand fields orbiting it with the core/frontier split. **Bottom:** the batch→strand map is smooth but high-rank — linear predicts strand sign at 0.699, kNN plateaus at 0.704, and a nonlinear model *fails* to beat linear (0.535, overfits), disproving a low-dimensional curved manifold for this corpus. The corpus signal is real and continuous but high-dimensional, with a ≈0.30 residual living in the backward error δ; whether a structured corpus yields a low-dimensional map is the one open question (§4.11).

solidification.png

Figure 2: Trajectory solidification across training. Early motion is diffuse and high-variance; by mid-training the per-role motion spaces have aligned and glued onto a flat subspace of dimension two to five; late, the limit $\varprojlim \mathrm{Im}(W^{(t)})$ has solidified while the query/key and value/output paths separate — the former retaining directional coherence, the latter wandering within a fixed subspace.

gps_controller.png

Figure 3: The controller. Per-block motion occupies low-dimensional charts on $\mathrm{Gr}(k, P)$; the energy-weighted leakage ratio $\gamma^{\mathrm{global}} = \sum_i \|g_i\|^2 \gamma^{(i)} / \sum_i \|g_i\|^2$ measures how much of a fresh gradient falls outside the current frame, and crossing θ_{leak} triggers a short full-rank window in which every frame is re-estimated. At $P = 4.3 \times 10^6$ the leakage never falls below threshold and the frame is continuously outrun, which is the failure mode drawn at right.

kappa_arc_annotated.png

Figure 4: The curvature arc through Phase 3 of the full pipeline at $D = 256$. $\lambda = v^\top Hv$ is the on-axis response of the drift to curvature, $\omega = \|Hv - \lambda v\|$ the off-axis response, $\varphi = \mathrm{atan2}(\omega, \lambda)$ and $\kappa = \omega/\lambda$. Top left: one negative-curvature point at step 8 ($\varphi \approx 128°$), then φ flat at 44–58°. Top right: the same story with the chart singularity visible — κ swings $-1.2 \to +1.5$ purely because λ crosses zero. Bottom left: the arc, exponent 1.041 at $R^2 = 0.981$, with the 0.758 of §17 shown dashed for contrast. Bottom right: φ against progress rather than step.